

HOME NURSING

Examination-oriented, fully covered, topper's notes

JKSSB JUNIOR PHARMACIST

Jammu & Kashmir Services Selection Board

24 CHAPTERS • 25 SECTIONS OF THE SYLLABUS • ONE BOOK

What this book covers — every topic of the Home Nursing section of the syllabus, in the same order as the syllabus:

- Introduction to Home Nursing • Nurse • Sick Room • Bed Making • Patient's Toilet • Observation of the Sick
- Infection • Surgical Techniques • Diet • Medicines • Special Conditions & Treatments • Bandaging
- Further Observations • Immunity & Infectious Diseases • Care of the Aged and the Long-term Patient
- Care of the Mentally Ill Patient • Special Drugs, their Control & Administration
- Preparation of the Patient for Operation and the After Care • Shock and Blood Transfusion
- Special Treatment • Nursing in Special Diseases • The Hospital Services
- Preparation for Special Treatment • Child Birth and Its Management + **Rapid Revision Appendix**

HOW TO USE THIS BOOK

- Read a chapter fully once, then revise only the coloured boxes and tables — they carry the examination-worthy facts.
- **Blue boxes** = definitions. **Yellow boxes** = high-yield/most-asked points. **Purple boxes** = memory aids.
- **Green boxes** = practical/clinical points. **Red boxes** = cautions and 'never do this'. **Grey boxes** = chapter recap.
- Every chapter ends with a '**Chapter at a Glance**' recap; the book ends with a **Rapid Revision** appendix (Appendix A).
- Tables are the highest-yield part of this subject — numbers, timings, temperatures, colour codes and schedules are asked directly.

Syllabus Coverage Map

Each topic printed in the official syllabus is mapped to the chapter of this book in which it is covered in full. Nothing in the syllabus has been left out.

Syllabus topic	Covered in
Introduction to Home Nursing	Chapter 1
Nurse	Chapter 2
Sick Room	Chapter 3
Bed Making	Chapter 4
Patient's Toilet	Chapter 5
Observation of the Sick	Chapter 6
Infection	Chapter 7
Surgical Techniques	Chapter 8
Diet	Chapter 9
Medicines	Chapter 10
Special Conditions & Treatments	Chapter 11
Bandaging	Chapter 12
Further Observations	Chapter 13
Immunity & Infectious Diseases	Chapter 14
Care of the Aged and Long-term Patient	Chapter 15
Care of the Mentally Ill / Healthy Patient	Chapter 16
Special Drugs, their Control & Administration	Chapter 17
Preparation of the Patient for Operation and the After Care	Chapter 18
Shock and Blood Transfusion	Chapter 19
Special Treatment	Chapter 20
Nursing in Special Diseases	Chapter 21
The Hospital Services	Chapter 22
Preparation for Special Treatment	Chapter 23
Child Birth and Its Management	Chapter 24
Rapid Revision — key facts, figures and one-liners of the whole subject	Appendix A

EXAM STRATEGY

A note on how the subject is examined. In the JKSSB Junior Pharmacist paper, Home Nursing questions are almost always **single-fact recall**: a normal value, a temperature, a position, a colour code, a time interval, a definition, the 'commonest' or 'best' answer, or the first step in a procedure. Every such fact in this book is printed in **bold navy** or inside a coloured box so that it can be picked out in a final revision.

Table of Contents

Chapter-wise contents with page numbers

Chapter 1 Introduction to Home Nursing	7
1.1 Nursing and Home Nursing — Meaning and Definitions	7
1.2 Aims and Objectives of Home Nursing	7
1.3 Need, Scope and Importance of Home Nursing	7
1.4 Advantages and Limitations of Home Nursing	8
1.5 Difference between Home Nursing and Hospital Nursing	8
1.6 Basic Principles of Nursing Care	8
1.7 Steps of the Nursing Process (ADPIE)	8
1.8 Maslow's Hierarchy of Human Needs (basis of care priority)	9
1.9 The Home Nursing Team and the Role of the Pharmacist	9
1.10 Historical Development of Nursing (frequently asked)	9
1.11 Common Nursing Terminology (must-know glossary)	10
1.12 Articles Required for Home Nursing (the home sick-room kit)	11
1.13 Ethical and Legal Aspects; Rights of the Patient	11
Chapter 2 The Nurse	12
2.1 Who is a Nurse?	12
2.2 Categories of Nursing Personnel in India	12
2.3 Essential Qualities of a Good Nurse	12
2.4 Duties and Responsibilities of the Nurse	13
2.5 Personal Hygiene, Grooming and Uniform of the Nurse	13
2.6 Nurse-Patient Relationship and Communication	13
2.7 Professional Ethics and Legal Responsibility	14
2.8 Records and Reports	15
2.9 The Nurse's Own Safety — Occupational Hazards	15
Chapter 3 The Sick Room	16
3.1 Meaning and Importance	16
3.2 Selection of the Sick Room — Ideal Requirements	16
3.3 Ventilation	16
3.4 Lighting	17
3.5 Temperature, Humidity and Comfort	17
3.6 Furniture and Equipment of the Sick Room	18
3.7 Cleanliness of the Sick Room	18
3.8 Disposal of Waste, Excreta and Soiled Linen at Home	19
3.9 Disinfection of the Sick Room	19
3.10 Control of Insects, Flies and Rodents	19
3.11 Safety in the Sick Room	20
3.12 Comfort and the Psychological Environment	20
3.13 Care of Equipment and Rubber Goods	20
Chapter 4 Bed Making	22
4.1 Definition, Purposes and Principles	22
4.2 Articles Required and the Parts of a Bed	22
4.3 Types of Beds and their Purposes	23
4.4 Procedure — Making a Simple / Closed Bed	23
4.5 Making an Occupied Bed / Changing Linen with the Patient in Bed	24
4.6 Bedside Appliances and their Uses	24
4.7 Positions of the Patient	25
4.8 Moving, Lifting and Transferring the Patient (Body Mechanics)	25
4.9 Pressure Sores (Bed Sores / Decubitus Ulcer)	26
4.10 Restraints (Protective Devices)	28
Chapter 5 Patient's Toilet (Personal Hygiene of the Patient)	29
5.1 Meaning, Importance and Principles	29
5.2 Bathing the Patient	29
5.3 Care of the Mouth (Oral Hygiene)	30
5.4 Care of the Hair	31
5.5 Care of the Eyes, Ears and Nose	31

5.6 Care of Hands, Feet and Nails.....	31
5.7 Back Care (Pressure Area Care).....	32
5.8 Meeting Elimination Needs.....	32
5.9 Care of Vomiting, Expectoration and Dressing the Patient.....	33
Chapter 6 Observation of the Sick.....	34
6.1 Meaning and Importance of Observation.....	34
6.2 General (Head-to-Foot) Observation.....	34
6.3 Vital Signs — Body Temperature.....	35
6.4 Vital Signs — Pulse.....	36
6.5 Vital Signs — Respiration.....	37
6.6 Vital Signs — Blood Pressure.....	38
6.7 The TPR Chart, Intake-Output Chart and Other Records.....	39
6.8 Level of Consciousness — Glasgow Coma Scale.....	39
6.9 Observation of Urine, Stool, Sputum and Vomit.....	40
6.10 Signs of Improvement, Deterioration and of Approaching Death.....	40
Chapter 7 Infection.....	42
7.1 Basic Terminology.....	42
7.2 The Chain of Infection.....	43
7.3 Modes of Transmission.....	43
7.4 Hospital-Acquired (Nosocomial) Infection.....	44
7.5 Hand Hygiene.....	44
7.6 Sterilisation.....	44
7.7 Disinfectants and Antiseptics.....	45
7.8 Standard (Universal) Precautions and Barrier Nursing.....	46
7.9 Bio-Medical Waste Management (BMW Rules, 2016 — India).....	46
7.10 Injection Safety and Post-Exposure Prophylaxis.....	47
Chapter 8 Surgical Techniques.....	49
8.1 Medical Asepsis and Surgical Asepsis.....	49
8.2 Surgical Hand Scrub, Gowning and Gloving.....	49
8.3 Wounds — Types, Healing and Complications.....	50
8.4 Dressing a Wound.....	51
8.5 Sutures, Staples and Drains.....	52
8.6 Common Surgical Instruments and their Use.....	52
8.7 Minor Surgical Procedures — Nursing Assistance.....	53
Chapter 9 Diet.....	54
9.1 Basic Concepts and Definitions.....	54
9.2 The Nutrients — Functions, Sources and Deficiency.....	54
9.3 Vitamins.....	55
9.4 Minerals.....	56
9.5 The Balanced Diet and the Food Groups.....	56
9.6 Therapeutic Diets — Modifications in Consistency.....	57
9.7 Therapeutic Diets in Disease.....	58
9.8 Serving Food and Feeding the Helpless Patient.....	59
9.9 Tube (Enteral) Feeding.....	59
9.10 Parenteral Nutrition and Fluids.....	60
9.11 Food Hygiene and Food Poisoning.....	60
Chapter 10 Medicines.....	61
10.1 Basic Terms.....	61
10.2 Routes of Administration.....	61
10.3 Dosage Forms.....	62
10.4 The 'Rights' of Drug Administration.....	63
10.5 Reading a Prescription — Abbreviations and Latin Terms.....	63
10.6 Dose Calculation.....	64
10.7 Administering Oral Medicines.....	65
10.8 Injections — Sites, Angles and Technique.....	65
10.9 Intravenous Infusion — Nursing Points.....	66
10.10 Storage, Stock and Disposal of Medicines.....	66
10.11 Adverse Drug Reactions, Interactions and Pharmacovigilance.....	67
Chapter 11 Special Conditions and Treatments.....	68

11.1 Application of Heat.....	68
11.2 Application of Cold.....	68
11.3 Inhalations.....	69
11.4 Mouth, Throat, Eye, Ear and Nose Treatments.....	70
11.5 Enema, Suppository and Flatus Tube.....	70
11.6 Urinary Catheterisation and Bladder Irrigation.....	71
11.7 Other Special Treatments.....	72
Chapter 12 Bandaging.....	73
12.1 Definition and Purposes.....	73
12.2 Types of Bandages and Materials.....	73
12.3 General Rules and Principles of Bandaging.....	74
12.4 Basic Bandage Turns.....	74
12.5 Slings.....	75
12.6 Splints, Strapping and Plaster of Paris.....	75
Chapter 13 Further Observations.....	77
13.1 Collection of Specimens — General Rules.....	77
13.2 Simple Bedside Tests.....	78
13.3 Normal Values Every Nurse Should Know.....	78
13.4 Fluid Balance, Dehydration and Oedema.....	79
13.5 Observation of Pain.....	79
13.6 Danger Signals — When to Send for the Doctor at Once.....	80
13.7 Recording and Reporting Observations.....	80
Chapter 14 Immunity and Infectious Diseases.....	81
14.1 Immunity — Definition and Types.....	81
14.2 Vaccines.....	81
14.3 The Cold Chain.....	82
14.4 National Immunization Schedule (India, UIP).....	82
14.5 Infectious Diseases — Comparative Profiles.....	83
14.6 Isolation, Quarantine, Notification and Disinfection.....	86
Chapter 15 Care of the Aged and the Long-Term Patient.....	88
15.1 Basic Concepts.....	88
15.2 Physiological Changes of Ageing (system-wise).....	88
15.3 Common Problems of the Aged — the 'Geriatric Giants'.....	89
15.4 Nursing Care of the Long-Term Bedridden Patient.....	90
15.5 Palliative and Terminal Care; Care of the Dying.....	90
15.6 Health Care Services and Schemes for the Elderly in India.....	91
Chapter 16 Care of the Mentally Ill and the Mentally Healthy Patient.....	92
16.1 Mental Health and Mental Hygiene.....	92
16.2 Causes and Classification of Mental Illness.....	92
16.3 Principles of Psychiatric Nursing and Therapeutic Communication.....	93
16.4 Safety of the Psychiatric Patient.....	94
16.5 Psychotropic Drugs.....	94
16.6 Other Treatments and Nursing Care.....	95
16.7 Mental Health Legislation and Programmes (India).....	96
Chapter 17 Special Drugs, Their Control and Administration.....	97
17.1 What Are 'Special' Drugs?.....	97
17.2 Laws Governing Drugs in India.....	97
17.3 Schedules of the Drugs and Cosmetics Rules (must know).....	97
17.4 Narcotic and Psychotropic Drugs — Control in the Ward and Pharmacy.....	99
17.5 High-Alert Medications.....	99
17.6 Handling Cytotoxic (Anticancer) Drugs.....	100
17.7 Storage and Administration of Other Special Preparations.....	100
17.8 Rational Drug Use, Antibiotic Stewardship and DOTS.....	101
17.9 The Pharmacist's and Nurse's Statutory Responsibilities — Summary.....	101
Chapter 18 Preparation of the Patient for Operation and the After Care.....	103
18.1 Classification of Surgery.....	103
18.2 Pre-operative Assessment and Investigations.....	103
18.3 Informed Consent.....	103
18.4 Pre-operative Preparation.....	104

18.5	The Operation Theatre and the Immediate Peri-operative Period.....	105
18.6	Post-operative (After) Care.....	105
18.7	Post-operative Complications.....	106
Chapter 19	Shock and Blood Transfusion.....	108
19.1	Shock — Definition and Types.....	108
19.2	Stages and Clinical Features of Shock.....	108
19.3	Management of Shock.....	109
19.4	Blood Groups and Compatibility.....	109
19.5	Blood Components, Storage and Indications.....	110
19.6	Procedure of Blood Transfusion — Nursing Responsibilities.....	111
19.7	Transfusion Reactions.....	111
Chapter 20	Special Treatment.....	113
20.1	Oxygen Therapy.....	113
20.2	Suction, Nebulisation and Chest Physiotherapy.....	113
20.3	Cardiopulmonary Resuscitation (CPR) and Basic Life Support.....	114
20.4	Tracheostomy and Airway Care.....	114
20.5	Care of Tubes, Drains and Stomas.....	115
20.6	Dialysis.....	115
20.7	Physiotherapy, Traction and Rehabilitation Modalities.....	116
20.8	Other Special Treatments.....	116
Chapter 21	Nursing in Special Diseases.....	118
21.1	Respiratory Diseases.....	118
21.2	Cardiovascular Diseases.....	118
21.3	Neurological Conditions.....	119
21.4	Endocrine, Renal and Gastro-intestinal Conditions.....	120
21.5	Surgical, Orthopaedic and Emergency Conditions.....	121
21.6	Paediatric Nursing Highlights.....	122
Chapter 22	The Hospital Services.....	124
22.1	The Hospital — Definition and Functions.....	124
22.2	Classification of Hospitals.....	124
22.3	Organisation and Departments of a Hospital.....	125
22.4	Hospital Pharmacy Services.....	125
22.5	Hospital Committees.....	126
22.6	Health Care Delivery System in India.....	126
22.7	National Health Programmes (quick list).....	127
22.8	Quality, Safety and Patient Rights.....	127
Chapter 23	Preparation of the Patient for Special Treatment and Investigations.....	128
23.1	General Principles of Preparing a Patient for Any Procedure.....	128
23.2	Blood, Urine and Other Laboratory Investigations.....	128
23.3	Radiological and Imaging Investigations.....	129
23.4	Endoscopic Procedures.....	129
23.5	Aspirations, Punctures and Biopsies.....	130
23.6	Cardiac, Neurological and Functional Tests.....	131
Chapter 24	Child Birth and Its Management.....	133
24.1	Basic Obstetric Terms.....	133
24.2	Diagnosis of Pregnancy and Calculation of the Due Date.....	133
24.3	Antenatal Care.....	134
24.4	Labour.....	134
24.5	Care of the Newborn.....	136
24.6	Postnatal (Puerperal) Care.....	136
24.7	Complications of Labour and the Puerperium.....	137
24.8	Family Planning (Contraception).....	138
Appendix A	Rapid Revision — The Last-Night Sheet.....	140
A.1	Vital Signs and Normal Values.....	140
A.2	Timings and Intervals.....	140
A.3	'Firsts', 'Commonests' and 'Bests'.....	140
A.4	Important Days and Landmarks.....	141
A.5	Colour Codes and Mnemonics.....	141

Introduction to Home Nursing

Meaning, aims, scope, principles, historical background and basic terminology

1.1 Nursing and Home Nursing — Meaning and Definitions

DEFINITION

Nursing (ICN) — Nursing encompasses autonomous and collaborative care of individuals of all ages, families, groups and communities, sick or well, in all settings; it includes promotion of health, prevention of illness and the care of ill, disabled and dying people.

Florence Nightingale's definition — "Nursing is the act of utilising the environment of the patient to assist him in his recovery."

Virginia Henderson's definition — "Nursing is assisting the individual, sick or well, in the performance of those activities contributing to health, its recovery or a peaceful death, that he would perform unaided if he had the necessary strength, will or knowledge."

Home Nursing — The care given to a sick, injured, convalescent, aged or disabled person *in his own home* by a member of the family or a trained/semi-trained attendant, under the guidance of a doctor or a qualified nurse.

- The word **nurse** comes from the Latin *nutrix* = 'to nourish / to nurture'; *nutricius* means 'nourishing'.
- Home nursing is also called **domiciliary nursing** or **domiciliary care**; when organised by an agency it is called *home health care*.
- Home nursing is the **oldest form of nursing** — historically all nursing was done at home by mothers and female relatives ('mother is the first nurse').
- In India, home nursing is taught as a part of **First Aid and Home Nursing** by the **Indian Red Cross Society** and **St. John Ambulance Association**.

1.2 Aims and Objectives of Home Nursing

1. To help the patient **regain health** in the shortest possible time with the least expense and discomfort.
2. To provide **comfort, safety and cleanliness** to the patient.
3. To prevent **complications** (bed sores, contractures, pneumonia, constipation, deformities) and **spread of infection** to other family members.
4. To carry out the **doctor's instructions** accurately — medicines, diet, rest, treatment.
5. To **observe and report** the patient's condition correctly.
6. To give **psychological support** — reduce fear, loneliness and anxiety of the patient and family.
7. To use **available household articles** economically by improvisation.
8. To promote **health education** and self-care so that the family becomes independent in caring.
9. To assist in **rehabilitation** and restore the patient to a useful place in family and society.
10. To relieve the **burden on hospitals** and to give dignified terminal (palliative) care at home.

1.3 Need, Scope and Importance of Home Nursing

Who needs home nursing?

- Patients with **chronic and long-term illness** — paralysis, arthritis, cancer, tuberculosis, diabetes, chronic kidney disease.
- **Convalescents** — persons recovering after acute illness, surgery or childbirth.
- **Aged and bedridden** persons; physically and mentally **disabled** persons.
- **Mothers and newborns** during the ante-natal and post-natal period.
- Patients with **minor ailments** (fever, cold, diarrhoea) not needing hospital admission.
- **Terminally ill/dying** patients who prefer to be cared for at home (palliative and hospice care).
- **Infectious disease** patients kept in home isolation (chickenpox, measles, mumps, COVID-19).

Importance

- Hospital beds are limited and costly — most illnesses in India are actually managed at home.
- The home environment gives **emotional security**; recovery is faster where the patient feels loved.

- Reduces **hospital acquired (nosocomial) infection** risk.
- Home nursing skill is a part of every citizen's **social responsibility** and of national health programmes (ASHA, ANM, MPW home visits).

1.4 Advantages and Limitations of Home Nursing

Table 1.1 Home nursing at a glance

Advantages	Limitations / Disadvantages
Familiar, homely, affectionate atmosphere; patient is with family	Trained personnel and equipment are not available
Much cheaper than hospital care	Emergency help may be delayed
Less risk of cross (hospital) infection	Lack of knowledge of the care-giver may lead to complications
Individual, undivided attention; diet as per patient's taste	Risk of spread of infection to other family members
Children and elderly do not feel separated (no separation anxiety)	Physical and mental strain, loss of wages for the family care-giver
Family learns health care and becomes self-reliant	Isolation, investigations, operations, oxygen, ICU care not possible at home
Rehabilitation is easier within the family and community	Improper diagnosis or self-medication may be dangerous

1.5 Difference between Home Nursing and Hospital Nursing

Table 1.2 Home nursing vs. hospital nursing

Point	Home Nursing	Hospital Nursing
Place	Patient's own home	Hospital ward / ICU
Care giver	Family member or attendant, guided by doctor/nurse	Qualified, registered nurse (24 h shift duty)
Equipment	Improvised household articles	Standard, sterile, modern equipment
Cost	Low	High
Supervision	Intermittent (doctor's visit / tele-advice)	Continuous medical supervision
Type of patient	Mild, chronic, convalescent, aged, terminal	Acute, serious, surgical, emergency
Records	Simple diary / TPR note	Elaborate legal case sheets and charts
Infection risk	Mainly to family members	Cross-infection between patients
Diet	Home-cooked, as per liking	Hospital diet as prescribed by dietician

1.6 Basic Principles of Nursing Care

- **Safety first** — protect the patient from injury, falls, burns, infection and wrong medicine.
- **Cleanliness / asepsis** — hand washing before and after every procedure is the single most important measure.
- **Comfort** — physical (position, clean bed, relief of pain) and mental (kind words, privacy).
- **Individualisation** — no two patients are alike — care is planned according to age, illness, habits and culture.
- **Economy** — of time, energy, money and material; improvise wherever possible.
- **Observation and reporting** — watch continuously, record accurately, report promptly.
- **Body mechanics** — correct posture and lifting technique protect the care-giver's back.
- **Privacy and dignity** — screen the patient, expose only the required part.
- **Self-care/rehabilitation** — encourage the patient to do whatever he can do himself.
- **Health education** — every contact is an opportunity to teach the family.

1.7 Steps of the Nursing Process (ADPIE)

Table 1.3 The five steps of the nursing process

Step	What is done
1. Assessment	Collect data — history, observation, vital signs, physical examination
2. Diagnosis (nursing diagnosis)	Identify the patient's problems / needs

Step	What is done
3. Planning	Set priorities and goals; write the nursing care plan
4. Implementation	Carry out the planned nursing care
5. Evaluation	Judge whether the goal is achieved; modify the plan

MEMORY AID

Remember **A-D-P-I-E** : Assessment → Diagnosis → Planning → Implementation → Evaluation.

1.8 Maslow's Hierarchy of Human Needs (basis of care priority)

Table 1.4 Maslow's need hierarchy

Level (lowest first)	Needs	Nursing example
1. Physiological	Air, water, food, elimination, rest, temperature, sex, activity	Oxygen, feeding, bedpan, sleep
2. Safety and security	Protection from injury and infection	Side rails, asepsis, correct drug
3. Love and belonging	Affection, acceptance, company	Allow visitors, talk to patient
4. Self esteem	Respect, recognition, independence	Privacy, praise, self-care
5. Self actualisation	Achieving one's full potential	Rehabilitation, vocation

HIGH-YIELD / MOST ASKED

Physiological needs are met first — of these, the need for **oxygen (air)** is the most urgent.

Order of urgency in an emergency: **Airway → Breathing → Circulation (ABC)**.

1.9 The Home Nursing Team and the Role of the Pharmacist

- **Team members** — patient and family, doctor, nurse/ANM, pharmacist, physiotherapist, dietician, social worker, ASHA/health worker, laboratory technician.
- The **patient is the centre** of the team; the family care-giver is the key worker in home nursing.

Role of the pharmacist in home nursing / home health care

1. **Dispensing** medicines accurately with clear labelling and instructions.
2. **Patient counselling** — dose, timing, relation to food, duration, expected side effects, what to avoid.
3. Advising on **storage** (cool dry place, refrigeration of insulin/vaccines, keeping away from children).
4. Detecting and preventing **drug interactions, duplication and polypharmacy** in chronic and elderly patients.
5. Supplying and teaching use of **surgical dressings, bandages, sick-room appliances, glucometers, BP instruments, nebulisers, inhalers, insulin pens**.
6. **Reporting adverse drug reactions** (pharmacovigilance) and discouraging self-medication and antibiotic misuse.
7. **Health education** — ORS preparation, immunisation, family planning, first aid, personal hygiene, safe disposal of expired drugs and sharps.
8. Maintaining statutory **records** for Schedule H/H1/X and narcotic drugs.

1.10 Historical Development of Nursing (frequently asked)

Table 1.5 Landmarks in the history of nursing

Person / Body	Contribution	Year
Florence Nightingale	'The Lady with the Lamp'; founder of modern nursing; worked in the Crimean War at Scutari Barrack Hospital; opened the first nursing school at St. Thomas' Hospital, London (1860); wrote <i>Notes on Nursing</i> ; environmental theory; reduced death rate from 42% to about 2%	1820-1910 (Crimean War 1854-56)
International Nurses Day	Celebrated on 12 May — Florence Nightingale's birthday	12 May
Nightingale Pledge / Lamp	Lamp = symbol of nursing; the pledge is taken by student nurses	-

Person / Body	Contribution	Year
Henri Dunant	Founder of the Red Cross after the Battle of Solferino; wrote <i>A Memory of Solferino</i> ; first Nobel Peace Prize (1901); World Red Cross Day 8 May (his birthday)	1863 (ICRC)
Indian Red Cross Society	Established by an Act of Parliament; teaches First Aid and Home Nursing	1920
St. John Ambulance Association (India)	First aid and home nursing training	1920
International Council of Nurses (ICN)	First international organisation of nurses; HQ Geneva	1899
Trained Nurses' Association of India (TNAI)	National professional body of nurses in India	1908
Indian Nursing Council (INC)	Statutory body constituted by the Indian Nursing Council Act 1947 to maintain uniform standards of nursing education; HQ New Delhi	Act 1947, constituted 1949
Dai / traditional birth attendant	Conducted deliveries at home in old India	-
Sister Dora, Mary Seacole, Clara Barton	Clara Barton founded the American Red Cross (1881)	19th century
WHO	World Health Organization established; World Health Day 7 April	7 April 1948

1.11 Common Nursing Terminology (must-know glossary)

Table 1.6 Basic terminology

Term	Meaning
Patient / client	A person receiving health care
Acute illness	Sudden onset, short duration, severe
Chronic illness	Slow onset, long duration (more than 3 months)
Convalescence	Period of recovery between the end of disease and return to full health
Prognosis	Prediction of the probable outcome of a disease
Diagnosis	Identification of the disease
Sign	Objective evidence seen by others (rash, swelling, fever)
Symptom	Subjective complaint felt by the patient (pain, nausea)
Syndrome	A group of signs and symptoms occurring together
Etiology	Cause of disease
Pathogenesis	Mechanism by which disease develops
Remission / Relapse	Temporary disappearance of symptoms / return of symptoms
Morbidity / Mortality	Rate of sickness / rate of death in a population
Epidemic / Endemic / Pandemic / Sporadic	Sudden excess in an area / constantly present in an area / worldwide spread / occasional isolated cases
Prophylaxis	Measure taken to prevent disease
Palliative care	Relief of symptoms without curing the disease
Rehabilitation	Restoring a person to maximum usefulness
Ambulatory patient	A patient who is able to walk about
Bedridden / Helpless patient	Confined to bed / unable to do anything for himself
Terminal illness	Illness from which recovery is not expected
Prone / Supine	Lying on abdomen (face down) / lying on back (face up)
Vital signs	Temperature, pulse, respiration, blood pressure ($\pm$ pain, SpO ₂)
NPO / Nil by mouth	Nothing to be given by mouth
Stat / SOS / PRN	At once / if necessary / as and when required
Aseptic technique	Practices that keep an area free from micro-organisms

1.12 Articles Required for Home Nursing (the home sick-room kit)

Table 1.7 Standard home nursing articles

Group	Articles
Measuring / observation	Clinical thermometer, watch with second hand, BP apparatus with stethoscope, weighing machine, measuring glass, torch, glucometer, pulse oximeter
Linen	Bed sheets, draw sheet, mackintosh/rubber sheet, pillow covers, blankets, towels, patient gowns, cotton pads
Elimination	Bedpan, urinal, kidney tray, sputum cup with lid, commode chair, bucket with lid, catheter and urine bag (if needed)
Dressing / first aid	Sterile gauze, cotton wool, roller and triangular bandages, adhesive plaster, scissors, forceps, antiseptic lotion (povidone iodine, spirit), band-aid, sterile gloves
Medicine administration	Medicine glass, spoon (5 mL), dropper, oral syringe, syringes and needles, nebuliser, inhaler with spacer
Comfort appliances	Backrest, air/rubber ring, bed cradle, hot water bag, ice bag, air cushion, footboard, sandbags, walker, wheelchair
Hygiene	Soap, basin, mug, mouth wash, nail cutter, comb, hand sanitiser, disinfectant (bleaching powder, phenol), waste bags

CLINICAL / PRACTICAL POINT

Improvisation is the heart of home nursing — using ordinary household articles in place of hospital equipment:

- Bed cradle ← wooden stool or cardboard box placed under the top sheet
- Back rest ← inverted chair, wooden plank or firm pillows
- Draw mackintosh ← thick polythene sheet or an old plastic table cover
- Air ring ← a ring made of cloth/cotton or a rubber tube
- Sand bag ← cloth bag filled with dry sand; Footboard ← wooden plank or a suitcase
- Kidney tray ← steel bowl; Sputum cup ← covered tin with newspaper lining
- Ice bag ← polythene bag with crushed ice in a towel; Hot water bag ← bottle wrapped in cloth
- Steam inhaler ← kettle with a paper funnel; Feeding cup ← spouted cup / straw

1.13 Ethical and Legal Aspects; Rights of the Patient

- **Confidentiality** — never disclose the patient's illness or secrets to outsiders.
- **Informed consent** — necessary before any procedure, operation or investigation; a written consent is taken from the patient (adult, conscious, sound mind) or from the guardian for minors (<18 years) and the unconscious/mentally ill.
- **Negligence** — failure to give the standard of care expected; **malpractice** = professional negligence. *Res ipsa loquitur* = 'the thing speaks for itself'.
- **Assault and battery** — threatening / touching or treating a patient without consent.
- **Patient's rights** — right to information, right to privacy and dignity, right to a second opinion, right to refuse treatment, right to records, right to safe care without discrimination.
- **Documentation rule** — 'If it is not recorded, it is not done'; records are legal documents. Never erase or use white ink; strike out with a single line, write 'error', and initial it.

CHAPTER AT A GLANCE

- Nursing = *nutrix* (to nourish). Home nursing = care of the sick in his own home.
- Florence Nightingale (1820-1910): founder of modern nursing, Crimean War, St. Thomas' Hospital school 1860, *Notes on Nursing*; Nurses' Day 12 May.
- Red Cross — Henri Dunant, 1863; Indian Red Cross Society 1920; World Red Cross Day 8 May.
- ICN 1899 (Geneva) • TNAI 1908 • INC Act 1947, constituted 1949 • WHO 7 April 1948.
- Nursing process = ADPIE. Maslow: physiological needs first; oxygen is the most urgent.
- Principles: safety, asepsis (hand washing = most important), comfort, privacy, economy, observation, health education.
- Home nursing advantages: cheap, homely, less cross-infection; limitation: no trained staff or emergency equipment.

The Nurse

Categories, qualifications, qualities, duties, ethics, communication, records and self-care

2.1 Who is a Nurse?

DEFINITION

Nurse — A person who is educated and trained in the art and science of nursing, is **registered with a State Nursing Council/Registration Council**, and is licensed to practise nursing.

Home nurse / attendant — a family member or semi-trained person who gives nursing care at home under guidance.

2.2 Categories of Nursing Personnel in India

Table 2.1 Nursing personnel and their preparation

Category	Course and duration	Chief function
ANM / MPW (Female) — <i>Auxiliary Nurse Midwife</i>	ANM: 2 years after 10+2	Village-level worker at the sub-centre; MCH care, immunisation, home visits, deliveries
GNM — <i>General Nursing and Midwifery</i>	3 years (+ 6 months internship) after 10+2	Bedside nursing in hospitals and community; registered nurse and midwife (RN, RM)
B.Sc. Nursing	4 years after 10+2 (PCB)	Staff nurse, ward in-charge, community health nurse, teacher
Post Basic B.Sc. Nursing	2 years for registered GNM nurses	Supervisory and teaching posts
M.Sc. Nursing / Ph.D.	2 years after B.Sc. / research	Nurse specialist, tutor, administrator, researcher
Nurse Practitioner in Midwifery / NPCC	Post-graduate residency	Advanced independent practice
Health Worker (Male) / MPW(M)	1 year	Communicable disease control, sanitation
Nursing aide / Ward attendant / Home attendant	Short certificate course	Assists the nurse — moving patients, cleanliness, feeding (works under supervision)
Dai (traditional birth attendant)	Trained by the health department	Assists home deliveries, refers complications

Regulatory and professional bodies

- **Indian Nursing Council (INC)** — statutory body, Act of 1947 (constituted 1949), New Delhi — prescribes and maintains uniform standards of nursing education, recognises qualifications.
- **State Nursing Council / Registration Council** — registers nurses and midwives in the State; issues the licence to practise; renews registration.
- **TNAI (1908)** — professional welfare association of nurses in India; publishes *The Nursing Journal of India*; member of ICN.
- **ICN (1899, Geneva)** — world federation of national nurses' associations; gives the *ICN Code of Ethics for Nurses*.
- **Nurses' Registration and Tracking System / NCA (2023)** — the National Nursing and Midwifery Commission Act, 2023 replaces the INC Act to regulate nursing education and practice.

2.3 Essential Qualities of a Good Nurse

Table 2.2 The five/six groups of nursing qualities

Group	Qualities expected
Physical	Good health, physical fitness and stamina, freedom from infection, clean and neat appearance, good posture, normal vision and hearing, pleasant voice, manual dexterity, immunised (Hepatitis B, tetanus, COVID, influenza)
Mental / intellectual	Intelligence, sound knowledge and skill, keen powers of observation, alertness, memory, resourcefulness, common sense, quick and correct decision-making, willingness to learn
Moral / ethical	Honesty, truthfulness, loyalty, integrity, sense of duty, discipline, punctuality, confidentiality, no discrimination, respect for life
Emotional	Emotional stability, patience, self-control, tolerance, cheerfulness, calmness in emergency, empathy (not mere sympathy), tactfulness

Group	Qualities expected
Social	Courtesy, good manners, co-operation, good communication, ability to work in a team, adaptability, sense of humour
Professional	Kindness and sympathy, gentleness, dependability, accuracy, neatness, initiative, accountability, keeping professional secrets, continuing education, leadership

MEMORY AID

The nurse's '5 C's': *Care, Compassion, Competence, Communication, Conscience* (some add Commitment and Courage).

Nightingale's triad: **the nurse must be a good observer, a good reporter and a good companion.**

2.4 Duties and Responsibilities of the Nurse

Towards the patient

- Provide safe, clean, comfortable environment; give bed bath, mouth care, back care and change position.
- Observe and record **temperature, pulse, respiration, blood pressure, intake-output** and general condition.
- Give medicines and treatments exactly as prescribed — right drug, dose, route, time and patient.
- Feed the helpless, attend to elimination needs, prevent bed sores and deformities.
- Give psychological support, listen to the patient, maintain privacy and dignity, explain procedures.
- Teach the patient and family about diet, drugs, hygiene, exercise and follow-up.
- Prepare the patient for investigations, operation and discharge; give after care.

Towards the doctor, the institution and the profession

- Carry out instructions accurately and report changes and errors immediately; never alter a prescription on one's own.
- Keep equipment and drugs in order; check the emergency tray, oxygen and suction daily; avoid wastage.
- Maintain accurate records and hand over the ward at the change of shift.
- Maintain professional standards, take part in continuing education, guide juniors and students.

Towards the family and community

- Keep the family informed, allay their anxiety and involve them in care.
- Give health education, promote immunisation, sanitation, family planning and nutrition.
- Report notifiable diseases; participate in national health programmes and disaster relief.

2.5 Personal Hygiene, Grooming and Uniform of the Nurse

- **Daily bath**, deodorant, clean under-clothing; hair washed regularly and **tied up / covered** so that it does not fall forward.
- **Nails short**, clean and unpolished; **no artificial nails**; hand jewellery removed (rings and wrist watch harbour organisms) — only a plain wedding band allowed.
- **Hand hygiene** before and after every patient contact — the most important single habit.
- Uniform: clean, well fitting, changed daily, worn only on duty; comfortable **low-heeled, closed, non-slip shoes** with soft soles (noiseless).
- Teeth brushed twice daily, no tobacco or strong-smelling food; no heavy perfume or make-up.
- **Cuts and abrasions covered** with waterproof dressing; a nurse with a respiratory or skin infection or diarrhoea should not attend patients.
- Adequate rest, balanced diet, exercise, periodic medical check-up and immunisation.

2.6 Nurse-Patient Relationship and Communication

Phases of the therapeutic (helping) relationship

- **Pre-interaction** — the nurse prepares herself and collects data.
- **Introductory / orientation** phase — introduce oneself, build trust, identify problems, set goals.
- **Working phase** — actual care, teaching, problem solving.
- **Termination phase** — evaluate, prepare the patient for discharge or independence.

Elements and types of communication

- Elements: **sender → message → channel → receiver → feedback** (the loop is completed only by feedback).
- **Verbal** — spoken and written words: clear, simple, in the patient's language, no medical jargon.
- **Non-verbal** — facial expression, eye contact, touch, posture, gestures, tone, silence, appearance. About **70-90%** of communication is non-verbal; *non-verbal cues are more reliable*.
- **Therapeutic techniques** — active listening, open-ended questions, silence, reflection, clarification, paraphrasing, summarising, giving information, offering self, touch.
- **Non-therapeutic (blocks)** — giving false reassurance ('everything will be fine'), giving advice, judging or moralising, changing the subject, why-questions, arguing, belittling feelings, stereotyped comments, defensive response.

Table 2.3 Barriers to communication

Barrier	Example / remedy
Physical	Noise, distance, poor lighting → choose a quiet, private place
Physiological	Deafness, aphasia, unconsciousness, pain → use writing, signs, hearing aid, pictures
Psychological	Fear, anxiety, anger, depression, prejudice → build trust, be patient
Social / cultural	Language, dialect, customs, illiteracy → use interpreter, simple local words, pictures
Semantic	Medical jargon, abbreviations → use lay terms and repeat

CLINICAL / PRACTICAL POINT

Talking to an unconscious patient: always speak to him by name and explain what you are doing — hearing may be intact. Never discuss the prognosis at the bedside.

Deaf patient: face the patient, speak slowly in good light so lips can be read, do not shout.

Blind patient: announce your arrival and departure, keep articles in fixed places, use touch.

Child: speak at eye level, use play, allow the parent to stay, never threaten with an injection.

2.7 Professional Ethics and Legal Responsibility

Basic ethical principles

Table 2.4 Ethical principles

Principle	Meaning
Autonomy	Respecting the patient's right to decide (consent, refusal)
Beneficence	Doing good, acting in the patient's best interest
Non-maleficence	'Primum non nocere' — above all, do no harm
Justice	Fair and equal treatment of all, no discrimination
Veracity	Truthfulness
Fidelity	Keeping promises, loyalty, accountability
Confidentiality	Keeping the patient's information secret

Legal terms every nurse must know

- **Negligence** — omission to do what a reasonable person would do; four elements — *duty, breach of duty, damage, direct causation*.
- **Malpractice** — negligence by a professional in the course of professional duty.
- **Assault / Battery** — threat of bodily harm / actual unauthorised touching or treatment.
- **Invasion of privacy** — exposing the patient or revealing information without consent.
- **Defamation** — libel (written) or slander (spoken) damaging reputation.
- **False imprisonment** — restraining or detaining a patient without proper authority (illegal use of restraints).
- **Informed consent** — explanation of procedure, risks, alternatives before signature; must be voluntary.
- **Vicarious liability** — the employer is responsible for the acts of the employee ('respondeat superior').

CAUTION / NEVER DO THIS

- Never give a medicine on a verbal order without written confirmation (except in a documented emergency).
- Never leave a prepared injection unlabelled, or administer a drug prepared by someone else.
- Never leave a helpless, confused or paediatric patient unattended with the side rails down.
- Never disclose the patient's diagnosis (HIV, cancer, psychiatric illness) to anyone not authorised.

2.8 Records and Reports

- **Purposes** — communication between team members, continuity of care, legal evidence, research and teaching, statistics, audit, billing.
- **Principles of recording** — *accuracy, brevity, legibility, completeness, chronological order (with date and time), objectivity (record facts not opinion), authenticity (sign with name and designation), confidentiality, no blank lines.*
- **Types of records** — admission sheet, case sheet/history, TPR-BP chart, intake-output chart, medication (treatment) chart, nurse's notes, nursing care plan, diet sheet, investigation reports, consent forms, referral and discharge summary, birth/death records, narcotic and stock registers.
- **Types of reports** — change-of-shift (handover) report, telephonic report, transfer/referral report, incident/accident report, 24-hour census, notifiable disease report.
- **Corrections: single line through the error, write 'error', initial and date** — never erase, overwrite or use correcting fluid; never use pencil.

2.9 The Nurse's Own Safety — Occupational Hazards

Table 2.5 Occupational hazards of nursing personnel

Hazard	Prevention
Biological — HBV, HCV, HIV, tuberculosis, COVID-19	Standard precautions, gloves, mask/N95, hepatitis-B vaccination, never recap needles, puncture-proof sharps box
Needle-stick injury	Do not squeeze; wash with soap and running water; do not scrub; report at once; test the source; start post-exposure prophylaxis (PEP) within 2 hours, ideally not later than 72 hours , for 28 days
Back injury / musculoskeletal	Correct body mechanics, bend knees not back, keep load close, use lifting aids and helpers, adjust bed height
Chemical / cytotoxic	Gloves, gown, mask, goggles, biosafety cabinet, spill kit
Radiation	Lead apron, distance, dosimeter badge, avoid in pregnancy
Psychological — burnout, stress, violence	Duty rotation, rest, recreation, counselling, de-escalation training

CHAPTER AT A GLANCE

- INC — Act 1947, constituted 1949, New Delhi; State Council registers the nurse; TNAI 1908; ICN 1899 Geneva.
- Qualities are grouped as physical, mental, moral, emotional, social, professional.
- Hand washing before and after every patient contact = most important duty; nails short, hair tied, no rings.
- Non-verbal communication makes up most of a message; false reassurance and advice-giving are blocks.
- Ethics: autonomy, beneficence, non-maleficence, justice, veracity, fidelity, confidentiality.
- Records: accurate, dated, signed, never erased; 'not recorded = not done'.
- Needle-stick: wash, report, PEP within 2 h (up to 72 h) for 28 days.

The Sick Room

Selection, ventilation, lighting, temperature, furniture, cleanliness, disinfection and safety

3.1 Meaning and Importance

DEFINITION

Sick room — the room in the house that is selected and prepared for the care of a sick person, so arranged that the patient gets maximum rest, comfort, fresh air and sunlight and the nurse can work with the least effort.

Nightingale's environmental theory states that a healthy environment — pure air, pure water, efficient drainage, cleanliness and light — is essential for recovery. The sick room must therefore be planned before the patient is shifted into it.

3.2 Selection of the Sick Room — Ideal Requirements

Table 3.1 Ideal features of a sick room

Feature	Ideal requirement	Reason
Situation	Quiet part of the house, away from the kitchen, staircase, street and children's play area; preferably on the ground floor	Rest and sleep; easy access for the doctor and stretcher
Aspect / direction	South or south-east facing with windows on two opposite walls	Gets morning sun and cross ventilation
Size	About 10-12 ft x 12-14 ft ; floor space not less than 50 sq ft (1.5 m x 3 m) per patient and air space about 500-1000 cubic feet	Space to move around the bed on three sides
Height of room	At least 10-12 feet	Adequate air space
Windows	Window area = 1/5 of floor area ; sill about 3 ft from floor	Light and ventilation
Floor	Smooth, hard, impervious, washable (mosaic/cement), no carpets or rugs	Easy cleaning; prevents dust and slipping
Walls and ceiling	Light coloured, smooth, washable distemper/oil paint; no cracks or pictures collecting dust	Reflects light; easily disinfected
Attached facilities	Bathroom and lavatory adjoining; water supply and electric points near the bed	Convenience, less exertion
Furniture	Minimum — bed, bedside table, chair, screen, cupboard	Less dust, more space
Doors	Wide enough to admit a stretcher/wheelchair; open outward if possible	Emergency removal

3.3 Ventilation

DEFINITION

Ventilation — the process of replacing foul, used air of an occupied room by fresh outside air, so as to maintain the physical and chemical quality of air (temperature, humidity, movement and purity).

Standards

- Air space required: **1000-1200 cubic feet per person** (minimum 500 cu ft); floor space **50-100 sq ft**.
- Air change required: **2-3 air changes per hour** for a sick room (up to 6 in hospitals; **12 or more** in operation theatres and isolation rooms).
- Rate of air movement should be about **5-10 ft/second (1-3 m/s)** — enough to feel refreshing but not a draught.
- **Carbon dioxide** of room air should not exceed **0.04-0.06%** (600 ppm); it is used as an *index of air pollution* (Pettenkofer's standard 0.06%).

Types of ventilation

Table 3.2 Methods of ventilation

Type	Sub-types / methods	Principle
Natural ventilation	Windows, doors, ventilators, skylight, roof ventilator, open verandah; <i>cross ventilation</i> = openings on opposite walls	Wind (perflation), diffusion, and 'stack effect' (warm air rises and escapes through high openings, fresh air enters low openings) — aided by the <i>inequality of temperature</i>
Artificial / mechanical	(a) Exhaust (extraction) — fan removes foul air, e.g. kitchens, mines (b) Plenum (propulsion) — fresh air blown in, e.g. theatres (c) Balanced — both exhaust and plenum (d) Air conditioning — control of temperature, humidity, purity and movement	Fans, blowers, ducts, filters

Practical points

- Ventilate **without draught** — never let cold air blow directly on the patient; protect with a screen, and place the bed so that the head is not in the line of the window.
- Air the room while giving the bed bath only after covering the patient properly; keep the room aired for a while every day.
- Windows are opened **from the top** in cold weather (warm foul air escapes, cold air is deflected upward).
- Do not use a kerosene stove/charcoal *sigri* in a closed sick room — risk of **carbon monoxide poisoning** and oxygen depletion.
- Foul air causes discomfort chiefly by **excess heat, humidity, body odour and lack of air movement**, not mainly by CO₂ (Hill's theory).

HIGH-YIELD / MOST ASKED

Effects of **inadequate ventilation**: stuffiness, headache, drowsiness, nausea, fainting, poor sleep, increased spread of droplet infection (TB, influenza, measles, COVID-19), and heat cramps/exhaustion.

3.4 Lighting

- **Natural light (sunlight)** is best — it is a natural disinfectant (ultraviolet rays), gives vitamin D, cheers the patient and helps detect abnormal colour (jaundice, cyanosis, pallor).
- Light should come from **behind or the side** of the patient, never directly into his eyes; diffuse it with a curtain or blind.
- A **shaded bedside lamp / dim night lamp** is required at night; a torch is kept ready for observation without disturbing sleep.
- Glare and flickering cause eye strain and headache; a patient with **measles, meningitis, migraine, eye injury or photophobia** needs a darkened room.
- Illumination of about **100-200 lux** is adequate for a sick room; more (300-500 lux) for reading and procedures.

3.5 Temperature, Humidity and Comfort

Table 3.3 Thermal comfort in the sick room

Factor	Desirable level	Notes
Room temperature (adult sick room)	20-22 °C (68-72 °F)	Measure at bed level with a room thermometer
For infants, the aged, and after operation	24-27 °C (75-80 °F)	They lose heat rapidly
Newborn / preterm nursery	28-32 °C (warmer)	Prevent hypothermia
Relative humidity	40-60% (ideal about 50%)	Dry air irritates the throat; damp air prevents sweat evaporation
Comfortable 'cooling power'	Measured by Kata thermometer	Effective temperature combines temperature, humidity and air movement

- **Humidity is raised** by keeping an open vessel of water, wet towel or steam kettle (useful in croup, bronchitis, dry cough, tracheostomy).
- **Humidity is lowered** and heat relieved by fans, air-cooler, cross ventilation, and by sprinkling water on the floor (not near the bed).
- Instruments: **dry and wet bulb thermometers / hygrometer** measure humidity; **sling psychrometer, globe thermometer** (radiant heat), **Kata thermometer** (air velocity and cooling power).

3.6 Furniture and Equipment of the Sick Room

Table 3.4 Sick room furniture and equipment

Article	Requirement / use
Bed	Single, narrow (about 3 ft/90 cm wide) , high (about 26-30 inch / 65-75 cm) to save the nurse's back, with a firm level mattress ; accessible from both sides; head of the bed away from the window
Bedside table / locker	For drinking water, medicines, tissues, call bell, kidney tray; keep on the patient's convenient side
Over-bed table	For meals, reading, writing
Chair / stool	For the nurse, visitors, and for the patient to sit out of bed
Screen (2-3 fold)	Privacy during procedures; protection from draught and light
Cupboard / shelf	Linen, dressings and equipment — kept outside the room if the disease is infectious
Waste bucket with lid and foot pedal	Soiled dressings and rubbish, lined with a plastic bag
Sink / wash basin, soap, towel, sanitiser	Hand hygiene
Bed pan, urinal, kidney tray, sputum cup	Elimination and expectoration, each with a lid
Clock, calendar, call bell / hand bell	Orientation and summoning help
Rubber goods	Mackintosh, air ring, hot water bag, ice bag, rubber sheet
Linen	Two sets of bed sheets, draw sheet, pillow covers, blankets, towels, gowns

CLINICAL / PRACTICAL POINT

Keep the room **simple and uncluttered**: 'Everything that is not needed must be taken out of the sick room.' Excess furniture, curtains, carpets, flowers in stale water and soft toys collect dust and organisms.

3.7 Cleanliness of the Sick Room

Principles

- Cleaning must be done **without raising dust** and with the least disturbance to the patient.
- **Damp dusting** is the rule — a cloth wrung in warm water with a disinfectant/soap; dry dusting and sweeping with a broom raise dust and organisms.
- Clean **from cleaner to dirtier** areas and **from above downwards** (ceiling → walls → furniture → floor).
- Use a **vacuum cleaner or wet mop** for the floor; wash the floor daily with soap and water and a disinfectant (phenol 1-2%, bleaching powder solution, sodium hypochlorite 1%).
- Cleaning should be finished before meals and before the doctor's visit; the patient is covered and the bed screened.

Daily and weekly routine

Table 3.5 Cleaning routine

Frequency	Work to be done
Several times a day	Ventilate the room; tidy the bed; damp dust the bedside table, bed rails and equipment; empty and disinfect bedpan, urinal, sputum cup and waste bucket; clean the wash basin
Daily	Wet mop / wash the floor and bathroom; change soiled linen; clean the thermometer; wipe the door handles and switches with disinfectant; remove dead flowers and leftover food

Frequency	Work to be done
Weekly	Wash windows, doors and window sills; clean under and behind the bed and cupboard; sun the mattress, pillows and blankets; scrub the bathroom; check and replenish stock
On discharge / recovery	Terminal cleaning and disinfection — see 3.9

3.8 Disposal of Waste, Excreta and Soiled Linen at Home

- **Excreta** — normally flushed down the water closet. Infectious excreta (cholera, typhoid, dysentery, hepatitis A) must first be **disinfected for 1-2 hours** with an equal volume of **bleaching powder (5%)** or **crude phenol (2-5%)** or freshly prepared **1% sodium hypochlorite**, and then flushed. In villages with no latrine, bury deep away from the water source.
- **Sputum** — collected in a covered sputum cup lined with paper/containing disinfectant; the paper and sputum are **burnt**; the cup is boiled.
- **Soiled dressings, cotton, paper tissues** — collected in a paper/plastic bag and **burnt or buried**.
- **Soiled linen** — handled with gloves, rolled with the soiled part inside, **never shaken**, soaked in disinfectant, then **boiled/washed separately** and dried in the sun.
- **Left-over food and vomit** of an infectious patient — disinfected and disposed; crockery boiled or washed with hot detergent water.
- **Sharps (needles, lancets, blades)** — never thrown in household waste; put in a **puncture-proof container** (tin/thick plastic bottle), needle destroyed/cut, then given to a health facility.
- Keep **waste covered** at all times to keep out flies and animals; wash hands after handling waste.

3.9 Disinfection of the Sick Room

Table 3.6 Concurrent, terminal and gaseous disinfection

Type	When done	How
Concurrent disinfection	Throughout the illness , immediately after the discharges (urine, faeces, sputum, vomit, pus, dressings) are produced	Disinfect excreta and articles at once; damp dust with disinfectant daily
Terminal disinfection	After recovery, death or removal of the patient	Remove and wash/boil/sun all linen; wash the floor, walls and furniture with disinfectant; sun the mattress, pillows and blankets for 6-8 hours; expose the room to sunlight and air for 24 hours; whitewash/paint if necessary; boil or autoclave instruments
Fumigation / gaseous disinfection	Now considered of little value for routine rooms; used rarely for special situations	Formaldehyde vapour (500 mL formalin + 1000 mL water per 1000 cu ft, room sealed 12 h, neutralised with ammonia), sulphur dioxide by burning sulphur; <i>today replaced by thorough cleaning with chemical disinfectants and UV lamps</i>

NUMBERS TO REMEMBER

- Bleaching powder should contain about **33% available chlorine** (not less than 30%).
- Household bleach for surface disinfection: **1% sodium hypochlorite** (1 part of 5% bleach + 4 parts water).
- Phenol (crude carbolic acid) **2-5%** for floors and excreta; **70% ethyl alcohol / 60-80% isopropyl** for skin and small surfaces.
- Boiling for **10 minutes (preferably 20-30 min)** kills all ordinary pathogens (not spores).

3.10 Control of Insects, Flies and Rodents

- **Flies** (carry typhoid, cholera, dysentery, diarrhoea, food poisoning, trachoma, polio) — keep all food covered, use wire mesh on windows, fly swat/fly paper, bury or cover refuse, remove breeding places (garbage, manure, open drains), use insecticide spray.
- **Mosquitoes** (malaria — *Anopheles*; dengue/chikungunya — *Aedes aegypti*, a day biter breeding in clean stored water; filariasis — *Culex*; Japanese encephalitis — *Culex tritaeniorhynchus*) — **bed nets (preferably insecticide-treated)**, screens, repellents, mosquito coil/mat, removal of stagnant water, weekly emptying of coolers and flower pots, larvicides (temephos), *Gambusia* fish, space spray.

- **Bed bugs, lice, fleas** — boiling and sunning of bedding, scrubbing the cot, insecticide application; pediculosis treated with **permethrin 1% lotion** or 5% dust in seams.
- **Cockroaches** (mechanical carriers) — block cracks, keep the kitchen dry, bait/insecticide.
- **Rats** (plague, leptospirosis, rat-bite fever, salmonellosis) — rat-proof storage, traps, rodenticides; handle with care.
- **Ants** — keep sugar/food in closed containers; keep the patient's water covered.

3.11 Safety in the Sick Room

Table 3.7 Accident prevention in the sick room

Hazard	Preventive measures
Falls	Bed at low height with brakes on, side rails up for the unconscious, restless, elderly and children; call bell within reach; dry non-slippery floor; adequate light at night; support while walking; slippers with grip; no loose wires or mats
Burns / scalds	Test hot water bag temperature (50-60 °C) and cover it; never leave hot drinks or a heater within reach of a confused patient; no smoking, especially with oxygen
Fire	No naked flame, spark, oil or grease near oxygen ; keep matches away; know the exit route; keep sand/water/extinguisher
Poisoning / wrong medicine	Keep all drugs labelled and locked away from children; read the label three times; never keep kerosene or acid in a drink bottle; never keep an external application near oral medicines
Electric shock	Earthed appliances, no wet hands, no over-loaded socket, no wires under the carpet
Suffocation / aspiration	No pillow for infants, no polythene near a child; position the unconscious patient on the side (recovery/lateral position) ; suction ready
Infection	Hand hygiene, isolation where needed, safe disposal, clean linen
Injury to the nurse	Correct body mechanics and lifting aids

3.12 Comfort and the Psychological Environment

- **Noise control** — soft shoes, oil hinges, low voices, no radio/TV noise, no whispering near the patient (it creates suspicion); noise should ideally be **below 35-45 dB** in a sick room.
- **Odour control** — ventilate, remove bedpans and soiled dressings at once, mouth care, clean linen; avoid strong deodorants.
- Allow the patient **personal belongings, photographs, a small radio, books**; permit **limited visitors** at fixed times for a fixed time.
- Maintain a **routine** for meals, medicines, sleep and toilet; group the nursing activities so that the patient gets an uninterrupted rest period.
- Avoid discussing the illness within the patient's hearing; answer questions honestly but with hope; respect religious practices.

3.13 Care of Equipment and Rubber Goods

Table 3.8 Care of sick room articles

Article	Care
Mackintosh / rubber sheet	Wash with soap and cold or lukewarm water, wipe dry, dust with talcum powder, roll (never fold sharply) ; keep away from heat, sunlight, oil and spirit
Hot water bag	Fill only two-thirds , expel air, screw tightly, invert to test for leak, cover with a flannel bag; store inflated with air and stopper closed
Ice bag / air cushion	Wash, dry, store slightly inflated
Clinical thermometer	Wipe from bulb upward, wash in cool soapy water (never hot water — it bursts), disinfect with spirit swab, store dry in individual container or in antiseptic solution
Catheters and tubing	Wash, flush, boil/autoclave or use disposable
Instruments (metal)	Clean off blood/pus, wash, dry, autoclave or boil 20-30 min
Glassware	Cold water first (blood is water soluble), then hot soapy water, rinse, dry
Linen	Soak stains (blood — cold water/hydrogen peroxide), wash, boil if infected, dry in the sun, iron
Mattress and pillows	Sun for 6-8 hours, protect with mackintosh, turn regularly

CHAPTER AT A GLANCE

- Ideal sick room: quiet, south/south-east, ground floor, 2 windows opposite (window = 1/5 of floor area), 50-100 sq ft floor and 500-1200 cu ft air space per patient, washable floor, minimum furniture.
- Ventilation: natural (perflation, diffusion, stack effect) and artificial (exhaust, plenum, balanced, air-conditioning); 2-3 air changes/hour; CO2 index of pollution 0.04-0.06%.
- Temperature 20-22 °C (higher for infants, aged, post-op); humidity 40-60%; Kata thermometer measures cooling power.
- Bed: narrow, firm, 26-30 inch high, accessible from both sides.
- Damp dusting only; clean from above downwards and clean to dirty.
- Concurrent disinfection = during illness; terminal = after recovery/death; fumigation is now of little value.
- Bleaching powder 33% available chlorine; 1% hypochlorite for surfaces; boil 20-30 min.
- Safety: side rails, call bell, locked medicines, no flame near oxygen, lateral position for the unconscious.

Bed Making

Types of beds, procedure, appliances, patient positions, moving and lifting, and prevention of bed sores

4.1 Definition, Purposes and Principles

DEFINITION

Bed making — the technique of preparing a bed in such a way that the patient gets maximum comfort, rest and safety, while the bed remains clean, dry, wrinkle-free and attractive.

Purposes

- To provide **comfort, rest and sleep** — a wrinkle-free, dry bed prevents pressure sores.
- To keep the bed and the patient **clean and neat** and to prevent infection and odour.
- To give the patient the **required position** for treatment, examination or recovery.
- To economise the **nurse's time and energy** and to keep the room tidy.

General principles of bed making

1. Collect **all articles** before starting and arrange them in order of use (saves time and energy).
2. **Wash hands** before and after the procedure; wear gloves if the linen is soiled.
3. Explain the procedure and ensure **privacy** (screen the bed); maintain the patient's dignity.
4. Work from **one side to the other** (complete one side fully, then go to the other side) — this saves steps.
5. **Never shake the linen** (it scatters dust and organisms); fold or roll soiled linen with the **soiled surface inside** and place it in the linen bag, *never on the floor or on another bed*.
6. Keep the bed **tight, smooth and wrinkle free**; the mackintosh must always be **completely covered** by the draw sheet so it never touches the patient's skin.
7. Use correct **body mechanics** — raise the bed to a comfortable height, keep the feet apart, bend the knees, keep the back straight.
8. Watch the patient's **condition, tubes, drains and IV line** throughout; do not tire him.
9. Finally: pillow comfortable, top sheet loose over the toes, **call bell, water and personal articles within reach**, side rails up if needed, and the bed lowered.

4.2 Articles Required and the Parts of a Bed

Table 4.1 Bed linen and equipment

Article	Size / purpose
Cot / hospital bed	Narrow (90 cm), firm, height 65-75 cm; may be plain, fowler (adjustable back), or electrically operated with side rails and wheels with brakes
Mattress	Firm, even, of uniform thickness; covered with a mattress cover/protector
Bottom (under) sheet	Large sheet spread over the mattress and tucked with mitred corners
Mackintosh / rubber sheet	Waterproof sheet placed across the middle of the bed to protect the mattress; must be covered by the draw sheet
Draw sheet	Half sheet placed over the mackintosh; can be 'drawn' through to give a fresh dry surface without complete bed making
Top sheet	Covers the patient; placed with the wrong side up and the wide hem at the top
Blanket	For warmth, covered by the top sheet or counterpane
Counterpane / bed spread	Outermost decorative cover
Pillow and pillow cover	Open end of the cover away from the door / entrance
Linen bag / hamper	For soiled linen
Extras	Bath towel, gown, screen, kidney tray, bedpan, disinfectant

4.3 Types of Beds and their Purposes

Table 4.2 Types of beds (a favourite table for objective questions)

Type of bed	Description	Used for
Closed bed	Top covers (counterpane) drawn right up to cover the whole bed including pillows	Unoccupied bed, ready for a new patient; keeps the bed free of dust
Open bed	Top covers folded back (fan-folded) to the foot end	A patient who is ambulant / who will return to bed shortly
Occupied bed	Bed made while the patient is in it, changing linen by turning the patient side to side	Bedridden, helpless, unconscious patients
Admission bed	Clean, warm bed prepared with extra blankets and hot water bags, top covers fan-folded	Receiving a new patient
Post-operative / anaesthetic / recovery bed	Bed stripped of pillows; mackintosh and draw sheet at the top and middle; top sheet fan-folded to one side lengthwise for easy transfer; warmed with hot water bags (removed before the patient arrives); kidney tray, mouth gag, tongue depressor, suction, oxygen and IV stand kept ready	Patient returning from the operation theatre / unconscious patient
Cardiac bed	Patient supported in a sitting position with a backrest and 4-5 pillows; knees slightly flexed (cardiac chair position); over-bed table in front	Heart failure, dyspnoea, severe asthma — reduces venous return and eases breathing (orthopnoea)
Fracture bed / orthopaedic bed	Firm mattress with fracture board (bed board) under it; blocks to raise the foot of the bed; may have a Balkan frame, traction and bed cradle	Fractures, traction, spinal injury
Divided bed	Bed made in two separate halves (top and bottom) leaving the affected part exposed	Amputation of a limb, plaster cast, perineal or rectal operations
Amputation bed	A divided bed with sandbags and a bed cradle; a tourniquet kept at the bedside for sudden bleeding	After amputation
Blanket bed (bed for blanket bath/pack)	Bed with mackintosh and blanket next to the patient instead of a sheet	Hot/cold packs, blanket bath, hyperpyrexia
Burn bed	Sterile bed with sterile linen, bed cradle, cradle to keep covers off the body; may be an air-fluidised bed	Extensive burns — asepsis and prevention of friction
Plaster bed	Firm bed with a bed board and extra pillows to support the wet cast, cast left uncovered to dry	Patient with a large plaster cast
Air / water bed, alternating pressure mattress, ripple bed	Mattress with air-filled cells that inflate and deflate alternately	Prevention and treatment of pressure sores in long-term bedridden patients
Fowler's bed	Head end raised 45-60 degrees (semi-Fowler 30-45 degrees, high Fowler 90 degrees)	Dyspnoea, chest disease, after abdominal surgery, feeding, unconscious with head injury
Sponge / TPR bed	Mackintosh with towels spread over the bed	Cold sponging in high fever

4.4 Procedure — Making a Simple / Closed Bed

1. Wash hands; collect linen in the order of use (bottom sheet, mackintosh, draw sheet, top sheet, blanket, counterpane, pillow cover) and carry it to the bedside on a chair/trolley.
2. Strip the bed: loosen the linen all round, remove the pillow, and remove the covers one by one, **folding each into three (or four)**; place reusable linen on the chair and soiled linen in the bag.
3. Turn the mattress if required; brush off crumbs; wipe the cot with a damp cloth from the head downwards.
4. Spread the **bottom sheet** with the centre fold in the middle of the bed and the wide hem at the head end; tuck in at the head end first, make a **mitred corner**, then tuck the side from head to foot.
5. Place the **mackintosh** across the middle of the bed (about 45 cm from the head end) and cover it completely with the **draw sheet**; tuck both in tightly.
6. Go to the other side and repeat steps 4-5 (this is the 'one side at a time' rule).
7. Spread the **top sheet** wrong side up, wide hem at the head end, and the **blanket** about 20-25 cm below the top; turn the top sheet over the blanket; tuck in at the foot end with mitred corners, leaving a **toe pleat (vertical or horizontal fold)** to prevent foot drop and pressure on the toes.

8. Spread the counterpane; put on a fresh pillow cover and place the pillow with its open end away from the door.
9. Leave the bed neat; clear the articles; wash hands and record.

CLINICAL / PRACTICAL POINT

Mitred (envelope) corner — tuck the sheet end under the mattress, lift the side of the sheet at about 30-45 cm from the corner to form a triangle on the bed, tuck the hanging part under the mattress, then drop the triangle and tuck it in. It keeps the sheet tight and wrinkle-free.

4.5 Making an Occupied Bed / Changing Linen with the Patient in Bed

1. Explain, screen the bed, close windows/fan; wash hands, wear gloves if soiled.
2. Remove the counterpane and blanket; keep the patient covered with the top sheet (never expose the patient).
3. Loosen all the bottom linen on the near side; turn the patient **gently to the far side** (with help, using the side rails for support).
4. Roll the soiled bottom sheet, mackintosh and draw sheet lengthwise towards the patient's back; brush and wipe the exposed mattress.
5. Spread the clean bottom sheet on the free half with the excess fan-folded/rolled close to the patient's back; place clean mackintosh and draw sheet; tuck in with mitred corners.
6. Turn the patient to the clean side (over the rolls); remove the soiled linen, place it in the bag *without shaking*; pull through and tuck the clean linen tightly.
7. Change the top sheet by placing the clean one over the soiled one and withdrawing the soiled one from beneath; change the pillow cover and gown as required.
8. Position the patient comfortably, tidy the bed, give a **back rub** while the back is exposed, lower the bed, put up side rails, keep the call bell within reach, remove the screen, wash hands and record.

HIGH-YIELD / MOST ASKED

Always make the bed **from one side completely, then move to the other side**. Two nurses working on opposite sides make an occupied bed most safely; *always turn the patient away from the side you are working on*.

4.6 Bedside Appliances and their Uses

Table 4.3 Bedside appliances

Appliance	Use
Bed cradle (hoop/frame over the bed)	Keeps the weight of the bed clothes off a painful part, burn, wound, plaster or ulcer; allows air circulation and inspection; may carry a light for warmth
Backrest	Supports the patient in a sitting position (dyspnoea, cardiac, after abdominal surgery, for meals)
Air ring / rubber ring / cotton ring	Relieves pressure on the sacrum and buttocks (<i>use with caution — it may itself impair circulation; a ripple mattress is better</i>)
Air/water mattress, alternating pressure (ripple) mattress	Distributes body weight; prevents pressure sores
Bed blocks / shock blocks	Raise the head or foot end of the bed (foot end raised in shock; head end raised in head injury, dyspnoea)
Sandbags	Immobilise and support a limb or the head; prevent external rotation
Footboard / foot rest	Keeps the foot at a right angle to prevent foot drop and toe pressure; gives something to push against
Bed board / fracture board	Placed under the mattress to give a firm surface (traction, spinal or orthopaedic cases)
Pillows, wedge and trochanter rolls	Support the head, back, limbs; maintain alignment; prevent external rotation of the hip
Bed rails / side rails	Prevent falls in the confused, restless, unconscious, aged and children
Over-bed (cardiac) table	For meals, reading, resting arms in dyspnoea
Mackintosh and draw sheet	Keep the bed dry; allow the sheet to be drawn through
Bedpan, urinal, commode	Elimination in bed or at the bedside
Balkan frame / Bohler-Braun frame, traction, pulleys	Continuous traction in fractures

Appliance	Use
Hand roll, splint	Prevent contracture in a paralysed hand
Trapeze bar (monkey bar)	Overhead bar the patient grasps to lift himself and change position
Cradle boot / heel protector	Prevents pressure on the heel and foot drop

4.7 Positions of the Patient

Table 4.4 Patient positions — very frequently asked

Position	How the patient lies	Indications / uses
Supine (dorsal recumbent)	Flat on the back, face upward, arms at the sides, one pillow	Rest, spinal injury, after spinal anaesthesia, examination of abdomen, CPR (on a hard surface)
Prone	On the abdomen, face turned to one side, arms at the sides or above the head	Drainage of the mouth after tonsillectomy, unconscious patient, back/spine surgery, prevents hip flexion contracture; prone ventilation improves oxygenation in ARDS/COVID-19
Lateral (side lying)	On one side, upper knee flexed and supported by a pillow	Rest, relieving pressure on the back, giving an enema/injection (IM gluteal), unconscious patient
Recovery / semi-prone (Sims') position	On the left side, lower arm behind the back, upper knee and hip flexed	Unconscious patient (prevents aspiration, keeps the airway open), enema, rectal and vaginal examination, during labour
Fowler's	Head end raised 45-60° , knees slightly flexed (Low/semi-Fowler 30-45°, High Fowler 90°)	Dyspnoea, heart failure, chest disease, after abdominal surgery, feeding, nasogastric tube insertion, head injury (raised ICP)
Orthopnoeic	Sitting upright, leaning forward on an over-bed table with arms supported	Severe dyspnoea — asthma, pulmonary oedema, COPD
Trendelenburg	Whole body tilted with the head lower than the feet	Pelvic surgery, insertion of a central line, <i>formerly in shock — now only passive leg raising is advised</i>
Reverse Trendelenburg	Head higher than feet, body straight	After head/neck surgery, gastro-oesophageal reflux
Lithotomy	On the back, legs raised and separated, hips and knees flexed, feet in stirrups	Vaginal examination, delivery, perineal repair, catheterisation in the female, cystoscopy
Knee-chest (genupectoral)	Kneeling with the chest on the bed, head turned to one side, arms above the head	Rectal and sigmoidoscopic examination, correction of a displaced uterus, prolapsed cord
Dorsal recumbent (with knees flexed)	On the back with knees drawn up and soles on the bed	Abdominal examination, perineal care, catheterisation
Left lateral (enema) position	Left side with the right knee flexed	Giving an enema/suppository (follows the direction of the colon)
Sitting (upright)	Sitting on the bed/chair with feet supported	Meals, exercises, physiotherapy, examination of the chest
Elevated limb position	Affected limb raised on pillows above heart level	Oedema, after limb surgery, venous ulcer, sprain

MEMORY AID

Fowler = sitting up (Fowl/bird sits up); Sims'/recovery = side (S for side, Safety of airway); Trendelenburg = head down; Lithotomy = legs up for delivery; Knee-chest = rectal examination.

4.8 Moving, Lifting and Transferring the Patient (Body Mechanics)

Rules of good body mechanics

- Keep the **back straight** and bend at the **hips and knees**, not at the waist; use the strong thigh and leg muscles.
- Keep the **feet apart (about 25-30 cm)** with one foot slightly forward to widen the base of support; keep the **centre of gravity low**.
- Hold the load **close to the body**; face the direction of movement; **pivot with the feet, never twist the spine**.
- **Push, pull or roll** rather than lift; use a draw sheet, sliding board, hoist or trolley.

- **Raise the bed** to hip height and lower the side rails on the working side.
- Work with a **rhythm and a count** ('on three') when two or more people lift; the tallest/strongest person takes the heaviest part (head and shoulders); one person gives the commands.
- Tell the patient what to do and encourage him to help.

Common techniques

- **Moving up in bed** — one or two nurses with a draw sheet; the patient flexes the knees and pushes with the feet while holding the trapeze bar or crossing the arms on the chest.
- **Turning to the side** — cross the far leg over the near, place the near arm across the chest, then roll the patient towards you with the hand on the shoulder and hip (log-rolling for spinal injury, keeping the head, spine and legs in a straight line).
- **Bed to chair/wheelchair** — chair locked at a 45° angle to the bed; the patient sits, dangles the legs, then pivots on the stronger leg; the nurse blocks the patient's knees and feet and holds the transfer belt.
- **Bed to stretcher** — three-person carry or a sliding board; stretcher wheels locked, patient covered and strapped, head end first through doors, feet first down a slope.
- **Lifting an unconscious patient** — always two or more persons; support the head and neck; keep the airway clear.

CAUTION / NEVER DO THIS

Never lift a patient alone if he is heavy or helpless. Never drag the patient over the sheet — friction and shearing force cause **pressure sores**. Never pull on a paralysed arm (risk of shoulder dislocation).

4.9 Pressure Sores (Bed Sores / Decubitus Ulcer)

DEFINITION

Pressure sore — a localised area of **ischaemic necrosis** of the skin and underlying tissue caused by **unrelieved pressure** over a bony prominence (usually between a bone and a hard surface), often combined with friction, shearing force and moisture. Also called *decubitus ulcer*, *bed sore* or *pressure injury*.

Causes and contributing factors

Table 4.5 Causes of pressure sores

Group	Factors
Mechanical (direct)	Sustained pressure (capillary closing pressure is about 32 mmHg ; tissue damage may begin within 1-2 hours), friction (dragging), shearing force (sliding down in Fowler's position), wrinkled sheets, crumbs, hard objects, tight bandage or plaster, splints and tubes
Moisture	Incontinence of urine and faeces, sweating, wound discharge, wet linen (maceration of skin)
Patient factors	Immobility (unconscious, paralysed, sedated, in traction, very ill), poor nutrition (low protein, vitamin C and zinc, anaemia, dehydration) , emaciation or obesity, old age, thin dry skin, diabetes, peripheral vascular disease, oedema, fever, loss of sensation, faecal/urinary incontinence
Nursing factors	Failure to change position, careless lifting, poor hygiene, wrinkled or damp bed, use of hard bedpans

Common sites

- **Supine position** — **sacrum (most common site overall)**, heels, elbows, occiput, scapulae, spinous processes.
- **Lateral position** — greater trochanter (hip), ear, shoulder (acromion), lateral knee and malleolus (ankle), inner knees.
- **Prone position** — forehead, cheek/ear, chin, shoulder, breast, iliac crest (hip bones), knees, toes, genitalia in men.
- **Sitting** — **ischial tuberosities** (the commonest site in a wheelchair patient), coccyx, heels.

Stages (grades) of pressure injury

Table 4.6 Staging of pressure sores

Stage	Features
Stage I	Intact skin with non-blanchable redness (erythema); warmth, oedema, pain; discolouration in dark skin — <i>reversible</i>
Stage II	Partial-thickness loss of the dermis — shallow open ulcer, abrasion or intact/ruptured blister
Stage III	Full-thickness skin loss; subcutaneous fat visible; slough may be present; bone, tendon and muscle are not exposed ; may undermine
Stage IV	Full-thickness tissue loss with exposed bone, tendon or muscle ; slough/eschar, undermining and tunnelling; risk of osteomyelitis
Unstageable	Base covered by slough or eschar so the depth cannot be determined
Deep tissue injury	Purple/maroon intact skin or a blood-filled blister from damage to underlying soft tissue

Prevention — the nurse's most important responsibility

1. **Change position at least every 2 hours** (every 1 hour if sitting in a chair); keep a **turning chart/clock** and follow a sequence (supine → right lateral → prone if permitted → left lateral).
2. Use a **pressure-relieving mattress** (alternating air/ripple mattress, water bed, foam or gel pads), pillows, heel protectors and a bed cradle; *avoid the doughnut-type ring*.
3. Keep the skin **clean and dry**; wash with mild soap and lukewarm water, pat dry (do not rub), apply a moisturiser; use barrier cream for incontinence; change wet linen at once.
4. Keep the bed **smooth, dry and free of crumbs**; use a draw sheet to move the patient; **never drag**.
5. Give **back care / pressure area care every 2-4 hours** — gentle massage of the surrounding area with a moisturiser; *do not massage directly over a reddened bony prominence* and do not use spirit or hot water (they dry the skin).
6. Provide a **high protein, high calorie, vitamin-C, zinc and iron rich diet** with adequate fluids (2-3 L/day if permitted).
7. Encourage **active and passive exercise, early ambulation and sitting out of bed**.
8. Keep the head of the bed at **30° or less** when possible to reduce shearing; raise the knee gatch slightly.
9. **Inspect the skin at every turn** and at every bath, especially over bony points; use a risk-assessment scale (**Braden scale, Norton scale, Waterlow scale**) on admission and at regular intervals.
10. Avoid tight clothing, hard splints, and pressure from tubes, catheters and oxygen masks.

Treatment

- **Relieve the pressure completely** over the sore — this is the first and most important step.
- Clean the ulcer with **normal saline** (avoid hydrogen peroxide, povidone-iodine and antiseptics on granulating tissue); **debride** slough (surgical, enzymatic, autolytic).
- Apply the appropriate dressing to keep the wound **moist and the surrounding skin dry** — hydrocolloid, hydrogel, alginate or foam dressing; gauze soaked in saline for cavities.
- Treat infection with systemic antibiotics if there is cellulitis, osteomyelitis or sepsis; take a swab for culture; topical silver sulphadiazine/metronidazole for odour.
- Improve **nutrition** (protein 1.2-1.5 g/kg/day, vitamin C, zinc), correct anaemia and control diabetes; relieve pain.
- Advanced measures: negative pressure wound therapy (vacuum dressing), skin graft or myocutaneous flap for Stage III-IV.

HIGH-YIELD / MOST ASKED

- **Sacrum is the most common site** of pressure sore; **ischial tuberosity** in sitting patients; **heel** is the second commonest.
- Turn the patient **every 2 hours**; capillary closing pressure **32 mmHg**.
- **Stage I** = non-blanchable erythema with intact skin; **Stage IV** = bone/tendon/muscle exposed.
- **Braden scale** (score ≤ 18 = at risk; 6 sub-scales: sensory perception, moisture, activity, mobility, nutrition, friction/shear) and **Norton scale** assess risk.
- Clean pressure sores with **normal saline**, not antiseptics.

4.10 Restraints (Protective Devices)

- **Types** — physical (jacket, wrist/ankle cuff, mitten, belt, mummy restraint for infants, side rails) and chemical (sedatives).
- **Indications** — to protect a confused, delirious, restless, psychotic or paediatric patient from pulling out tubes or falling; only when all other measures have failed.
- **Rules** — written doctor's order (except in an emergency), **informed consent**, use the least restrictive type, apply with a **quick-release (clove hitch) knot to the bed frame, not the side rail**, pad the bony points, release and exercise the limb **every 2 hours** (or at least every 30 min observation), check circulation and skin, record time, reason and response, remove as soon as possible.
- Improper use = **false imprisonment** and can cause injury, strangulation, nerve damage and pressure sores.

CHAPTER AT A GLANCE

- Bed making: work one side at a time, never shake linen, mackintosh always covered by the draw sheet, mitred corners, toe pleat to prevent foot drop.
- Post-operative bed: no pillow, top sheet fan-folded lengthwise, suction/oxygen/kidney tray ready; cardiac bed = sitting with backrest; divided bed = amputation/perineal surgery; fracture bed = bed board.
- Positions: Fowler 45-60°, semi-Fowler 30-45°, high Fowler 90°; Sims'/lateral = unconscious; Trendelenburg = head low; lithotomy = delivery; knee-chest = rectal exam.
- Body mechanics: bend knees, load close, pivot with feet, push rather than lift, count together.
- Pressure sore: pressure > 32 mmHg; sacrum commonest; turn 2-hourly; Braden/Norton risk scales; clean with saline; high protein diet.
- Restraints need a written order, quick-release knot to the bed frame, release every 2 hours.

Patient's Toilet (Personal Hygiene of the Patient)

Bed bath, mouth, hair, eye, ear, nail and back care, elimination needs and care of the incontinent patient

5.1 Meaning, Importance and Principles

DEFINITION

Personal hygiene (patient's toilet) — the self-care measures by which a person maintains cleanliness of the body, mouth, hair, nails, skin and genitalia. When the patient cannot do it himself, the nurse performs it for him.

Importance / purposes

- Removes **dirt, sweat, sebum, dead skin and micro-organisms** — prevents skin infection and body odour.
- **Stimulates circulation** and the nerve endings; relaxes the muscles; promotes sleep and a sense of well-being.
- Gives the nurse the chance for **complete observation** of the skin, pressure areas, oedema, rashes, lumps, ranges of joint movement and the patient's mental state.
- Prevents **pressure sores, contractures, dental caries, gum disease, parasitic infestation and constipation**.
- Improves **self-esteem, appetite** and the patient's relationship with the nurse.

General principles

- **Privacy** (screen the bed, close the door) and **dignity** at all times — expose only the part being cleaned.
- Prevent **chilling** — close windows and fans, use water at **40-46 °C (105-115 °F)**, cover the rest of the body with a bath blanket.
- **Hand hygiene** and gloves for contact with secretions; clean **from the cleanest to the dirtiest area** and from **above downwards**.
- Follow the patient's **habits, culture and wishes** (time of bath, oil, soap); encourage self-care as much as possible.
- Work **quickly but gently**, with long firm strokes towards the heart; do not tire the patient; observe his condition throughout.
- Keep the used water, linen and articles away from the clean ones; clean and replace all equipment after use.

5.2 Bathing the Patient

Table 5.1 Types of baths

Type of bath	Description / indication
Self / shower / tub bath	For the ambulant patient; the nurse keeps the articles ready, ensures a non-slip floor, a low-height stool and that the door is not bolted from inside ; stays within call
Assisted bath	Patient bathes at the wash basin or in the bathroom with the nurse's help
Bed bath (complete/sponge bath)	Given in bed to a helpless, unconscious, very weak, paralysed or post-operative patient
Partial bath	Only the face, hands, axillae, back and perineum ('the parts that matter') are washed
Therapeutic / medicated bath	Medicated solution added — potassium permanganate (KMnO ₄ 1:5000-1:10 000) for infected skin; sodium bicarbonate/oatmeal/starch for itching; <i>sitz bath</i> for perineal conditions; cold sponging for hyperpyrexia; emollient bath for dry skin; sulphur/benzyl benzoate applications for scabies

Bed bath — articles required

Two basins/buckets of warm water, mug, soap in a dish, two wash cloths/sponges, two towels (face and body), bath blanket, clean gown/clothes, clean bed linen, comb, nail cutter, oil/moisturiser, talcum powder, mackintosh, screen, kidney tray, paper bag, gloves, linen bag; bedpan and urinal kept ready.

Bed bath — procedure (sequence is important)

1. Explain the procedure; offer the **bedpan/urinal first**; screen the bed; close windows and switch off the fan; wash hands; collect articles; raise the bed.
2. Remove the top covers, keeping the patient covered with a **bath blanket**; remove the patient's clothes (remove from the **unaffected/uninjured limb first** and put on to the affected limb first).

3. **Face** — wash the eyes with a soft wet cloth (*no soap*), from the **inner to the outer canthus**, using a separate corner of the cloth for each eye; then the face, ears and neck; dry.
4. **Upper limbs** — place a towel under the arm, wash the far arm first with long strokes **from the wrist towards the shoulder (distal to proximal)**, wash the axilla, then the near arm; soak the hands in the basin, clean the nails, dry well between the fingers.
5. **Chest and abdomen** — expose, wash, rinse and dry; special attention to **under the breasts and the umbilicus**; observe the skin.
6. **Lower limbs** — wash the far leg first from ankle to thigh; flex the knee and place the foot in the basin; clean between the toes; dry thoroughly (important in diabetes).
7. **Back and buttocks** — turn the patient to the lateral position; wash, rinse and dry the back and buttocks; give **back care with a gentle massage** and inspect all pressure points; change the bottom linen at this stage.
8. **Perineum (genitalia)** — **last of all**, with fresh water, a separate cloth and gloves; clean **from front to back** in the female (to prevent faecal contamination of the urethra); in the male retract the foreskin, clean the glans in a circular motion and **replace the foreskin**.
9. Change the water **whenever it becomes cool, soapy or dirty** (at least 2-3 times).
10. Apply oil/moisturiser and powder in the skin folds (*use powder sparingly — caking causes irritation*); comb the hair; cut the nails; dress the patient; make the bed.
11. Position comfortably, keep the call bell and water within reach, clear and disinfect articles, wash hands, and **record the procedure, the condition of the skin and pressure areas**.

CAUTION / NEVER DO THIS

- **Do not give a bed bath** immediately after a meal, during severe haemorrhage, in acute myocardial infarction/unstable condition, or when the patient is in severe pain or shock, without the doctor's permission.
- Never leave a helpless patient alone in the bathroom, and never let a bathroom door be bolted from inside.
- Avoid soap on the face and eyes, and on broken or inflamed skin.

5.3 Care of the Mouth (Oral Hygiene)

- **Purposes** — clean, moist, odour-free mouth; prevents **dental caries, gingivitis, stomatitis, halitosis, parotitis, glossitis, oral thrush and aspiration pneumonia**; improves appetite and taste.
- The conscious patient brushes **twice daily** (morning and bedtime) with a soft brush, using the correct technique and flossing; a helpless patient is helped in the **Fowler's/sitting** position with a kidney tray under the chin.
- **Special mouth care** is required for the unconscious, seriously ill, dehydrated, high fever, NPO, oxygen therapy, post-operative, nasogastric tube, diabetic, cancer/chemotherapy and terminal patient — given **every 2-4 hours** (2-hourly for the unconscious).

Special (unconscious) mouth care — procedure

1. Articles: sterile tray with artery/sponge-holding forceps, gauze swabs, tongue depressor, mouth gag, kidney tray, mouth wash solution, torch, lubricant (soft paraffin/glycerine), suction apparatus, gloves, towel.
2. Position the patient in the **lateral/side-lying position with the head turned to one side** and the head of the bed lowered (to prevent aspiration); place a towel and kidney tray.
3. Keep the mouth open with a gag/tongue depressor; using swabs on forceps (one swab per stroke, discarded after use) clean in order: **teeth (outer then inner surfaces), gums, palate, inner cheeks, tongue and finally the lips**.
4. **Never put a large amount of fluid** in the mouth of an unconscious patient; use suction to remove secretions.
5. Apply a lubricant to the lips; leave the mouth clean; inspect with a torch for ulcers, thrush, bleeding gums and loose teeth.
6. Clean dentures separately with a brush over a basin of water, store in a labelled container of **plain cold water** when out of the mouth, and never leave them loose in the bed.

Table 5.2 Common mouth washes and oral applications

Mouth wash / agent	Strength and use
Normal saline (0.9%)	Safest, general cleansing and after surgery
Sodium bicarbonate solution	1 teaspoon in 500 mL water; dissolves thick mucus and sordes

Mouth wash / agent	Strength and use
Hydrogen peroxide	Diluted 1:4 or 1:8 with water; loosens sordes and debris (<i>not for long use — it harms granulating tissue and enamel</i>)
Chlorhexidine gluconate 0.2%	Antiseptic; prevents plaque, gingivitis and ventilator-associated pneumonia
Povidone-iodine gargle 2%	Sore throat, before oral surgery
Potassium permanganate 1:5000-1:8000	Mild antiseptic gargle
Nystatin suspension / clotrimazole	Oral thrush (candidiasis) — white curd-like patches
Glycerine with lemon / soft paraffin / lip balm	Moistens dry lips (<i>glycerine alone dries the mucosa, so it is now avoided</i>)
Benzocaine / lignocaine viscous gel	Painful stomatitis and mucositis
Warm saline gargle	Sore throat, tonsillitis, after tooth extraction

DEFINITION

Sordes — dry, brown-black crusts of dried mucus, food, epithelial cells and bacteria on the teeth, lips and tongue of a dehydrated, febrile or unconscious patient with neglected mouth care. **Halitosis** = foul breath. **Stomatitis** = inflammation of the oral mucosa. **Parotitis** (surgical mumps) = infection of the parotid gland from a dry, dirty mouth.

5.4 Care of the Hair

- Comb and brush the hair **twice daily**; part it in small sections; for tangled hair apply oil, coconut oil or dilute vinegar/conditioner and comb from the ends upwards; plait long hair loosely at the side (not at the back of the head where it presses).
- **Bed shampoo** — done weekly or when needed: articles are a mackintosh with a trough/kelly pad, bucket, jug of warm water (40 °C), shampoo, cotton balls for the ears, pad for the eyes, towels; the patient lies with the head at the edge of the bed, neck supported, the trough draining into the bucket; massage the scalp with the finger tips (not the nails), rinse thoroughly, dry and comb.
- **Shaving** — soften the beard with warm water and soap/cream, shave **in the direction of hair growth** with short strokes, holding the skin taut; *take great care in patients on anticoagulants or with bleeding disorders — use an electric razor.*
- **Pediculosis (lice)** — types: *Pediculus capitis* (head), *P. corporis* (body, vector of **epidemic typhus, relapsing fever and trench fever**) and *Phthirus pubis* (pubic, 'crab louse'). Treatment: **permethrin 1% lotion** (leave 10 min), 5% permethrin cream, malathion 0.5%, or ivermectin; comb out nits (eggs) with a fine **nit comb** soaked in vinegar; repeat after **7 days**; treat all contacts; boil/hot-iron clothes and bedding, or seal them in a bag for 2 weeks.
- **Dandruff** — ketoconazole or selenium sulphide shampoo; keep the scalp clean.

5.5 Care of the Eyes, Ears and Nose

Table 5.3 Care of the eyes, ears and nose

Part	Care
Eyes	Clean with sterile cotton/gauze moistened with normal saline or cool boiled water, wiping from the inner canthus to the outer canthus , a separate swab for each eye (<i>clean the less infected eye first</i>). For the unconscious patient with an absent blink reflex: instil artificial tears/methylcellulose drops , apply an antibiotic eye ointment and close the lids with a pad or tape to prevent corneal ulcer and exposure keratitis . Care of spectacles, contact lenses and an artificial eye (remove, clean with saline, store in saline)
Ears	Clean only the outer ear (pinna) and the visible part of the canal with a soft cloth; never insert a bud, pin, matchstick or hairpin into the canal. Soften hard wax with warm olive oil/ soda glycerine drops for 2-3 days and let the doctor syringe it with water at body temperature (37 °C). Care of a hearing aid — switch off when not in use, keep dry, check the battery
Nose	Clean with a moist cotton swab or a cotton applicator moistened with saline; do not go deep . Soften dried secretions with saline drops. For a patient with a nasogastric or oxygen tube, clean the nostril and change the tape daily and observe for pressure sores at the nostril. Teach the patient to blow the nose gently with both nostrils open

5.6 Care of Hands, Feet and Nails

- Soak the hands in warm soapy water for 5-10 minutes (feet 10-20 min) to soften the nails; clean under the free edge; **cut the finger nails in a curve and the toe nails straight across**; file the edges; avoid injury to the cuticle.

- Dry well **between the toes**; apply moisturiser to the soles and heels but **not between the toes**; cotton socks; well-fitting footwear.
- **Diabetic foot care** (very high yield) — inspect the feet daily (use a mirror), wash with lukewarm water and **test the temperature with the elbow** (because of neuropathy), never walk barefoot, never use hot water bottles or corn-removing plasters, **never cut corns or nails too short** (get a podiatrist/nurse to do it), report any blister, ulcer, colour change or numbness at once; control blood sugar.
- Look for **fungal infection (tinea pedis)**, ingrowing toe nail, callosities, corns, cracks and oedema.

5.7 Back Care (Pressure Area Care)

1. Given **after the bath and every 2-4 hours**, and after each episode of incontinence, to all bedridden patients.
2. Articles: warm water, soap, towel, moisturising lotion/oil, talcum powder, gloves, clean linen.
3. Turn the patient to the lateral or prone position; expose only the back; wash, rinse and pat dry.
4. Pour a little lotion in the palm, warm it, and massage the back with **long, firm, upward strokes from the buttocks to the shoulders** and circular strokes over the shoulders for **3-5 minutes** — techniques: *effleurage* (long stroking), *petrissage* (kneading), *tapotement* (tapping).
5. Inspect all pressure points for redness; **do not massage over a reddened or broken area**; do not use spirit (drying) or hot water.
6. Change position, straighten the linen, and record the condition of the skin.

5.8 Meeting Elimination Needs

Giving the bedpan and urinal

- Warm the bedpan (rinse with warm water), dry it, and **powder the rim** (do not powder if a specimen is needed); cover it while carrying.
- Screen the bed, wear gloves; flex the patient's knees, ask him to raise the hips (or turn him to the side and roll the pan into place); place a mackintosh under the buttocks.
- Raise the head end slightly to give a **near-normal squatting position**; leave the toilet paper, call bell and water within reach and **give privacy** — but never leave a weak patient alone for long.
- Afterwards: clean the perineum from **front to back**, remove the pan covered, give water to wash the hands, ventilate the room.
- **Observe and record** the amount, colour, consistency, odour and abnormal contents (blood, mucus, pus, worms, undigested food) of stool and urine before disposal; measure the urine if an output chart is being kept; send a specimen if ordered.
- Clean and disinfect the bedpan/urinal after each use; each patient should have his own.
- A **fracture pan** (with a shallow flat end) is used for patients in traction, after hip surgery or with a spinal injury; a **commode chair** for those who can sit out of bed.

Perineal care (vulval/genital toilet)

- Indicated after childbirth, after perineal surgery, with an indwelling catheter, in incontinence, vaginal discharge and for the unconscious patient — given **twice daily and after each soiling**.
- Position: dorsal recumbent with knees flexed (female) / supine (male); mackintosh and bedpan under the buttocks; sterile swabs and warm antiseptic solution (savlon/normal saline) poured from a jug.
- Clean **from above downwards and from the centre outwards, from the cleanest to the dirtiest area**: labia majora, labia minora, urethral meatus, vaginal orifice, then the perineum and anus **last**; use one swab for one stroke and discard it; **never wipe from the anus towards the vagina**.
- Dry, apply the prescribed ointment or a sterile pad, and record any discharge, swelling, redness or stitch line condition.

Care of the incontinent patient

- **Types** — stress, urge, overflow, functional, total incontinence; faecal incontinence.
- Keep the skin **clean, dry and protected** with a barrier cream (zinc oxide, dimethicone); change pads/diapers frequently; never leave the patient on wet linen.
- Offer the bedpan/urinal **2-hourly (timed voiding)**; establish a **bladder and bowel training programme** (fixed times, pelvic floor/**Kegel exercises**, fluid 1.5-2 L in the day but restricted 2 hours before bed, avoid caffeine).
- Use a condom (Uridom) drainage for men; an **indwelling catheter only as a last resort** (infection risk).

- Prevent **pressure sores, urinary tract infection and dermatitis**; protect the patient's dignity — never scold or shame him.

Constipation and its management

- **Causes in bed patients** — immobility, low-fibre diet, inadequate fluid, ignoring the urge, lack of privacy, opioids/anticholinergics/iron/antacids (aluminium), painful piles, depression, hypothyroidism, dehydration.
- **Management** — high-fibre diet (whole grains, fruit, vegetables), 2-3 L of fluid, warm drinks in the morning, exercise, privacy and a regular time, sitting position; then bulk laxative (isabgol/psyllium), osmotic (lactulose, milk of magnesia, PEG), stimulant (bisacodyl, senna), stool softener (liquid paraffin, docusate), glycerine suppository, and lastly an **enema** or manual evacuation of **faecal impaction**.
- **Never give a laxative** in undiagnosed abdominal pain, suspected appendicitis, or intestinal obstruction.

5.9 Care of Vomiting, Expectoration and Dressing the Patient

- **Vomiting** — turn the head to the **side** (prevent aspiration), support the head, hold the kidney tray, loosen tight clothes, give mouth wash afterwards, ventilate the room, remove the vomitus at once; **observe and record** the amount, colour (bile-stained, 'coffee ground' = old blood, faecal = intestinal obstruction), content, smell, force (projectile), relation to food and any accompanying pain; preserve a sample if poisoning is suspected.
- **Sputum** — a **covered sputum cup** with a disinfectant or paper lining is kept, emptied and disinfected 2-3 times a day; teach the patient to **cough covering the mouth** and never to spit on the floor; record the amount, colour, consistency and odour (rusty in pneumonia, copious foul in bronchiectasis/abscess, blood-stained in TB/cancer, pink frothy in pulmonary oedema).
- **Changing the patient's clothes** — remove the garment from the **unaffected arm first** and put it on the **affected (or IV) arm first**; for a patient with an IV line, thread the bottle and tubing through the sleeve without disconnecting; keep the patient covered; use front-open gowns for the helpless.
- **Care of linen** — remove soiled linen at once, do not shake, carry it away from the uniform, soak blood/faecal-stained linen in cold water first (hot water fixes protein stains), then wash with hot soapy water; boil/soak infected linen in disinfectant; dry in the sun.

CHAPTER AT A GLANCE

- Bed bath water 40-46 °C; sequence: face (eyes inner→outer canthus, no soap) → arms (distal to proximal) → chest/abdomen → legs → back (back care) → **perineum last**; change water 2-3 times.
- Special mouth care 2-hourly for the unconscious, in the lateral position with the head turned; clean teeth→gums→palate→cheeks→tongue→lips; sordes = crusts of neglect.
- Eyes of the unconscious: artificial tears + closure to prevent corneal ulcer; never insert anything in the ear canal.
- Finger nails cut curved, toe nails straight; diabetic foot — never barefoot, never hot water, test with the elbow.
- Back care 2-4 hourly with long upward strokes; never massage a reddened bony prominence.
- Perineal care: front to back, centre outwards, one swab one stroke, anus last.
- Pediculosis: permethrin 1%, repeat after 7 days; body louse transmits epidemic typhus.
- Vomiting: head to one side; observe colour — coffee ground = bleeding, faecal = obstruction.

Observation of the Sick

General observation, vital signs, TPR charting, intake-output and interpretation of findings

6.1 Meaning and Importance of Observation

DEFINITION

Observation — the deliberate, systematic use of all the senses (sight, hearing, smell, touch) to collect information about the patient's condition.

Nightingale: 'The most important practical lesson to be given to nurses is to teach them what to observe, how to observe, which symptoms indicate improvement and which the reverse.'

- **Purposes** — to know the patient's condition and progress, to detect complications early, to judge the effect of treatment, to plan nursing care, to give accurate reports and to keep legal records.
- **Methods** — *inspection* (look), *palpation* (feel), *percussion* and *auscultation* (listen), *olfaction* (smell), plus talking to the patient and measuring with instruments.
- Observation must be **continuous, systematic, objective (facts, not guesses), accurate and promptly reported**.
- **Subjective data (symptoms)** = what the patient says (pain, nausea, giddiness); **objective data (signs)** = what the nurse sees and measures (rash, vomiting, BP 150/90).

6.2 General (Head-to-Foot) Observation

Table 6.1 Systematic observation of the patient

What to observe	Points to note / abnormal findings and meaning
General appearance and build	Well/ill-looking, obese or emaciated (cachexia), posture, hygiene, dress
Level of consciousness	Alert → drowsy → confused → delirious → stuporous → semi-comatose → comatose; restlessness is an early sign of hypoxia, pain, a full bladder or shock
Face (facies)	Anxious, pinched (peritonitis), flushed (fever), puffy (nephritis, myxoedema), 'risus sardonicus' (tetanus), mask-like (Parkinsonism)
Skin colour	Pallor (anaemia, shock, haemorrhage); cyanosis — bluish, central on the tongue/lips (heart-lung disease) or peripheral on the fingers (cold, poor circulation) — appears when SpO ₂ is below about 85% or 5 g/dL of reduced Hb; jaundice — yellow of skin, sclera, urine (liver, bile duct, haemolysis; visible when bilirubin is over 2-3 mg/dL); redness/erythema , pigmentation, cherry-red (carbon monoxide)
Skin condition	Dryness, sweating, rash, petechiae/purpura, bruising, ulcer, oedema (press over the tibia for 5 s — pitting), turgor (pinch test — poor in dehydration), temperature, itching
Eyes	Sunken (dehydration), puffy lids (renal), yellow sclera, pallor of conjunctiva, pupil size and reaction (pin-point in opioid poisoning/pontine lesion; dilated fixed in death, atropine, brain damage), squint, discharge, vision
Mouth and tongue	Dry/coated/furred tongue (dehydration, fever), sordes, ulcers, thrush, bleeding gums, smooth red tongue (vitamin B deficiency), cracked lips, halitosis; smell of breath — acetone (ketoacidosis), ammonia/urine (uraemia), foul (lung abscess), alcohol
Neck	Stiffness (meningitis), raised jugular veins (heart failure), goitre, lymph nodes
Chest and breathing	Rate, rhythm, depth, use of accessory muscles, in-drawing, wheeze, cough and sputum, chest pain, symmetry
Abdomen	Distension, rigidity, tenderness, visible peristalsis, bowel sounds, girth, hernia, stoma
Limbs	Power, movement, deformity, contracture, swelling, calf tenderness (deep vein thrombosis), colour, peripheral pulses, capillary refill (normal <2-3 s)
Elimination	Urine — amount, colour, deposit, smell, frequency, burning, retention or incontinence; stool — frequency, consistency, colour, blood, mucus, worms
Sleep and behaviour	Insomnia, disturbed sleep, irritability, depression, anxiety, hallucination, co-operation
Appetite, thirst, weight	Anorexia, nausea, excessive thirst (diabetes), loss or gain of weight
Pain	Site, onset, character, radiation, severity (0-10 scale), duration, aggravating and relieving factors

6.3 Vital Signs — Body Temperature

DEFINITION

Body temperature = balance between heat produced (metabolism, muscle activity, food, hormones) and heat lost (radiation, conduction, convection, evaporation). It is regulated by the **hypothalamus** (the body's thermostat).

Table 6.2 Sites for recording temperature

Site	Normal average	Normal range	Time to keep the thermometer
Oral (mouth)	37 °C / 98.6 °F	36.4-37.2 °C (97.6-99.0 °F)	2-3 minutes
Axilla (armpit)	36.5 °C / 97.6 °F (~0.5 °C less than oral)	35.9-36.7 °C	3-5 minutes (safest for children and the unconscious)
Rectal	37.5 °C / 99.6 °F (~0.5 °C more than oral)	37.0-38.1 °C	1-2 minutes (most accurate; lubricate and insert 2-4 cm)
Tympanic (ear)	37.5 °C	Near core temperature	2-5 seconds (infrared)
Forehead/temporal, non-contact	36.5-37.3 °C	Screening only	Instant

Conversion formulae (asked very often)

NUMBERS TO REMEMBER

$$^{\circ}\text{F} = (^{\circ}\text{C} \times 9/5) + 32 \quad ^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9$$

Useful values: 36.1 °C = 97 °F • 37 °C = 98.6 °F • 38 °C = 100.4 °F • 39 °C = 102.2 °F • 40 °C = 104 °F • 41 °C = 105.8 °F

Terms and grades of temperature

Table 6.3 Types and patterns of fever

Term	Meaning
Pyrexia / fever	Temperature above 37.2-38 °C (99-100.4 °F)
Low grade / moderate / high fever	Up to 38.3 °C / 38.4-39.4 °C / above 39.5 °C
Hyperpyrexia	Above 41 °C (105.8 °F) — a medical emergency; may cause convulsions and brain damage
Hypothermia	Below 35 °C (95 °F) — newborn, elderly, exposure, drowning, hypothyroidism
Afebrile / apyrexia	Absence of fever
Onset, fastigium (stadium), defervescence	Beginning, highest plateau and fall of fever
Crisis / Lysis	Sudden fall of temperature to normal (with sweating) / gradual fall over days
Constant (continuous) fever	Remains high, fluctuates less than 1 °C, never touches normal — lobar pneumonia, typhoid (2nd week), UTI
Remittent fever	Fluctuates more than 2 °C but never reaches normal — typhoid, infective endocarditis, sepsis
Intermittent fever	Rises and falls to normal daily; quotidian = daily, tertian = every 48 h (<i>P. vivax/ovale</i>), quartan = every 72 h (<i>P. malariae</i>) — malaria, kala-azar, pyaemia
Relapsing fever	Febrile periods alternating with 1-2 afebrile days (<i>Borrelia</i>); Pel-Ebstein fever in Hodgkin's lymphoma
Inverse fever	Temperature higher in the morning than the evening — tuberculosis

Clinical thermometer and precautions

- A mercury clinical thermometer is graduated **35-42 °C (94-108 °F)**; it has a **constriction/kink** above the bulb so the mercury does not fall back; it must be **shaken down below 35 °C** before use.

- A **rectal thermometer has a short blunt (pear-shaped) bulb**; an oral thermometer has a long slender bulb; each patient (or each site) should have a separate thermometer, kept in a dry container or disinfectant, and wiped **from the stem towards the bulb** before use.
- **Never use hot water** to clean it (mercury expands and the bulb bursts); if a mercury thermometer breaks in the mouth, do not induce vomiting — remove the pieces, wash the mouth and inform the doctor. Digital and infrared thermometers have now replaced mercury ones (mercury is toxic and is being phased out).
- **Do not take an oral temperature** in: infants and children under 5-6 years, the unconscious, confused or uncooperative, mouth breathers, oxygen mask/nasal tube, after mouth surgery or with sores in the mouth, convulsive or shivering patients. Wait **15-30 minutes** after hot/cold drinks, food, smoking or chewing.
- **Avoid the rectal route** in rectal surgery, diarrhoea, piles, cardiac patients (vagal stimulation) and newborns (risk of perforation).
- **Avoid the axilla** when it is sweaty or there is local inflammation; dry the axilla and place the bulb in the centre with the arm across the chest.

Factors affecting body temperature

- **Raise it** — exercise, hot bath/hot weather, meals, emotion/stress, ovulation and pregnancy, infection, dehydration, hyperthyroidism, drugs (atropine), heat stroke, brain injury (hypothalamic).
- **Lower it** — sleep and early morning (temperature is **lowest between 2 and 6 a.m.**, **highest between 4 and 8 p.m.**), starvation, old age, shock, haemorrhage, hypothyroidism, cold exposure, alcohol, sedatives.
- **Age** — newborns and the aged have poor temperature regulation.

Nursing care in fever and in hyperpyrexia

- Bed rest, light clothing and light bed covers, good ventilation, cool room.
- **Plenty of fluids (3 L/day if allowed)** — ORS, fruit juice, barley water; light, bland, high-calorie, easily digested food in small frequent feeds.
- **Tepid sponging / cold sponging** for temperature above 39.5-40 °C: water at 27-32 °C (or tepid), long strokes to the limbs, cold compresses to the forehead, axillae and groins (over the large vessels), stop if shivering occurs; recheck the temperature after 30 minutes.
- Antipyretic as prescribed (**paracetamol 10-15 mg/kg per dose** in children; maximum 4 g/day in adults); tepid sponging is an adjunct, *never use ice-cold water or alcohol rubs*.
- Mouth care 2-4 hourly (dry mouth, sores), skin care and change of damp linen after sweating, observe for **convulsions, delirium, dehydration and herpes on the lips**.
- Record temperature 4-hourly (2-hourly or continuously in hyperpyrexia); watch the urine output.

6.4 Vital Signs — Pulse

DEFINITION

Pulse — the wave of expansion and recoil felt over a superficial artery each time the left ventricle contracts and forces blood into the already full aorta.

Table 6.4 Normal pulse rate by age

Age group	Normal pulse (beats/min)
Newborn (0-1 month)	120-160 (up to 180 when crying)
Infant (1-12 months)	100-160
Toddler (1-3 years)	90-140
Pre-school (3-6 years)	80-120
School child (6-12 years)	75-110
Adolescent / adult	60-100 (average 72)
Elderly	60-100 (may be slower, less elastic vessels)
Trained athlete	45-60

Sites for feeling the pulse

- **Radial** (wrist, thumb side) — the site of routine counting; **brachial** (inner elbow) — used for BP and for the *infant's pulse in CPR*; **carotid** (side of the neck) — used in *adult CPR / collapse*; **femoral** (groin); **popliteal** (behind the knee); **posterior tibial and dorsalis pedis** (foot — checked in diabetes and peripheral vascular disease); **temporal** (in front of the ear, used in children); **apical (apex beat)** — heard with a stethoscope at the 5th left intercostal space just inside the mid-clavicular line; **umbilical cord/brachial** in newborns.
- Count for **one full minute** (especially if irregular; 30 s × 2 is acceptable if regular) with the **first two or three fingers (never the thumb — it has its own pulse)**.

Characteristics to record

Table 6.5 Characteristics of the pulse

Character	Description and abnormalities
Rate	Beats/min. Tachycardia = above 100 (fever — the pulse rises about 10 beats per 1 °F/18 beats per °C, exercise, pain, anxiety, haemorrhage, shock, anaemia, thyrotoxicosis, heart failure, atropine, adrenaline). Bradycardia = below 60 (sleep, athletes, myxoedema, jaundice, raised intracranial pressure, heart block, digoxin, beta blockers, opioids)
Rhythm	Regular or irregular. Arrhythmia/dysrhythmia ; regularly irregular (sinus arrhythmia of children), irregularly irregular (atrial fibrillation); ectopic/dropped beat
Volume / amplitude	Full and bounding (exercise, hypertension, fever), weak, thready, feeble (shock, haemorrhage, heart failure, dehydration); imperceptible ; water-hammer/collapsing (aortic regurgitation); pulsus paradoxus (fall in pulse volume with inspiration — cardiac tamponade, severe asthma); pulsus alternans (alternating strong and weak — LV failure)
Tension / character	Compressibility of the vessel wall; hard, tortuous vessel in arteriosclerosis
Equality	Compare both sides; an absent or unequal pulse suggests embolism, aortic dissection, coarctation or a tight bandage/plaster
Pulse deficit	Apical (heart) rate minus radial rate — present in atrial fibrillation

6.5 Vital Signs — Respiration

Table 6.6 Normal respiratory rate

Age	Normal respiratory rate (breaths/min)
Newborn	30-60
Infant (1-12 months)	30-50
1-3 years	24-40
3-6 years	22-34
6-12 years	18-30
Adolescent/adult	12-20 (average 16-18)
Elderly	16-24

- Count the respirations **without the patient knowing** (while apparently still feeling the pulse), for one full minute, by watching the rise and fall of the chest or abdomen.
- Normal pulse : respiration ratio is about **4-5 : 1**.
- Observe **rate, rhythm, depth, character, sound, effort, symmetry, and the colour of the patient**.

Table 6.7 Terms describing respiration

Term	Meaning
Eupnoea	Normal quiet breathing
Tachypnoea / Bradypnoea	Abnormally rapid (over 20-24) / abnormally slow (under 12) breathing
Apnoea	Absence of breathing
Dyspnoea	Difficult or laboured breathing (the patient is 'short of breath')
Orthopnoea	Breathlessness on lying flat, relieved by sitting up — left heart failure
Paroxysmal nocturnal dyspnoea	Sudden night-time breathlessness waking the patient — heart failure
Hyperpnoea / hyperventilation	Deep, rapid breathing → may cause dizziness and tingling (respiratory alkalosis)

Term	Meaning
Hypoventilation	Shallow, slow breathing → CO ₂ retention
Cheyne-Stokes respiration	Rhythmic waxing and waning of depth with periods of apnoea — heart failure, uraemia, brain damage, terminal illness, and normally in infants
Kussmaul's breathing	Deep, sighing, rapid 'air hunger' — diabetic ketoacidosis and metabolic acidosis
Biot's / ataxic breathing	Irregular breaths with irregular apnoea — meningitis, brain stem damage
Stertorous / snoring	Noisy breathing due to obstruction of the upper airway — unconscious patient, apoplexy
Stridor	Harsh, high-pitched crowing noise on inspiration — laryngeal obstruction, croup, foreign body (an emergency)
Wheeze / rhonchi	Musical whistling, mainly expiratory — asthma, bronchitis
Crepitations (crackles/rales)	Fine bubbling sounds — pneumonia, pulmonary oedema, bronchiectasis
Grunting, nasal flaring, chest retraction	Signs of respiratory distress in an infant (pneumonia)
Asphyxia / anoxia / hypoxia / hypoxaemia	Failure of air exchange / absence of oxygen in tissue / reduced tissue oxygen / reduced oxygen in blood
Cyanosis	Bluish discolouration when reduced haemoglobin exceeds 5 g/dL (SpO ₂ under about 85%)
Haemoptysis / Epistaxis	Coughing up blood / bleeding from the nose

NUMBERS TO REMEMBER

Pulse oximetry (SpO₂) — normal **95-100%**; below 94% needs attention; **below 90% = hypoxia** requiring oxygen; below 85% = severe hypoxia. Falsely low readings occur with cold hands, nail polish, poor perfusion and movement; falsely high in carbon monoxide poisoning and anaemia.

WHO fast-breathing cut-offs for childhood pneumonia: **under 2 months: 60/min or more; 2-12 months: 50/min or more; 1-5 years: 40/min or more.**

6.6 Vital Signs — Blood Pressure

DEFINITION

Blood pressure — the lateral force exerted by circulating blood on the walls of the arteries, expressed as *systolic/diastolic* in mmHg.

Systolic = maximum pressure during ventricular contraction; **diastolic** = minimum pressure during ventricular relaxation.

Pulse pressure = systolic – diastolic (normally **30-40 mmHg**). **Mean arterial pressure (MAP)** = diastolic + 1/3 of the pulse pressure (normal **70-105 mmHg**; a MAP of at least 60-65 mmHg is needed to perfuse the organs).

Table 6.8 Classification of blood pressure

Category (adults, ACC/AHA and JNC)	Systolic (mmHg)	Diastolic (mmHg)
Normal	Below 120	and below 80
Elevated / high-normal	120-129	and below 80
Pre-hypertension (JNC-7)	120-139	or 80-89
Hypertension Stage 1	130-139 (JNC: 140-159)	or 80-89 (90-99)
Hypertension Stage 2	140 or more (JNC: 160+)	or 90 or more (100+)
Hypertensive crisis / emergency	Above 180	and/or above 120
Hypotension	Below 90	or below 60

Technique and sources of error

- Instruments: **sphygmomanometer** (mercury, aneroid or digital) and a stethoscope; the pressure is heard as **Korotkoff sounds** — phase 1 = systolic, disappearance (phase 5) = diastolic.
- The patient should rest **5 minutes**, sit or lie with the arm supported at **heart level**, feet on the floor, no talking; no smoking, caffeine or exercise for 30 minutes; the **cuff bladder should cover 80% of the arm circumference and about 40% of its width** and be placed 2.5 cm above the elbow crease, with the mercury column at eye level.

- Inflate 20-30 mmHg above the point where the radial pulse disappears, then deflate slowly at **2-3 mmHg per second**; wait 1-2 minutes before repeating; record the arm used and the position.
- **Common errors** — cuff too narrow or loose (falsely **high**), cuff too wide (falsely low), arm below heart level (high), arm above heart level (low), rapid deflation (systolic low, diastolic high), full bladder/pain/anxiety ('white-coat' hypertension) and the *auscultatory gap* (missing 20-40 mmHg) leading to under-reading if the cuff is not inflated high enough.
- **Avoid** the arm with an intravenous infusion, arteriovenous fistula, lymphoedema/after mastectomy, fracture, plaster, burn or paralysis.
- **Postural (orthostatic) hypotension** = a fall of at least 20 mmHg systolic or 10 mmHg diastolic on standing — check in the elderly and in patients on antihypertensives or diuretics.
- **Causes of hypertension** — essential (90-95%), renal disease, endocrine (phaeochromocytoma, Cushing's, Conn's), pregnancy (pre-eclampsia), coarctation, drugs (steroids, oral pills, NSAIDs), obesity, high salt, alcohol, stress.
- **Causes of hypotension** — haemorrhage, shock, dehydration, myocardial infarction, sepsis, anaphylaxis, Addison's disease, over-treatment with antihypertensives, prolonged bed rest, vasovagal attack.

6.7 The TPR Chart, Intake-Output Chart and Other Records

- The **TPR chart (vital signs chart)** is a graphic sheet: **temperature is plotted as a dot joined by a blue/black line, pulse in red, and respiration as a figure or a separate line**; the time scale is usually 4-hourly (6 a.m., 10 a.m., 2 p.m., 6 p.m., 10 p.m., 2 a.m.).
- Routine frequency: **twice daily** for ordinary patients, **4-hourly** in fever, and **1-4 hourly or continuous** for critical, post-operative and unconscious patients.
- **Never guess or copy previous readings**; report at once a temperature above 39 °C, pulse below 60 or above 120, respiration below 10 or above 30, systolic BP below 90, SpO2 below 90% or any sudden change.
- **Intake-output chart** — intake: oral fluids, IV fluids, blood, tube feeds and medicines given in fluid; output: urine, vomit, aspirate, drainage, liquid stool, and estimated loss in sweat. Totalled every shift and over **24 hours**.
- **Normal fluid balance**: intake about **2500 mL/day** (1500 mL drink + 800 mL food + 250 mL metabolic water); output — urine **1500 mL**, insensible loss through skin and lungs 800-900 mL, faeces 100-200 mL.
- **Normal urine output = 1-2 mL/kg/hour in adults (about 30-60 mL/hour; 1500 mL/day)**; **oliguria** = less than 400 mL/day (or under 0.5 mL/kg/h), **anuria** = less than 100 mL/day, **polyuria** = more than 2500-3000 mL/day.
- Other records: weight chart (daily in cardiac, renal and oedema patients; a **1 kg gain = about 1 L of retained fluid**), diabetic chart, fluid-balance sheet, pain score chart, Glasgow coma scale chart, neurological observation chart, drug chart.

6.8 Level of Consciousness — Glasgow Coma Scale

Table 6.9 Glasgow Coma Scale (GCS)

Response	Score and criteria
Eye opening (E) — 4	4 spontaneous • 3 to speech • 2 to pain • 1 no response
Verbal response (V) — 5	5 oriented • 4 confused conversation • 3 inappropriate words • 2 incomprehensible sounds • 1 none
Motor response (M) — 6	6 obeys commands • 5 localises pain • 4 withdraws from pain • 3 abnormal flexion (decorticate) • 2 extension (decerebrate) • 1 none

NUMBERS TO REMEMBER

Total GCS = 3 (worst) to 15 (fully conscious). Mild head injury 13-15 • Moderate 9-12 • **Severe / coma 8 or less** (a score of 8 or below usually needs intubation).

Also observe **pupils (size, equality, reaction to light)**, limb power, vital signs; **rising BP with a slow pulse and irregular breathing (Cushing's reflex)** indicates rising intracranial pressure.

6.9 Observation of Urine, Stool, Sputum and Vomit

Table 6.10 Observation of excreta and secretions

Specimen	Normal	Abnormal findings and significance
Urine	1200-1500 mL/day, pale straw/amber, clear, faint aromatic smell, acidic (pH 4.6-8), specific gravity 1.010-1.025, no protein, sugar, ketone, blood or pus	• Concentrated dark — dehydration, fever • Red/smoky — haematuria (stones, tumour, glomerulonephritis); dark brown/'tea coloured' — jaundice/bile, myoglobin • Frothy — albuminuria (nephrotic syndrome) • Milky/cloudy — pus (UTI), chyluria (filariasis) • Sweet/acetone smell — ketosis; fishy/ammoniacal — infection • Painful (dysuria), frequency, urgency, retention, incontinence, dribbling, nocturia • Red with rifampicin, orange with pyridium, dark with metronidazole (harmless drug colouring)
Stool	1-2 formed brown stools/day, characteristic odour	• Black tarry (melaena) — upper GI bleeding (also harmless black with iron, bismuth, charcoal) • Fresh red blood — piles, fissure, dysentery, lower GI bleed • Clay/putty coloured with dark urine — obstructive jaundice • Rice-watery — cholera; blood and mucus with tenesmus — bacillary dysentery/amoebiasis • Bulky, greasy, foul, floating (steatorrhoea) — malabsorption, coeliac disease • Green — enteritis in infants; worms, segments of tapeworm visible • Constipation, diarrhoea, incontinence, ribbon-like stool (obstruction)
Sputum	Nil or a little clear mucoid in the morning	• Thick yellow-green — bacterial infection • Rusty/blood-tinged — lobar pneumonia; copious foul, in layers — bronchiectasis/lung abscess • Blood-stained (haemoptysis) — tuberculosis, bronchial carcinoma, bronchiectasis • Pink frothy — acute pulmonary oedema • Currant jelly — Klebsiella; thick tenacious plugs — asthma
Vomit	Nil	• Coffee-ground / red — haematemesis (peptic ulcer, varices) • Bile stained green-yellow — persistent vomiting, intestinal obstruction • Faecal smelling — low intestinal obstruction, peritonitis • Projectile (forceful, without nausea) — raised intracranial pressure; in infants — congenital pyloric stenosis • Undigested food hours after a meal — gastric outlet obstruction

6.10 Signs of Improvement, Deterioration and of Approaching Death

Table 6.11 Judging the course of illness

Improvement	Deterioration (report at once!)
Temperature, pulse, respiration and BP returning to normal; skin warm and dry	Rising or falling temperature, weak thready rapid pulse, fall in BP, rapid shallow or irregular breathing
Alert, oriented, interested in surroundings, sleeps well	Restlessness, confusion, drowsiness, convulsion, unconsciousness
Appetite returning, tongue clean and moist, bowels regular	Persistent vomiting, abdominal distension, no urine (under 30 mL/h), bleeding from any site
Good urine output, clear urine, less oedema	Increasing pain, cold clammy skin, cyanosis, pallor, sweating, gasping (<i>air hunger</i>)

Improvement	Deterioration (report at once!)
Wound clean, dry and healing; pain decreasing	Foul, discharging wound; spreading redness; sudden severe chest pain or breathlessness

Signs of approaching death and last offices

- **Signs** — sunken eyes with a fixed stare and loss of the corneal reflex, dilated non-reacting pupils, pinched cold clammy skin with 'hippocratic facies', mottled and cold extremities, incontinence, weak irregular imperceptible pulse, falling BP, **Cheyne-Stokes** then gasping respiration with noisy secretions ('death rattle'), restlessness followed by coma, loss of sphincter control; **hearing is usually the last sense to go**.
- **Signs of death** — no pulse, heart sounds or respiration for a stated period, no response to any stimulus, pupils fixed and dilated, loss of all reflexes, falling body temperature (*algor mortis*), post-mortem lividity (*livor mortis*), stiffening (*rigor mortis* begins in 2-4 hours, complete in 12 hours, passes off in 24-48 hours). **Death must be certified by a doctor**.
- **Last offices (care of the body)** — record the time of death; give privacy and inform relatives kindly; wear gloves; straighten the body in the supine position with one pillow, close the eyes and mouth, remove tubes, catheters and jewellery (make an inventory before witnesses), clean the body and dress the wounds, pad the orifices, tie the limbs and jaw with a bandage, label the body (two identification tags), wrap it in a sheet as per the family's religious custom, hand over valuables against signature, complete the death record and inform the mortuary; disinfect the bed and unit (terminal disinfection). Handle the body of an infectious case as per the biosafety protocol.

CHAPTER AT A GLANCE

- Normal adult vital signs: T 37 °C (98.6 °F), P 60-100, R 12-20, BP under 120/80, SpO2 95-100%.
- Rectal temperature 0.5 °C above oral, axillary 0.5 °C below; hyperpyrexia above 41 °C, hypothermia below 35 °C; °F = (°C × 9/5) + 32.
- Fever raises the pulse about 10 beats per °F; count the pulse for 1 full minute, never with the thumb; apex beat at the 5th left intercostal space.
- Fever types: continuous, remittent, intermittent (quotidian/tertian/quartan), relapsing, inverse (TB); crisis = sudden fall, lysis = gradual.
- Cheyne-Stokes = heart failure/brain damage; Kussmaul = diabetic ketoacidosis; stridor = upper airway obstruction.
- Urine output 1-2 mL/kg/h (30-60 mL/h); oliguria under 400 mL/day, anuria under 100 mL/day.
- GCS 3-15 (E4 V5 M6); 8 or less = coma; Cushing's reflex (rising BP + slow pulse) = raised ICP.
- Black tarry stool = upper GI bleed; clay stool + dark urine = obstructive jaundice; pink frothy sputum = pulmonary oedema; rusty sputum = lobar pneumonia.

Infection

Terminology, chain of infection, transmission, sterilisation and disinfection, isolation, universal precautions and waste management

7.1 Basic Terminology

Table 7.1 Terminology of infection

Term	Definition
Infection	Entry, multiplication and establishment of a pathogenic organism in the tissues of a host, with a harmful (pathological) response
Contamination	Presence of organisms on an inanimate article, body surface or in food/water — no invasion of tissue
Colonisation	Organisms present and multiplying on a body surface without causing disease or immune response
Infestation	Lodgement and development of an <i>animal parasite</i> on or in the body (lice, scabies, worms)
Pathogen / Virulence	Organism capable of causing disease / the degree of its disease-producing power
Sepsis / Asepsis / Antisepsis	Presence of pus-forming organisms in blood or tissue / absence of disease-producing organisms / destruction of organisms on living tissue
Incubation period	Interval between the entry of the organism and the appearance of the first sign or symptom
Prodromal period	Interval between the earliest vague symptoms and the specific (characteristic) signs, e.g. the rash
Period of communicability (infectivity)	The time during which the patient can transmit the infection to others
Carrier	An apparently healthy person who harbours and excretes the organism. Types: healthy, incubatory, convalescent, chronic (e.g. 'Typhoid Mary'), temporary, paradoxical, contact
Reservoir / Source	Natural habitat where the organism lives and multiplies (man, animal, soil) / the place from which it immediately passes to the host
Zoonosis	Disease transmissible from animal to man — rabies, plague, brucellosis, anthrax, leptospirosis, bird flu, Nipah, COVID-19
Fomites	Inanimate contaminated objects — bed linen, towels, toys, utensils, door handles, stethoscope, thermometer, mobile phone
Vector	A living carrier (insect or animal) — mosquito, housefly, louse, flea, tick, sandfly; <i>mechanical</i> (housefly) or <i>biological</i> (mosquito)
Nosocomial / Health-care-associated infection (HAI)	Infection acquired in hospital (appearing 48 hours or more after admission) that was not present or incubating at admission
Opportunistic infection	Infection by a normally harmless organism in an immunocompromised host (candidiasis, <i>Pneumocystis pneumonia</i> in AIDS)
Endogenous / Exogenous	Infection from the patient's own flora / from outside the body
Local / Systemic infection	Confined to one part (abscess) / spread through the blood stream (septicaemia)
Bacteraemia / Septicaemia / Toxaemia / Pyaemia	Bacteria in the blood without multiplying / bacteria multiplying in the blood with illness / toxins in the blood / pus-forming organisms with multiple abscesses
Immunity / Susceptible host	Power of the body to resist infection / a person lacking resistance
Quarantine / Isolation	Restriction of the movement of an <i>apparently healthy contact</i> for the longest incubation period / separation of an <i>infected person</i> for the period of communicability
Surveillance / Notifiable disease	Continuous watch over disease occurrence / a disease that must be reported to the health authority by law
Disinfection / Sterilisation	Destruction of pathogenic organisms (<i>not spores</i>) on inanimate objects / destruction of all forms of microbial life including spores
Sanitisation / Decontamination	Reduction of microbial numbers to a safe level / removal of contamination so an article is safe to handle

7.2 The Chain of Infection

Table 7.2 The six links in the chain of infection

Link	Explanation	How the nurse breaks it
1. Infectious agent	Bacteria, virus, fungus, protozoa, helminth, prion	Correct diagnosis and treatment; antibiotics; disinfection and sterilisation
2. Reservoir	Man (patient, carrier), animals, soil, water, food, equipment	Treat cases and carriers, chlorinate water, control animals, clean equipment
3. Portal of exit	Respiratory secretions, faeces, urine, blood, wound discharge, genital secretions, breast milk, placenta	Cover cough, safe disposal of excreta and dressings, cover wounds
4. Mode of transmission	Contact, droplet, airborne, vehicle, vector	Hand hygiene , gloves, mask, isolation, vector control, food and water hygiene
5. Portal of entry	Mouth, nose, respiratory tract, broken skin, mucous membrane, urethra, placenta, injection site, surgical wound	Aseptic technique, wound care, catheter and IV care, safe injections
6. Susceptible host	Very young, aged, malnourished, immunosuppressed, diabetic, post-operative, on steroids/chemotherapy	Immunisation , nutrition, control of diabetes, early ambulation, shortening hospital stay

HIGH-YIELD / MOST ASKED

Hand hygiene is the single most effective measure to prevent the spread of infection — it breaks the chain at the 'mode of transmission' link. The transmission link is the easiest one to break.

7.3 Modes of Transmission

Table 7.3 Modes of transmission with classic examples

Mode	Sub-type / mechanism	Examples
Contact — direct	Skin-to-skin, touching, kissing, sexual contact, droplet spread within 1 metre, transplacental (vertical)	Scabies, impetigo, syphilis, gonorrhoea, HIV, hepatitis B, infectious mononucleosis, congenital rubella
Contact — indirect	Through fomites, contaminated hands, instruments, endoscopes, linen	Staphylococcal infection, MRSA, <i>Clostridioides difficile</i> , conjunctivitis
Droplet (large, over 5 micron, travel under 1-2 m)	Coughing, sneezing, talking, suctioning	Influenza, COVID-19, pertussis, diphtheria, mumps, rubella, meningococcus, common cold
Airborne (droplet nuclei under 5 micron, remain suspended; also dust)	Inhalation over a long distance; needs a negative-pressure room and an N95 mask	Tuberculosis, measles, chickenpox (varicella) , aspergillus, smallpox
Vehicle-borne	Water, food, milk, blood and blood products, IV fluids, drugs	Cholera, typhoid, hepatitis A and E, polio, food poisoning, amoebiasis, brucellosis (milk), hepatitis B/C and HIV (blood)
Vector-borne — mechanical	Organism carried on the vector's body	Housefly: typhoid, diarrhoea, dysentery, trachoma, polio
Vector-borne — biological	Organism multiplies/develops inside the vector	Anopheles mosquito: malaria • Aedes aegypti : dengue, chikungunya, yellow fever, Zika • Culex : filariasis, Japanese encephalitis • Sandfly : kala-azar • Tsetse fly : sleeping sickness • Rat flea (Xenopsylla) : plague • Louse : epidemic typhus, relapsing fever • Tick : KFD, Lyme disease • Mite : scrub typhus • Blackfly : onchocerciasis
Soil-borne	Spores and larvae in soil	Tetanus, gas gangrene, anthrax, mycetoma, hookworm, ascariasis
Inoculation / parenteral	Needle-stick, sharps, transfusion, tattoo, unsafe injection, animal bite	Hepatitis B and C, HIV, rabies, tetanus

7.4 Hospital-Acquired (Nosocomial) Infection

- **Commonest sites (in order)** — urinary tract infection (about 40%, mostly catheter-associated), surgical site infection, respiratory tract/ventilator-associated pneumonia, and blood stream (IV cannula) infection.
- **Commonest organisms** — *Escherichia coli*, *Staphylococcus aureus* (including MRSA), *Pseudomonas aeruginosa*, *Klebsiella*, *Enterococcus*, *Candida*, *Clostridioides difficile* (after antibiotics), *Acinetobacter*.
- **Risk factors** — indwelling catheter and cannula, ventilator, surgery, prolonged stay, broad-spectrum antibiotics, steroids, extremes of age, diabetes, malnutrition, overcrowding, poor hand hygiene.
- **Prevention** — hand hygiene, aseptic technique, minimum use and early removal of catheters and cannulae, closed drainage systems, sterile dressings, isolation of infected patients, antibiotic stewardship (rational use), surveillance by the **hospital infection control committee**, staff training and immunisation, environmental cleaning, and safe waste disposal.

7.5 Hand Hygiene

- **WHO 'My 5 Moments for Hand Hygiene'** — (1) before touching a patient, (2) before a clean/aseptic procedure, (3) after body-fluid exposure risk, (4) after touching a patient, (5) after touching patient surroundings.
- **Types** — *social/routine hand wash* with soap and water for **20-40 seconds** (40-60 s including wetting and drying); *hygienic hand rub* with **alcohol-based hand rub (60-80% alcohol) for 20-30 seconds**; *surgical scrub* with an antiseptic (chlorhexidine 4%, povidone-iodine) for **2-5 minutes** including the forearms up to the elbow, with a sterile brush for the nails.
- **WHO 6/7 steps** — palm to palm; right palm over left dorsum and vice versa; palm to palm with fingers interlaced; backs of fingers to opposing palms; rotational rubbing of each thumb; finger tips/nails in the opposite palm; (then the wrists).
- **Soap and water is essential (alcohol rub is NOT enough)** when the hands are visibly soiled, after using the toilet, and after caring for a patient with **~C. difficile~ diarrhoea or norovirus** (spores are not killed by alcohol).
- Keep the nails short, remove rings and watch, dry the hands with a clean/disposable towel, and use elbow or foot-operated taps.

7.6 Sterilisation

Table 7.4 Methods of sterilisation (a favourite examination table)

Method	Details, temperature and time	Used for
Red heat	Heating to red hot in a flame	Inoculation loops, wires, needle points, forceps tips
Flaming	Passing through a flame	Scalpels, mouth of test tubes, slides
Incineration (burning)	Complete burning to ash	Soiled dressings, cotton, pathological waste, animal carcasses, plastic disposables
Hot air oven (dry heat)	160 °C for 2 hours (or 170 °C for 1 h) — holding time	Glassware (syringes, petri dishes, test tubes), metal instruments, powders, oils, greases, sharp instruments (does not blunt them)
Boiling (disinfection, not true sterilisation)	100 °C for 20-30 minutes (add 2% sodium carbonate to raise the boiling point and prevent rusting); <i>does not kill spores or hepatitis virus reliably</i>	Home use — metal instruments, syringes, rubber goods, feeding bottles
Autoclave (moist heat under pressure) — the most reliable, commonest hospital method	121 °C at 15 lb/sq inch (1.05 kg/cm²) for 15-20 minutes , or 134 °C at 30 lb for 3-5 minutes (flash); prions need 134 °C for 18 min	Surgical instruments, dressings, linen, gloves, syringes, culture media, rubber tubing; NOT for oils, powders, sharp cutting edges (blunts them) or heat-labile plastics
Pasteurisation	Holder method 63 °C for 30 min; flash (HTST) 72 °C for 15-20 seconds ; ultra-high temperature 125-150 °C for 1-4 s; then rapid cooling	Milk — kills <i>M. tuberculosis</i> , <i>Brucella</i> , <i>Salmonella</i> , <i>Coxiella</i> (the test of adequacy is the <i>phosphatase</i> test)
Tyndallisation (intermittent)	100 °C for 20 min on 3 successive days	Media with sugar, gelatin, egg
Inspissation	80-85 °C for 30 min on 3 days	Loeffler's serum, Lowenstein-Jensen medium
Filtration	Membrane filter (0.22 micron), candle, asbestos, sintered glass, HEPA filters for air	Serum, vaccines, antibiotic solutions, IV fluids, air in theatres and laminar flow
Radiation — ionising (gamma, cobalt-60)	'Cold sterilisation', industrial	Disposable syringes, needles, gloves, catheters, sutures, plastic ware

Method	Details, temperature and time	Used for
Radiation — non-ionising (UV, 254 nm)	Poor penetration; surface and air only	Operation theatre, laboratory hoods, water treatment (does not penetrate glass or dust)
Gas — ethylene oxide (ETO)	Needs humidity and long aeration; toxic, inflammable and carcinogenic	Heat-sensitive items: plastics, endoscopes, heart-lung machine, ventilator tubing, sutures, pacemakers
Gas — formaldehyde, hydrogen peroxide plasma, peracetic acid	Low-temperature methods	Endoscopes, delicate instruments, room fumigation
Chemical (cold) sterilisation	Glutaraldehyde 2% (Cidex) — 20 min for disinfection, 6-10 h for sterilisation; ortho-phthalaldehyde	Endoscopes, cystoscopes, plastic and rubber goods, anaesthetic tubes

NUMBERS TO REMEMBER

Autoclave = 121 °C, 15 lb/sq in, 15-20 minutes (moist heat, most efficient). **Hot air oven = 160 °C for 2 hours** (dry heat).

Sterilisation controls: *physical* (thermocouple, gauge), *chemical* (Browne's tube, autoclave tape, Bowie-Dick test), *biological* — *~Geobacillus stearothermophilus~* for the autoclave and *~Bacillus atrophaeus~ (subtilis)* for the hot air oven and ETO.

Order of resistance (most resistant first): **prions > bacterial spores > mycobacteria > non-lipid viruses > fungi > vegetative bacteria > lipid viruses.**

7.7 Disinfectants and Antiseptics

Table 7.5 Common disinfectants and antiseptics

Agent	Strength used	Uses and remarks
Alcohol — ethyl 70%, isopropyl 60-70%	60-80%	Skin before injection, thermometers, hand rub; <i>not sporicidal</i> ; needs 30 s contact; inflammable
Povidone-iodine (Betadine)	5-10% solution, 7.5% scrub	Skin preparation before surgery, wounds, gargle; stains; avoid in newborns and iodine allergy
Chlorhexidine gluconate	0.5% in alcohol, 2-4% scrub, 0.2% mouth wash	Surgical scrub, skin preparation (best for central lines), oral care; <i>keep out of the eyes and ears</i>
Sodium hypochlorite (bleach)	0.5-1% for surfaces and blood spills; 0.05% for clean surfaces	Floors, spills, linen, excreta, dialysis; corrodes metal; inactivated by organic matter; <i>never mix with acid or ammonia (toxic gas)</i>
Bleaching powder	33% available chlorine ; 5% solution	Excreta, latrines, wells (1 ppm residual chlorine in drinking water), drains
Phenol (carbolic acid) / cresol / Lysol	Phenol 1-2%, Lysol 2-5%	Floors, drains, excreta; irritant, absorbed through the skin, avoid in nurseries
Hydrogen peroxide	3% (6% for disinfection)	Wound cleaning (effervescence lifts debris), mouth wash diluted; unstable in light
Potassium permanganate (KMnO₄)	1:5000 to 1:10 000	Sitz bath, wet dressings, gargle, fungal infections of the foot, well disinfection
Formaldehyde / formalin	10% solution, 40% formalin	Fumigation, preservation of specimens; irritant, carcinogenic
Glutaraldehyde 2%	20 min-10 h	Endoscopes, plastic and rubber; irritant to eyes and skin
Quaternary ammonium compounds (cetrimide, benzalkonium)	0.1-1%	Cleaning skin, floors; <i>inactivated by soap</i> , poor against <i>Pseudomonas</i> and spores
Ethylene oxide gas	-	Heat-sensitive articles (see Table 7.4)
Silver sulphadiazine / mupirocin	1% cream / 2% ointment	Burns / staphylococcal skin infection
Boric acid, gentian violet, soap and water	-	Eye wash (2%), oral thrush, plain mechanical cleaning

DEFINITION

Spaulling classification of instruments — *Critical* items enter sterile tissue or the blood stream (surgical instruments,

implants, needles) and need **sterilisation**; *Semi-critical* items touch mucous membranes (endoscopes, laryngoscope blades, thermometers) and need **high-level disinfection**; *Non-critical* items touch intact skin only (BP cuff, stethoscope, bedpan) and need **low-level disinfection/cleaning**.

7.8 Standard (Universal) Precautions and Barrier Nursing

- **Standard precautions** apply to **every patient, at all times, irrespective of diagnosis**, treating all blood, body fluids, secretions, excretions (except sweat), non-intact skin and mucous membranes as potentially infectious.
- Components: **hand hygiene; personal protective equipment (PPE); safe injection practice; safe handling and disposal of sharps; respiratory hygiene and cough etiquette; cleaning of equipment and environment; safe handling of linen; and waste management**.
- **Never recap a used needle** (if unavoidable use the one-handed scoop technique); never bend, break or hand-pass a needle; drop it directly into a **puncture-proof container** that is sealed at three-quarters full.
- **Transmission-based precautions** are added to standard precautions: *Contact* (gown and gloves, single room, dedicated equipment — MRSA, C. difficile, scabies); *Droplet* (surgical mask within 1 m, patient wears a mask when moved — influenza, pertussis, mumps, meningococcus); *Airborne* (**N95 respirator, negative-pressure single room with 6-12 air changes, door closed** — tuberculosis, measles, chickenpox).

Personal protective equipment — order of use

Table 7.6 Sequence of donning and removing PPE

Step	Donning (putting on)	Doffing (removing)
1	Hand hygiene	Gloves (most contaminated — remove first)
2	Gown/apron	Goggles/face shield
3	Mask / N95 respirator (fit check)	Gown
4	Goggles or face shield	Mask/respirator (by the strings, from behind)
5	Gloves (last)	Hand hygiene (and again after leaving the room)

Barrier nursing and isolation

- **Barrier nursing** — nursing an infectious patient in a separate room or cubicle with a 'barrier' of gowns, masks, gloves and separate equipment, so that infection does not pass out to others.
- **Reverse (protective) barrier nursing** — protecting a **highly susceptible patient** (neutropenic, on chemotherapy, transplant recipient, extensive burns, severe combined immunodeficiency, preterm baby) from organisms brought in by staff and visitors: laminar-flow positive-pressure room, sterile linen, cooked food, no flowers or fresh fruit, no visitor with any infection.
- **Requirements of an isolation unit** — single room (preferably with an ante-room and attached toilet), door kept closed with a notice, separate gowns/masks/gloves outside the door, separate linen, crockery, thermometer, BP cuff and bedpan, a bowl of disinfectant, a covered bucket for soiled linen, a hand-washing basin and a chart kept outside the room.
- **Concurrent disinfection** throughout the illness and **terminal disinfection** after discharge (see Chapter 3).
- Psychological care of the isolated patient is important — he feels lonely, rejected and anxious; explain the reason, visit frequently, provide a phone/TV/books, and allow limited screened visitors.

7.9 Bio-Medical Waste Management (BMW Rules, 2016 — India)

Table 7.7 Bio-medical waste colour coding (very frequently asked)

Colour of bag/container	Category of waste	Examples	Treatment and disposal
Yellow	Human and animal anatomical waste, soiled waste, expired/discarded medicines, chemical waste, microbiology and laboratory waste, blood bags, used linen soiled with blood/body fluid	Placenta, body parts, dressings, cotton, plaster casts, expired drugs, blood bags, laboratory cultures	Incineration / plasma pyrolysis / deep burial (only in rural areas); chemical disinfection for laboratory waste

Colour of bag/container	Category of waste	Examples	Treatment and disposal
Red	Contaminated recyclable plastic waste	Syringes without needles, IV sets and tubing, catheters, urine bags, gloves, vacutainers	Autoclave/microwave/hydroclave, then shred and send for recycling to a registered recycler
White (translucent, puncture-proof)	Sharps	Needles, needles with fixed syringes, scalpels, blades, broken glass ampoules	Autoclave or dry-heat sterilise, then shred/encapsulate/concrete pit; final disposal to a metal foundry
Blue (cardboard box with blue marking)	Glassware and metallic implants	Broken/discarded glass bottles and vials, medicine vials, ampoules, metallic body implants	Disinfection (1% hypochlorite) or autoclaving , then recycling
Black (general, not bio-medical)	General non-infectious municipal waste	Paper, packaging, kitchen and office waste, flowers	Ordinary municipal solid waste disposal
Cytotoxic waste	Cytotoxic drugs and their containers, contaminated items	Anticancer drug vials, syringes	Yellow bag with 'cytotoxic' label — incineration at 1200 °C or encapsulation; <i>never landfill</i>

NUMBERS TO REMEMBER

- Bags must not be filled beyond **three-quarters**, must be labelled with the biohazard/cytotoxic symbol, and must be stored for **not more than 48 hours** before treatment.
- Liquid infectious waste is disinfected with **1-2% sodium hypochlorite** before discharge into the drain.
- Needles are destroyed at the point of generation with a **needle destroyer/hub cutter**.
- Every health-care facility must keep records for **5 years**, report annually to the State Pollution Control Board by **30 June**, and immunise all staff against **hepatitis B and tetanus**.
- Deep burial pit: **at least 2 metres deep**, away from habitation and water sources — permitted only for towns with under 5 lakh population.

7.10 Injection Safety and Post-Exposure Prophylaxis

- **One needle, one syringe, one patient, one time** — never re-use a syringe, needle or lancet, and never re-enter a single-dose vial.
- Reconstitute with a new needle and syringe; clean the vial top with alcohol and let it dry; avoid touching the needle or the plunger shaft.
- **After exposure** (needle stick, splash to mucosa or broken skin): do not squeeze or suck the wound; **wash with soap and running water** (mucosa and eyes with plenty of water/saline); do not use bleach or spirit on the wound; report immediately and record; test the source and the exposed person.
- **HIV PEP** — start as soon as possible, ideally **within 2 hours (not later than 72 hours)**, continue **28 days** (usually tenofovir + lamivudine + dolutegravir); test at 6 weeks, 3 and 6 months.
- **Hepatitis B** — if unvaccinated, give **hepatitis B immunoglobulin (HBIG) plus the vaccine within 24 hours** (preferably) and complete the schedule; **hepatitis C** — no prophylaxis available, only follow-up testing.
- Approximate risk per percutaneous exposure: **hepatitis B 6-30%, hepatitis C 1.8%, HIV 0.3%** (mucous membrane splash 0.09%).

CHAPTER AT A GLANCE

- Disinfection kills pathogens (not spores); sterilisation kills everything including spores.
- Chain of infection: agent → reservoir → exit → transmission → entry → susceptible host; hand hygiene is the best single break.
- Airborne (needs N95 + negative pressure): TB, measles, chickenpox. Droplet (surgical mask): influenza, pertussis, mumps, COVID.
- Autoclave 121 °C/15 lb/15-20 min; hot air oven 160 °C/2 h; boiling 20-30 min; ETO for heat-sensitive items; glutaraldehyde 2% for endoscopes.
- Biological indicators: *G. stearothermophilus* (autoclave), *B. atrophaeus* (hot air oven/ETO).
- Commonest nosocomial infection = catheter-associated urinary tract infection.
- BMW 2016: Yellow = anatomical/soiled/expired drugs (incinerate); Red = recyclable plastics (autoclave + shred); White =

sharps; Blue = glass and metal implants; Black = general.

- PPE: gloves last on, gloves first off. Never recap a needle.
- PEP for HIV within 2 h (max 72 h) for 28 days; HBIG + vaccine within 24 h for hepatitis B.

Surgical Techniques

Asepsis, sterile technique, wounds and healing, dressings, sutures, drains and minor surgical procedures

8.1 Medical Asepsis and Surgical Asepsis

Table 8.1 Medical vs surgical asepsis

Point	Medical asepsis (clean technique)	Surgical asepsis (sterile technique)
Aim	Reduce the number and spread of organisms	Complete absence of all organisms including spores
Used in	Routine ward work — bed making, bathing, giving oral drugs, handling bedpans	Operation theatre, dressings, catheterisation, injections, IV cannulation, lumbar puncture, labour room
Practices	Hand washing, gloves, clean linen, disinfection, isolation	Sterile gloves, sterile gown and drapes, sterile field, sterilised instruments, mask and cap
Rule	'Clean' and 'dirty' areas are kept separate	Once contaminated, the article is no longer sterile and must be discarded

Principles of the sterile field (learn these rules)

1. A sterile object touched by an unsterile object becomes **unsterile**; when in doubt, consider it unsterile.
2. Only sterile articles may be placed in a sterile field; the **outer 2.5 cm (1 inch) margin** of a sterile drape is considered unsterile.
3. A sterile field must be kept **in view** and **above waist level**; anything below the waist or out of sight is unsterile.
4. **Never reach across a sterile field**, never turn your back on it, and never talk, cough or sneeze over it.
5. **Moisture carries organisms** — a wet sterile field (strike-through) is contaminated.
6. Air currents carry organisms — do not fan, shake linen or allow fans over the field; keep doors closed and traffic minimal.
7. Sterile persons keep their hands **in front, between the shoulders and the waist**, and pass each other back-to-back; the gown is considered sterile only in front from the chest to the waist, and the sleeves to 5 cm above the elbow.
8. Pour lotions without touching the container to the bowl; hold the bottle lip away from the sterile field and discard the first few drops.
9. Check the **sterility indicator, packing integrity and expiry date** before opening; open the first flap away from you and the last towards you.
10. Use a **sterile transfer forceps (cheatle forceps)** kept in a disinfectant, with the tips down; never touch the sterile item with the hands.

8.2 Surgical Hand Scrub, Gowning and Gloving

- **Surgical scrub** — remove all jewellery and nail polish; wet the hands and forearms; apply an antiseptic (**chlorhexidine 4% or povidone-iodine 7.5%**); scrub with a sterile brush and follow the anatomical time or count method for **3-5 minutes** (first scrub of the day 5 min, subsequent 3 min), cleaning the nails, all four surfaces of each finger, the palm, the back and the forearm **up to 5 cm above the elbow**; keep the **hands above the elbows** at all times so that water runs from the clean hands towards the elbows; rinse from the finger tips downwards; dry with a sterile towel from the fingers to the elbow, using a different side for each arm.
- Alcohol-based surgical hand rub (with persistent activity) is an accepted alternative and is less damaging to the skin.
- **Closed gloving** — the hands stay inside the gown sleeves until the gloves are drawn over the cuffs (preferred in the theatre). **Open gloving** — the folded cuff is handled by the inside surface only (used in the ward for dressings and catheterisation).
- Gowning: hold the gown at the neck, let it unfold away from the body without touching anything, slip the arms in, and let the circulating nurse tie it from behind.
- **If a glove is punctured or touched by anything unsterile, change it at once** — with a fresh scrub if necessary.

8.3 Wounds — Types, Healing and Complications

Table 8.2 Classification of wounds

Classification	Types
By skin break	Closed (contusion/bruise, haematoma) and open (skin broken)
By nature of injury	Incised (clean cut by a sharp blade — bleeds freely, heals best); Lacerated (torn, ragged edges); Abrasion (graze of superficial layers); Puncture/stab (small entry, deep track, high infection risk — tetanus); Penetrating and perforating (enters / passes through); Contused (blunt, bruised); Gunshot ; Crush ; Avulsion/deglowing ; Bite (highly infected — rabies risk)
By degree of contamination	Clean (elective, no inflammation, no entry into the respiratory, GI or genito-urinary tract — infection rate 1-2%); Clean-contaminated (controlled entry into those tracts); Contaminated (fresh traumatic wound, spillage); Dirty/infected (old traumatic wound, pus, perforated viscus)
By depth	Superficial, partial-thickness, full-thickness, deep (involving muscle, tendon, bone)

Phases of wound healing

Table 8.3 Phases of wound healing

Phase	Time	Events
1. Haemostasis	Immediate (minutes)	Vasoconstriction, platelet plug and clot formation
2. Inflammatory (lag/exudative) phase	Day 0-4 (up to 5)	Vasodilatation, neutrophils and macrophages clean the wound; redness, heat, swelling, pain and loss of function
3. Proliferative (fibroblastic/granulation) phase	Day 3-21	Fibroblasts lay collagen, new capillaries form (angiogenesis), granulation tissue and epithelialisation; wound contraction
4. Maturation (remodelling) phase	Day 21 to 1-2 years	Collagen re-organised, scar becomes pale and strong — reaching only about 70-80% of the original strength

- **Healing by first intention (primary union)** — clean, sutured wound with apposed edges — minimal granulation, fine scar, fastest.
- **Healing by second intention** — wide, infected or gaping wound left open — heals by granulation, contraction and epithelialisation, leaving a broad ugly scar.
- **Healing by third intention (delayed primary closure)** — wound left open for a few days until clean, then sutured.

Factors influencing wound healing

Table 8.4 Factors affecting wound healing

Favourable	Unfavourable (delay healing)
Young age, good blood supply, adequate oxygen	Old age, ischaemia (peripheral vascular disease), anaemia, hypoxia, smoking
Good nutrition — protein, vitamin C, vitamin A, zinc, iron	Malnutrition, protein and vitamin C deficiency (defective collagen), obesity
Clean, moist wound; approximated edges; immobilisation and rest	Infection (the commonest cause of delay) , foreign body, dead tissue/slough, haematoma, repeated trauma, movement
Control of diabetes; absence of systemic disease	Diabetes mellitus , uraemia, jaundice, malignancy, immunosuppression, radiotherapy
No interfering drugs	Corticosteroids , cytotoxic drugs, NSAIDs, anticoagulants

Complications of wounds

- **Haemorrhage** — primary (at the time of injury), **reactionary** (within 24 h, as BP rises or a ligature slips), **secondary** (after **7-14 days**, due to infection eroding a vessel).
- **Infection** — appears 3-7 days after surgery with pain, redness, swelling, heat, purulent discharge and fever; commonest organism *Staphylococcus aureus*.
- **Wound dehiscence** (bursting open of the layers, typically on the **5th-10th post-operative day**, often heralded by a serosanguineous 'pink' discharge) and **evisceration** (protrusion of viscera — a surgical emergency: place the patient supine with knees flexed, cover the bowel with a **sterile saline-soaked dressing**, do not push it back, nil by mouth, inform the surgeon at once).

- Sinus, fistula, keloid and hypertrophic scar, contracture, incisional hernia, adhesions, gangrene, tetanus, gas gangrene.
- Chronic wounds/ulcers — pressure sore, venous ulcer, diabetic foot ulcer, arterial ulcer.

8.4 Dressing a Wound

Purposes of a dressing

- To protect the wound from injury and contamination; to absorb discharge; to apply medication; to give support and immobilisation; to arrest bleeding by pressure; to keep the wound **moist and at body temperature** (which speeds healing) and to hide disfigurement.

Articles on the dressing trolley

Table 8.5 Dressing trolley

Top (sterile) shelf	Bottom (unsterile) shelf
Sterile dressing pack — 2 pairs of artery/dissecting forceps, sponge-holding forceps, scissors, kidney tray, gallipots, cotton balls, gauze pieces, pads, drape/towel, sterile gloves, probe, sinus forceps	Mackintosh and towel, bandages and adhesive plaster, scissors, bowl of antiseptic lotion (normal saline, povidone-iodine, hydrogen peroxide), prescribed ointment, paper bag/kidney tray for soiled dressings, hand sanitiser, gloves, spirit swabs, specimen container for culture, extra bandages and a waste bag, sharps container

Procedure of a simple aseptic dressing

1. Verify the order, explain to the patient, give an analgesic 30 min before if the dressing is painful; screen the bed; close fans and windows; do not dress during ward cleaning or bed making.
2. **Wash hands**; clean the trolley from top to bottom with an antiseptic; arrange the articles; open the sterile pack without touching the inside.
3. Position the patient, expose only the wound, place a mackintosh and towel under the part.
4. Wear clean gloves and remove the outer dressing; **moisten the adherent inner layer with saline** and remove it gently in the direction of hair growth, *never dry-pull*; observe the amount, colour and odour of the discharge; discard into the paper bag; remove the gloves.
5. Perform hand hygiene; wear **sterile gloves** and arrange the sterile field.
6. Clean the wound with saline on a swab **from the centre outwards / from the cleanest to the dirtiest area**, one swab per stroke, discarding each swab; for a drain, clean **around the drain from the inner to the outer area**. In a **sutured wound**, swab along the incision line first, then the surrounding skin.
7. Dry, apply the prescribed medication, cover with sterile gauze and a pad **extending at least 2.5 cm beyond the wound margin**, and secure with adhesive tape (applied across, not along, the limb) or a bandage.
8. Position the patient comfortably; remove the gloves; discard the waste in the correct colour-coded bag; wash and disinfect instruments; wash hands.
9. **Record** the date and time, the appearance of the wound and surrounding skin, the type and amount of discharge, the medication used, the condition of sutures/drain, the amount of drainage and the patient's response to the procedure.

CAUTION / NEVER DO THIS

- Dress **clean wounds before infected wounds**; a patient with an infected wound is dressed last, and the trolley is re-sterilised afterwards.
- Never use the same swab twice; never use the same forceps for a clean and a dirty wound (use the '2 forceps technique' — one to hold the swab that touches the wound, one to hand over the material).
- Do not cough, sneeze or talk over an open wound; wear a mask if you have a cold.
- Never use cotton wool directly on an open wound (fibres stick); use gauze.

Types of dressings

Table 8.6 Types of dressings

Dressing	Description / indication
Dry gauze / gauze and pad	Clean, sutured, dry wounds
Wet-to-dry saline dressing	Mechanical debridement of sloughy wounds

Dressing	Description / indication
Hydrocolloid (e.g. Duoderm)	Pressure sores, minimal-exudate wounds; keeps the wound moist and can stay 3-7 days
Hydrogel	Dry, necrotic wounds; rehydrates and promotes autolytic debridement
Alginate (seaweed)	Heavily exuding wounds and cavities; also haemostatic
Foam	Moderate-to-heavy exudate; cushions pressure areas
Transparent film (Tegaderm)	IV sites, superficial wounds — allows inspection
Paraffin gauze/tulle gras, silver or honey dressing	Burns, skin graft donor sites, infected wounds
Pressure dressing	Controls bleeding and oozing after surgery
Negative pressure (vacuum) dressing	Large cavity wounds, dehiscence, diabetic foot

8.5 Sutures, Staples and Drains

Table 8.7 Suture materials

Suture material	Type	Notes
Catgut (plain 7-10 days, chromic 14-21 days)	Absorbable, natural	Ligating vessels, subcutaneous tissue; now largely replaced
Polyglactin (Vicryl), polyglycolic acid (Dexon), poliglecaprone (Monocryl), PDS	Absorbable, synthetic	Deep layers, bowel, subcuticular closure
Silk, linen, cotton	Non-absorbable, natural (braided)	Skin, bowel, ligation; braided sutures hold knots well but harbour organisms
Nylon (Ethilon), polypropylene (Prolene), stainless steel	Non-absorbable, synthetic (monofilament)	Skin closure, hernia mesh, tendon and vascular repair; low infection risk
Skin staples, adhesive strips (Steri-strips), tissue glue	-	Quick skin closure; strips used after removing alternate sutures

- Suture **sizes** run 5, 4, 3, 2, 1, 0, 2-0, 3-0 ... 10-0; the **more zeros, the finer the thread** (10-0 is used for eye and microvascular surgery; 1 or 2 for the abdominal wall).
- **Time of removal of skin sutures:** face and neck **3-5 days**; scalp, arm and hand 7-10 days; trunk/abdomen and chest **7-10 days**; back and over joints, leg and foot **10-14 days**; retention sutures 14-21 days. Children heal faster; diabetics, steroid users and the malnourished need longer.
- **Removing sutures** — sterile technique with stitch scissors/blade and forceps: clean the line, lift the knot with forceps, cut the suture **close to the skin on the side away from the knot** so that no part that was outside the skin is drawn through the tissue; pull out towards the wound; remove **alternate sutures first** and inspect for gaping; support the wound with adhesive strips; count and record the number removed.
- **Drains** — remove blood, pus, serum, bile or air and prevent collection: *open/passive* (corrugated rubber, Penrose, gauze wick), *closed* (Redivac/suction, chest tube with an underwater seal, T-tube for the bile duct, Malecot/Foley, Jackson-Pratt, Ryle's tube, Infant feeding tube). Nursing care: keep the bag **below the level of the wound** (never raise it above), never let it kink, measure and record the drainage every shift (**report over 100 mL/hour of fresh blood or a sudden stop of drainage**), keep the site dressed and dry, never empty a chest drain bottle or clamp a chest tube without an order, and clamp before moving only if instructed.

8.6 Common Surgical Instruments and their Use

Table 8.8 Basic surgical instruments

Instrument	Use
Scalpel (BP handle No. 3/4 with blade No. 10, 11, 15, 22)	Incising tissue
Artery (haemostatic) forceps — Spencer Wells, Kocher, Mosquito	Clamping bleeding vessels
Dissecting forceps (toothed and non-toothed), Allis, Babcock	Holding tissue; Babcock for bowel
Sponge-holding forceps (Rampley)	Holding swabs to paint the skin
Cheate/transfer forceps	Lifting sterile articles from a drum
Needle holder (Mayo-Hegar), suture needles (straight, curved, cutting, round)	Suturing

Instrument	Use
Scissors — Mayo (tissue), Metzenbaum (fine), stitch scissors	Cutting tissue and sutures
Retractors — Langenbeck, Deaver, Morris, self-retaining (Balfour)	Holding the wound open
Towel clips (Backhaus)	Fixing sterile drapes
Sinus forceps, probe, grooved director	Exploring a sinus or abscess cavity
Trocar and cannula, aspirating needle, syringes	Drainage of fluid, aspiration
Autoclave drum, sterilisation indicator tape, kidney tray, gallipot, instrument tray	Sterilisation and holding

8.7 Minor Surgical Procedures — Nursing Assistance

- **Incision and drainage of an abscess** — explain and obtain consent; prepare local anaesthetic (lignocaine 2%), scalpel, sinus forceps, drain/wick, specimen bottle for culture and sensitivity, dressings; after care — observe bleeding, change the dressing as ordered, complete the antibiotic course, warm compresses.
- **Suturing of a laceration** — clean with saline, check tetanus immunisation status, assist with local anaesthesia and suturing, give wound care instructions and the date of suture removal, watch for signs of infection.
- **Aspiration (pleural/ascitic/joint)** — position the patient correctly, keep the specimen bottles labelled, monitor vital signs during and after, apply a sterile dressing, strict asepsis (see Chapter 23).
- **Removal of a foreign body, nail avulsion, biopsy, catheterisation, venesection** — sterile tray, good light, correct position, specimen handling, documentation.
- **Tetanus prophylaxis in a wound** — clean wound with complete immunisation and the last dose under 5 years → nothing needed; unimmunised or dirty/deep/puncture wound → **tetanus toxoid (Td) plus tetanus immunoglobulin (TIG 250-500 IU)** and thorough surgical toilet with antibiotics.

CHAPTER AT A GLANCE

- Medical asepsis = clean technique; surgical asepsis = sterile technique; the outer 1 inch of a sterile drape and anything below the waist is unsterile; moisture contaminates.
- Surgical scrub 3-5 min with chlorhexidine 4% or povidone-iodine 7.5%; hands held above the elbows.
- Healing phases: haemostasis → inflammatory (0-4 d) → proliferative (3-21 d) → maturation (up to 2 years, 70-80% strength).
- First intention = clean sutured wound; second intention = granulation; third = delayed closure.
- Infection is the commonest cause of delayed healing; vitamin C and protein are essential.
- Clean the wound from the centre outwards, one swab one stroke; dress clean wounds before infected ones.
- Secondary haemorrhage occurs after 7-14 days from infection; dehiscence typically on day 5-10.
- Sutures: face 3-5 days, abdomen/trunk 7-10 days, over joints/leg 10-14 days; more zeros = finer suture.
- Keep drainage bags below the wound; evisceration — cover with sterile saline gauze, do not push back.

Diet

Nutrients, balanced diet, therapeutic diets, feeding the helpless patient, tube and parenteral feeding

9.1 Basic Concepts and Definitions

Table 9.1 Nutritional terminology

Term	Definition
Food	Any edible substance that supplies the body with nourishment
Nutrient	A chemical component of food required for growth, repair, energy and regulation — carbohydrate, protein, fat, vitamins, minerals, water (and dietary fibre)
Nutrition	The science of food and its relationship to health — the sum of the processes of ingestion, digestion, absorption, metabolism, utilisation and excretion
Dietetics	The practical application of the principles of nutrition to feeding individuals and groups
Diet	The usual food and drink of a person
Therapeutic (modified) diet	A normal diet modified in consistency, nutrients or energy to suit a disease condition
Balanced diet	A diet that contains all nutrients in the correct proportions and amounts to meet the body's requirements, with a small margin for lean periods
Malnutrition	Any disorder of nutrition — under-nutrition, over-nutrition or imbalance
RDA	Recommended Dietary Allowance — the amount of a nutrient sufficient for nearly all (97.5%) healthy persons in a group (given by ICMR-NIN in India)
Basal metabolic rate (BMR)	Energy needed at complete rest to maintain the vital functions — about 1 kcal/kg/hour (roughly 1600-1800 kcal/day in an adult male)
Anorexia / Cachexia	Loss of appetite / extreme wasting

9.2 The Nutrients — Functions, Sources and Deficiency

Table 9.2 The macronutrients

Nutrient	Energy value & requirement	Functions	Good sources	Deficiency
Carbohydrate	4 kcal/g; should give 55-60% of total calories	Chief and cheapest source of energy; spares protein; provides fibre; needed for fat metabolism	Cereals (rice, wheat, maize), millets, potato, sugar, jaggery, honey, fruits, milk (lactose)	Weakness, weight loss, ketosis, marasmus
Protein	4 kcal/g; 10-15% of calories; RDA about 0.8-1 g/kg/day (more in illness, pregnancy, lactation, burns: 1.2-2 g/kg)	Growth and repair of tissue, enzymes, hormones, antibodies, haemoglobin, plasma proteins (oncotic pressure), wound healing	Class I (complete, animal) — milk, egg (the reference protein, biological value 100), meat, fish, liver; Class II (incomplete, vegetable) — pulses/dal, soya, nuts, cereals	Kwashiorkor (oedema, moon face, flaky-paint dermatosis, fatty liver, apathy, hair changes), marasmus (severe wasting, under 60% weight, old-man face), poor healing, anaemia, hypoalbuminaemia, frequent infection
Fat	9 kcal/g (the most concentrated); 20-30% of calories (visible fat 20-30 g/day)	Energy store, insulation, protects organs, carries the fat-soluble vitamins A, D, E, K, supplies essential fatty acids (linoleic, alpha-linolenic), palatability	Ghee, butter, oils, nuts, oil seeds, milk, egg yolk, meat, fish oil	Growth retardation, dermatitis (essential fatty acid deficiency), deficiency of vitamins A, D, E, K; excess → obesity, atherosclerosis
Water	Nil calories; 2-3 L/day (30-35 mL/kg)	Solvent and transport medium, regulates temperature, lubricates, excretes waste; 60-70% of body weight	Drinking water, milk, juice, soup, fruits, vegetables	Dehydration — thirst, dry mouth, sunken eyes, poor skin turgor, concentrated scanty urine, tachycardia, fall of BP, confusion

Nutrient	Energy value & requirement	Functions	Good sources	Deficiency
Dietary fibre	Nil (non-digestible); 25-40 g/day	Adds bulk, prevents constipation, lowers cholesterol and blood sugar, gives satiety, prevents colon cancer, piles and diverticulosis	Whole grains, bran, oats, pulses, green leafy vegetables, fruit with skin, guava, isabgol	Constipation, piles, diverticulosis, obesity, raised cholesterol

NUMBERS TO REMEMBER

Energy values: carbohydrate 4 kcal/g • protein 4 kcal/g • fat 9 kcal/g • alcohol 7 kcal/g.

Egg protein has the highest **biological value (100)** and is the reference protein; egg is a complete food except for **vitamin C and iron** (and fibre).

1 g of nitrogen = 6.25 g of protein. Ideal ratio of calories: carbohydrate : protein : fat = 60 : 15 : 25 (approx).

9.3 Vitamins

Table 9.3 Vitamins — a very high-yield table

Vitamin	Functions	Sources	Deficiency disease
A (retinol / beta-carotene) — fat soluble; RDA 600-1000 mcg	Vision in dim light (rhodopsin), integrity of epithelium, immunity, growth	Liver, fish liver oil, milk, butter, egg yolk; carrot, papaya, mango, green leafy vegetables (carotene)	Night blindness → conjunctival xerosis → Bitot's spots → corneal xerosis → keratomalacia (blindness) ; follicular hyperkeratosis; <i>excess is teratogenic</i>
D (calciferol) — the 'sunshine vitamin'; 400-600 IU	Absorption of calcium and phosphorus; bone mineralisation	Sunlight on skin (chief source), fish liver oil, egg yolk, fortified milk	Rickets in children (bow legs, rickety rosary, craniotables, pot belly, delayed closure of fontanelle), osteomalacia in adults, tetany
E (tocopherol)	Antioxidant, protects cell membranes and red cells, fertility	Vegetable oils, wheat germ, nuts, green leafy vegetables	Haemolytic anaemia in preterm babies, neuropathy
K (phylloquinone)	Synthesis of prothrombin and clotting factors II, VII, IX, X	Green leafy vegetables, liver; synthesised by gut bacteria	Bleeding tendency, haemorrhagic disease of the newborn (hence 1 mg IM at birth), prolonged prothrombin time
B1 (thiamine)	Coenzyme in carbohydrate metabolism; nerve function	Whole grains, rice polishings, pulses, nuts, pork, yeast (lost in polishing rice and in cooking soda)	Beri-beri — dry (peripheral neuritis, burning feet, wrist drop) and wet (oedema, heart failure); Wernicke-Korsakoff psychosis in alcoholics
B2 (riboflavin)	Flavoprotein coenzymes in oxidation	Milk, egg, liver, green leafy vegetables, pulses	Angular stomatitis, cheilosis, glossitis, magenta tongue, photophobia, scrotal dermatitis
B3 (niacin / nicotinic acid)	NAD/NADP coenzymes; can be made from tryptophan	Liver, meat, fish, groundnut, whole grains (maize-eaters are at risk — niacin is bound)	Pellagra — the 3 D's: Dermatitis (in sun-exposed areas, Casal's necklace), Diarrhoea, Dementia (4th D = death)
B6 (pyridoxine)	Amino acid metabolism, haem synthesis	Meat, liver, whole grains, banana, nuts	Peripheral neuritis (<i>isoniazid therapy — hence give pyridoxine with INH</i>), convulsions in infants, sideroblastic anaemia
B9 (folic acid) — 200-500 mcg (pregnancy 400-600 mcg)	DNA synthesis, red cell maturation, prevents neural tube defects	Green leafy vegetables, liver, pulses, nuts (destroyed by cooking)	Megaloblastic anaemia, glossitis, neural tube defect (spina bifida) in the fetus
B12 (cyanocobalamin) — 1-2 mcg	DNA synthesis, myelin formation; needs intrinsic factor from the stomach for absorption	Only animal foods — liver, meat, fish, egg, milk (strict vegans are at risk)	Pernicious (megaloblastic) anaemia with subacute combined degeneration of the cord (glossitis, paraesthesia, ataxia)
C (ascorbic acid) — 40-80 mg	Collagen formation and wound healing , antioxidant, enhances iron absorption , immunity	Amla (richest), guava, citrus fruits, tomato, green chilli, sprouts, cabbage (easily destroyed by heat, light and cooking soda)	Scurvy — spongy bleeding gums, petechiae, subperiosteal haemorrhage, painful swollen joints, delayed wound healing, anaemia

9.4 Minerals

Table 9.4 Important minerals

Mineral	Functions and requirement	Sources	Deficiency / excess
Calcium (RDA adult about 600-1000 mg; pregnancy/lactation 1200 mg)	Bone and teeth (99%), clotting, muscle contraction, nerve conduction, enzyme action	Milk and milk products, ragi, til (sesame), green leafy vegetables, small fish with bones	Rickets, osteomalacia, osteoporosis , tetany (carpopedal spasm, positive Trousseau and Chvostek signs)
Phosphorus	Bone, ATP, nucleic acids, buffers	Milk, cereals, pulses, meat, egg	Rickets, weakness
Iron (RDA man 17-19 mg, woman 21-29 mg, pregnancy 27-35 mg)	Haemoglobin, myoglobin, cytochromes; haem iron (animal) is much better absorbed than non-haem (plant) ; absorption is increased by vitamin C and decreased by phytates, tannin (tea), calcium and antacids	Liver, meat, fish, egg yolk, green leafy vegetables, jaggery, dates, ragi, pulses	Iron deficiency (microcytic hypochromic) anaemia — pallor, weakness, koilonychia, glossitis, pica; the commonest nutritional deficiency in the world
Iodine (RDA 150 mcg; pregnancy 250 mcg)	Thyroid hormones T3 and T4 (growth and metabolism)	Iodised salt , sea fish, sea weed, milk	Goitre, hypothyroidism, cretinism (in children — mental and physical retardation, deaf-mutism), still birth, myxoedema; <i>India: iodised salt must contain at least 15 ppm iodine at the consumer level</i>
Sodium / Chloride	Extracellular fluid volume, osmotic pressure, nerve conduction; need under 5 g of salt (2 g sodium)/day	Common salt, processed and preserved food, papad, pickle	Hyponatraemia (confusion, fits) with vomiting/diuretics; excess → hypertension, oedema
Potassium	Main intracellular cation; cardiac and muscle function	Banana, citrus fruits, coconut water, potato, tender coconut, dal, dry fruits	Hypokalaemia (diarrhoea, vomiting, diuretics) → weakness, arrhythmia, ileus; hyperkalaemia (renal failure) → cardiac arrest
Zinc	Wound healing, immunity, growth, taste, insulin	Meat, liver, egg, whole grains, pulses, nuts	Poor healing, growth retardation, diarrhoea, loss of taste, hypogonadism, acrodermatitis
Magnesium, copper, selenium, fluoride, chromium	Enzyme cofactors; fluoride prevents dental caries	Green vegetables, nuts, whole grains, water	Fluoride: under 0.5 ppm → dental caries; over 1.5-2 ppm → dental and skeletal fluorosis (optimum in drinking water about 0.5-0.8 ppm in India)

9.5 The Balanced Diet and the Food Groups

Table 9.5 The five food groups

Food group (ICMR)	Examples	Chief nutrients supplied
1. Cereals, millets and pulses	Rice, wheat, jowar, bajra, ragi, dal, rajma	Energy, protein, B vitamins, iron, fibre
2. Vegetables and fruits	Green leafy vegetables, roots and tubers, other vegetables, all fruits	Vitamins A and C, folate, minerals, fibre, antioxidants
3. Milk and milk products; egg, meat and fish	Milk, curd, paneer, egg, chicken, fish	Good-quality protein, calcium, vitamin B12, vitamin A
4. Oils, fats, nuts and oilseeds	Groundnut oil, mustard oil, ghee, butter, almond, til	Energy, essential fatty acids, fat-soluble vitamins
5. Sugars	Sugar, jaggery, honey	Energy only ('empty calories')

Approximate daily requirement of an adult (sedentary, ICMR guidelines)

Table 9.6 Daily food guide for an adult

Food	Amount per day
Cereals and millets	300-400 g (sedentary man 375 g, woman 270 g)
Pulses	60-80 g (at least 30 g if non-vegetarian)
Milk / curd	300-500 mL
Green leafy vegetables	100 g
Other vegetables and roots/tubers	200 g + 100 g
Fruits	100-200 g
Visible fat / oil	20-30 g (up to 25 g for sedentary work)
Sugar / jaggery	Under 20-25 g
Nuts and oilseeds	20-30 g
Egg / meat / fish (non-vegetarian)	1 egg or 50-75 g
Salt	Under 5 g
Water	8-12 glasses (2-3 L)

NUMBERS TO REMEMBER

Approximate energy requirement: **sedentary adult man 2100-2320 kcal, moderate 2700 kcal, heavy 3500 kcal**; adult woman 1900-2100 / 2200 / 2900 kcal.

Pregnancy: + 350 kcal and + 15-23 g protein (2nd and 3rd trimester). **Lactation (0-6 months): + 600 kcal and + 19-25 g protein.**

Body Mass Index **BMI = weight (kg) / height (m)²**: under 18.5 = underweight • **18.5-22.9 normal (Asian)** • 23-24.9 overweight • 25 or more = obese (WHO: 25-29.9 overweight, 30+ obese).

Waist circumference of concern: **men over 90 cm, women over 80 cm** (Asian).

9.6 Therapeutic Diets — Modifications in Consistency

Table 9.7 Diets modified in consistency

Diet	Composition	Indications
Clear liquid diet	Water, clear broth/soup without fat, strained fruit juice, barley water, whey water, coconut water, black tea, glucose water, ORS, jelly — no milk or solids; about 400-500 kcal only	First 24-48 h after surgery, acute vomiting/diarrhoea, before bowel investigations, on return of bowel sounds
Full liquid diet	Everything liquid at body temperature — milk, custard, ice cream, strained soups, egg-flip, lassi, porridge, soft cereal gruel, fruit juice; can be nutritionally adequate with supplements	Difficulty in chewing/swallowing, jaw wiring, oral surgery, high fever, unconscious/tube feeding, after a clear liquid diet
Soft / bland diet	Soft-cooked, low-fibre, non-spicy food — khichri, dalia, curd, banana, boiled potato, soft chapatti, custard, cooked vegetables without skin	Convalescence, after surgery, peptic ulcer, gastritis, diarrhoea, elderly, small children, no teeth
Light / semi-solid diet	Between soft and normal diet, easily digestible	Convalescent and ambulant patients
Regular (normal/full) diet	Balanced diet of the household, no restriction	Patients with no dietary restriction
Mechanically altered/pureed/thickened	Minced, mashed, pureed or thickened with a commercial thickener	Dysphagia (stroke, motor neuron disease, Parkinsonism) — thin liquids are the most dangerous for aspiration

9.7 Therapeutic Diets in Disease

Table 9.8 Therapeutic diets in common diseases

Condition	Diet principles	Allow	Avoid / restrict
Fever / infection	High calorie (2500-3000 kcal), high protein, high fluid (3 L), plenty of vitamin C; small frequent bland feeds; extra 7% calories for every 1 °F rise	Milk, curd, soft khichri, egg, soup, juice, glucose water, ORS	Heavy fried and spicy food, high fibre
Diabetes mellitus	Calorie-controlled, high complex carbohydrate and fibre (30-40 g) , low glycaemic index, 3 meals plus 2-3 snacks at fixed times, carbohydrate distributed evenly, weight reduction if obese; protein 15-20%, saturated fat under 7%	Whole grains, millets, oats, pulses, green vegetables, fenugreek, salads, bitter gourd, whole fruit in measured amounts, artificial sweeteners	Sugar, jaggery, honey, sweets, glucose, soft drinks, fruit juice, dry fruits, banana/mango in excess, refined flour, fried food, alcohol; no skipping or delaying meals
Hypertension / heart disease	Low salt (under 5 g; 2-3 g in heart failure) , low saturated fat and cholesterol, high potassium, high fibre, weight reduction, DASH diet	Fruits, vegetables, whole grains, low-fat milk, fish, garlic, unsaturated oils	Salt at the table, papad, pickles, pappadam, chips, processed and canned food, ghee/butter/vanaspati, red meat, organ meat, egg yolk (limit), coffee, alcohol, smoking
Myocardial infarction (first days)	Soft, low-calorie, low-salt, small frequent feeds, no hot/cold extremes, no straining or gas-forming food	Liquids first, then soft diet	Caffeine, heavy meals, constipating food
Chronic kidney disease / renal failure	Low protein (0.6-0.8 g/kg, high biological value) , low sodium, low potassium, low phosphate , fluid restriction as per urine output, adequate calories from carbohydrate	Rice, sago, arrowroot, apple/papaya, gourd vegetables, leached (double-boiled) vegetables	Banana, citrus, coconut water, dry fruits, potato, tomato, dal in excess, milk in excess, salt, nuts, chocolate, cola
Nephrotic syndrome	Normal-to-high protein (1 g/kg), low salt , low fat	Egg white, fish, pulses	Salt, fried food
Renal stones	High fluid (3-4 L/day) ; for calcium oxalate stones, low oxalate	Plenty of water, citrus (citrate)	Spinach, tomato seeds, beetroot, chocolate, tea, nuts, excess salt and animal protein
Liver disease / hepatitis	High carbohydrate, moderate protein, low fat , no alcohol; small frequent feeds; vitamin supplements	Glucose, fruit juice, rice, potato, curd, egg white	Fried and fatty food, alcohol strictly , red meat
Hepatic coma (encephalopathy)	Protein restricted (20-40 g) or vegetable protein , high calorie, lactulose for constipation	Glucose drinks, vegetable protein	Animal protein, salt (if ascites), constipation
Peptic ulcer / gastritis	Bland, soft, small frequent meals (every 2-3 h), adequate protein, no long fasting	Milk, curd, banana, boiled vegetables, soft rice	Chilli, spices, pickles, fried food, tea/coffee, aerated drinks, alcohol, smoking, aspirin/NSAIDs
Diarrhoea	Fluid and electrolyte replacement with ORS , continue feeding (never starve), low fibre, low fat, small frequent feeds; continue breast feeding	ORS, rice water (kanji), curd, banana, apple, boiled potato, khichri, buttermilk, coconut water	Milk (if lactose intolerant), high fibre, fried, spicy, sugary drinks
Constipation	High fibre (30-40 g) and high fluid , regular meals and exercise	Whole grains, bran, papaya, guava, figs, prunes, green vegetables, isabgol, warm water in the morning	Refined flour, polished rice, low fluid, excess tea
Obesity	Calorie deficit of 500-1000 kcal/day for 0.5 kg/week loss; high fibre and protein, low fat and sugar; behaviour change and exercise (150 min/week)	Salads, whole grains, low-fat milk, sprouts, plenty of water	Sugar, fried food, sweets, ghee, refined flour, alcohol, snacking
Underweight / tuberculosis / cancer / HIV / burns	High calorie (35-45 kcal/kg), high protein (1.5-2 g/kg) , vitamins and minerals, small frequent energy-dense feeds	Milk shakes, banana, egg, ghee, nuts, paneer, chicken/fish, supplements	Bulky low-calorie food that fills the stomach

Condition	Diet principles	Allow	Avoid / restrict
Gout	Low purine, high fluid, weight reduction	Milk, egg, vegetables, cereals	Organ meat (liver, kidney, brain), red meat, sardine, shellfish, yeast extract, alcohol especially beer, pulses in excess
Celiac disease	Gluten-free	Rice, maize, millets, potato, soya	Wheat, barley, rye and oats (gluten)
Hyperthyroidism / Hypothyroidism	High calorie, high protein / calorie-controlled with adequate iodine	-	Goitrogens in excess (cabbage, cauliflower) in hypothyroidism
Anaemia	Iron and folate rich, vitamin C with meals to aid absorption	Green leafy vegetables, jaggery, dates, liver, meat, ragi, amla, sprouts	Tea and coffee with meals (tannins), antacids and calcium with iron
Pre-eclampsia in pregnancy	Adequate protein, moderate salt restriction, plenty of fluids, calcium	Milk, fruits, vegetables	Excess salt, processed food

9.8 Serving Food and Feeding the Helpless Patient

- **Before the meal** — remove bedpans, urinals and soiled dressings; ventilate the room; give mouth care and help with hand washing; place the patient in a **sitting or high Fowler's (at least 30-45°) position**; do not carry out treatments or investigations at meal times.
- Serve food **attractively, in the right quantity, at the right temperature** (hot food hot, cold food cold), on clean crockery, with the tray checked against the diet order and the patient's name.
- **Feeding** — sit at the patient's eye level, do not rush, offer small mouthfuls, tell a blind patient what is on the plate ('use the clock method'), allow the patient to hold the cup or spoon if he can, offer fluids between mouthfuls with a straw or feeding cup, give the food **on the unaffected side** in stroke patients.
- **Watch for aspiration** — coughing, choking, wet voice, drooling; keep suction ready for high-risk patients; **keep the patient sitting up for 30-60 minutes after the meal**.
- Encourage self-feeding with adapted spoons, plate guards and non-slip mats (rehabilitation); praise and do not hurry.
- **After the meal** — mouth care, hand washing, note and record the **amount actually eaten**, and report anorexia, nausea, dysphagia or refusal; record the intake on the fluid chart.

CLINICAL / PRACTICAL POINT

Never feed an **unconscious patient or a patient without a gag reflex by mouth** — the risk of aspiration pneumonia is very high. Use a nasogastric tube or intravenous route on the doctor's order.

9.9 Tube (Enteral) Feeding

- **Routes** — **nasogastric (Ryle's tube, the commonest)**, nasoduodenal/nasojejunal, **gastrostomy (PEG)** and jejunostomy for long-term feeding (over 4-6 weeks), and oro-gastric in newborns.
- **Indications** — unconscious patient, stroke with dysphagia, head and neck or oesophageal cancer, oesophageal stricture or burns, jaw fracture, severe anorexia, extensive burns, prematurity, tetanus, coma, post-operative feeding.
- **Tube sizes** — adults 12-18 Fr; children 8-10 Fr; infants 5-8 Fr. Length to insert = the **NEX measurement** (from the tip of the Nose to the Ear lobe to the Xiphisternum, about 45-55 cm in an adult).
- **Confirming the position (before every feed)** — best method is **measuring the pH of the aspirate (pH 5.5 or less = gastric)**; X-ray confirmation is the gold standard at the first insertion; also check the external length mark and inspect the mouth; *the old 'auscultate while pushing air' (whoosh) test is now considered unreliable*.
- **Procedure of feeding** — sit the patient at **30-45° and keep him so for 30-60 min after the feed**; check the position and the **gastric residual volume** (withhold and inform if it exceeds about 200-250 mL or the ordered limit; return the aspirate); feed at room/body temperature by gravity over 15-20 minutes (never force with a plunger); **flush with 30-50 mL of water before and after** the feed and after medicines; keep the feed volume to 200-400 mL per feed, 3-6 hourly, or use a continuous pump.
- Give crushed/liquid medicines separately, flushing between each; never mix drugs with the feed unless permitted.

- **Complications** — aspiration pneumonia (the most serious), tube blockage, displacement or coiling, diarrhoea or constipation, abdominal distension and cramps, hyperglycaemia, electrolyte imbalance, refeeding syndrome, nasal and oesophageal erosion, sinusitis and otitis.
- **Nursing care** — oral and nasal hygiene twice daily, change the fixing tape daily and alternate the nostril position, watch the skin at the nostril, keep the tube capped when not in use, change the tube as per policy, and record the type, amount and tolerance of each feed.

CAUTION / NEVER DO THIS

If the patient **coughs, chokes, becomes cyanosed or cannot speak** during insertion, the tube may be in the trachea — **withdraw it at once**. Never instil anything into a tube whose position has not been confirmed.

9.10 Parenteral Nutrition and Fluids

- **Total parenteral nutrition (TPN)** — all nutrients (glucose, amino acids, lipid emulsion, electrolytes, vitamins and trace elements) given **intravenously**, usually through a **central line** because the solution is hypertonic; peripheral partial nutrition (PPN) may be used for short periods.
- **Indications** — non-functioning gut: intestinal obstruction, high-output fistula, short bowel syndrome, severe pancreatitis, prolonged ileus, severe malabsorption, major burns, hyperemesis gravidarum.
- **Nursing care** — strict aseptic technique with the central line (the greatest risk is **catheter-related blood stream infection**), use a dedicated lumen, **never interrupt suddenly** (risk of rebound hypoglycaemia — taper off), monitor blood sugar 4-6 hourly, daily weight, strict intake-output, electrolytes and liver function, watch for fluid overload, hyperglycaemia, electrolyte disturbance and **refeeding syndrome** (low phosphate, potassium and magnesium), change the giving set every 24 hours, keep the bag refrigerated and use within 24 hours of hanging.
- **'If the gut works, use it'** — enteral feeding is always preferred as it is cheaper, safer and maintains the gut mucosa.

9.11 Food Hygiene and Food Poisoning

- **Rules of food hygiene** — wash hands before cooking and eating; use safe water; wash fruits and vegetables; cook food thoroughly (**above 70 °C in the centre**); eat food **freshly cooked (within 2 hours)**; store cooked food either **above 60 °C or below 5 °C** and reheat thoroughly; keep raw and cooked food separate; keep food covered and away from flies and pets; clean kitchen surfaces; do not let an ill person (diarrhoea, skin sores) cook for others; check expiry dates and do not use swollen, dented or leaking cans.
- **Food poisoning** — *Staphylococcus aureus* (toxin, 1-6 h, vomiting, from cream/milk/meat handled by a person with a skin lesion), *Bacillus cereus* (reheated rice), *Clostridium perfringens* (meat dishes), *Clostridium botulinum* (canned food — paralysis, diplopia, dysphagia; a medical emergency), *Salmonella* (12-36 h, egg and poultry), *Vibrio parahaemolyticus* (seafood), *E. coli*, mushroom and chemical poisoning.
- Danger zone for bacterial multiplication in food: **5-60 °C**.
- **Food-borne diseases** also include typhoid, cholera, hepatitis A and E, amoebiasis, giardiasis, taeniasis, trichinosis and brucellosis.

CHAPTER AT A GLANCE

- Carbohydrate 4, protein 4, fat 9 kcal/g; carbohydrate 55-60%, protein 10-15%, fat 20-30% of calories.
- Egg = reference protein (biological value 100), lacks vitamin C and iron.
- Vitamin A → night blindness/Bitot's spots; B1 → beri-beri; B2 → angular stomatitis; B3 → pellagra (3 D's); B12/folate → megaloblastic anaemia; C → scurvy; D → rickets/osteomalacia; K → bleeding.
- Iodine deficiency → goitre/cretinism; iodised salt at least 15 ppm; fluoride over 1.5 ppm → fluorosis.
- BMI = kg/m²; Asian normal 18.5-22.9. Pregnancy + 350 kcal and + 15-23 g protein; lactation + 600 kcal.
- Diets: clear liquid → full liquid → soft → normal; low protein in renal failure, low purine in gout, gluten-free in coeliac, low salt in hypertension.
- NG tube: NEX measurement, confirm by aspirate pH 5.5 or less, feed at 30-45°, flush 30-50 mL water, watch residual volume; aspiration is the most serious complication.
- 'If the gut works, use it' — enteral nutrition is preferred over TPN; TPN needs a central line and strict asepsis.
- Keep cooked food above 60 °C or below 5 °C; danger zone 5-60 °C.

Medicines

Routes, dosage forms, the rights of administration, prescription reading, dose calculation, injections and storage

10.1 Basic Terms

Table 10.1 Pharmacological terms

Term	Meaning
Drug / Medicine	Any substance used in the diagnosis, prevention, treatment or cure of disease (a 'medicine' is the drug in its final dosage form)
Pharmacology / Pharmacy	Science of drugs and their action on the body / art and science of preparing and dispensing medicines
Pharmacokinetics	What the body does to the drug — Absorption, Distribution, Metabolism, Excretion (ADME)
Pharmacodynamics	What the drug does to the body — mechanism of action, receptors, dose-response
Bioavailability	Fraction of an administered dose reaching the systemic circulation unchanged; 100% for the intravenous route
Generic / Brand (proprietary) name	Official non-proprietary name (paracetamol) / manufacturer's trade name (Crocin)
Therapeutic dose / Maximum dose / Minimum dose	Usual effective dose / largest safe dose / smallest effective dose
Loading dose / Maintenance dose	Large initial dose to reach the effective level quickly / dose to maintain that level
Therapeutic index	Ratio of toxic dose to therapeutic dose — a narrow therapeutic index (digoxin, phenytoin, lithium, warfarin, theophylline, aminoglycosides) needs monitoring
Side effect / Adverse drug reaction (ADR) / Toxic effect	Unwanted but predictable effect at normal dose / any noxious unintended response / effect of over-dosage
Idiosyncrasy / Allergy / Anaphylaxis	Genetically determined abnormal reaction / immune-mediated reaction / severe, life-threatening allergic reaction
Tolerance / Dependence / Addiction / Withdrawal	Need for a larger dose for the same effect / physical or psychological need / compulsive drug-seeking / symptoms on stopping
Cumulation	Accumulation of the drug when excretion is slower than administration (digoxin, heavy metals)
Synergism / Antagonism / Potentiation	Combined effect greater than the sum / one drug opposing another / one drug increasing the effect of another
Contraindication	A condition in which the drug must not be given
Placebo	An inert substance given as a medicine
Prophylactic / Palliative / Curative drug	Prevents disease / relieves symptoms / cures the disease
Antidote	A substance that counteracts a poison
First-pass effect	Metabolism of an oral drug in the liver/gut wall before it reaches the circulation (avoided by the sublingual, rectal and parenteral routes)

10.2 Routes of Administration

Table 10.2 Routes of drug administration

Route	Examples / method	Advantages	Disadvantages
Oral (per os, PO)	Tablets, capsules, syrups, suspensions, powders	Safest, cheapest, most convenient, self-administered, no asepsis needed	Slow onset, unsuitable if vomiting/unconscious/NPO, gastric irritation, first-pass metabolism, unpleasant taste, cannot be used in emergency
Sublingual / Buccal	Tablet under the tongue or in the cheek — nitroglycerine , isosorbide dinitrate, buprenorphine, ondansetron	Very rapid absorption, avoids first-pass metabolism , can be spat out	Only a few drugs, irritation, must not be swallowed or chewed

Route	Examples / method	Advantages	Disadvantages
Rectal	Suppositories, enema, ointment — diazepam, paracetamol, bisacodyl, indomethacin	Useful in vomiting, unconscious and uncooperative patients and children; partly avoids first pass	Embarrassing, erratic absorption, local irritation
Intramuscular (IM)	Deltoid, vastus lateralis, ventrogluteal, dorsogluteal; 1-5 mL	Rapid absorption, suitable for depot/oily preparations, can give irritant drugs	Painful, risk of nerve/vessel injury, abscess, contraindicated in bleeding disorders and shock
Subcutaneous (SC)	Outer upper arm, abdomen, anterior thigh — insulin, heparin, adrenaline, vaccines ; up to 1-2 mL	Slow, uniform absorption; can be self-administered; depot implants possible	Small volume only, irritant drugs cause necrosis, slower than IM
Intradermal (ID)	Inner forearm, deltoid — Mantoux test, BCG, allergy testing ; 0.1 mL	Very small dose, for diagnostic and immunisation purposes	Tiny volume, painful, needs skill
Intravenous (IV)	Bolus, infusion, via cannula or central line	Fastest action (bioavailability 100%) , exact dose, large volumes, drugs for emergency, irritant solutions can be given diluted	Most dangerous — no recall once given; risk of embolism, thrombophlebitis, infiltration, extravasation, infection, anaphylaxis; needs skill and asepsis
Inhalation	Metered dose inhaler with spacer, dry powder inhaler, nebuliser, oxygen, anaesthetic gases — salbutamol, budesonide, ipratropium	Direct action on the airway with a small dose and few systemic effects, rapid onset	Needs correct technique and coordination, local irritation, oral thrush with steroids (rinse the mouth)
Topical / transdermal	Ointment, cream, lotion, gel, paste, dusting powder, patch (nitroglycerine, fentanyl, hormones, nicotine)	Local action, steady blood level with patches, avoids first pass	Slow, may stain, systemic absorption if applied over a large or broken area
Instillation / local	Eye, ear and nose drops, ointment, gargle, throat paint, vaginal pessary, urethral, intrathecal, intra-articular, intracardiac, intraosseous	Direct local effect where needed	Needs asepsis; risk if the wrong strength or wrong site is used
Others	Intraperitoneal (dialysis), intrapleural, epidural/spinal, intra-arterial, intra-ocular, implant	Specialised indications	Performed by the doctor only

HIGH-YIELD / MOST ASKED

Order of speed of onset (fastest first): IV → inhalation/intraosseous → IM → SC → sublingual → rectal → oral → topical.
The IV route is the fastest and the most dangerous; the oral route is the safest and the commonest.

10.3 Dosage Forms

Table 10.3 Dosage forms

Group	Dosage forms
Solid oral	Tablet (plain, coated, enteric-coated, sustained/controlled release , chewable, dispersible, effervescent, sublingual), capsule (hard and soft gelatin, spansule), powder, granules, pill, lozenge/troche, sachet
Liquid oral	Syrup, elixir (hydro-alcoholic), linctus (for cough), suspension (shake well), emulsion, mixture, drops, tincture, spirit, solution, oral rehydration solution
Parenteral	Injection in ampoule, vial, pre-filled syringe, large-volume infusion; lyophilised powder for reconstitution; depot injection
Topical	Ointment (oily), cream (water-washable), gel/jelly, paste (thick), lotion, liniment (rubbed in), paint, dusting powder, spray, aerosol, medicated patch, medicated shampoo
Rectal and vaginal	Suppository, enema, rectal ointment, pessary, vaginal cream/tablet
Ophthalmic / otic / nasal	Eye drops and ointment (sterile, single patient, discard 4 weeks after opening), ear drops, nasal drops and spray, inhaler
Inhalational	MDI, dry powder inhaler, respirator (nebuliser) solution, oxygen, volatile anaesthetics

CAUTION / NEVER DO THIS

- **Never crush, break or chew an enteric-coated or sustained-release tablet or open a spansule capsule** (dose dumping / gastric irritation / loss of action).
- **Shake suspensions and emulsions well** before pouring; do not refrigerate a suspension unless directed.
- Keep a **separate marked measure/spoon** for external applications, and never store external preparations with oral medicines.
- Reconstituted syrups (many antibiotics) are usually stable only **7-14 days**, often needing refrigeration — write the date of reconstitution on the label.

10.4 The 'Rights' of Drug Administration

1. **Right patient** — check the identity by name band, asking the name and date of birth, and the bed number (never by bed number alone).
2. **Right drug** — read the label three times (when taking it out, while preparing, and before returning/discarding); check the generic name, not just the appearance.
3. **Right dose** — check the strength and calculate carefully; get high-risk calculations double-checked by a second person.
4. **Right route** — as written; never change the route on your own.
5. **Right time** — as prescribed (within 30 minutes of the scheduled time), noting before/after food.
6. **Right documentation** — record **immediately after** giving (never before); record refusal, omission and the reason.
7. **Right to refuse** — the informed patient may refuse; record and report it.
8. **Right assessment and right evaluation** — check the pulse before digoxin, BP before an antihypertensive, blood sugar before insulin, respiration before an opioid, and the response afterwards.
9. **Right education** — tell the patient what the drug is, why it is given, and what to expect.

CLINICAL / PRACTICAL POINT

General rules while giving medicines:

- **Never give a drug prepared by someone else**, and never leave a prepared dose unlabelled or unattended.
- **Never give a drug from an unlabelled, illegible or damaged container** — return it to the pharmacy.
- Check the **expiry date** and look for discolouration, precipitate, cloudiness, cracks and leakage.
- Check for **allergies** before every new drug; question any order that appears unusual, illegible or a wrong dose; do not accept a verbal/telephone order except in an emergency, and then write it down and get it signed.
- Stay with the patient until the drug is **actually swallowed** (especially in psychiatry and tuberculosis DOTS).
- Report and record any **medication error immediately** — the patient's safety comes before self-protection.
- Remember **LASA (look-alike, sound-alike)** drugs: *dopamine/dobutamine, hydralazine/hydroxyzine, chlorpromazine/chlorpropamide, cefotaxime/ceftazidime, metoprolol/misoprostol, insulin types* — read the full name and store them apart.

10.5 Reading a Prescription — Abbreviations and Latin Terms

Table 10.4 Prescription abbreviations

Abbreviation	Meaning	Abbreviation	Meaning
Rx	Recipe — 'take thou'	ac / pc	Before food / after food
OD / BD (BID) / TDS (TID) / QID	Once / twice / three times / four times a day	hs / nocte / mane	At bedtime / at night / in the morning
q4h, q6h, q8h	Every 4, 6 or 8 hours	SOS / PRN	If necessary / as and when required
STAT	Immediately, at once	NPO / NBM	Nothing by mouth
PO / PR / PV	By mouth / per rectum / per vagina	SL	Sublingual
IM / IV / SC / ID / IVI	Intramuscular / intravenous / subcutaneous / intradermal / IV infusion	OU / OD (eye) / OS	Both eyes / right eye / left eye
gtt / gtts	Drop / drops	AD / AS / AU	Right ear / left ear / both ears
mg / mcg (ug) / g / mL / L	Milligram / microgram / gram / millilitre / litre	IU / U	International unit / unit (<i>always write 'unit' in full — 'U' is misread as 0</i>)

Abbreviation	Meaning	Abbreviation	Meaning
tab / cap / syp / inj / oint	Tablet / capsule / syrup / injection / ointment	q / qs / ad lib	Each / a sufficient quantity / as desired
1/2, 1/4 tab, ss	Half, quarter tablet, one half	c / s (c̄ / s̄)	With / without
DNS / NS / RL / D5	Dextrose-normal saline / normal saline / Ringer lactate / 5% dextrose	O2, NG, RT	Oxygen, nasogastric, Ryle's tube

Parts of a prescription

- **Date; name, age, sex, weight and address of the patient; superscription (Rx); inscription** (name, strength and quantity of each drug); **subscription** (directions to the pharmacist); **signatura / transcription (Sig.)** — directions for the patient; **renewal/refill instructions**; and the **prescriber's signature, registration number and address**.
- A prescription for a **Schedule H, H1 or X** drug must be a written, signed prescription of a registered medical practitioner; **Schedule X** needs a prescription in duplicate, one copy retained by the pharmacist for **2 years**.

CAUTION / NEVER DO THIS

Error-prone abbreviations to be avoided (as per the 'Do Not Use' list):

- 'U' for unit (misread as 0 or 4) → write **'unit'**; 'IU' → write **'international unit'**.
- **A trailing zero** (1.0 mg read as 10 mg) — write '1 mg'; always use a **leading zero** ('0.5 mg', never '.5 mg').
- 'OD' for 'once daily' can be read as 'right eye'; 'µg' can be read as 'mg' → write **'microgram'**.
- 'MS/MSO4/MgSO4' — write **morphine sulphate** and **magnesium sulphate** in full.

10.6 Dose Calculation

NUMBERS TO REMEMBER

Basic formula: Dose to be given = (Dose ordered ÷ Dose on hand) × Quantity on hand (volume or number of tablets).

Example 1: Order 250 mg; stock is a 500 mg tablet → 250/500 = **half a tablet**.

Example 2: Order amikacin 375 mg; stock 500 mg in 2 mL → (375/500) × 2 = **1.5 mL**.

Example 3 (syrup): Order paracetamol 180 mg; syrup is 125 mg in 5 mL → (180/125) × 5 = **7.2 mL**.

IV drip rate (drops/minute) = (Total volume in mL × drop factor) ÷ (Time in minutes). Drop factor: **macro-drip 15 or 20 drops/mL (adult sets), micro-drip/paediatric 60 drops/mL**.

Example 4: 1000 mL over 8 hours with a 20 drops/mL set → (1000 × 20) ÷ (8 × 60) = **41-42 drops/min**.

Infusion time (hours) = Total volume ÷ mL per hour.

Body surface area (BSA, m²) = square root of [height (cm) × weight (kg) ÷ 3600] (Mosteller formula).

Paediatric dose calculation

Table 10.5 Paediatric dose rules

Rule	Formula
Body weight (the preferred method)	Dose = mg/kg × weight in kg (e.g. paracetamol 15 mg/kg/dose)
Body surface area (most accurate, used for cytotoxics)	Child dose = (BSA of child in m ² ÷ 1.73) × adult dose
Young's rule (1-12 years)	Child dose = adult dose × age in years ÷ (age + 12)
Clark's rule	Child dose = adult dose × weight in lb ÷ 150
Fried's rule (infants under 1 year)	Dose = adult dose × age in months ÷ 150
Dilling's rule	Child dose = adult dose × age in years ÷ 20

Household and metric measures

- 1 teaspoon = 5 mL • 1 dessertspoon = 10 mL • 1 tablespoon = 15 mL • 1 cup = 240 mL • 1 glass/tumbler = 200-250 mL • 15-16 drops = 1 mL (standard dropper, 20 drops for water).
- 1 g = 1000 mg; 1 mg = 1000 micrograms; 1 L = 1000 mL; 1 kg = 2.2 lb; 1 inch = 2.54 cm; 1 grain = 60-65 mg; 1 minim = 0.06 mL; 1 ounce = about 30 mL.
- 1 unit of insulin = 0.01 mL of U-100 insulin — always use an **insulin syringe marked in units**, never a tuberculin or ordinary syringe.

10.7 Administering Oral Medicines

1. Check the order, the patient's name, the drug, dose, route, time and allergies; wash hands.
2. Pour tablets into the cap/medicine cup **without touching them with the hand**; pour liquids with the **label uppermost** (so drips do not spoil the label), hold the measure at **eye level on a flat surface and read the lower meniscus**, and wipe the neck of the bottle.
3. Give with a **full glass of water** (unless restricted) in the **sitting or Fowler's position**; never give oral medicines to a lying-flat or drowsy patient.
4. Stay until the medicine is swallowed; check the mouth of a psychiatric or non-compliant patient.
5. Special points — give irritant drugs (iron, NSAIDs, steroids, metformin) **after food**; give sucralfate, ampicillin, levothyroxine, rifampicin and iron **on an empty stomach**; do not give tetracycline or iron with milk, antacids or calcium; do not crush enteric-coated/SR tablets; dilute and give liquid iron and potassium with a straw; give cough linctus **undiluted and without water afterwards**; sublingual tablets are **not swallowed**.
6. Record immediately and observe for the effect and for side effects.

10.8 Injections — Sites, Angles and Technique

Table 10.6 Injection sites and technique

Injection	Site	Needle & angle	Volume
Intradermal (ID)	Inner (flexor) surface of the forearm, upper chest, scapular area; BCG: left upper arm over the deltoid insertion	26-27 G, 10-15 mm, 10-15 degrees , bevel up — a bleb/wheel must form	0.1 mL (BCG 0.05 mL in newborns, 0.1 mL after 1 month)
Subcutaneous (SC)	Outer aspect of the upper arm, anterior thigh, abdomen (insulin — rotate sites, avoid 5 cm around the umbilicus) , upper back	25-26 G, 45 degrees (90 degrees with a short insulin needle or if the skin is pinched) ; <i>do not massage after heparin or insulin</i>	0.5-1 mL (max 2 mL)
Intramuscular (IM)	Deltoid (2-3 finger breadths below the acromion — vaccines, max 1-2 mL); Vastus lateralis (middle third of the outer thigh — the site of choice in infants); Ventrogluteal (safest in adults — palm on the greater trochanter, index finger on the anterior superior iliac spine, middle finger along the iliac crest, inject in the V); Dorsogluteal (upper outer quadrant of the buttock — risk of sciatic nerve injury, avoid in children under 3 years)	21-23 G, 1-1.5 inch, 90 degrees ; Z-track technique for irritant drugs (iron dextran, hydroxyzine, promethazine)	Adult 1-3 mL (up to 5 mL in the gluteal); deltoid 1 mL; infant 0.5-1 mL
Intravenous (IV)	Dorsum of the hand, forearm (cephalic, basilic), median cubital (for sampling), great saphenous, scalp veins in infants; central: subclavian, internal jugular, femoral	18-24 G cannula; 15-30 degrees , bevel up, tourniquet 10-15 cm above the site, released before injecting	Any volume, at the prescribed rate

General rules for injections

- Strict asepsis: wash hands, clean the top of the vial and the skin with a **spirit/alcohol swab in a circular motion from the centre outwards, and let it dry**; use a new sterile syringe and needle for each injection.
- Expel air completely; **aspirate before an IM injection** to make sure the needle is not in a vessel (*aspiration is no longer recommended for vaccines and for the ventrogluteal site*); if blood appears, withdraw and start again with a fresh syringe.
- Inject **slowly**, withdraw quickly, and apply gentle pressure with a dry swab (**do not rub after insulin, heparin or a vaccine**).
- Never inject into an inflamed, infected, scarred, oedematous, paralysed or hardened area; **rotate sites** in patients on repeated injections (insulin, heparin) to prevent lipodystrophy.
- Dispose of the needle and syringe **immediately into a puncture-proof container without recapping**.
- Keep **adrenaline (1:1000), hydrocortisone, an antihistamine, oxygen and an airway** ready whenever giving an injection or a vaccine — and observe the patient for **15-30 minutes** afterwards.

Complications of injections

- **Local** — pain, bleeding, bruising, abscess, cellulitis, sterile abscess, tissue necrosis, nodule/fibrosis, lipodystrophy, nerve injury (sciatic, radial), accidental intra-arterial injection (severe pain, blanching — an emergency), needle breakage.
- **IV therapy** — **infiltration** (fluid into the tissue: swelling, coolness, pallor — stop, remove, elevate, warm/cold compress), **extravasation** (leakage of a vesicant drug — stop at once, aspirate, inform the doctor, consider an antidote), **phlebitis/thrombophlebitis** (redness, warmth, a palpable cord along the vein — remove the cannula, apply a warm compress), air embolism, catheter embolism, speed shock, circulatory overload (dyspnoea, raised JVP, crepitations — stop or slow the drip, sit the patient up, give oxygen, inform the doctor), sepsis.
- **Systemic** — anaphylaxis, fainting, drug reaction, transmission of hepatitis B/C or HIV through unsafe practice.

CAUTION / NEVER DO THIS

Anaphylaxis — recognise and act in seconds: itching, urticaria, flushing, swelling of the lips/tongue/throat, hoarseness, stridor, wheeze, severe dyspnoea, abdominal cramps, fall of BP, tachycardia, collapse.

Treatment: stop the drug → call for help → **adrenaline (epinephrine) 1:1000, 0.5 mL (0.5 mg) IM into the mid-outer thigh** (child 0.01 mg/kg), repeat every 5-15 min if needed → lay the patient flat with the legs raised (sit up if breathless) → high-flow **oxygen** → secure the airway → **IV fluids** → antihistamine (chlorpheniramine/pheniramine) and **hydrocortisone 100-200 mg IV** → nebulised salbutamol for wheeze → observe for **6-24 hours** (biphasic reaction).

10.9 Intravenous Infusion — Nursing Points

- Check the fluid: name, strength, expiry, clarity (no particles, no turbidity), intact seal; label the bottle with the patient's name, the added drug, the date, time and rate, and the nurse's initials.
- Prime the set to remove all air; maintain a closed system; change the giving set every **72-96 hours (24 h for blood, lipid and TPN)**, and the peripheral cannula as per policy/when indicated.
- Keep the site **visible and dressed with a transparent film**; check hourly for pain, swelling, redness, leakage, the backflow of blood and the drip rate.
- Never 'catch up' a slow infusion by opening the roller clamp fully (risk of overload, especially in the elderly, infants, cardiac and renal patients); **use an infusion pump for children, potent drugs and precise rates**.
- Common IV fluids: **normal saline 0.9% (isotonic, for dehydration, shock, with blood)**, **Ringer lactate (balanced — the fluid of choice in burns and surgical loss)**, **5% dextrose (free water and calories, ~not for resuscitation~)**, **DNS, isolyte-P/M (paediatric)**, **mannitol (osmotic diuretic)**, **colloids (dextran, gelatin, albumin, hydroxyethyl starch)**.
- **Never give a hypotonic or dextrose solution with blood** — it causes haemolysis; flush the line with normal saline only.

10.10 Storage, Stock and Disposal of Medicines

- Store in a **cool, dry, dark place away from direct sunlight**; 'store below 25-30 °C' means room temperature, '**cold chain**' **2-8 °C** (refrigerator, never the freezer for vaccines/insulin, never the refrigerator door), deep freeze for polio vaccine (–20 °C).
- **Vaccines and insulin must never be frozen** (adsorbed vaccines — DPT, TT, hepatitis B — are destroyed by freezing); the **shake test** detects a frozen adsorbed vaccine; use the **vaccine vial monitor (VVM)** — discard at **stage 3 or 4**.
- Keep medicines in the **original labelled container**, tightly closed; light-sensitive drugs in amber bottles; store **external preparations separately from oral ones**; store liquids below powders/tablets to prevent damage from leakage.
- **Narcotics and Schedule X drugs** in a **double-locked cupboard** with a register and daily count (see Chapter 17); keep the keys with the nurse-in-charge.
- Follow **FEFO (first expiry, first out)** and FIFO for stock rotation; check expiry dates monthly; maintain a stock/bin card, indent and issue register, and an emergency (crash) tray checked and sealed daily.
- Keep all medicines **out of the reach of children**; never store medicines with food; never transfer a drug to an unlabelled or food container.
- **Disposal** — return expired and unused drugs to the pharmacy for documented destruction; *never flush drugs down the toilet or throw them in household waste* (antibiotics and cytotoxics are environmental hazards); expired drugs go in the **yellow bag for incineration**, cytotoxics in a labelled cytotoxic (yellow) container; narcotics are destroyed only before a witnessing committee with a record.

10.11 Adverse Drug Reactions, Interactions and Pharmacovigilance

Table 10.7 Adverse reactions, interactions and antidotes

Type / topic	Details
ADR types	Type A — augmented, dose-related and predictable (bleeding with warfarin, hypoglycaemia with insulin); Type B — bizarre, unpredictable, not dose-related (penicillin anaphylaxis); Type C — chronic/cumulative; Type D — delayed (carcinogenesis, teratogenesis); Type E — end-of-treatment (withdrawal); Type F — failure of therapy
Common ADRs to watch	Aspirin/NSAIDs — gastric bleeding; steroids — hyperglycaemia, Cushingoid features, infection, osteoporosis; aminoglycosides — nephrotoxicity and ototoxicity; chloroquine — retinopathy; isoniazid — hepatitis and neuritis; rifampicin — orange urine and hepatitis; ethambutol — optic neuritis; phenytoin — gum hypertrophy; metoclopramide/haloperidol — dystonia and extrapyramidal effects; ACE inhibitors — dry cough and hyperkalaemia; opioids — respiratory depression and constipation
Important interactions	Warfarin + aspirin/NSAID (bleeding); warfarin + vitamin-K rich food (reduced effect); digoxin + diuretic (hypokalaemia → toxicity); tetracycline/iron/fluoroquinolone + antacid, milk or calcium (chelation); MAO inhibitor + tyramine-rich food (hypertensive crisis); sildenafil + nitrate (severe hypotension); oral contraceptive + rifampicin/antiepileptics (failure); alcohol + metronidazole (disulfiram reaction); statin + grapefruit juice; methotrexate + NSAID
Antidotes	Paracetamol → N-acetylcysteine ; opioids → naloxone ; benzodiazepines → flumazenil ; organophosphate → atropine + pralidoxime ; warfarin → vitamin K (+ FFP) ; heparin → protamine sulphate ; iron → desferrioxamine ; methanol/ethylene glycol → ethanol/fomepizole; digoxin → digoxin-specific antibody; cyanide → sodium thiosulphate/hydroxocobalamin; methotrexate → folinic acid (leucovorin); lead → EDTA/DMSA; snake bite → polyvalent anti-snake venom
Pharmacovigilance	PvPI — Pharmacovigilance Programme of India (National Coordination Centre: IPC, Ghaziabad); report any suspected ADR on the Suspected ADR reporting form (Form for reporting to PvPI) , or by the ADR mobile app / toll-free number 1800-180-3024; WHO centre: Uppsala Monitoring Centre, Sweden

CHAPTER AT A GLANCE

- Rights of administration: patient, drug, dose, route, time, documentation, refusal, assessment, education.
- IV is fastest (100% bioavailability); oral is safest; sublingual and rectal avoid first-pass metabolism.
- Angles: ID 10-15°, SC 45° (or 90° with short needle), IM 90°, IV 15-30°.
- IM sites: deltoid (vaccines), vastus lateralis (infants), ventrogluteal (safest adult), dorsogluteal (sciatic nerve risk).
- Dose = (ordered / on hand) × quantity; drip rate = (volume × drop factor) / minutes; micro-drip = 60 drops/mL.
- 1 tsp = 5 mL; 1 tbsp = 15 mL; 1 g = 1000 mg; insulin only in an insulin syringe.
- Never crush enteric-coated or sustained-release tablets; shake suspensions; record after giving, never before.
- Anaphylaxis: adrenaline 1:1000, 0.5 mL IM in the mid-outer thigh, then oxygen, fluids, hydrocortisone, antihistamine.
- Cold chain 2-8 °C; never freeze DPT/TT/hepatitis B; discard a vaccine at VVM stage 3/4; FEFO for stock.
- Antidotes: paracetamol → N-acetylcysteine; opioid → naloxone; organophosphate → atropine + pralidoxime; heparin → protamine.

Special Conditions and Treatments

Local applications of heat and cold, inhalations, gargles, irrigations, enema, catheterisation and other nursing treatments

11.1 Application of Heat

DEFINITION

Local application of heat causes **vasodilatation** → increased blood supply, increased leucocyte and nutrient delivery, muscle relaxation and softening of exudate. It **relieves pain and spasm, hastens suppuration ('ripening' of an abscess), and promotes absorption and healing.**

Table 11.1 Methods of applying heat

Method	Technique and temperature	Uses	Precautions
Hot water bag/bottle (dry heat)	Fill two-thirds , expel air, screw the cap, invert to test for leaks, wrap in a flannel cover; water at 50-60 °C (adults), 40.5-46 °C for children, the elderly and the unconscious ; apply 20-30 min	Abdominal colic, backache, muscle spasm, warming a cold patient, dysmenorrhoea	Never place directly on the skin ; never use for the unconscious, paralysed, anaesthetised, diabetic or confused patient, on a baby, or on an area with poor sensation; inspect the skin every 5-10 min for redness (risk of burns)
Electric heating pad / infrared lamp	Low setting; lamp 45-60 cm away for 15-20 min	Muscular pain, stiff joints, bed sores (to dry the area)	No pins, no wet dressing (electrocution), never lie on the pad
Hot moist compress / fomentation (moist heat — penetrates better)	Gauze or flannel wrung out in hot water (40-46 °C), applied and covered with a mackintosh and towel; changed every 5 min for 15-20 min; may be sterile for a wound	Boil, abscess, whitlow, stye, infected wound, breast engorgement, injection site induration, thrombophlebitis	Aseptic technique for open wounds; protect the surrounding skin with petroleum jelly
Poultice (cataplasm)	Thick paste of kaolin/linseed/magnesium sulphate spread on gauze, heated and applied warm, covered to retain heat for 2-4 h	Boils, abscesses, carbuncle — to localise infection and relieve pain	Test the temperature on the back of your hand; do not apply too hot
Hot soaks / arm and foot bath	Immerse the part in water at 40-43 °C for 15-20 min	Infected hand/foot, sprain, ingrowing nail, diabetic foot (with great care)	Maintain the temperature by adding hot water at the side, away from the limb
Sitz (hip) bath	Patient sits in a tub with hot water (40-43 °C) covering the hips and buttocks for 15-20 min, feet outside; may add potassium permanganate or salt	After childbirth/episiotomy, haemorrhoids, fissure, perineal or rectal surgery, prostatitis — relieves pain, cleans and promotes healing	Watch for faintness and giddiness; keep the shoulders covered; never leave the patient alone; check the temperature
Steam and hot air bath, wax bath, short-wave diathermy, ultrasound	Physiotherapy modalities	Arthritis, chronic joint and muscle pain	Given by a physiotherapist; diathermy is contraindicated with a metal implant or pacemaker

11.2 Application of Cold

DEFINITION

Local cold causes **vasoconstriction** → reduced blood flow, reduced metabolism, reduced nerve conduction and reduced inflammation. It **checks bleeding and oedema, relieves pain and itching, limits inflammation and reduces temperature.**

Table 11.2 Methods of applying cold

Method	Technique	Uses	Precautions
Ice bag / cold pack	Fill one-third to two-thirds with crushed ice (small pieces), expel air, cover with a towel; apply 20-30 min, then rest	Sprain, contusion, fracture, headache, toothache, after tonsillectomy (ice collar), insect sting, sports injury, after an injection, to reduce swelling and bleeding	Never apply ice directly on the skin (frost bite); remove if the skin becomes pale, blue, mottled or numb; avoid over an open wound, in Raynaud's disease and peripheral vascular disease
Cold compress	Gauze wrung out in cold water or iced water, changed every 2-3 min for 15-20 min	Black eye, eye injury, epistaxis, headache, sprain, bleeding	Do not let it become warm
Cold (tepid) sponging	Water at 27-32 °C, long strokes to the limbs, compresses in the axillae, groins and forehead, for about 20-30 min	Hyperpyrexia (over 39.5-40 °C) , heat stroke	Stop if shivering , cyanosis or collapse occurs; take the temperature after 30 min; never use ice water or alcohol
Ice cradle / ice collar / cold blanket	Ice bags suspended over the part / collar round the neck	After tonsillectomy and thyroid surgery, hyperthermia	Watch the skin; change the position
Evaporating lotion / cooling lotion	Lotion (spirit, calamine, lead lotion) applied on gauze and left uncovered to evaporate	Sprains, inflamed joints, itching	Do not cover with mackintosh (prevents evaporation)

HIGH-YIELD / MOST ASKED

RICE for a fresh soft-tissue injury (first 24-48 hours) — **R~est~, I~ce~, C~ompression~, E~levation~**; heat is applied only *after* 48 hours, when bleeding has stopped.

Never apply heat to: an acute injury within 48 h, an undiagnosed abdominal pain (*possible appendicitis — heat can cause rupture*), active bleeding, or a malignant swelling.

11.3 Inhalations

Table 11.3 Inhalation therapy

Type	Method	Indication
Simple steam inhalation	Boiling water in a jug/inhaler with a paper funnel or a steam inhaler; the patient inhales steam through the mouth/nose for 10-15 minutes , with a towel over the head; done 2-4 times daily	Common cold, sinusitis, laryngitis, dry cough, croup, loosening thick secretions
Medicated steam inhalation	Add tincture benzoin (5 mL to 500 mL water) or menthol, eucalyptus or camphor ('Nelson's inhaler')	Sinusitis, bronchitis, nasal congestion
Dry/medicated inhalation	A few drops of a volatile oil on a handkerchief or in a bowl	Nasal blockage
Nebulisation	Drug in solution converted to a fine mist by a jet/ultrasonic nebuliser and breathed through a mask or mouth-piece over 10-15 minutes : salbutamol, ipratropium, budesonide, adrenaline, hypertonic saline, antibiotics	Acute asthma, bronchiolitis, croup, COPD, cystic fibrosis , and any patient too breathless to use an inhaler
Metered dose inhaler (MDI) with spacer / dry powder inhaler	Shake, exhale, seal the lips, press and inhale slowly and deeply, hold the breath for 10 seconds , wait 30-60 s before the next puff; use a spacer for children and the elderly; rinse the mouth after a steroid inhaler	Maintenance treatment of asthma and COPD

CAUTION / NEVER DO THIS

Steam inhalation carries a real risk of **scalding** — keep the jug on a low, stable surface, never on the patient's lap or bed, and never leave a child or a confused patient alone with it. Watch for giddiness and stop if the patient feels faint.

11.4 Mouth, Throat, Eye, Ear and Nose Treatments

Table 11.4 Local treatments of the head and neck

Treatment	Procedure and points
Gargle	Warm solution (normal saline, salt water, povidone-iodine 2%, chlorhexidine, potassium permanganate 1:8000) held at the back of the throat and agitated by exhaling, then spat out, not swallowed ; 2-4 times daily. For sore throat, tonsillitis, after oral surgery
Throat paint / swab	Mandl's paint (compound iodine glycerine) applied to the tonsils and pharynx with a swab on a holder, using a tongue depressor and good light; the patient must not eat or drink for 30 minutes afterwards; watch for gagging and vomiting
Eye drops (instillation)	Wash hands, check the drug, strength, eye (right/left) and expiry; clean the lids from inner to outer canthus; the patient looks up; pull down the lower lid to form a pouch and drop the medicine into the lower conjunctival sac (never on the cornea) holding the dropper 1-2 cm above without touching the eye or lashes; ask the patient to close the eye gently (not to squeeze) and press the inner canthus (<i>punctal occlusion</i>) for 1 min to reduce systemic absorption; wait 5 minutes between two different drops ; apply ointment in a thin line from the inner to the outer canthus after the drops; one dropper per patient per eye; use for a maximum of 4 weeks after opening
Eye irrigation (eye wash)	Normal saline or the prescribed lotion at body temperature, poured from the inner canthus to the outer canthus with the head turned to the side, kidney tray below; used for chemical burns (irrigate copiously for at least 15-20 min), discharge and foreign bodies
Eye pad and bandage	Sterile pad over the closed lid, secured obliquely (never press on the eyeball); after surgery or injury
Ear drops	Warm the bottle to room/body temperature (cold drops cause giddiness and vertigo); position the patient with the affected ear uppermost ; pull the pinna up and back in adults, down and back in children under 3 years ; instil the drops on the wall of the canal; keep the position for 5-10 minutes and press the tragus gently; plug loosely with cotton only if ordered
Ear irrigation (syringing)	Done by a doctor/trained nurse for wax (softened first with olive oil or soda-glycerine for 2-3 days); fluid at 37 °C directed at the upper posterior wall; contraindicated if the drum is perforated or there is discharge/infection ; watch for pain, giddiness and vomiting
Nasal drops / spray	Blow the nose first; for the sinuses use the ' Proetz ' head-low position (head extended back over the edge of the bed) or lateral head-low position; instil the drops and keep the position for 1-2 minutes; breathe through the mouth; do not share the dropper; limit decongestant drops to under 5-7 days (rebound congestion — rhinitis medicamentosa)

11.5 Enema, Suppository and Flatus Tube

DEFINITION

Enema — the introduction of a solution into the rectum and lower colon through a rectal tube, for cleansing, evacuation, retention or diagnostic purposes.

Table 11.5 Types of enema

Type of enema	Solution and amount	Purpose / indication
Cleansing (evacuant) enema — soap and water, normal saline, tap water	Adult 500-1000 mL at 40.5-43 °C (105-110 °F) ; child 250-500 mL; infant 50-150 mL; soap solution: 5 mL of soft soap per 1000 mL	Constipation and faecal impaction; before surgery, delivery, sigmoidoscopy/colonoscopy or an X-ray of the abdomen
Retention enema (oil retention)	Olive/arachis oil or glycerine 100-200 mL , retained for 1-3 hours (or as long as possible)	Softening hard impacted faeces; often followed later by a cleansing enema
Medicated / therapeutic enema	Prescribed drug — steroid (ulcerative colitis), neomycin, lactulose, sodium polystyrene sulphonate (for high potassium), paraldehyde/diazepam (for convulsions)	To give a drug when the oral route is impossible or for local action
Carminative (flatus) enema	Milk and molasses, or magnesium sulphate + glycerine + water (MGW) 30:60:90 mL	To expel flatus and relieve abdominal distension
Astringent enema	Cold normal saline, alum or tannic acid solution	To check bleeding and reduce inflammation
Emollient / sedative / stimulant / nutrient enema	Starch, thin gruel / paraldehyde / coffee / glucose saline (<i>obsolete</i>)	Soothing, sedation, stimulation, nutrition (historical)

Type of enema	Solution and amount	Purpose / indication
Barium enema	Barium sulphate suspension given under X-ray control	Diagnostic — X-ray of the large bowel
Disposable phosphate (Enema Fleet) / glycerine suppository	Small volume (about 118 mL) ready-to-use	Quick bowel evacuation; commonly used today

Procedure of giving a cleansing enema

1. Check the order; explain; screen the bed; keep a bedpan/commode and toilet paper ready close by; wash hands, wear gloves.
2. Position: **left lateral (Sims') position with the right knee flexed** — this follows the anatomical direction of the sigmoid colon (the knee-chest position is an alternative; infants are placed supine with the legs raised).
3. Place a mackintosh and towel under the buttocks; hang the enema can **45-50 cm (18 inches) above the anus** (a lower height and slower flow for a child); expel the air from the tubing.
4. Lubricate the rectal tube (12-16 Fr for an adult) and insert it gently **7.5-10 cm (3-4 inches) in an adult** (5 cm in a child, 2.5-3.75 cm in an infant), directing it towards the umbilicus; *never force against resistance*.
5. Allow the fluid to run in **slowly over 5-10 minutes**; ask the patient to breathe deeply through the mouth; stop temporarily if there is cramping; clamp the tube before it empties completely (to avoid air entry) and withdraw it gently, holding a pad against the anus.
6. Ask the patient to **retain the fluid for 5-15 minutes** (oil retention 1-3 h); then give the bedpan or assist to the toilet; do not leave a weak patient alone.
7. Observe and record the **amount, colour, consistency and content of the returned flow**, the patient's tolerance, and the result (whether the bowel was cleared). Clean and disinfect the equipment; wash hands.

CAUTION / NEVER DO THIS

Contraindications to an enema: undiagnosed abdominal pain, suspected appendicitis or peritonitis, intestinal obstruction, recent rectal/colonic surgery, rectal bleeding, severe piles/fissure, pregnancy near term (unless ordered), raised intracranial pressure, and a patient with a very low platelet count or neutropenia.

Dangers: perforation of the rectum, vagal stimulation (bradycardia, fainting), water intoxication and electrolyte imbalance (repeated tap-water enemas, especially in children and renal patients), dependence with repeated use, and mucosal damage.

- **Suppository** — a bullet-shaped solid inserted into the rectum: **glycerine/bisacodyl (constipation), paracetamol or diclofenac (fever and pain), diazepam (convulsions), antithaemorrhoidal**. Position: left lateral; wear a glove, lubricate the tip, insert **past the internal sphincter (about 5-7.5 cm in an adult, 2.5 cm in a child) against the rectal wall**; ask the patient to retain it for **20-30 minutes**. Store suppositories in a **cool place/refrigerator**.
- **Flatus tube** — a wide, soft rectal tube (24-32 Fr) inserted 10-15 cm with the outer end under water in a bowl, kept for **not more than 20-30 minutes**, to relieve **abdominal distension (flatulence) after operation**; bubbling shows the escape of gas. Alternatives: hot fomentation, early ambulation, and a prescribed prokinetic.

11.6 Urinary Catheterisation and Bladder Irrigation

- **Indications** — acute or chronic **retention of urine**, accurate output measurement in a critical patient, before and after certain surgeries and during labour, bladder irrigation, instillation of a drug, urinary incontinence with pressure sores, collection of a sterile specimen, and to keep the bladder empty in pelvic surgery.
- **Types** — intermittent (Nelaton/simple, plain catheter), **indwelling (Foley — 2-way with a balloon, 3-way for irrigation)**, suprapubic, condom (Uridom) drainage in men. Sizes: **adult female 14-16 Fr, adult male 16-18 Fr, children 6-10 Fr**; 1 Fr = 1/3 mm.
- **Strict aseptic technique** — sterile gloves, sterile catheter and lubricant (2% lignocaine jelly), antiseptic cleaning of the meatus, sterile drape. Position: female — dorsal recumbent with knees flexed and separated; male — supine with the legs slightly apart.
- Clean the **female** perineum from front to back and identify the **urethral meatus above the vaginal opening** (insert 5-7.5 cm until urine flows, then 2.5 cm more); in the **male**, retract the foreskin, hold the penis at **60-90 degrees**, insert **17-22.5 cm** until urine flows, then advance further before inflating; **replace the foreskin afterwards**.
- **Inflate the balloon only after urine flows** (5-10 mL of sterile water, *not saline* — it can crystallise); if the patient complains of pain while inflating, stop at once (the balloon may be in the urethra).

- In **chronic retention**, drain gradually (some authorities advise clamping after 300-500 mL and releasing after 15-30 min) to prevent **haematuria, hypotension and post-obstructive diuresis**.
- **Care of an indwelling catheter** — keep the **drainage bag below the level of the bladder but off the floor**, never disconnect the closed system, prevent kinking and loops, secure the catheter to the thigh/abdomen, do a **meatal toilet with soap and water daily**, encourage **fluid intake of 2-3 L**, empty the bag when two-thirds full using a separate container for each patient, measure and record output, take specimens from the sampling port with aseptic technique, and **remove the catheter as early as possible** (the risk of infection rises by about 5% per day).
- **Bladder irrigation/washout** — continuous with a 3-way catheter (after prostate/bladder surgery to prevent clot retention) or intermittent with sterile normal saline at room temperature; record the volume in and out separately and **subtract the irrigation fluid from the total output**; report bright-red drainage, clots or no outflow at once.
- **Complications** — catheter-associated UTI (the commonest hospital-acquired infection), urethral trauma and stricture, bleeding, false passage, paraphimosis (foreskin not replaced), bladder spasm, blockage, and encrustation.

11.7 Other Special Treatments

Table 11.6 Miscellaneous nursing treatments

Treatment	Description and nursing points
Vaginal douche / irrigation	Warm antiseptic solution (40-43 °C) run into the vagina in the dorsal recumbent position with a bedpan; for pre-operative preparation and discharge; <i>avoid in pregnancy and menstruation</i>
Vaginal pessary/cream	Inserted at bedtime in the lying position with an applicator; the patient is told it may stain and to use a pad
Medicated (therapeutic) bath	KMnO ₄ 1:5000-1:10 000 for infected skin; sodium bicarbonate/oatmeal/starch for itching; emollient bath for eczema; 15-30 min, water at 37-40 °C
Wet packs and cold/hot wet sheet packs	Whole-body application for hyperpyrexia or skin disease; watch the pulse and for shivering
Counter-irritants	Mustard plaster/poultice (1 part mustard to 4-6 parts flour, applied for 15-20 minutes only , protecting the skin with gauze and oil — remove at once if there is burning), turpentine stupe, liniment rubs, capsaicin ointment; used for chest congestion and muscular pain; never apply to broken skin or to children/the unconscious
Cupping, blistering, leeching, venesection	Old (largely historical) methods of counter-irritation and blood letting; leech therapy is still occasionally used in plastic surgery and traditional medicine; venesection is used today for polycythaemia and haemochromatosis
Local applications	Lotion (cooling), liniment (rubbed in for pain), paint, ointment/cream (applied thinly in the direction of hair growth with a spatula or gloved hand), paste (thick protective layer), dusting powder (dry, on intact skin), plaster; wear gloves for hormone, steroid and cytotoxic preparations
Ryle's tube aspiration (gastric lavage/decompression)	For poisoning (only within 1 hour, with a protected airway; <i>contraindicated in corrosive and hydrocarbon poisoning</i>), pyloric obstruction, paralytic ileus, before emergency surgery; use warm normal saline/tap water 200-300 mL at a time, save the first aspirate for analysis, measure and record the aspirate
Cold and hot applications to the eye, ear and joints	As described above; never apply heat to an inflamed eye without an order

CHAPTER AT A GLANCE

- Heat = vasodilatation (relieves spasm, hastens suppuration); cold = vasoconstriction (checks bleeding, oedema, pain). RICE for fresh injury; heat only after 48 h.
- Hot water bag: fill two-thirds, 50-60 °C for adults, 40.5-46 °C for children/elderly/unconscious, always covered.
- Sitz bath 40-43 °C for 15-20 min — episiotomy, piles, fissure, perineal surgery.
- Steam inhalation 10-15 min with tincture benzoin; nebulisation for acute asthma; rinse the mouth after steroid inhalers.
- Eye drops in the lower conjunctival sac, inner→outer canthus for cleaning; ear drops warmed, pinna up-and-back in adults, down-and-back in children.
- Enema: left lateral (Sims') position, can 45-50 cm high, tube inserted 7.5-10 cm, solution 500-1000 mL at 40.5-43 °C, retain 5-15 min; contraindicated in suspected appendicitis and obstruction.
- Catheter: female 14-16 Fr / male 16-18 Fr; inflate the balloon only after urine flows, with sterile water; bag below bladder level, closed system, remove early.
- Flatus tube for not more than 20-30 minutes; mustard plaster for 15-20 minutes only.

Bandaging

Purposes, principles, types of bandages, basic turns, slings, splints and strapping

12.1 Definition and Purposes

DEFINITION

Bandage — a strip or roll of cloth or other material used to hold a dressing in place, to support, immobilise or compress a part, or to correct a deformity. **Binder** — a broad bandage applied to a large area such as the chest, abdomen, breast or perineum.

Purposes of bandaging

- To **keep a dressing or splint in position** and protect the wound from injury and infection.
- To give **support** to a part (sprain, muscle strain, hernia, pendulous breast).
- To apply **pressure** — to arrest haemorrhage, to reduce or prevent swelling and oedema, and to prevent varicose veins and deep vein thrombosis (compression bandage/stocking).
- To **immobilise and rest** a part — fracture, dislocation, after tendon repair.
- To **restrict movement**, to maintain a limb in a required position, and to correct a deformity.
- To **hold an appliance** (splint, traction, plaster) in place and to hold packs.

12.2 Types of Bandages and Materials

Table 12.1 Types of bandages

Type	Description	Common uses
Roller bandage	A long strip rolled up; made of cotton, gauze, flannel, crepe, elastic (crepe/elastocrepe), plaster of Paris or a self-adhesive material; has a head (the roll) and a tail (the free end)	Most commonly used bandage for limbs, head and trunk
Triangular bandage	A square (about 1 m x 1 m) of cloth cut across diagonally, giving a base, a point and two ends; can be used open, as a broad fold or a narrow fold	Slings, head, hand, foot, chest, hip and knee dressings; first aid — the most versatile first-aid bandage
T-bandage (single and double T)	A horizontal band with one or two vertical tails	Holding a perineal or rectal dressing; double-T for men
Four-tailed bandage	A strip with both ends split into two tails	Chin/jaw, nose, head and elbow dressings
Many-tailed (Sculptetus) bandage	Several overlapping strips fixed to a central piece	Abdomen and chest — easy to change without moving the patient
Suspensory bandage	A bag with a waist band	Support of the scrotum (after hydrocele or vasectomy surgery, orchitis)
Tubular / tubegauze bandage	Seamless woven tube applied with an applicator	Fingers, toes, limbs, head — quick and comfortable
Capeline / Barton / Gibson bandage	Special double-headed roller bandages of the head and jaw	Scalp and jaw injuries
Adhesive strapping (plaster) and tape	Zinc oxide, micropore, dynaplast, elastic adhesive	Fixing dressings, strapping ribs, holding splints
Crepe / elastic compression bandage	Stretchable, self-clinging	Sprains, varicose veins, oedema, venous ulcer, after a fracture
Plaster of Paris (POP) bandage and fibreglass cast	Gauze impregnated with gypsum, dipped in water	Immobilising fractures
Improvised bandages	Clean handkerchief, dupatta, towel, sari strip, belt, stockings, necktie	Emergency and home nursing

NUMBERS TO REMEMBER

Standard widths of a roller bandage (learn these):

- Finger and toe — 1.25-2.5 cm (0.5-1 inch)

- Hand, wrist and head — 5 cm (2 inches) (head 5-7.5 cm)
 - Arm and forearm — 5-6.5 cm (2-2.5 inches)
 - Leg and foot — 7.5 cm (3 inches) • Thigh — 9-10 cm
 - Trunk, chest and abdomen — 10-15 cm (4-6 inches)
- Usual length: 5-6 metres for limb bandages.

12.3 General Rules and Principles of Bandaging

1. Explain the procedure and place the patient in a **comfortable position**; the nurse stands **facing the part** to be bandaged and supports the limb.
2. Bandage the part in the **position in which it is to remain** — joints slightly flexed, the ankle at a right angle, the fingers slightly flexed; place soft pads between skin surfaces and over bony points.
3. Work generally from **below upwards and from within outwards (medial to lateral)**, and from the **distal to the proximal end** (this aids venous return).
4. Hold the **roll (head) uppermost in the right hand** with the outer surface applied to the part; unroll only a little at a time, keeping the roll close to the skin.
5. Fix the bandage with **two circular turns** at the start, and finish with a circular turn secured by a **safety pin, adhesive tape, a clip or a tie** — the fastening must be placed on the **outer side, away from the wound, and never over a bony point, a joint or the inner side of a limb**.
6. Apply with **even, firm but not tight pressure**; each turn should cover **one-half to two-thirds** of the previous one, and no skin should show between the turns.
7. Keep the bandage **neat, smooth and free of wrinkles, creases and gaps**; it should be comfortable and light.
8. **Leave the finger tips and toes exposed whenever possible** so that the circulation can be watched.
9. **Never bandage over a wet dressing**, and never apply a wet bandage that will shrink on drying.
10. **Check the circulation** after applying, and at intervals; re-bandage if the patient complains of pain, numbness, tingling or throbbing; re-apply a limb bandage at least once a day and keep the part elevated if there is swelling.
11. Remove a bandage by cutting with blunt-nosed scissors (away from the wound) or by unwinding it, gathering it loosely from hand to hand — *never drag it across a wound*.

CAUTION / NEVER DO THIS

Signs that a bandage is too tight (impaired circulation) — remove or loosen at once:

- Pain, throbbing or a feeling of tightness; **numbness, tingling (pins and needles) or loss of sensation**
- **Coldness, pallor, then blueness (cyanosis) of the fingers or toes**; swelling below the bandage
- **Capillary refill slower than 3 seconds**; absent or weak pulse beyond the bandage; inability to move the fingers/toes
- Remember the **5 P's of impaired circulation: Pain, Pallor, Paraesthesia, Pulselessness, Paralysis** — the warning signs of **compartment syndrome**, a surgical emergency.

12.4 Basic Bandage Turns

Table 12.2 The basic bandage turns

Turn	Technique	Where used
Circular turn	Each turn exactly over the previous one	To begin and end every bandage; wrist, neck, forehead, finger
Spiral turn	Turns ascend at a slight angle, each overlapping two-thirds of the previous	Parts of uniform thickness — finger, upper arm, trunk, wrist
Spiral reverse turn	The bandage is reversed (folded on itself) halfway through each turn, with the thumb steadying the fold	Parts of varying thickness — forearm and leg (used with a non-elastic bandage)
Figure-of-eight turn	Oblique turns alternately above and below the joint, crossing in front like the figure 8	Joints — elbow, knee, ankle, wrist ; also for firm support and compression
Spica (a modified figure-of-eight)	Alternate ascending and descending turns overlapping to look like an ear of corn	Shoulder, hip, thumb, groin, breast (shoulder spica, hip spica, thumb spica)
Recurrent (capeline) turn	Bandage passed back and forth over the tip of the part, then fixed with circular turns	Finger tip, toe, stump after amputation, head (scalp)

Turn	Technique	Where used
Divergent / convergent turn	Turns spread out from or converge towards the middle of a flexed joint	Elbow and knee in a flexed position

Bandaging particular parts — key points

Table 12.3 Bandaging specific parts

Part	Bandage and method
Finger	2.5 cm roller — circular at the wrist, oblique to the finger, recurrent over the tip, spiral down the finger, then figure-of-eight back to the wrist
Hand and palm	5 cm roller — figure-of-eight round the hand and wrist, leaving the thumb free
Thumb	Thumb spica
Forearm and leg	Spiral reverse turns from the wrist/ankle upwards
Elbow and knee	Figure-of-eight with the joint slightly flexed (divergent turns)
Shoulder and hip	Spica
Foot and ankle	Figure-of-eight, starting at the instep, with the ankle at 90 degrees; the heel and toes may be left exposed
Heel	Figure-of-eight with locking turns
Head / scalp	Capeline (double-headed) or recurrent turns , or a triangular bandage as a head cap
Eye	Oblique turns round the head and under the ear on the affected side (never over the other eye if it can be avoided)
Ear / jaw / chin	Four-tailed or a barrel bandage
Chest and abdomen	Broad roller, many-tailed (Scultetus) bandage or a binder
Breast	Breast binder or a spica; applied for support and after surgery
Stump (after amputation)	Recurrent turns with firm even pressure to shape the stump

12.5 Slings

Table 12.4 Types of slings

Sling	Method	Indication
Arm (large/St John) sling	Triangular bandage: point towards the injured elbow, one end over the sound shoulder and round the neck, tied in the hollow above the clavicle; the hand is supported slightly higher than the elbow , the fingers visible, the point tucked in and pinned	Fracture/injury of the forearm, wrist and hand ; after a dressing or plaster; support in rib injury
Collar-and-cuff (triangular) sling	A clove-hitch round the wrist with a narrow-fold bandage, tied round the neck, the elbow hanging free	Fracture of the humerus/upper arm and of the clavicle, where the weight of the arm gives traction
Elevation (St John) sling	Fingers placed on the opposite shoulder with the forearm across the chest; the bandage is brought round the back and tied, so the hand is well raised	To control bleeding of the hand/forearm , with a hand injury or infection, and in a fractured clavicle or crushed hand (elevation reduces swelling)
Figure-of-eight bandage of the shoulders	Both shoulders braced back	Fractured clavicle
Triangular bandage as a broad/narrow fold	Fixation of splints, tying limbs together	Fracture immobilisation in first aid

12.6 Splints, Strapping and Plaster of Paris

- **Splint** — a rigid or semi-rigid appliance used to immobilise and support a fractured or injured part. Types: wooden (Bohler), metal, plastic, air/inflatable, vacuum, moulded thermoplastic, Thomas splint (for the femur),

Cramer wire, Kramer, POP slab, cardboard/newspaper and improvised (rolled magazine, umbrella, stick, even the patient's own body).

- **Rules for splinting** — immobilise the **joint above and the joint below** the fracture; pad the splint well and place padding over bony points; splint **in the position found** (do not attempt to reduce); check the **pulse, colour, warmth, sensation and movement before and after** splinting; tie the bandages **firmly but not tightly, and never directly over the fracture site**; remove rings and bangles; elevate afterwards; do not delay transport for perfect splinting in a life-threatening emergency.
- **Strapping (adhesive taping)** — used for rib injury, sprained ankle, fixing a dressing, and to approximate wound edges (Steri-strips). Clean and dry the skin, shave the hair, apply a protective spray or under-wrap, and apply the strips at right angles to the direction of pull, never in a complete circle around a limb.
- **Care of a plaster cast** — keep the limb **elevated** for 24-48 h; leave the wet cast **uncovered and exposed to the air to dry (24-72 h)**, handle it with the **palms, never the finger tips** (dents cause pressure sores), do not rest it on a hard edge; keep the cast **dry and clean**, do not insert anything (knitting needle, pencil) inside to scratch, do not remove the padding or cut the cast without an order; encourage exercise of the free joints and isometric exercises of the muscles under the cast.
- **Observe and report at once** — pain (especially increasing or on passive stretch), swelling, **coldness, blueness or numbness of the toes/fingers, inability to move them**, a foul smell or a discharge stain on the cast (infection), a hot spot, or a loose or cracked cast. **Window or bivalve the cast** if the surgeon orders; keep a cast cutter/shears available.
- **Complications of a cast** — compartment syndrome, pressure sores, nerve palsy (peroneal, ulnar), joint stiffness, muscle atrophy, disuse osteoporosis, deep vein thrombosis, and plaster dermatitis.

CHAPTER AT A GLANCE

- Bandage = holds dressings, supports, compresses, immobilises. Binder = broad bandage for the trunk.
- Widths: finger 1.25-2.5 cm, hand/head 5 cm, arm 5-6.5 cm, leg 7.5 cm, thigh 9-10 cm, trunk 10-15 cm.
- Bandage from distal to proximal, medial to lateral, roll uppermost, each turn covering two-thirds of the previous; start and finish with circular turns; fasten on the outer side away from the wound.
- Turns: circular (start/finish), spiral (uniform part), spiral reverse (forearm/leg), figure-of-eight (joints), spica (shoulder/hip/thumb), recurrent (finger tip/stump/head).
- Slings: arm sling = forearm and wrist injury; collar-and-cuff = fractured humerus; elevation sling = bleeding hand, crushed hand, fractured clavicle.
- Too tight: pain, pallor, paraesthesia, pulselessness, paralysis (5 P's) — loosen at once; leave the finger tips exposed.
- Splint the joints above and below; never dent a wet plaster — handle with the palms; keep it elevated, uncovered and dry.

Further Observations

Specimen collection, bedside tests, fluid balance, danger signals and documentation

13.1 Collection of Specimens — General Rules

1. Check the doctor's order and the **correct container** (with or without preservative/anticoagulant); explain the procedure to the patient and obtain co-operation.
2. **Label the container before or immediately after collection** with the patient's name, age, sex, hospital/bed number, the specimen, the site, the date and **time of collection**, and the test required; fill and attach the requisition form.
3. Collect the **right specimen, in the right amount, at the right time** (for example, a fasting sample before breakfast, a sputum sample early in the morning, a blood culture **at the onset of a fever spike and before starting antibiotics**).
4. Use **standard precautions** — gloves for every specimen; collect with aseptic technique where sterility is needed; never contaminate the outside of the container; place it in a leak-proof plastic bag ('double bagging').
5. **Send it to the laboratory at once**; if delayed, follow the preservation rule (refrigerate urine; keep a blood culture bottle at room temperature or in an incubator; never refrigerate cerebrospinal fluid for culture).
6. **Record** the collection and dispatch in the nurse's notes, and **follow up the report** and inform the doctor of an abnormal or critical value.

Table 13.1 Collection of common specimens

Specimen	How to collect	Key points
Routine (random) urine	10-20 mL in a clean, dry, wide-mouthed container	Preferably the first morning sample (most concentrated)
Mid-stream clean-catch urine (for culture)	Clean the meatus with soap and water/antiseptic (female: front to back); pass the first part into the toilet, collect the middle part in a sterile container without touching the rim, and let the last part go	Never take a culture sample from the urine bag or bedpan; send within 1 hour or refrigerate (up to 24 h); the sample must not be taken while on antibiotics if avoidable
Catheter specimen	Aspirate with a sterile syringe from the sampling port after cleaning it with spirit (clamp the tube briefly below the port)	Never disconnect the closed system or take urine from the bag
24-hour urine	Discard the first voiding at the start time and note it; collect every drop for the next 24 hours , including the voiding at the end time, in a large container with the correct preservative, kept cool/refrigerated	If one sample is missed, the whole collection must be restarted; used for creatinine clearance, protein, VMA, catecholamines, 17-ketosteroids
Stool	A walnut-sized/2-3 spoonful sample from a clean, dry bedpan in a clean container, using a spatula; include portions with blood or mucus	Must not be contaminated with urine or water ; send a fresh warm sample within 30 minutes for amoebic trophozoites; 3 samples on alternate days for ova/parasites; use a sterile container for culture; occult blood test needs a meat-free, iron-free diet for 3 days
Sputum	Early morning, on rising, before eating or brushing , after rinsing the mouth with plain water; take a deep cough from the chest (not saliva) into a sterile wide-mouthed container; 5-10 mL; ~2 samples (spot and morning) for tuberculosis__	Explain the difference between sputum and saliva; wear a mask and collect in a well-ventilated place/away from others; send at once or refrigerate; for cytology, send in 70% alcohol
Throat swab	Good light, tongue depressor, sterile swab rubbed over both tonsils and the posterior pharynx avoiding the tongue, teeth and cheeks ; the patient says 'aah'	Best before food or gargling and before antibiotics; may cause gagging
Blood	Venepuncture with aseptic technique — the correct vacutainer and the correct order of draw ; the tourniquet must not stay on for more than 1 minute; mix anticoagulated tubes by gentle inversion (never shake — it haemolyses the sample); apply pressure, not massage, afterwards	Never draw blood from the arm with an IV infusion (or take it below the site with the drip stopped); fasting samples need 8-12 h of fasting (water allowed); label at the bedside in the patient's presence

Specimen	How to collect	Key points
Blood culture	2 sets from 2 different sites, at the onset of a chill/fever and before antibiotics , 10 mL per bottle in an adult; clean the skin with 70% alcohol then chlorhexidine/povidone-iodine and let it dry; disinfect the bottle top	Skin preparation is critical to avoid contamination; do not refrigerate
Wound swab / pus	Sterile swab rotated over the base and edge of the wound (not the surrounding skin or the dry crust) after removing the old dressing and cleaning off the slough with saline; aspirated pus in a sterile syringe is better	Take the sample before applying any antiseptic and before starting antibiotics
Vomit / gastric aspirate	Whole specimen in a clean covered container	Essential in suspected poisoning — preserve and hand over to the doctor/police
Cerebrospinal fluid, pleural, ascitic and joint fluid	Collected by the doctor at lumbar puncture or aspiration; the nurse labels 3 numbered sterile tubes and sends them at once	Handle as highly infectious; never delay or refrigerate a CSF culture sample

13.2 Simple Bedside Tests

Table 13.2 Bedside/point-of-care tests

Test	Method	Interpretation
Urine sugar (Benedict's test)	5 mL Benedict's qualitative reagent + 8 drops of urine , boil for 2 min and cool	Blue = nil • green = + (0.5%) • yellow = ++ (1%) • orange = +++ (1.5%) • brick red = ++++ (2% or more)
Urine ketones (Rothera's / strip)	Saturate urine with ammonium sulphate, add sodium nitroprusside and ammonia	A purple ring = ketones present (diabetic ketoacidosis, starvation, vomiting)
Urine protein (heat and acetic acid test)	Boil the upper part of the urine in a test tube and add 2-3 drops of acetic acid	Persistent cloudiness (turbidity) = albuminuria (renal disease, pre-eclampsia)
Reagent (dip) strip	Dip, remove excess, read against the chart at the exact time	Reads pH, specific gravity, protein, glucose, ketone, blood, nitrite, leucocyte esterase, bilirubin, urobilinogen
Capillary blood glucose (glucometer)	Clean the side of the finger tip with spirit and let it dry, prick with a sterile lancet, wipe the first drop, apply the second drop to the strip	Fasting normal 70-100 mg/dL (diabetes 126 or more); random/post-prandial under 140 (diabetes 200 or more); hypoglycaemia under 70 mg/dL — an emergency
Pulse oximetry	Probe on a clean, warm finger (remove nail polish), read after the waveform stabilises	Normal 95-100% ; under 90% = hypoxia needing oxygen
Peak expiratory flow (PEFR)	Standing, maximum inspiration, then a fast hard blow; best of 3 readings	Compare with the personal best/predicted — under 80% = poor control, under 50% = severe attack
Weight and height, BMI, mid-arm circumference	Same time of day, same clothes, calibrated scale	A gain of 1 kg = about 1 L of fluid; MUAC under 11.5 cm (6-59 months) = severe acute malnutrition
Pregnancy test (urine hCG)	First morning urine on the card/strip	Two lines = positive; detectable about 10-14 days after conception

13.3 Normal Values Every Nurse Should Know

Table 13.3 Common normal laboratory values

Investigation	Normal value
Haemoglobin	Men 13-17 g/dL, women 12-15 g/dL ; WHO anaemia cut-off: men under 13, women under 12, pregnancy under 11, children 6-59 months under 11 g/dL
Total leucocyte count / differential	4000-11 000/mm³ ; neutrophils 40-75%, lymphocytes 20-45%, eosinophils 1-6%, monocytes 2-10%, basophils 0-1%
Platelets	1.5-4.5 lakh/mm³ (150 000-450 000) — bleeding risk below 50 000, spontaneous bleeding below 20 000
ESR	Men 0-15 mm, women 0-20 mm in the 1st hour
Blood sugar	Fasting 70-100 mg/dL ; 2-hour post-prandial under 140; HbA1c under 5.7% (diabetes 6.5% or more; target under 7%)

Investigation	Normal value
Blood urea / Serum creatinine	Urea 15-40 mg/dL (BUN 7-20); creatinine 0.6-1.2 mg/dL
Serum electrolytes	Sodium 135-145 mEq/L; potassium 3.5-5.0 mEq/L; chloride 96-106; bicarbonate 22-28
Serum calcium / uric acid	Calcium 8.5-10.5 mg/dL; uric acid — men 3.5-7.2, women 2.6-6.0 mg/dL
Total bilirubin	0.3-1.2 mg/dL (jaundice becomes visible above 2-3 mg/dL)
Liver enzymes	SGPT/ALT 7-45 U/L; SGOT/AST 8-40 U/L; alkaline phosphatase 40-130 U/L
Serum protein / albumin	Total protein 6-8 g/dL; albumin 3.5-5.5 g/dL
Total cholesterol / LDL / HDL / triglycerides	Under 200 / under 100 / above 40-50 / under 150 mg/dL
Prothrombin time / INR	PT 11-13.5 s; INR 0.8-1.2 (therapeutic on warfarin 2-3)
Cerebrospinal fluid	Clear, pressure 60-150 mm water, protein 15-45 mg/dL, glucose 50-80 (two-thirds of blood), 0-5 lymphocytes/mm ³
Blood volume / cardiac output	About 5-6 L (70-80 mL/kg); cardiac output 4-6 L/min

13.4 Fluid Balance, Dehydration and Oedema

Table 13.4 Assessment of dehydration (WHO/IMNCI)

Degree of dehydration	Signs	Fluid deficit
No dehydration	Alert, normal eyes, drinks normally, skin pinch retracts at once	Under 3% of body weight
Some (mild-moderate) dehydration	Restless or irritable, sunken eyes, thirsty and drinks eagerly, skin pinch retracts slowly, dry mouth, reduced urine , sunken fontanelle in infants	3-9% (about 5%)
Severe dehydration	Lethargic or unconscious, very sunken eyes, unable to drink, skin pinch retracts very slowly (over 2 s), weak/absent pulse, cold clammy extremities, low BP, no urine for 6-8 h, deep rapid breathing	10% or more — needs immediate IV fluids (Ringer lactate)

- **ORS (WHO low-osmolality formula, per litre)** — sodium chloride **2.6 g**, glucose (anhydrous) **13.5 g**, potassium chloride **1.5 g**, trisodium citrate dihydrate **2.9 g**; total osmolality **245 mOsm/L**. Dissolve one packet in **1 litre of clean drinking water**, use within **24 hours**, and never add sugar, salt or milk to it.
- Give ORS **75 mL/kg over 4 hours** for 'some dehydration'; after each loose stool give **50-100 mL (under 2 years), 100-200 mL (2-10 years)** and as much as wanted above 10 years. Add **zinc 10 mg (under 6 months) or 20 mg (over 6 months) daily for 14 days** in childhood diarrhoea, and continue feeding and breast feeding.
- **Homemade ORS** — 1 litre of clean water + **1 level teaspoon of salt + 6 level teaspoons of sugar** (or salt-and-sugar solution); rice kanji/lemon water are useful fluids.
- **Oedema** — abnormal collection of fluid in the tissue spaces: **pitting** (heart failure, renal disease, hypoproteinaemia, pregnancy) or **non-pitting** (myxoedema, lymphoedema/filariasis); **dependent** (feet when sitting, sacrum when lying), **periorbital** (renal, early morning), **ascites** (abdomen), **anasarca** (generalised), **pulmonary oedema** (breathlessness, pink frothy sputum, crepitations — an emergency).
- Nursing care in oedema: daily **weight at the same time**, strict intake-output, **salt and fluid restriction** as ordered, elevation of the limbs, careful skin care (oedematous skin breaks down easily), measuring the abdominal girth at a marked level, watching for breathlessness, and monitoring electrolytes with diuretic therapy.
- **Fluid overload** — raised JVP, puffy face, breathlessness, crepitations, bounding pulse, rapid weight gain: stop or slow the infusion, sit the patient up, give oxygen and inform the doctor.

13.5 Observation of Pain

- Pain is the '**fifth vital sign**' and is **whatever the patient says it is** — believe the patient's report.
- Assess by **PQRST**: *Provoking/Palliating factors, Quality (burning, cramping, stabbing, dull), Region and Radiation, Severity, Timing and duration* (or **SOCRATES**).
- **Scales** — numerical rating (0-10), visual analogue, **Wong-Baker FACES scale (for children 3 years and older)**, **FLACC** and **neonatal/infant scales (NIPS, CRIES)** for those who cannot speak; behavioural signs in the non-verbal patient are grimacing, guarding, restlessness, moaning, rigidity, refusal to move, tachycardia, sweating and raised BP.

- **Types** — acute vs chronic; somatic, visceral (deep, poorly localised, with nausea and sweating), **referred** (gall bladder → right shoulder tip, myocardial infarction → left arm and jaw, appendix → umbilicus then right iliac fossa, diaphragm → shoulder), neuropathic (burning, shooting — diabetes, shingles), phantom limb pain.
- **Management** — the **WHO analgesic ladder** (step 1 paracetamol/NSAID, step 2 weak opioid ± adjuvant, step 3 strong opioid such as morphine) with drugs given '**by the clock, by the mouth, by the ladder**'; plus non-drug measures — position, rest, splinting, heat/cold, massage, distraction, relaxation, music, TENS, and reassurance.
- Always record the pain score **before and 30-60 minutes after** giving an analgesic, and watch for constipation, sedation and **respiratory depression** (respiration under 10-12/min — **hold the drug and inform the doctor; antidote naloxone**) with opioids.

13.6 Danger Signals — When to Send for the Doctor at Once

CAUTION / NEVER DO THIS

- **Airway/breathing:** noisy breathing, stridor, choking, severe breathlessness, respiration under 10 or over 30/min, SpO₂ under 90%, cyanosis, gasping, apnoea.
- **Circulation:** systolic BP under 90 mmHg or a fall of over 40 mmHg, pulse under 50 or over 130, thready or absent pulse, chest pain, cold clammy skin, any active bleeding (haematemesis, malaena, haemoptysis, post-partum haemorrhage, bleeding through a dressing).
- **Neurological:** sudden fall in the level of consciousness (a drop of 2 or more in the GCS), convulsion, sudden severe headache with vomiting, unequal or fixed pupils, sudden weakness of one side, severe confusion or delirium.
- **Others:** temperature above 40 °C or below 35 °C, no urine for 6-8 hours (under 30 mL/h), persistent vomiting, sudden severe abdominal pain with a rigid abdomen, a reaction during transfusion or injection, a new rash with fever, wound dehiscence, a fall or an injury, and any sudden unexplained change in the patient's condition or 'a feeling that something is wrong'.

Never wait for the routine round to report a danger signal. Stay with the patient, start the appropriate emergency measure (position, oxygen, pressure on a bleeding point), and send someone else for help.

13.7 Recording and Reporting Observations

- Record **immediately** at the bedside or as soon as possible; never record in advance, and never record an observation you have not made.
- Use **objective, descriptive language** with **measurements, not impressions**: write 'drainage 30 mL, thick, yellow, offensive' rather than 'wound looks bad'; write 'passed 150 mL of dark concentrated urine at 10 a.m.' rather than 'urine output poor'.
- Use the **SBAR** format when reporting to a doctor: *Situation* (who and what is wrong), *Background* (relevant history), *Assessment* (vital signs and your judgement), *Recommendation* (what you want done).
- Every entry must have the **date, time, signature, name and designation**; corrections by a single line with 'error' and initials; use only approved abbreviations.
- **Hand-over (change of shift) report** — identification, diagnosis, current condition and vital signs, treatments due, investigations pending, special observations required, and any patient or family concern.
- Document **patient teaching, refusals, incidents and the notification of the doctor** (with the time and the instruction received) — these are the entries most often examined in a legal enquiry.

CHAPTER AT A GLANCE

- Label the container, send at once, record and follow up; use standard precautions for all specimens.
- Mid-stream urine for culture in a sterile container; sputum = early morning deep cough, 2 samples for TB; blood culture before antibiotics at the fever spike; 24-h urine — discard the first voiding, keep the last.
- Benedict's test: blue = nil, green +, yellow ++, orange +++, brick red ++++ (8 drops of urine in 5 mL reagent).
- Fasting sugar 70-100 mg/dL; Hb men 13-17, women 12-15; platelets 1.5-4.5 lakh; potassium 3.5-5.0 mEq/L; creatinine 0.6-1.2 mg/dL; INR on warfarin 2-3.
- ORS (low osmolarity, 245 mOsm/L): NaCl 2.6 g, glucose 13.5 g, KCl 1.5 g, trisodium citrate 2.9 g per litre; use within 24 h; zinc for 14 days in child diarrhoea.
- Severe dehydration (10%+): unconscious, unable to drink, skin pinch over 2 s, no urine — give IV Ringer lactate.
- Pain is the fifth vital sign; assess with PQRST and the 0-10 or FACES scale; WHO ladder 'by the clock, by the mouth, by the ladder'.
- Report using SBAR; 'if it is not recorded, it is not done'.

Immunity and Infectious Diseases

Types of immunity, vaccines and the cold chain, immunisation schedule, and disease-wise profiles

14.1 Immunity — Definition and Types

DEFINITION

Immunity — the ability of the body to resist or overcome infection or the effects of a toxin. It is provided by the immune system through non-specific (innate) and specific (acquired) mechanisms.

Table 14.1 Types of immunity — a very frequently asked table

Type	Sub-type	How acquired	Examples	Onset & duration
Innate (natural / non-specific) immunity	Present from birth; no memory	Barriers — intact skin, mucus and cilia, tears (lysozyme), gastric acid, normal flora, cough reflex; cells — neutrophils, macrophages, NK cells; chemicals — complement, interferon, inflammation, fever	Species, racial and individual resistance	Immediate; lifelong but non-specific
Acquired — active — natural	Body makes its own antibodies	After suffering from the disease (clinical or subclinical infection)	Measles, chickenpox, mumps, typhoid	Slow onset (days-weeks); long-lasting, often lifelong ; has memory
Acquired — active — artificial	Body makes its own antibodies after a vaccine	Vaccination — live attenuated, killed, toxoid, subunit, mRNA	BCG, OPV, measles, DPT, hepatitis B	Slow onset; long-lasting (may need boosters)
Acquired — passive — natural	Ready-made antibodies received	Mother to child — IgG across the placenta, IgA in colostrum and breast milk	Protection of the newborn for the first 3-6 months	Immediate ; temporary (weeks-months)
Acquired — passive — artificial	Ready-made antibodies injected	Antiserum, immunoglobulin or antitoxin — ATS/TIG, anti-rabies immunoglobulin, anti-snake venom, anti-D, HBIG, convalescent plasma	Used when immediate protection is needed or the person is already exposed	Immediate ; short lived (2-3 weeks); risk of serum sickness/anaphylaxis
Herd (community) immunity	Protection of the whole community	When a large proportion (usually 80-95%) is immune, transmission stops and even the unimmunised are protected	Measles needs about 95%, polio about 80-85% coverage	Depends on coverage being maintained

HIGH-YIELD / MOST ASKED

Active immunity = the body works (slow, long lasting, has memory, given *before* exposure). **Passive immunity** = ready-made antibodies (immediate, short lasting, no memory, given *after* exposure or when there is no time to wait).

Combined (simultaneous) active and passive immunisation is given in: **tetanus-prone wounds (TT/Td + TIG)**, **rabies (vaccine + RIG)**, **hepatitis B needle-stick (vaccine + HBIG)**, and **diphtheria**.

Immunoglobulins: **IgG** — most abundant (about 75%), the only one crossing the placenta; **IgA** — secretory, in milk/tears/saliva/mucosa; **IgM** — largest ('millionaire'), the **first antibody produced** in an infection and a marker of **recent/acute infection**; **IgE** — allergy and helminth infection; **IgD** — on B-cell surfaces.

14.2 Vaccines

Table 14.2 Classification of vaccines

Class of vaccine	Examples	Points
Live attenuated	BCG, OPV, measles, MMR, rotavirus, varicella, yellow fever, oral typhoid, JE (live), intranasal influenza	Give a strong, long-lasting immunity, often with a single dose; contraindicated in pregnancy and immunodeficiency (HIV with low CD4, leukaemia, steroids, chemotherapy) ; heat and light sensitive; two live vaccines are given either on the same day or at least 4 weeks apart

Class of vaccine	Examples	Points
Killed / inactivated (whole organism)	IPV (injectable polio), whole-cell pertussis, rabies, hepatitis A, cholera (oral killed), JE (inactivated), influenza, typhoid (Vi killed)	Safer but weaker — need multiple doses and boosters and usually an adjuvant
Toxoid	Tetanus toxoid, diphtheria toxoid (in DPT/DT/Td)	Modified, harmless toxin; needs boosters; adsorbed on alum → never freeze
Subunit / polysaccharide / conjugate / recombinant	Hepatitis B (recombinant DNA), HPV, Hib, pneumococcal (PCV), meningococcal, acellular pertussis, typhoid Vi conjugate	Very safe and purified; conjugation to a protein makes them effective in infants under 2 years
Viral vector and mRNA (new generation)	Covishield/ChAdOx (vector), Covaxin (inactivated), Comirnaty and Spikevax (mRNA)	mRNA vaccines need ultra-cold storage (–20 to –70 °C)

14.3 The Cold Chain

- **Cold chain** — the system of storing and transporting vaccines at the recommended low temperature from the manufacturer to the point of use.
- **Storage temperatures** — walk-in cooler and ILR (Ice-Lined Refrigerator): +2 to +8 °C; deep freezer: –15 to –25 °C (for OPV and measles at the district and state level); vaccine carrier and day carrier: with ice packs, for 1-2 days of field use.
- In an ILR the **top shelf/basket holds the most heat-sensitive vaccines (OPV, measles, BCG)** and the lower part holds **DPT, TT, Td, hepatitis B, PCV and diluents** (which must *never be frozen*).
- **Most heat-sensitive vaccine: OPV** (then measles and BCG). **Most sensitive to freezing: hepatitis B, DPT, TT/Td, PCV and IPV** (adsorbed/ liquid vaccines).
- **Vaccine vial monitor (VVM)** — a heat-sensitive square on the label: use while the **inner square is lighter than the outer circle (stages 1 and 2); discard at stage 3 or 4** (inner square the same colour as or darker than the circle).
- **Shake test** — used on a suspected frozen adsorbed vaccine (DPT/TT/hepatitis B): if it shows rapid sedimentation with flakes compared with a control vial, it has been frozen and **must be discarded**.
- **Open vial policy** — opened multi-dose vials of **OPV, DPT, TT/Td, hepatitis B and IPV may be used for up to 28 days** if the VVM is usable, the expiry date has not passed, the septum is not submerged in water, and aseptic technique was used; but **BCG, measles/MR, JE and rotavirus (reconstituted or without preservative) must be discarded within 4 hours** of opening/reconstitution.
- Reconstitute only with the **matching diluent of the same manufacturer, cooled to the same temperature**; never store a reconstituted vaccine overnight.
- Never keep vaccines in the **refrigerator door**; keep a thermometer inside and maintain a **twice-daily temperature record**; do not store food, drugs or specimens with vaccines.

14.4 National Immunization Schedule (India, UIP)

Table 14.3 National Immunization Schedule

Age	Vaccine	Dose, route and site
At birth (within 24 h for hep B; BCG and OPV-0 up to 1 year/15 days)	BCG, OPV-0 (zero dose), Hepatitis B birth dose	BCG 0.05 mL (under 1 month) / 0.1 mL, intradermal, left upper arm ; OPV 2 drops oral; Hep B 0.5 mL IM antero-lateral thigh
6 weeks	OPV-1, Pentavalent-1 (DPT + hepatitis B + Hib), Rotavirus-1, fIPV-1, PCV-1	OPV oral; Pentavalent 0.5 mL IM left thigh; Rotavirus 5 drops oral; fIPV 0.1 mL intradermal right upper arm ; PCV 0.5 mL IM right thigh
10 weeks	OPV-2, Pentavalent-2, Rotavirus-2	As above
14 weeks	OPV-3, Pentavalent-3, Rotavirus-3, fIPV-2, PCV-2	As above
9-12 months	Measles-Rubella (MR)-1, JE-1 (in endemic districts), PCV-booster, Vitamin A 1st dose (1 mL = 1 lakh IU)	MR 0.5 mL subcutaneous, right upper arm ; JE SC left upper arm

Age	Vaccine	Dose, route and site
16-24 months	MR-2, DPT booster-1, OPV booster, JE-2, Vitamin A 2nd dose (2 mL = 2 lakh IU)	DPT 0.5 mL IM left upper arm; MR SC right upper arm
5-6 years	DPT booster-2	0.5 mL IM left upper arm
10 years and 16 years	Td (Tetanus-diphtheria)	0.5 mL IM upper arm
Pregnant woman	Td-1 early in pregnancy, Td-2 four weeks later, Td booster if the last dose was within 3 years	0.5 mL IM upper arm
Vitamin A	9 doses in all — 1st at 9 months, then every 6 months up to 5 years	1 lakh IU at 9 months, then 2 lakh IU orally

NUMBERS TO REMEMBER

BCG — *Bacillus Calmette-Guerin*, *Mycobacterium bovis* (attenuated 'Danish 1331' strain); diluent **normal saline**; gives a **papule → ulcer → scar in 6-12 weeks**; no dressing or antiseptic on the site.

Measles and MR vaccine is reconstituted with **distilled/sterile water** and given **subcutaneously**; discard within 4 hours.

Vitamin A — 1 lakh IU at 9 months; 2 lakh IU every 6 months up to 5 years (total 9 doses).

AEFI (adverse event following immunisation) must be reported — minor (fever, local pain, swelling), severe or serious (anaphylaxis, seizure, abscess, toxic shock); keep **adrenaline ready** at every session and observe for 30 minutes.

Immunisation programme milestones: **EPI 1974 (WHO) → EPI India 1978 → Universal Immunization Programme (UIP) 1985 → Mission Indradhanush 2014 → Intensified Mission Indradhanush**; India was declared **polio-free on 27 March 2014**; **smallpox eradicated globally in 1980** (last natural case in India 1975); **maternal and neonatal tetanus eliminated in India in 2015**.

14.5 Infectious Diseases — Comparative Profiles

Table 14.4 Infectious diseases at a glance (learn incubation periods and modes of spread)

Disease & agent	Incubation period	Mode of spread	Main features	Prevention / treatment
Measles (<i>Measles/rubeola virus</i>)	10-14 days (rash on day 14)	Droplet, airborne (highly infectious)	Fever, cough, coryza, conjunctivitis (the 3 C's), Koplik's spots (pathognomonic, on the buccal mucosa), then a maculopapular rash starting behind the ears/hairline and spreading downwards , with staining on fading	MR vaccine ; vitamin A; isolate 4 days before to 5 days after the rash; complications — pneumonia (commonest cause of death), otitis media, diarrhoea, blindness, SSPE, encephalitis
Chickenpox (varicella) (<i>Varicella-zoster virus</i>)	14-16 days (10-21)	Droplet, airborne, contact with vesicle fluid	Mild fever then a centripetal (trunk first) rash , all stages present together (pleomorphic): macule → papule → vesicle → pustule → crust; <i>dew-drop on a rose petal</i> appearance; intense itching	Varicella vaccine; infectious 1-2 days before the rash until all lesions crust (about 6 days) ; acyclovir for the severe/immunocompromised; <i>herpes zoster (shingles)</i> is a reactivation
Mumps (<i>Mumps virus</i>)	14-18 days	Droplet, saliva	Painful swelling of the parotid gland(s) , fever, pain on chewing; complications — orchitis (sterility), meningitis, pancreatitis, deafness	MMR vaccine; isolate until the swelling subsides; symptomatic treatment, soft bland food, no sour food
Rubella (German measles)	14-21 days	Droplet, transplacental	Mild fever, fine pink rash, posterior auricular and occipital lymphadenopathy ; in the 1st trimester causes congenital rubella syndrome — cataract, deafness, heart defect	MR/MMR vaccine ; avoid pregnancy for 1 month after the vaccine ; keep away from pregnant women

Disease & agent	Incubation period	Mode of spread	Main features	Prevention / treatment
Diphtheria (<i>Corynebacterium diphtheriae</i>)	2-5 days	Droplet, contact, fomites, infected milk	Sore throat with a grey-white adherent pseudo-membrane that bleeds on removal, bull neck, toxin causing myocarditis and neuritis , respiratory obstruction	DPT/Td; treatment: antitoxin (ADS) at once + penicillin/erythromycin ; strict isolation and bed rest; Schick test (historical) for susceptibility
Whooping cough (pertussis) (<i>Bordetella pertussis</i>)	7-14 days	Droplet	Catarrhal stage (most infectious) → paroxysmal stage with bouts of coughing ending in an inspiratory 'whoop' and vomiting → convalescence; may last 6-10 weeks ('the 100-day cough'); lymphocytosis	DPT/pentavalent; erythromycin/azithromycin; small frequent feeds after a bout; avoid dust and smoke
Tetanus (lockjaw) (<i>Clostridium tetani</i> toxin)	6-10 days (3-21; shorter = worse prognosis)	Spores from soil/dust/manure entering a wound; unclean delivery and cord cutting (neonatal tetanus)	Trismus (lockjaw), risus sardonicus, neck and back stiffness, opisthotonus, board-like abdomen, painful generalised spasms provoked by light, noise and touch , with a fully conscious patient	Td/TT immunisation, 5 cleans at delivery, wound toilet, TIG/ATS ; treatment in a dark, quiet room with diazepam/muscle relaxants, metronidazole, ventilation; <i>one attack gives no immunity</i>
Poliomyelitis (<i>Poliovirus</i> 1, 2, 3)	7-14 days (3-35)	Faeco-oral (and droplet)	Fever, headache, then asymmetrical flaccid paralysis with no sensory loss , commonly of the leg; the vast majority of infections are subclinical	OPV and IPV; AFP surveillance of every case under 15 years ; no specific treatment — physiotherapy, splints, prevention of deformity
Tuberculosis (<i>Mycobacterium tuberculosis</i>)	4-12 weeks to develop a lesion (years to disease)	Airborne droplet nuclei (an untreated case infects 10-15 persons a year)	Cough over 2 weeks, evening rise of temperature, night sweats, loss of weight and appetite, haemoptysis , chest pain; extra-pulmonary forms — lymph node, spine, meninges, abdomen	BCG; sputum microscopy/NAAT (CBNAAT); DOTS — 2 months intensive (HRZE) + 4 months continuation (HRE) under the NTEP ; isolate/ventilate, cough etiquette, contact tracing, nutrition support (Ni-kshay Poshan Yojana)
Typhoid (enteric fever) (<i>Salmonella typhi</i>)	10-14 days (7-21)	Faeco-oral — contaminated water, food, milk, flies; carriers (gall bladder)	Step-ladder fever with a relative bradycardia, headache, coated tongue, rose spots on the abdomen (2nd week), splenomegaly, constipation then pea-soup diarrhoea; complications in the 3rd week — intestinal perforation and haemorrhage	Typhoid Vi/conjugate vaccine , safe water and food, sanitation, handwashing; blood culture (1st week), Widal test (2nd week), stool culture (3rd week) ; ceftriaxone/azithromycin; absolute bed rest and a soft low-fibre diet
Cholera (<i>Vibrio cholerae</i> O1 El Tor/O139)	Hours to 5 days (usually 1-2)	Faeco-oral — contaminated water and food	Sudden profuse painless 'rice-water' stools with fishy odour and effortless vomiting → rapid severe dehydration, muscle cramps, washerwoman's hands, sunken eyes, anuria, shock	ORS is the mainstay + IV Ringer lactate in severe cases ; doxycycline/azithromycin shortens shedding; safe water, chlorination, sanitation, oral cholera vaccine; notifiable
Bacillary dysentery / amoebiasis	1-7 days / 2-4 weeks	Faeco-oral	Frequent small stools with blood and mucus, tenesmus and colic ; amoebic — gradual, with liver abscess possible	ORS, ciprofloxacin (shigella), metronidazole for amoebiasis ; hand washing, safe water and food

Disease & agent	Incubation period	Mode of spread	Main features	Prevention / treatment
Hepatitis A and E	A: 15-45 days (about 4 weeks); E: 2-9 weeks	Faeco-oral (E often water-borne epidemics)	Prodromal fever, anorexia, nausea, distaste for smoking, then jaundice with dark urine and pale stools , tender hepatomegaly; hepatitis E is dangerous in pregnancy (up to 20% mortality)	Safe water and food, hand hygiene; hepatitis A vaccine ; rest, no alcohol, no fatty food; <i>no specific antiviral</i>
Hepatitis B and C	B: 45-180 days (2-6 months); C: 2 weeks-6 months	Blood, sharps, unsafe injections, transfusion, sexual contact, mother to child	May be asymptomatic; jaundice, and chronic infection → cirrhosis and hepatocellular carcinoma	Hepatitis B vaccine (0, 1, 6 months) and HBIG after exposure; screening of blood; safe injections and universal precautions ; antivirals (tenofovir; direct-acting antivirals cure hepatitis C)
HIV / AIDS (Human immunodeficiency virus)	Weeks (seroconversion 3-12 weeks); AIDS after 7-10 years	Sexual contact (the commonest, 80-85%), blood and blood products, contaminated needles, mother to child (in utero, at birth, breast milk); not by touch, air, food, water, insects, sharing utensils or swimming	Acute flu-like illness → long asymptomatic phase → weight loss, chronic fever and diarrhoea, oral thrush, herpes zoster, tuberculosis, <i>Pneumocystis pneumonia</i> , Kaposi's sarcoma; CD4 falls (AIDS when CD4 under 200/mm³)	ART for all (test and treat), universal precautions, safe blood, condoms, PPTCT/PMTCT with ART in pregnancy, PEP and PrEP ; counselling and confidentiality (ICTC); NACO programme
Malaria (<i>Plasmodium vivax, falciparum, malariae, ovale, knowlesi</i>)	<i>P. vivax</i> 12-17 days; <i>P. falciparum</i> 9-14 days	Bite of an infected female Anopheles mosquito (night biter); also transfusion and transplacental	Paroxysms of cold stage (rigor) → hot stage → sweating stage , splenomegaly, anaemia; <i>P. falciparum</i> causes cerebral malaria, renal failure and black-water fever (the dangerous species)	Blood smear/rapid diagnostic test ; treatment as per the National Drug Policy (chloroquine for vivax + primaquine 14 days; ACT — artemisinin combination therapy for falciparum + single-dose primaquine); prevention — insecticide-treated bed nets, indoor residual spraying, source reduction, larvicides, <i>Gambusia</i> fish
Dengue (Dengue virus 1-4)	4-10 days (3-14)	<i>Aedes aegypti</i> — a day-biting mosquito breeding in clean stored water (coolers, tyres, flower pots)	Sudden high fever, severe retro-orbital headache, muscle and joint pain ('break-bone fever') , rash, positive tourniquet test; warning signs — severe abdominal pain, persistent vomiting, bleeding, restlessness, falling platelet count with rising haematocrit (dengue haemorrhagic fever/shock syndrome)	No specific treatment — fluids and paracetamol only; AVOID aspirin, NSAIDs and IM injections (bleeding risk); source reduction, personal protection; monitor platelets and haematocrit
Chikungunya / Zika	3-7 days / 3-14 days	<i>Aedes</i> mosquito	Chikungunya: fever with severe prolonged joint pain ; Zika: mild illness but causes microcephaly in the fetus	Symptomatic; vector control; Zika — avoid pregnancy/travel advisory
Filariasis (<i>Wuchereria bancrofti</i>)	8-16 months	<i>Culex quinquefasciatus</i> mosquito (breeds in dirty water)	Recurrent fever with lymphangitis and lymphadenitis, elephantiasis of the legs, scrotum and breast, hydrocele, chyluria ; <i>night blood smear</i> for microfilariae	Mass drug administration — DEC + albendazole (with ivermectin in triple therapy) ; limb hygiene and elevation, vector control

Disease & agent	Incubation period	Mode of spread	Main features	Prevention / treatment
Kala-azar (visceral leishmaniasis)	2-6 months	Bite of the female Phlebotomus (sandfly) breeding in cracks of mud walls	Prolonged fever with a double rise, gross splenomegaly and hepatomegaly, darkening of the skin, wasting, pancytopenia	Liposomal amphotericin B (single dose) / miltefosine ; indoor residual spraying, plastering of cracks
Rabies (Rabies virus)	1-3 months (4 days to years; shorter with bites on the face and hands)	Bite, scratch or lick on broken skin by a rabid dog, cat, monkey, bat, mongoose, jackal (saliva)	Prodrome with pain/tingling at the bite site, then hydrophobia, aerophobia, hypersalivation, agitation, spasms of the throat (furious) or paralysis (dumb); 100% fatal once symptoms appear	Immediate wound washing with soap and running water for 15 minutes , then povidone-iodine; anti-rabies vaccine (0, 3, 7, 14, 28 days IM, or intradermal schedule) + rabies immunoglobulin infiltrated around the wound for category III ; never suture tightly; pre-exposure vaccination for those at risk; <i>animal is observed for 10 days</i>
Scabies / Pediculosis	2-6 weeks (scabies)	Close personal contact, shared clothes and bedding	Intense night itching with burrows in the finger webs, wrists, axillae, groin and genitals; family members affected	Permethrin 5% cream (whole body below the neck, overnight, repeat after 7 days) or ivermectin; treat all family members at the same time and wash/sun all clothes and bedding
Leprosy (Hansen's disease) (M. leprae)	2-5 years (up to 20)	Prolonged close contact, droplets from the nose of untreated multibacillary cases	Hypopigmented or erythematous patch with loss of sensation , thickened nerves, nodules, claw hand, foot drop, trophic ulcers, loss of eyebrows	Multidrug therapy (MDT) — rifampicin, dapsone (+ clofazimine for MB) ; free under the NLEP; disability care and no stigma — <i>a patient becomes non-infectious within days of starting MDT</i>
Influenza / COVID-19	Influenza 1-4 days; COVID-19 2-14 days (median 5)	Droplet, aerosol, contact	Fever, cough, sore throat, myalgia; COVID-19 — loss of smell and taste, breathlessness, hypoxia, ARDS	Annual influenza vaccine, COVID-19 vaccine, mask, hand hygiene, distancing, ventilation ; oseltamivir for influenza; oxygen and steroids for severe COVID-19
Plague (Yersinia pestis)	Bubonic 2-7 days; pneumonic 1-4 days	Rat flea (Xenopsylla cheopis) bite; pneumonic by droplet	Bubonic — fever with a very painful bubo (lymph node); pneumonic — fulminant pneumonia, highly fatal and highly contagious	Notifiable internationally ; streptomycin/doxycycline; rat and flea control, insecticide before rodenticide
Tetanus neonatorum / Ophthalmia neonatorum	3-14 days / 2-5 days	Unclean cord care / infected birth canal (gonococcus, chlamydia)	Refusal to suck, stiffness and spasms / purulent eye discharge in the newborn	5 cleans, Td in pregnancy / eye toilet and antibiotic eye drops at birth

14.6 Isolation, Quarantine, Notification and Disinfection

- **Isolation** — separation of an **infected** person for the **period of communicability**; **quarantine** — limitation of the movement of an apparently **healthy contact** for the **longest incubation period** of the disease.
- **Notifiable diseases (India, varies by State)** — cholera, plague, yellow fever (internationally notifiable under the IHR), plus tuberculosis, HIV/AIDS, malaria, dengue, chikungunya, kala-azar, leprosy, measles, diphtheria, whooping cough, tetanus, polio/AFP, viral hepatitis, meningitis, rabies, typhoid, food poisoning, Japanese encephalitis, COVID-19; also maternal and infant deaths. Reporting is a **legal duty** of the health worker (Form-P/IDSP reporting).
- **Disinfection** — concurrent during illness and terminal after recovery (Chapter 3); the excreta of enteric fever, cholera, dysentery and hepatitis A must be disinfected before disposal.

- **Contacts** — identify, examine, give chemoprophylaxis or vaccine where indicated (rifampicin for meningococcal and leprosy contacts, azithromycin for pertussis contacts, immunoglobulin for measles/hepatitis A contacts), and educate them about the early signs.
- **Surveillance** — IDSP (Integrated Disease Surveillance Programme) weekly reporting of S, P and L forms; outbreak investigation; AFP and measles surveillance.

CHAPTER AT A GLANCE

- Active immunity = slow, long-lasting, has memory (disease/vaccine); passive = immediate, short-lived (maternal antibody, immunoglobulin, antitoxin).
- IgG crosses the placenta and is the most abundant; IgM is the first formed and indicates recent infection; IgA is in breast milk.
- Live vaccines: BCG, OPV, measles/MR, MMR, rotavirus, varicella, yellow fever — contraindicated in pregnancy and immunodeficiency.
- Cold chain 2-8 °C (ILR); deep freezer -15 to -25 °C for OPV/measles; never freeze DPT, TT, hepatitis B, PCV, IPV; discard at VVM stage 3/4; shake test for a frozen vaccine.
- Open vial policy: OPV, DPT, TT, hepatitis B, IPV = 28 days; BCG, measles/MR, JE, rotavirus = 4 hours.
- BCG: 0.05/0.1 mL intradermal, left upper arm, saline diluent. Measles/MR: 0.5 mL subcutaneous, right upper arm.
- Incubation: measles 10-14 d, chickenpox 14-16 d, mumps 14-18 d, rubella 14-21 d, diphtheria 2-5 d, pertussis 7-14 d, tetanus 6-10 d, polio 7-14 d, typhoid 10-14 d, cholera 1-2 d, hepatitis A 2-6 wk, hepatitis B 2-6 months, rabies 1-3 months, leprosy 2-5 years.
- Koplik's spots = measles; pseudomembrane = diphtheria; rose spots and step-ladder fever = typhoid; rice-water stools = cholera; hydrophobia = rabies; night itching = scabies.
- Dengue: no aspirin/NSAIDs/IM injections; Aedes bites by day; malaria — Anopheles bites at night.

Care of the Aged and the Long-Term Patient

Ageing changes, geriatric problems, prevention of complications, rehabilitation and palliative care

15.1 Basic Concepts

Table 15.1 Terms in geriatric nursing

Term	Meaning
Geriatrics	The branch of medicine dealing with the diseases and care of old age
Gerontology	The scientific study of the process and problems of ageing (biological, psychological and social)
Elderly / Senior citizen (India)	A person aged 60 years and above (WHO also uses 60+; 'old old' 75-84, 'oldest old' 85+)
Senescence	The normal process of growing old
Long-term (chronic) patient	A patient needing care for more than 3 months — paralysis, arthritis, dementia, cancer, chronic kidney/lung disease
Rehabilitation	Restoring a disabled person to the fullest physical, mental, social and vocational usefulness of which he is capable
Palliative care	Active total care of a patient whose disease is not curable, aimed at the best possible quality of life for the patient and family
Activities of daily living (ADL)	Bathing, dressing, toileting, transferring, continence, feeding (assessed by the Katz index/Barthel index)

15.2 Physiological Changes of Ageing (system-wise)

Table 15.2 Changes of ageing and their nursing implications

System	Changes with age	Nursing implication
Skin and hair	Thin, dry, inelastic, wrinkled skin; loss of subcutaneous fat; fragile capillaries (easy bruising); grey, thin hair; brittle nails; reduced sweating	Gentle handling, moisturiser, avoid strong soap and hot water, protect from pressure and from injury, watch for pressure sores and hypothermia
Musculoskeletal	Loss of muscle mass and strength (sarcopenia), osteoporosis (especially after menopause), thinning discs and loss of height, stiff joints, kyphosis, osteoarthritis	High risk of fracture (neck of femur, wrist, vertebra) from a minor fall ; calcium, vitamin D, protein, weight-bearing exercise, fall prevention
Nervous system	Fewer neurons, slower reflexes and reaction time, reduced short-term memory, altered sleep (less deep sleep, early waking), reduced thirst and temperature sensation, unsteady gait	Speak slowly, give one instruction at a time, allow extra time, orient frequently, prevent falls and dehydration
Special senses	Presbyopia, cataract, glaucoma, reduced night vision and glare tolerance; presbycusis (high-tone deafness) , reduced taste and smell	Good non-glare lighting, large print, face the patient and speak clearly in a low pitch, check spectacles and hearing aids, food may need more flavour (not more salt)
Cardiovascular	Stiff arteries and valves, reduced cardiac output and reserve, postural hypotension , varicose veins	Change position slowly, watch for dizziness and syncope, avoid sudden exertion, monitor BP in both sitting and standing
Respiratory	Reduced elastic recoil and vital capacity, weak cough and ciliary action, rigid chest wall	High risk of pneumonia and atelectasis — deep breathing, early ambulation, position change, influenza and pneumococcal vaccine
Gastro-intestinal	Loss of teeth, dry mouth, reduced saliva and gastric acid, slow peristalsis, reduced liver mass and enzymes	Constipation (fibre, fluids, activity), indigestion, dysphagia and aspiration risk, altered drug metabolism, dental care
Urinary	Reduced renal blood flow and GFR, reduced bladder capacity, weak sphincter, prostatic enlargement in men	Nocturia, frequency, urgency, incontinence, retention ; drug doses must be reduced; watch for UTI (which may present as confusion)
Endocrine and metabolic	Reduced glucose tolerance, lower basal metabolic rate, menopause/andropause, reduced thyroid function	Risk of diabetes and hypothyroidism; adjust calories; avoid weight gain
Immune system	Reduced immunity (immunosenescence)	Infections are common, often without fever ; vaccinate; watch for atypical presentation

System	Changes with age	Nursing implication
Psychosocial	Retirement, loss of income and status, death of spouse and friends, loneliness, dependence, fear of death	Respect, allow decision-making, encourage social contact, recreation and spiritual needs

15.3 Common Problems of the Aged — the 'Geriatric Giants'

Table 15.3 Common geriatric problems and their management

Problem	Causes	Nursing management
Falls and instability	Poor vision, weakness, postural hypotension, dizziness, arthritis, neuropathy, sedatives and antihypertensives, poor footwear, wet or cluttered floors, poor lighting, unfamiliar surroundings	Fall risk assessment (Morse scale) ; clear pathways, non-slip dry floor, adequate night lighting, grab bars and raised toilet seat in the bathroom, bed at low height with brakes, call bell in reach, well-fitting non-slip footwear, walking aid at the correct height, review of medicines, vitamin D and balance exercises, no rushing, teach to sit at the edge of the bed for a minute before standing
Immobility	Paralysis, arthritis, fracture, fear of falling, deconditioning, depression	Complications of immobility affect every system — pressure sores, contractures and foot drop, muscle atrophy, osteoporosis, constipation and faecal impaction, urinary stasis and infection and stones, pneumonia and atelectasis, deep vein thrombosis and pulmonary embolism, postural hypotension, depression and confusion ; prevent with 2-hourly position change, passive and active exercises, deep breathing, early mobilisation, adequate fluid and fibre, and a footboard/splints
Incontinence (urinary and faecal)	Weak sphincter, prostatic enlargement, UTI, constipation, immobility, dementia, diuretics	Never scold; 2-hourly toileting and bladder training, pelvic floor (Kegel) exercises, easy-to-remove clothing, commode near the bed, barrier cream and pads, catheter only as a last resort, treat constipation and infection
Intellectual impairment — dementia and delirium	Dementia: Alzheimer's disease (commonest), vascular dementia — <i>slow, progressive, irreversible</i> . Delirium: infection (especially UTI and pneumonia), dehydration, drugs, retention of urine, constipation, hypoxia, pain, electrolyte imbalance — <i>sudden, fluctuating, reversible</i>	Any sudden confusion in an elderly person is delirium until proved otherwise — look for a treatable cause. Keep a calm, well-lit, familiar environment with a fixed routine; orient with a clock and calendar; use short simple sentences and the patient's name; avoid restraints and multiple sedatives; ensure safety (wandering, gas, stairs); involve the family; support the care-giver
Malnutrition and dehydration	Poor appetite, loss of teeth, dysphagia, poverty, loneliness, depression, reduced thirst sensation, drugs	Small frequent energy- and protein-dense meals, soft food, dentures, company at meals, supplements, weigh monthly, offer fluids every 1-2 hours (do not wait for thirst) , watch for a dry tongue and concentrated urine
Insomnia	Pain, nocturia, anxiety, daytime napping, caffeine, dyspnoea, noise	Sleep hygiene — fixed bedtime, no daytime naps, no tea/coffee in the evening, warm milk, back rub, quiet dark room, empty bladder, comfortable position; avoid routine hypnotics (they cause falls and confusion)
Depression and social isolation	Bereavement, loneliness, disability, chronic pain, dependence, financial worry	Listen, allow the expression of grief, encourage hobbies and social/religious contact, involve in decisions, watch for suicidal ideas (report at once) , refer for treatment; depression in the elderly often presents as physical complaints or 'pseudodementia'
Polypharmacy and adverse drug reactions	Multiple diseases and prescribers, self-medication, altered kinetics (reduced renal and hepatic clearance, reduced albumin, altered body composition)	'Start low, go slow' ; review the whole drug list regularly (including over-the-counter and herbal products); simplify the regimen and use a pill box/calendar; watch specially for sedatives, anticholinergics, NSAIDs, digoxin, warfarin, oral hypoglycaemics and antihypertensives (the Beers criteria list drugs to avoid in the elderly); teach the patient and family in writing with large print
Sensory deprivation and communication difficulty	Deafness, blindness, aphasia	Hearing aid and spectacles kept clean and working, good light, face the patient, write, use gestures and pictures
Elder abuse and neglect	Care-giver stress, dependence, financial motives	Look for unexplained bruises, fear, poor hygiene, malnutrition, unfilled prescriptions; report as per the Maintenance and Welfare of Parents and Senior Citizens Act, 2007

15.4 Nursing Care of the Long-Term Bedridden Patient

1. **Prevent pressure sores** — 2-hourly turning with a turning chart, pressure-relieving mattress, skin inspection at every turn, keep dry and clean, high protein and vitamin C diet (see Chapter 4).
2. **Prevent contractures and deformities** — maintain **functional position** (joints in mid-position, ankle at 90 degrees with a footboard, hand roll to prevent claw hand, trochanter roll to prevent external rotation of the hip); give **passive range-of-motion exercises to every joint 2-3 times a day, 5-10 repetitions each**; encourage active and isometric exercise; do not force a painful joint.
3. **Prevent chest complications** — semi-Fowler's position, **deep breathing and coughing exercises hourly while awake, incentive spirometry, blow a balloon/bubbles**, chest physiotherapy and postural drainage as ordered, adequate fluids, oral hygiene, and change of position.
4. **Prevent deep vein thrombosis** — leg and ankle exercises, early mobilisation, adequate hydration, elastic compression stockings, prescribed anticoagulant prophylaxis; **never massage the calf** if DVT is suspected (unilateral calf pain, swelling, warmth) — report at once.
5. **Prevent urinary problems** — 2-3 L of fluid daily, regular voiding, upright position for voiding, perineal hygiene, avoid unnecessary catheterisation.
6. **Prevent constipation and impaction** — fibre, fluids, activity, regular time, privacy, prescribed laxative; check for impaction if there is spurious diarrhoea or no stool for 3 days.
7. **Nutrition** — high protein, high calorie, high fibre, vitamin- and mineral-rich diet with adequate fluid; assist or feed sitting up; weigh weekly.
8. **Maintain hygiene, grooming and dignity** — daily bed bath, mouth care 2-3 times a day, hair care, nail care, clean clothes; let the patient wear his own clothes and be well groomed — it improves morale.
9. **Psychological and social care** — talk to the patient, keep him informed and involved in decisions, provide radio/TV/books/phone, encourage visitors, maintain a day-night routine, respect religious practice, and **encourage maximum self-care (never do for the patient what he can do himself)**.
10. **Support and teach the family care-giver** — demonstrate lifting, bathing, feeding, drug administration and the signs to report; arrange respite; recognise care-giver burnout.

CLINICAL / PRACTICAL POINT

Rehabilitation aids and appliances — walking stick (held in the **hand opposite the affected leg**, elbow flexed 15-30°, height to the wrist crease), walker, crutches (2-3 finger breadths below the axilla with the weight taken on the **hands, not the axilla**), wheelchair, tripod, hand rails, raised toilet seat, commode chair, long-handled reacher, adapted cutlery, button hook, non-slip mat, hospital cot with side rails, air mattress, hoist.

Teaching a patient to walk with a stick after a stroke/hip surgery: stick → affected leg → sound leg; **going up the stairs: good leg first; coming down: bad (affected) leg and the stick first** ('up with the good, down with the bad').

15.5 Palliative and Terminal Care; Care of the Dying

- **Aim** — 'to add life to the days, not days to the life': relief of pain and distressing symptoms, psychological, social and spiritual support for the patient **and the family**, neither hastening nor postponing death.
- **Symptom control** — pain (WHO ladder, **regular oral morphine 'by the clock' with a laxative and an antiemetic**; there is no maximum dose of morphine in terminal pain and **addiction is not a concern**), breathlessness (position, fan, oxygen, low-dose morphine), nausea, constipation, dry mouth (frequent sips, ice chips, mouth care), bed sores, anorexia (do not force food), 'death rattle' secretions (position and suction gently, anticholinergic), restlessness and fear.
- **Communication** — answer questions honestly and simply, allow the patient to talk about dying and to express anger or fear, never take away hope, maintain privacy and never whisper at the bedside, let the family stay, allow the religious rites the family wishes.
- **Kubler-Ross stages of grief** — **Denial → Anger → Bargaining → Depression → Acceptance** (they may not occur in order, and some patients never reach acceptance).
- **Hospice care** — specialised palliative care for the terminally ill, given in a hospice or at home by a team.
- **Advance directives, do-not-resuscitate orders and the patient's right to refuse treatment** must be respected; euthanasia is illegal in India, but the Supreme Court has recognised **passive euthanasia and a living will (2018)** in strictly defined circumstances.
- **Care of the body after death (last offices)** and support of the bereaved family — see Chapter 6.

15.6 Health Care Services and Schemes for the Elderly in India

- **National Programme for the Health Care of the Elderly (NPHCE), 2010** — geriatric OPD and wards at district hospitals, regional geriatric centres, rehabilitation units at CHC/PHC and domiciliary visits by health workers.
- **Maintenance and Welfare of Parents and Senior Citizens Act, 2007** — legally obliges children to maintain their parents; provides for old age homes and protection from abuse and abandonment.
- **National Policy on Older Persons (1999); Integrated Programme for Senior Citizens; Rashtriya Vayoshri Yojana** (free aids and appliances); **Vayoshreshtha Samman**; travel and income-tax concessions; Ayushman Bharat/PM-JAY coverage for those aged 70+.
- **International Day of Older Persons — 1 October; World Alzheimer's Day — 21 September.**

CHAPTER AT A GLANCE

- Elderly = 60 years and above in India; geriatrics = clinical care, gerontology = the study of ageing.
- Ageing: presbyopia and presbycusis, osteoporosis, reduced GFR (so lower drug doses), postural hypotension, weak cough, constipation, reduced thirst and immunity.
- Geriatric giants: falls/instability, immobility, incontinence, intellectual impairment (dementia/delirium), plus malnutrition, insomnia, depression and polypharmacy.
- Sudden confusion in an elderly patient = delirium — look for UTI, pneumonia, dehydration, drugs, retention or constipation.
- Prescribing: 'start low, go slow'; review the drug list; Beers criteria list drugs to avoid.
- Complications of immobility: pressure sores, contractures, foot drop, pneumonia, DVT, constipation, UTI, osteoporosis, depression — turn 2-hourly, passive exercises, deep breathing, leg exercises.
- Stick in the hand opposite the affected leg; crutch weight on the hands, not the axilla; stairs — 'up with the good, down with the bad'.
- Palliative care: regular oral morphine by the clock with a laxative; Kubler-Ross — denial, anger, bargaining, depression, acceptance.
- NPHCE 2010; Maintenance and Welfare of Parents and Senior Citizens Act 2007; Older Persons' Day 1 October.

Care of the Mentally Ill and the Mentally Healthy Patient

Mental health, classification of mental illness, therapeutic communication, safety, psychotropic drugs and ECT

16.1 Mental Health and Mental Hygiene

DEFINITION

Mental health (WHO) — a state of well-being in which the individual realises his own abilities, can cope with the normal stresses of life, can work productively and fruitfully, and is able to make a contribution to his community.

Mental hygiene — the science of promoting and preserving mental health and preventing mental illness.

Psychiatry — the branch of medicine dealing with the diagnosis, treatment and prevention of mental disorders. **Psychiatric nursing** — nursing care of a person with a mental disorder, using the self as a therapeutic tool.

Characteristics of a mentally healthy person

- Has a **realistic self-concept** and accepts himself and others as they are; accepts his own limitations and strengths.
- Is in **touch with reality**, orientated in time, place and person; has insight.
- Can **form and maintain satisfying relationships** and can give and receive affection.
- Can **handle stress, frustration and emotion** appropriately and recovers from a setback (resilience).
- Is **independent, self-directing and responsible**; can take decisions and accept their consequences.
- Works **productively**, has a purpose in life, uses his abilities, and can enjoy leisure.
- Uses **mature coping mechanisms** (sublimation, humour, altruism) rather than immature ones.

Promotion of mental health

- A secure, loving childhood; consistent discipline; success experiences and encouragement.
- Balanced routine of work, rest, recreation, exercise, sleep and nutrition; avoidance of tobacco, alcohol and drugs.
- Healthy relationships and social support; the ability to express feelings and ask for help.
- Sex education, marriage and family counselling, guidance at times of transition (school, adolescence, marriage, retirement, bereavement).
- Early recognition and treatment of mental illness; removal of stigma; mental health education in schools and the community.

16.2 Causes and Classification of Mental Illness

Table 16.1 Causes of mental illness

Group of causes	Examples
Predisposing (biological)	Heredity and genes, neurotransmitter imbalance (dopamine excess in schizophrenia; low serotonin and noradrenaline in depression; GABA in anxiety), birth injury, brain damage, endocrine disorder, epilepsy, intellectual disability, ageing
Physical / organic	Head injury, infection (meningitis, encephalitis, neurosyphilis, HIV), tumour, stroke, dementia, hypothyroidism, vitamin B12 deficiency, uraemia, hepatic failure, hypoxia, drugs (steroids), alcohol and substance abuse, high fever (delirium)
Psychological	Faulty upbringing, maternal deprivation, childhood abuse, over-protection or rejection, unresolved conflict, poor self-esteem, faulty learning, personality type
Social / environmental (precipitating)	Bereavement, failure in examination or love, unemployment, poverty, debt, marital discord, migration, disaster, war, isolation, retirement, chronic physical illness, stigma

Table 16.2 Classification of mental disorders

Category	Main features
Neurosis (minor psychiatric illness)	Insight and contact with reality are retained ; no hallucination or delusion; the patient is distressed and seeks help; includes anxiety disorder, panic disorder, phobia, obsessive-compulsive disorder , hysteria/conversion and dissociative disorder, somatoform disorder, post-traumatic stress disorder, mild depression
Psychosis (major psychiatric illness)	Loss of insight and of contact with reality, delusions and hallucinations , disorganised thought and behaviour; includes schizophrenia, bipolar (manic-depressive) disorder, severe depression with psychotic features, delusional disorder, organic psychosis/delirium

Category	Main features
Schizophrenia	Commonest psychosis; onset in adolescence/young adulthood; delusions (especially of persecution and reference), auditory hallucinations, thought disorder and echo, disorganised speech and behaviour (positive symptoms) with apathy, social withdrawal, poverty of speech, blunted affect and loss of volition (negative symptoms); subtypes: paranoid, catatonic, hebephrenic, simple, residual
Mood (affective) disorders	Depression — persistent sadness, loss of interest (anhedonia), fatigue, guilt and worthlessness, poor concentration, early morning waking, loss of appetite and weight, loss of libido, psychomotor retardation, suicidal ideas . Mania — elevated mood, over-activity, over-talkativeness (pressure of speech), grandiose ideas, flight of ideas, reduced need for sleep, over-spending, irritability, poor judgement. Bipolar disorder — alternating episodes
Anxiety disorders	Excessive worry with palpitation, sweating, tremor, dry mouth, breathlessness, dizziness, chest discomfort, frequency of urine and a feeling of impending doom; panic attack — sudden intense episode; phobia — irrational fear of a specific object or situation
Organic brain syndromes	Delirium (acute, fluctuating, with clouded consciousness, disorientation, visual hallucinations — a medical emergency) and dementia (chronic, progressive loss of memory and intellect with clear consciousness)
Substance use disorders	Alcohol (dependence, withdrawal: tremor, sweating, insomnia, seizures and delirium tremens 48-72 h after stopping; Wernicke's encephalopathy needs thiamine), opioids (heroin, injectable — pin-point pupils, withdrawal cramps and yawning; treated with naloxone for overdose and buprenorphine/methadone substitution), cannabis, tobacco, sedatives, inhalants, stimulants
Child and adolescent disorders	Intellectual disability, autism spectrum disorder, ADHD, conduct disorder, enuresis, learning disability, school refusal, adolescent behaviour problems
Personality disorders and others	Antisocial, borderline, paranoid, dependent; eating disorders (anorexia nervosa, bulimia); sexual dysfunction; puerperal (post-partum) psychosis and post-natal depression; epilepsy-related behaviour problems; suicide

DEFINITION

Terms to know: **Delusion** — a false, firm, unshakeable belief not in keeping with the person's culture (of persecution, grandeur, reference, guilt, nihilism, infidelity, hypochondriasis). **Hallucination** — a sense perception **without any external stimulus** (auditory is the commonest in schizophrenia; visual in delirium and alcohol withdrawal; tactile, olfactory, gustatory). **Illusion** — a **misinterpretation** of a real stimulus (a rope seen as a snake). **Obsession** — a recurrent, unwanted, intrusive thought; **compulsion** — the repetitive act that relieves it (hand washing, checking). **Insight** — awareness of one's own illness. **Affect/Mood** — observed emotional expression / sustained inner feeling. **Euphoria, elation, apathy, ambivalence, flight of ideas, echolalia, echopraxia, mutism, negativism, waxy flexibility, stupor, catatonia, confabulation, neologism, word salad, perseveration, tangentiality.**

16.3 Principles of Psychiatric Nursing and Therapeutic Communication

1. **Accept the patient as a person** — separate the person from the behaviour; be non-judgemental; never ridicule, argue or laugh at him.
2. Build **trust** through consistency, honesty and keeping promises; be reliable and punctual; explain everything.
3. Use the **self therapeutically** — a calm, confident, unhurried manner; be aware of your own feelings and reactions.
4. **Listen actively** and allow the patient to express feelings; use open-ended questions, reflection, clarification and silence; sit at a safe, comfortable distance at eye level.
5. **Do not agree with or reinforce a delusion or hallucination, and do not argue about it either** — acknowledge the feeling, present reality gently and simply: *'I do not hear the voices, but I understand they are frightening for you. I am here with you.'*
6. Set **clear, consistent, firm limits** on unacceptable behaviour, explaining the reason; the whole team must apply the same limits.
7. Encourage **self-care, activity, occupational therapy, recreation and group participation** — structure the day with a fixed routine.
8. Give recognition for positive behaviour; involve the patient in planning his own care wherever possible.
9. Maintain **confidentiality, dignity and rights**; obtain consent; avoid discussing the patient in his presence.
10. Involve and **educate the family**; work for adherence to medication and for removal of stigma; arrange follow-up and rehabilitation.

16.4 Safety of the Psychiatric Patient

Suicide risk and prevention

- **Warning signs** — talk of death or hopelessness ('there is no point in living'), a previous attempt (the **single strongest predictor**), giving away belongings, writing a will or a note, sudden calmness after severe depression, collecting tablets or a rope, social withdrawal, severe insomnia, alcohol use, recent loss, chronic painful illness.
- **Never ignore or dismiss a threat, and never promise to keep suicidal thoughts secret** — ask directly about plans, inform the doctor and the team at once and record it.
- **Nursing measures** — **constant observation (one-to-one, within arm's length for a high-risk patient)**, especially at night, at change of shift, in the early morning and during the early recovery phase (when energy returns); remove **sharp, blades, glass, belts, ropes, ties, cords, plastic bags, matches, poisons and drugs**; check the room, bathroom (no bolt, no hook), windows (grill/shatter-proof) and locker; supervise medication and **check the mouth to ensure the tablet is swallowed**; count and account for sharps and cutlery; accompany to the bathroom; do not allow the patient to be alone with visitors who may bring harmful articles; encourage him to talk about his feelings and instil hope.

The aggressive or violent patient

- **Early warning signs** — restlessness and pacing, clenched fists and jaw, loud abusive speech, staring, invasion of others' space, refusal to co-operate, throwing objects, a history of violence.
- **De-escalation** — remain calm and speak softly and slowly; keep a safe distance and **an exit behind you (never let the patient come between you and the door)**; do not turn your back; remove other patients from the area; never confront alone — **call for help and have adequate staff visible**; offer choices and food/medication; do not threaten, argue, shout or touch him suddenly; remove your own stethoscope, chains, dangling earrings and spectacles.
- **Restraint or seclusion is the last resort** — used only when there is imminent danger and all else has failed: the doctor's order, adequate trained staff (usually 4-5, one for each limb and one for the head), correct technique, continuous observation, release and exercise of the limb and check of circulation **every 15-30 minutes with 2-hourly release**, attention to food, fluids and toileting, and full documentation with the reason, time and the patient's response. **Restraints must never be used for punishment or staff convenience** (false imprisonment).
- **Rapid tranquillisation** as prescribed (e.g. haloperidol with promethazine or lorazepam IM) with monitoring of vital signs, airway and the level of consciousness.

16.5 Psychotropic Drugs

Table 16.3 Psychotropic drugs and nursing responsibilities

Group	Examples	Main side effects and nursing points
Typical antipsychotics (neuroleptics)	Chlorpromazine, haloperidol, trifluoperazine, fluphenazine decanoate (depot)	Extrapyramidal symptoms — acute dystonia, akathisia (restlessness), parkinsonism, tardive dyskinesia (treated/prevented with anticholinergics such as trihexyphenidyl or promethazine); postural hypotension (rise slowly), sedation, dry mouth, constipation, blurred vision, weight gain, photosensitivity (use sunscreen), galactorrhoea and amenorrhoea, jaundice, lowered seizure threshold; neuroleptic malignant syndrome — high fever, muscle rigidity, altered consciousness, unstable BP, raised CPK: STOP the drug and treat as an emergency
Atypical antipsychotics	Risperidone, olanzapine, quetiapine, aripiprazole, clozapine	Fewer extrapyramidal effects but weight gain, diabetes and dyslipidaemia — monitor weight, sugar and lipids. Clozapine causes agranulocytosis — a mandatory regular white cell count; report any sore throat or fever at once
Antidepressants — SSRIs	Fluoxetine, sertraline, escitalopram, paroxetine	Nausea, insomnia or drowsiness, sexual dysfunction, hyponatraemia; take 2-4 weeks (up to 6) to act — the suicide risk may increase early in treatment when energy returns before mood improves ; never stop abruptly; serotonin syndrome if combined with another serotonergic drug (agitation, tremor, hyperthermia, diarrhoea)
Tricyclic antidepressants	Amitriptyline, imipramine, nortriptyline	Anticholinergic effects (dry mouth, constipation, retention of urine, blurred vision), postural hypotension, cardiac arrhythmia; dangerous in overdose ; useful for nocturnal enuresis and neuropathic pain
MAO inhibitors	Phenelzine, tranylcypromine	Hypertensive crisis with tyramine-rich food — cheese, wine, beer, pickled and fermented food, yeast extract, broad beans — strict dietary teaching

Group	Examples	Main side effects and nursing points
Mood stabilisers	Lithium carbonate, valproate, carbamazepine, lamotrigine	Lithium has a narrow therapeutic range (0.6-1.2 mEq/L) — monitor levels, renal and thyroid function; maintain a steady salt and fluid intake (2-3 L) ; toxicity (over 1.5 mEq/L): coarse tremor, vomiting, diarrhoea, ataxia, slurred speech, drowsiness, convulsions — withhold the drug and report at once ; avoid dehydration, diuretics, NSAIDs and a low-salt diet; teratogenic
Anxiolytics / hypnotics	Diazepam, lorazepam, alprazolam, clonazepam, zolpidem	Sedation, falls in the elderly, dependence and withdrawal (Schedule H1) — for short-term use only ; never combine with alcohol; taper gradually; antidote flumazenil
Anti-epileptics and others	Phenytoin, carbamazepine, sodium valproate, levetiracetam; donepezil and memantine for dementia; disulfiram, acamprosate and naltrexone for alcohol dependence; methadone and buprenorphine for opioid substitution	Phenytoin — gum hypertrophy, hirsutism, ataxia; carbamazepine — rash, low sodium, blood dyscrasia; valproate — weight gain, hair loss, hepatotoxicity, teratogenic; disulfiram + alcohol = a severe flushing reaction (the patient must be warned and consenting)

CAUTION / NEVER DO THIS

Nursing responsibilities with psychotropic drugs: give the drug under **direct observation and check the mouth** (patients commonly hide tablets in the cheek or under the tongue); watch for early extrapyramidal reactions and report them; check BP before and after antipsychotics; warn about drowsiness (no driving or machinery); teach the patient and family that **the drug must be continued even after improvement** and never stopped suddenly, as relapse is common; maintain the record of controlled drugs.

16.6 Other Treatments and Nursing Care

Table 16.4 Treatments in psychiatry

Treatment	Description	Nursing care
Electroconvulsive therapy (ECT)	A brief controlled electrical current passed through the brain under general anaesthesia and a muscle relaxant, producing a modified seizure; a course of 6-12 treatments given 2-3 times a week. Indications: severe depression with a high suicide risk or refusal to eat, catatonia, puerperal psychosis, and drug-resistant schizophrenia/mania	Before: written informed consent, explain and reassure (dispel myths), nil by mouth for 6-8 hours , remove dentures, ornaments, hairpins, spectacles, contact lenses and nail polish, empty the bladder, loose clothing, record baseline vital signs, withhold the morning dose as ordered, keep the emergency tray, oxygen and suction ready. During: assist the anaesthetist, position the patient, support the joints and jaw (do not restrain the limbs forcibly). After: lateral (recovery) position , maintain the airway, oxygen and suction, observe respiration and vital signs until fully awake, side rails up, never leave alone, orientate repeatedly on waking (transient confusion and short-term memory loss are expected), offer food after the swallowing reflex returns, and record
Psychotherapy	Individual, group, family, cognitive behaviour therapy (CBT), behaviour therapy (systematic desensitisation, token economy, aversion therapy, relaxation), supportive therapy, play therapy for children	Prepare the patient, keep appointments, maintain confidentiality, reinforce the learning between sessions
Occupational and recreational therapy	Purposeful graded activity — crafts, gardening, music, yoga, games, sports	Raises self-esteem, gives structure and social contact, reduces preoccupation with symptoms
Milieu therapy / therapeutic community	The whole ward environment is used as the treatment	Consistent rules, ward meetings, patient responsibility
Rehabilitation and community psychiatry	Day-care centre, half-way home, sheltered workshop, vocational training, family and community education, follow-up clinic, District Mental Health Programme	Prevents relapse and institutionalisation; combats stigma; ensures drug adherence

16.7 Mental Health Legislation and Programmes (India)

- **Mental Healthcare Act, 2017** (replaced the Mental Health Act, 1987) — gives **every person the right to access mental health care**, the right to live in the community, to protection from cruel and degrading treatment, to confidentiality, to legal aid and to make an **advance directive** and appoint a **nominated representative**; provides for Central and State Mental Health Authorities and Mental Health Review Boards; regulates admission (independent, supported), the use of restraint and seclusion, and **prohibits unmodified ECT and ECT in minors without special sanction; decriminalises attempted suicide**.
- **Rights of Persons with Disabilities Act, 2016** — includes mental illness and intellectual disability among the recognised disabilities; certification and benefits.
- **Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985** — see Chapter 17.
- **National Mental Health Programme (1982)** and the **District Mental Health Programme (1996, Bellary model)** — integration of mental health into general health care, training of medical officers and health workers, and community care; **Tele-MANAS** (national tele-mental-health helpline, 14416) launched in 2022.
- **World Mental Health Day — 10 October**; World Suicide Prevention Day — 10 September; National Institute of Mental Health and Neurosciences (**NIMHANS**), Bengaluru is the apex institute.

CHAPTER AT A GLANCE

- Neurosis: insight retained, no hallucination/delusion. Psychosis: insight lost, hallucinations and delusions present.
- Delusion = false fixed belief; hallucination = perception without a stimulus (auditory in schizophrenia, visual in delirium); illusion = misinterpretation of a real stimulus.
- Never argue with or reinforce a delusion — acknowledge the feeling and present reality.
- Suicide: a previous attempt is the strongest predictor; constant observation, remove harmful articles, check the mouth after medication, never promise secrecy; risk rises early in recovery.
- Violence: stay calm, keep an exit behind you, call for help, restraint only as a last resort with a doctor's order and 15-30 minute checks.
- Antipsychotics → extrapyramidal symptoms and neuroleptic malignant syndrome; clozapine → agranulocytosis; lithium therapeutic 0.6-1.2 mEq/L (toxic over 1.5) — keep salt and fluid steady; SSRIs take 2-4 weeks; MAOI + tyramine = hypertensive crisis.
- ECT: consent, NPO 6-8 h, remove dentures/ornaments, empty bladder; after — lateral position, airway, expect transient confusion and memory loss.
- Mental Healthcare Act 2017: advance directive, nominated representative, decriminalises suicide attempt, bans unmodified ECT; NMHP 1982, DMHP 1996; World Mental Health Day 10 October.

Special Drugs, Their Control and Administration

Narcotics and psychotropics, schedules of the Drugs and Cosmetics Rules, storage, records, high-alert and cytotoxic drugs

17.1 What Are 'Special' Drugs?

- **Special (controlled) drugs** are those which, because of their **liability to abuse**, **their toxicity**, **their narrow therapeutic index** or the **need for special storage**, are subject to statutory control over their manufacture, sale, storage, prescription, administration, record keeping and disposal.
- They include: **narcotic (opioid) analgesics**, **psychotropic substances**, **barbiturates and other hypnotics**, **drugs of dependence**, **poisons**, **high-alert medicines**, **cytotoxic (anticancer) drugs**, **biologicals and vaccines**, **blood products**, **anaesthetic gases and medical oxygen**, **radiopharmaceuticals**, and **drugs with a narrow therapeutic index**.

17.2 Laws Governing Drugs in India

Table 17.1 Principal drug laws in India

Act / Rule	Year	What it controls
Drugs and Cosmetics Act and Rules	Act 1940, Rules 1945	Import, manufacture, distribution and sale of drugs and cosmetics; standards of quality; licensing; the Schedules (A to Y) ; defines misbranded, adulterated and spurious drugs
Pharmacy Act	1948	Regulates the profession and practice of pharmacy; constitutes the Pharmacy Council of India (PCI) and State Pharmacy Councils; registration of pharmacists; a drug may be compounded/dispensed only by or under the supervision of a registered pharmacist
Drugs and Magic Remedies (Objectionable Advertisements) Act	1954	Prohibits advertisement of drugs claiming to cure specified diseases and of magic remedies
Narcotic Drugs and Psychotropic Substances (NDPS) Act	1985 (amended 1988, 2001, 2014, 2021)	Prohibits and controls the cultivation, production, manufacture, possession, sale, purchase, transport, use, import and export of narcotic drugs and psychotropic substances ; provides for very stringent punishment (including up to 10-20 years' rigorous imprisonment and fine; death penalty possible for repeat offences in some cases) and for the treatment and rehabilitation of addicts; the Narcotics Control Bureau (NCB) is the enforcing agency; 'Essential Narcotic Drugs' (including morphine) were made easier to access for medical use by the 2014 amendment
Poisons Act	1919	Possession and sale of poisons
Drug Price Control Order (DPCO) / NPPA	DPCO 2013	Price control of scheduled (essential) medicines by the National Pharmaceutical Pricing Authority
Medical Termination of Pregnancy Act	1971 (amended 2021)	Conditions and places for the legal termination of pregnancy
Transplantation of Human Organs Act; Clinical Establishments Act; BMW Rules 2016; Food Safety and Standards Act 2006; Mental Healthcare Act 2017	-	Related statutes affecting hospital and pharmacy practice

17.3 Schedules of the Drugs and Cosmetics Rules (must know)

Table 17.2 Schedules of the Drugs and Cosmetics Rules, 1945 (learn H, H1, X, G, C, M, N, P, Y by heart)

Schedule	Subject matter
Schedule A	Specimen forms for applications and licences
Schedule B	Fees for the test or analysis of drugs by the Central Drugs Laboratory
Schedule C and C1	Biological and special products — sera, vaccines, toxoids, antitoxins, insulin, hormones, blood products (C) and other special products such as digitalis, ergot and antibiotic preparations (C1); they require special licensing and cold storage
Schedule D	Drugs exempted from the import licence requirement
Schedule E1	List of poisonous substances under the Ayurvedic, Siddha and Unani systems

Schedule	Subject matter
Schedule F, F1, FF	Requirements for blood banks and blood products (F), vaccines and sera (F1), ophthalmic solutions (FF)
Schedule G	Drugs to be used only under medical supervision ; the label must carry the warning: ' <i>Caution: it is dangerous to take this preparation except under medical supervision</i> ' — most anticancer (cytotoxic) drugs , and some antidiabetics such as metformin and glibenclamide
Schedule H	Prescription drugs — to be sold by retail only on the prescription of a registered medical practitioner ; the label must bear the symbol ' Rx ' and the warning ' <i>To be sold by retail on the prescription of a registered medical practitioner only</i> ' (most antibiotics, antihypertensives, hormones, sedatives)
Schedule H1	A sub-list of Schedule H (<i>about 46 drugs</i>) needing a stricter record: third and higher generation antibiotics (cephalosporins, carbapenems), anti-tubercular drugs, and habit-forming benzodiazepines (alprazolam, diazepam, nitrazepam), tramadol etc. The label has a red vertical band on the left with the Rx symbol ; the pharmacist must maintain a separate register with the name and address of the prescriber and the patient, the name of the drug and the quantity supplied, to be retained for 3 years
Schedule J	List of diseases and conditions for which no drug may claim to be a remedy (e.g. baldness, cancer, cataract, deafness, AIDS, blindness, diabetes, obesity — in advertisements)
Schedule K	Drugs and classes of drugs exempted from certain provisions (for example, household remedies sold in a village, drugs supplied by a registered practitioner to his own patient, and supply to a hospital)
Schedule M	Good Manufacturing Practices (GMP) — requirements for premises, plant and equipment for the manufacture of drugs (M1 homoeopathic, M2 cosmetics, M3 medical devices)
Schedule N	Minimum equipment required for the efficient running of a pharmacy (dispensing counter, balance, refrigerator, first-aid facilities, water supply etc.)
Schedule P	Life period (shelf life) and storage conditions of drugs
Schedule P1	Pack sizes of drugs
Schedule Q	Permitted colouring agents and dyes in cosmetics and soaps
Schedule R and R1	Standards for mechanical contraceptives (condoms) and for medical devices
Schedule S	Standards for cosmetics (as per BIS specifications)
Schedule T	GMP for Ayurvedic, Siddha and Unani medicines
Schedule U and U1	Records of manufacture to be maintained (drugs / cosmetics)
Schedule V	Standards for patent and proprietary medicines
Schedule W	Drugs to be sold only under generic names (largely superseded)
Schedule X	The most strictly controlled schedule — narcotic and psychotropic drugs, barbiturates and amphetamines (e.g. morphine, pethidine, fentanyl, methadone, barbiturates, amphetamine, methylphenidate, ketamine, pentazocine, buprenorphine, secobarbital). Requires a special licence , sale only against a prescription in duplicate (one copy retained by the pharmacist for 2 years), storage under lock and key in a separate cupboard , and maintenance of a register for 2 years; the label carries the symbol ' NRx ' in red
Schedule Y	Requirements and guidelines for clinical trials , import and manufacture of new drugs (now largely replaced by the New Drugs and Clinical Trials Rules, 2019)

MEMORY AID

Quick recall: **G** = 'Guard' (under medical supervision) • **H** = prescription (Rx) • **H1** = red band, separate register, 3 years • **X** = NRx, duplicate prescription, double-locked, 2 years • **C/C1** = biologicals/cold chain • **J** = no cure claims • **M** = manufacturing (GMP) • **N** = pharmacy equipment • **P** = shelf life • **Y** = clinical trials.

17.4 Narcotic and Psychotropic Drugs — Control in the Ward and Pharmacy

Table 17.3 Control of narcotic and psychotropic drugs

Requirement	Details
Storage	In a separate, substantially built cupboard/almirah with a double lock , fixed to the wall, inside another locked cupboard or room; the key is held personally by the nurse-in-charge/pharmacist on duty and handed over at each shift change; never left in a drawer or with a student
Stock (narcotic) register	A bound register with serially numbered pages (no loose sheets, no pages torn out); a separate page for each drug and strength; columns for the date, time, patient's name, bed/hospital number, prescriber, dose given, balance in stock, the signature of the nurse who administered and of the witness
Checking	The balance is physically counted and signed by two nurses at every shift change (three times a day); any discrepancy must be reported immediately to the nursing superintendent and the pharmacist and investigated
Administration	Only against a written, signed and dated prescription ; two nurses check the drug, strength, dose, patient and time; the entry is made immediately after giving; if only part of an ampoule is used, the remainder is wasted in the presence of a witness and both sign the register; empty ampoules/used vials are usually retained for accounting
Indent and issue	Issued by the pharmacy only against a requisition signed by an authorised person, in exchange for the empty ampoules/a completed account; separate purchase records with the licence number are kept
Records retention	Prescriptions and registers for 2 years (Schedule X); NDPS records as prescribed; available for inspection by the Drugs Inspector
Disposal / destruction	Expired, damaged or unused narcotics are never thrown away or flushed ; they are returned and destroyed only in the presence of an authorised committee , with a written record signed by all witnesses
Patient safety points	Check the respiratory rate before every opioid dose (withhold if below 10-12/min) , the level of sedation, the BP and the pain score; watch for constipation (give a laxative prophylactically), nausea, urinary retention and pruritus; keep naloxone available as the antidote; do not give an opioid with alcohol or other CNS depressants without an order

CAUTION / NEVER DO THIS

Any loss, theft, breakage or discrepancy of a narcotic drug must be **reported at once in writing**; unexplained loss is a **criminal offence under the NDPS Act** and also a serious professional misconduct. Never borrow, lend or 'adjust' narcotic stock between wards. Never leave a prepared narcotic injection unattended.

17.5 High-Alert Medications

Table 17.4 High-alert medications and their safeguards

Group	Examples	Safety measures
Insulins	All insulins	Insulin syringe only; write 'unit' in full ; double-check the type (regular/NPH/mix) and dose with a second nurse; check the blood sugar before giving; keep the meal ready; store the in-use pen at room temperature and stock in the refrigerator (2-8 °C, never frozen); know the signs of hypoglycaemia
Anticoagulants	Heparin, low molecular weight heparin, warfarin	Independent double check of the dose and the concentration (heparin comes in many strengths); monitor aPTT/INR and platelets ; watch for bleeding (gums, urine, stool, bruises); keep protamine sulphate (heparin) and vitamin K (warfarin) available; teach the patient about diet, interacting drugs, a soft toothbrush and an electric razor
Concentrated electrolytes	Potassium chloride , hypertonic saline, magnesium sulphate	Never give concentrated KCl as an IV bolus/push — it causes cardiac arrest ; always dilute (usually not more than 40 mEq/L peripherally) and infuse slowly with a pump, preferably with cardiac monitoring; store separately from other ampoules
Opioids and sedatives	Morphine, fentanyl, pethidine, midazolam	As in 17.4; monitor respiration and sedation; antidotes naloxone and flumazenil
Chemotherapy / cytotoxics	Methotrexate, vincristine, cyclophosphamide, cisplatin, 5-FU	Prescribed by a specialist; verified by two persons against the protocol and body surface area; vincristine is fatal if given intrathecally — it must be labelled 'for intravenous use only' ; methotrexate for arthritis is weekly, not daily

Group	Examples	Safety measures
Others	Digoxin, phenytoin, theophylline, lithium, aminoglycosides, thrombolytics, neuromuscular blockers, IV anaesthetics, oxytocin, adrenaline, magnesium sulphate	Check the pulse before digoxin (withhold if below 60/min in an adult or 90-100 in an infant); monitor drug levels for narrow-index drugs; neuromuscular blockers must be stored separately with a warning label — they cause respiratory arrest if given by mistake

17.6 Handling Cytotoxic (Anticancer) Drugs

- Prepare only in a **biological safety cabinet/laminar air flow (Class II B) in a designated area**, never on an open ward table; use a **closed-system transfer device** and a luer-lock syringe.
- **PPE** — double chemotherapy gloves (changed every 30-60 min), a long-sleeved impervious gown, mask/N95, cap, goggles and shoe covers.
- **Pregnant, breast-feeding and immunocompromised staff must not handle cytotoxic drugs**; staff should be rotated and have periodic health checks (blood count).
- Keep a **spill kit** ready; in case of a spill, restrict the area, wear PPE, absorb with the pads provided (do not sweep or wipe widely), wash the area three times, and dispose of everything as **cytotoxic waste**; report and record the incident. For skin contact, wash with plenty of soap and water; for eye contact, irrigate with water/saline for 15 minutes and see the ophthalmologist.
- **Waste** — all vials, syringes, needles, gowns, gloves and tubing go into a **labelled cytotoxic (yellow) container for incineration at high temperature**; the patient's excreta and linen are handled with gloves for **48 hours** after administration.
- **Extravasation** of a vesicant (doxorubicin, vincristine) — stop the infusion immediately, leave the cannula in place, aspirate the residual drug, mark the area, apply cold or warm compresses as per the drug protocol, give the specific antidote, elevate the limb, inform the doctor and document with a photograph.
- **Nursing care of the patient** — antiemetics before and after, mouth care for mucositis (soft brush, saline rinses), monitor the blood count (**neutropenia — protective isolation and report any fever at once as it is an emergency**), hydration and output for nephrotoxic drugs, care of alopecia and body image, contraception advice and fertility counselling.

17.7 Storage and Administration of Other Special Preparations

Table 17.5 Storage and administration of special preparations

Preparation	Storage	Administration points
Vaccines and sera (Schedule C/C1)	2-8 °C cold chain ; never freeze adsorbed vaccines; protect from light; VVM check	Correct site, route and dose (Chapter 14); keep adrenaline ready; observe for 30 min; record the batch number and expiry
Insulin	Unopened: refrigerator 2-8 °C ; in-use vial/pen: room temperature (below 30 °C) for 28 days ; never freeze ; do not shake vigorously (roll cloudy insulin gently)	Insulin syringe/pen only; rotate sites; give regular insulin 30 min before a meal , rapid analogues just before; when mixing, draw clear (regular) before cloudy (NPH) ; keep glucose/glucagon ready for hypoglycaemia
Blood and blood products	Whole blood/red cells 2-6 °C in a blood-bank refrigerator (never a ward refrigerator); platelets 20-24 °C with agitation; FFP frozen at -30 °C	See Chapter 19 — start within 30 minutes of issue, complete within 4 hours, use a blood set with a filter, saline only
Medical oxygen and anaesthetic gases	Cylinders upright, chained, in a cool, well-ventilated room away from oil, grease and any flame; colour codes: oxygen — black body with a white shoulder (white in ISO), nitrous oxide — blue, carbon dioxide — grey, medical air — grey/white-and-black, entonox — blue with blue-and-white quarters	No smoking, no naked flame, no oil or grease on fittings or on the patient's face; check the contents and the flow meter; humidify for flows above 4 L/min; treat oxygen as a prescribed drug (dose, flow, target saturation)
Eye preparations	Sterile; store as labelled	One container per patient; discard 4 weeks after opening ; never touch the dropper to the eye
Reconstituted antibiotics and suspensions	As labelled — many need refrigeration after reconstitution	Write the date and time of reconstitution and the discard date on the vial/bottle; shake before use

Preparation	Storage	Administration points
Sublingual nitroglycerine	Tight, amber glass container away from heat and light; potency lost in about 8 weeks after opening	Patient sits or lies down (it causes a fall in BP and headache); one tablet under the tongue, repeat up to 3 doses 5 min apart, then seek emergency help
Radiopharmaceuticals	Lead-shielded container in a designated area with radiation warning signs	Handled by trained personnel with dosimeters; the patient's urine and linen may need special precautions

17.8 Rational Drug Use, Antibiotic Stewardship and DOTS

- **Rational use of drugs (WHO)** — 'patients receive medications appropriate to their clinical needs, in doses that meet their own individual requirements, for an adequate period of time, and at the lowest cost to them and their community'.
- **Irrational practices** — unnecessary antibiotics for viral fever and simple diarrhoea, injections where oral drugs would do, irrational fixed-dose combinations, polypharmacy, self-medication, incomplete courses, and prescribing by brand where cheaper generics exist.
- **Antimicrobial stewardship** — prescribe an antibiotic only when indicated, ideally after culture; follow the hospital antibiotic policy and the **WHO AWaRe classification (Access, Watch, Reserve)**; give the right dose for the right duration; de-escalate on the basis of culture; review IV to oral switch; complete the course; and educate patients that **antibiotics do not cure viral illness** — this is central to controlling **antimicrobial resistance (AMR)**.
- **DOTS / NTEP (tuberculosis) — Directly Observed Treatment, Short-course**: every dose in the intensive phase is **swallowed in the presence of a treatment supporter; 2 months of HRZE (intensive) followed by 4 months of HRE (continuation)** in daily fixed-dose combinations by weight band; drug-resistant TB is treated under a separate regimen; the nurse/pharmacist explains the side effects (orange urine with rifampicin, neuritis with isoniazid — give pyridoxine, joint pain with pyrazinamide, visual change with ethambutol — report at once), ensures adherence, does contact tracing, and registers the patient on **Ni-kshay** (with nutritional support of Rs 1000 per month under Ni-kshay Poshan Yojana).
- **Essential medicines** — the **National List of Essential Medicines (NLEM)** guides procurement; prescribe generically; report ADRs to **PvPI**; report a suspected **spurious or substandard drug** to the Drugs Inspector.

17.9 The Pharmacist's and Nurse's Statutory Responsibilities — Summary

1. Dispense only against a valid prescription; verify the prescriber, the patient, the drug, the dose and the interactions; **refuse and refer back** a prescription that is illegible, incomplete or unsafe.
2. Maintain the statutory **registers — prescription register, Schedule H1 register (3 years), Schedule X register and prescriptions (2 years), narcotic register, purchase and sale records, and temperature records of the refrigerator**.
3. Store drugs as per **Schedule P** conditions, keep the pharmacy equipped as per **Schedule N**, and display the licence and the registered pharmacist's certificate.
4. Keep the controlled-drug cupboard **double-locked** and account for every dose; do the shift count with a witness.
5. **Label** every dispensed medicine with the patient's name, the drug and strength, the dose and directions, the date, the expiry, storage instructions and any warning ('shake well', 'for external use only', 'complete the course').
6. **Counsel** the patient in a language he understands and check his understanding by asking him to repeat the instructions.
7. Never substitute, alter or add to a prescription without the prescriber's consent; never dispense a Schedule X or narcotic drug without the prescribed formalities.
8. **Report** adverse drug reactions, medication errors, near misses, suspected spurious drugs and any loss of controlled drugs.
9. Dispose of expired, damaged and returned drugs by the approved route with documentation.
10. Maintain **confidentiality** of the prescription and the patient's condition, and practise within the limits of the Pharmacy Act and the Code of Pharmaceutical Ethics.

CHAPTER AT A GLANCE

- Drugs and Cosmetics Act 1940/Rules 1945; Pharmacy Act 1948 (PCI); NDPS Act 1985 (NCB); Poisons Act 1919; DPCO 2013.

- Schedule G = 'dangerous except under medical supervision'; H = Rx; H1 = red band + separate register kept 3 years; X = NRx, duplicate prescription, double lock, records 2 years.
- Schedule C/C1 = biologicals (cold chain); J = no cure claims; M = GMP; N = pharmacy equipment; P = shelf life; Y = clinical trials.
- Narcotics: double-locked cupboard, key with the nurse-in-charge, bound serially numbered register, count and sign at every shift by two nurses, waste witnessed, destruction only before a committee.
- Check respiration before an opioid (hold if under 10-12/min); antidote naloxone. Check the pulse before digoxin (hold if under 60).
- Never give concentrated potassium chloride as an IV push; never give vincristine intrathecally; methotrexate for arthritis is weekly.
- Insulin: unopened 2-8 °C, in-use at room temperature for 28 days, never frozen; draw clear before cloudy.
- Oxygen cylinder: black with a white shoulder, kept away from oil, grease and flame; oxygen is a prescribed drug.
- Cytotoxics: laminar flow cabinet, double gloves, spill kit, cytotoxic yellow waste, no handling by pregnant staff.
- DOTS: 2 months HRZE + 4 months HRE, every dose observed; rifampicin → orange urine; isoniazid → neuritis (give pyridoxine); ethambutol → optic neuritis.

Preparation of the Patient for Operation and the After Care

Pre-operative assessment and preparation, consent, the operation theatre, post-operative care and complications

18.1 Classification of Surgery

Table 18.1 Classification of surgical procedures

Basis	Types
Urgency	Emergency (immediately, to save life — ruptured spleen, perforation, obstructed labour, extradural haematoma); Urgent (within 24–48 h — acute appendicitis, fracture fixation); Elective/planned (at a convenient time — hernia, cataract, cholelithiasis); Optional (at the patient's choice — cosmetic surgery)
Purpose	Diagnostic (biopsy, laparotomy), Curative/ablative (appendicectomy), Reconstructive/restorative (skin graft, joint replacement), Palliative (colostomy for inoperable cancer, debulking), Cosmetic , Transplant
Degree of risk	Major (long duration, body cavity opened, high blood loss, vital organ involved — gastrectomy, craniotomy, open heart surgery) and Minor (short, local or regional anaesthesia, low risk — incision and drainage, biopsy, dilatation and curettage)
Access	Open, minimally invasive (laparoscopic/endoscopic — less pain, small scar, early discharge) , robotic, day-care surgery

18.2 Pre-operative Assessment and Investigations

- **History** — the presenting illness; previous surgery and anaesthesia (and any problem with it); **allergies (drug, latex, iodine, food)**; current medicines (especially **anticoagulants, antiplatelets, insulin, steroids, antihypertensives, oral contraceptives, herbal products**); tobacco, alcohol and drug use; co-morbidity (diabetes, hypertension, heart, lung, kidney, liver or thyroid disease, epilepsy, bleeding disorder); last menstrual period and possible pregnancy; family history of anaesthetic problems (malignant hyperthermia, pseudocholinesterase deficiency).
- **Physical assessment** — weight and height, vital signs and SpO₂, general condition, nutrition and hydration, level of consciousness, examination of the chest and heart, condition of the mouth and teeth (**loose teeth, dentures, caps — a risk during intubation**), the operative site and skin condition, peripheral veins, mobility, and pressure areas.
- **Routine investigations** — haemoglobin, total and differential count, platelets, **blood grouping and cross-matching**, bleeding and clotting time/PT-INR, blood sugar, blood urea and creatinine, serum electrolytes, urine routine, chest X-ray, **ECG (over 40 years or with cardiac disease)**, HIV/HBsAg as per policy; plus special tests — pulmonary function tests, echocardiogram, liver function tests, ultrasound, CT, thyroid profile as indicated.
- **Anaesthetic assessment** — ASA physical status grading (I healthy → V moribund; E for emergency), airway assessment (**Mallampati grade**), and the anaesthesia plan (general, regional/spinal/epidural, local, sedation).

18.3 Informed Consent

- Consent must be **informed, written, voluntary, and obtained before any sedation**; it is the **surgeon's/doctor's duty** to explain the diagnosis, the procedure, the alternatives, the benefits and the risks. The **nurse's role is to witness the signature, confirm that the patient has understood, and ensure the form is complete** and in the file.
- Valid consent requires an adult of **18 years or more** who is **conscious and of sound mind**. For a **minor (under 18)**, the **parent or legal guardian** signs; for an unconscious or mentally incapable patient, the next of kin; in a **life-threatening emergency with no relative available, life-saving treatment may be given without consent** (documented as such by two doctors).
- Special consents are needed for **sterilisation (with the spouse's knowledge as per programme rules)**, **medical termination of pregnancy (the woman's own consent; guardian if under 18 or mentally ill)**, **organ donation, high-risk surgery, blood transfusion, anaesthesia, photography and research**.
- Consent may be **withdrawn at any time**; a refusal must be documented and reported. Consent must be signed, dated and timed, with the procedure written in full (no abbreviations) and the side/site specified.

18.4 Pre-operative Preparation

A. Psychological preparation

- Allow the patient to **express his fears** (fear of death, pain, disfigurement, anaesthesia, cancer, dependence, loss of job); answer honestly and simply; never dismiss the fear.
- **Explain the routine** — the time of surgery, fasting, pre-medication, where he will wake up, tubes and drains he may find, the likely pain and its relief, the recovery room, and the expected length of stay.
- **Pre-operative teaching (proved to reduce complications)** — **deep breathing and coughing exercises with splinting of the wound, incentive spirometry, leg and ankle exercises, turning in bed, early ambulation, use of the pain scale and how to ask for analgesia.**
- Involve the family; arrange a visit by the anaesthetist and a religious/spiritual visit if desired; ensure a good night's sleep (a sedative may be prescribed).

B. Physical preparation (the day before and the day of surgery)

1. **Nutrition and hydration** — correct anaemia, hypoproteinaemia and dehydration; an IV line/fluids as ordered.
2. **Fasting (nil by mouth)** — modern guidelines: **clear fluids up to 2 hours, breast milk 4 hours, light meal/formula 6 hours, and fatty/fried food or a heavy meal 8 hours before anaesthesia**; traditionally 'nothing after midnight'. Put up an 'NPO' board, remove the water jug, and inform the patient and family why (risk of aspiration/Mendelson's syndrome).
3. **Bowel preparation** — a laxative the night before or an enema/suppository as ordered (essential for bowel, rectal and gynaecological surgery); full mechanical bowel preparation with polyethylene glycol for colorectal surgery; not routinely needed for other operations.
4. **Bladder** — ask the patient to **void immediately before going to theatre**; catheterise only if ordered (pelvic, prolonged or major surgery).
5. **Skin preparation** — bath or shower with soap (or chlorhexidine) **the night before and on the morning of surgery**; **hair removal only if necessary and then with a clipper or depilatory cream, immediately before surgery, never with a razor the night before** (shaving causes micro-abrasions and raises the infection rate); clean the umbilicus; the site is painted with antiseptic in the theatre.
6. **Removal of prostheses and personal items** — dentures, contact lenses, spectacles, hearing aid (may be retained till the anaesthetic room if needed), wigs, hair pins, **all jewellery (a ring that will not come off is taped)**, nail polish and make-up (they hide cyanosis and interfere with the pulse oximeter); valuables are listed and handed over to the relatives or the security custody **against a signature.**
7. **Dress** — clean theatre gown, cap; anti-embolism stockings if ordered.
8. **Medication** — give the **prescribed pre-medication at the exact time** (sedative/anxiolytic, analgesic, anticholinergic to dry secretions, H₂ blocker or proton-pump inhibitor and antacid to reduce gastric acid, antiemetic, **prophylactic antibiotic within 60 minutes of the incision**); continue essential drugs (antihypertensives, antiepileptics, thyroid, cardiac drugs) with a sip of water as advised; **stop/adjust as ordered: warfarin and antiplatelets (5-7 days before, bridged with heparin if needed), metformin, insulin (dose adjusted with sugar monitoring), oral contraceptives (4 weeks before major surgery)**; steroid cover for a patient on long-term steroids.
9. **Final check-list before transfer** — identity band, consent form signed, correct site marked, allergies noted, investigation reports and cross-matched blood available, vital signs recorded, fasting confirmed, bladder emptied, prostheses removed, pre-medication given and recorded, case notes and X-rays with the patient, and a completed **WHO Surgical Safety Checklist ('Sign in')**.
10. Transfer on a trolley with side rails up, covered warmly, accompanied by the nurse and the notes; hand over verbally to the theatre nurse and sign the transfer record.

HIGH-YIELD / MOST ASKED

The WHO Surgical Safety Checklist has three parts: 'Sign In' (before anaesthesia), 'Time Out' (before the skin incision — the whole team stops and confirms the patient, site and procedure), and 'Sign Out' (before the patient leaves the theatre — instrument, sponge and needle counts, specimen labelling, equipment problems). It is designed to prevent wrong patient, wrong site and wrong procedure surgery, and retained instruments.

18.5 The Operation Theatre and the Immediate Peri-operative Period

- **Zones of the theatre** — *outer/protective zone* (reception, changing rooms, offices), *clean zone* (pre-anaesthetic room, recovery, scrub area, CSSD interface), *aseptic/sterile zone* (operating room) and *disposal/dirty zone*. Traffic and clothing rules are stricter as one moves inwards.
- Theatre environment — **positive pressure, laminar air flow with HEPA filtration, 15-25 air changes/hour, temperature 20-24 °C, humidity 50-60%**, seamless washable floors and walls, shadowless light, no fans, minimum staff and minimum door opening.
- **Theatre team** — surgeon, assistant, anaesthetist, **scrub nurse (sterile — handles instruments, counts sponges and needles), circulating nurse (unsterile — fetches supplies, positions the patient, documents, connects equipment, counts with the scrub nurse)**, technician and attendant.
- **Nursing responsibilities in the theatre** — check the patient's identity, consent and site; safe **positioning** (protect the eyes, nerves and pressure points, pad the bony points, secure the limbs, avoid hyperextension — *brachial plexus and common peroneal nerve injury and pressure sores are real risks*); apply the diathermy plate correctly to dry, shaved, clean skin; prevent **hypothermia** (warm blankets, warmed fluids, forced-air warmer) and maintain sterile technique; **count and record sponges, instruments, needles and blades before, during and at the end**; label and despatch specimens correctly; record the blood loss, fluids and drugs given; complete the operation and anaesthesia records.
- **Types of anaesthesia** — **general** (unconscious: induction, maintenance, reversal; airway secured with a tube or supraglottic device), **regional** (spinal, epidural, nerve block — the patient is awake; watch for hypotension, headache, and the level of the block; the patient must lie flat as instructed after a spinal anaesthetic), **local** (infiltration, topical) and **sedation**. Complications of anaesthesia: aspiration, laryngospasm, hypoxia, hypotension, arrhythmia, **malignant hyperthermia (a rare emergency — dantrolene)**, awareness, nerve injury, post-dural-puncture headache, and post-operative nausea and vomiting.

18.6 Post-operative (After) Care

A. Immediate care — in the recovery room / on return to the ward

1. Receive the patient with a full **handover** from the anaesthetist/theatre nurse: the procedure done, the anaesthetic used, blood loss and replacement, drugs given, drains and tubes, the condition during surgery and any special instruction.
2. **Position** — for the unconscious patient, the **lateral or semi-prone (recovery) position with the head turned to one side and no pillow**, to maintain the airway and prevent aspiration; once awake, position as per the surgery (semi-Fowler's after abdominal or chest surgery; flat after spinal anaesthesia; head elevated after craniotomy or thyroid surgery; affected limb elevated after limb surgery).
3. **A-B-C first** — patency of the **airway** (look for snoring, stridor, obstruction by the tongue or secretions; use suction and an airway; jaw thrust), **breathing** (rate, depth, SpO₂, oxygen by mask/nasal prongs), and **circulation** (pulse, BP, colour, capillary refill, bleeding).
4. **Monitor and record vital signs every 15 minutes until stable (then half-hourly, hourly and 4-hourly)**: temperature, pulse, respiration, BP, SpO₂, level of consciousness, pain score, and pupils in neurosurgical cases.
5. **Check the wound and dressings** for bleeding or soakage (mark the extent and note the time), the drains (type, patency, character and amount of drainage), the IV line and infusion, the urinary catheter and the hourly output, and the nasogastric tube and aspirate.
6. Keep the patient **warm**, keep the side rails up, **never leave an unconscious patient alone**, and keep suction and oxygen at the bedside.
7. Assess and relieve **pain** (scheduled analgesia, patient-controlled analgesia or epidural as ordered) and **nausea**; observe for the return of the gag, swallowing and cough reflexes before giving anything orally.
8. Discharge from recovery only when the patient is awake and responding, with a stable airway, stable vital signs, controlled pain and no active bleeding (a scoring system such as the **Aldrete score** is used).

B. Continuing post-operative care

- **Respiratory** — **deep breathing and coughing every 1-2 hours with the wound splinted, incentive spirometry, early mobilisation**, nebulisation and chest physiotherapy as ordered, adequate analgesia so that the patient can breathe deeply, oxygen as prescribed.
- **Circulatory** — **leg and ankle exercises hourly, early ambulation, anti-embolism stockings and prescribed heparin prophylaxis**; observe for calf pain and swelling (DVT); change position slowly (postural hypotension).

- **Fluid and electrolytes** — strict intake-output record, IV fluids as prescribed, oral fluids started when bowel sounds return and the patient is fully awake, progressing **clear fluids → full fluids → soft diet → normal diet**; report a urine output of less than 30 mL/hour.
- **Nutrition** — high protein, high calorie, vitamin C rich diet for healing once oral feeding is allowed; supplements or enteral/parenteral feeding for major surgery.
- **Elimination** — watch for **retention of urine** (should void within 6-8 hours; a distended bladder, restlessness and frequent small voiding suggest retention — try privacy, a warm bedpan, the sound of running water, standing/sitting position, then catheterise if ordered); prevent constipation (fluids, fibre, mobility, prescribed laxative); note the first passage of flatus and stool (return of peristalsis).
- **Wound care** — keep the dressing dry and intact, aseptic dressing change as ordered, observe for bleeding, discharge, redness, gaping and smell; care of drains, tubes and the stoma; suture removal at the proper time (Chapter 8).
- **Comfort, hygiene and mobility** — mouth care, bed bath, back care and 2-hourly position change, progressive ambulation (sit up on day 0-1, stand and walk as permitted), physiotherapy.
- **Psychological support and health education** — explain the progress, involve the family, prepare for discharge.
- **Discharge advice (write it down)** — wound and stitch care and the date of removal, hygiene, bathing, the medicines with doses and duration, diet, activity restriction (lifting, driving, work, sexual activity), exercises, the danger signs to return for (**fever, increasing pain, bleeding, discharge, swelling, breathlessness, vomiting, no urine or stool, gaping of the wound**), the follow-up date, and whom to contact in an emergency.

18.7 Post-operative Complications

Table 18.2 Post-operative complications

Complication	When it occurs	Signs	Prevention and nursing action
Airway obstruction / hypoxia	Immediate	Snoring or stridor, in-drawing, restlessness, cyanosis, falling SpO ₂	Lateral position, suction, jaw thrust, oral airway, oxygen; call for help
Haemorrhage / reactionary bleeding	Within 24 h (secondary after 7-14 days from infection)	Soaked dressing, blood in the drain, rising pulse with falling BP , cold clammy pale skin, restlessness, thirst, low urine output	Check the wound and drain frequently, apply pressure, raise the foot end, keep flat, oxygen, fast IV fluids, inform the surgeon at once , send for blood, prepare for re-exploration
Shock	Immediate to 48 h	See Chapter 19	Monitor vital signs, fluids, oxygen, treat the cause
Post-operative nausea and vomiting	First 24-48 h	Nausea, retching, vomiting	Antiemetics, avoid early oral intake, head to one side, mouth care
Atelectasis and chest infection/pneumonia	Day 1-3 (the commonest cause of early post-operative fever)	Fever, tachypnoea, cough, reduced breath sounds, falling SpO ₂	Deep breathing, coughing, spirometry, early ambulation, position change, no smoking, good analgesia , chest physiotherapy, antibiotics as ordered
Wound infection	Day 3-7	Fever, throbbing pain, redness, swelling, heat, purulent discharge, raised white cell count	Aseptic dressing, hand hygiene, prophylactic antibiotic, nutrition, control of blood sugar; send pus for culture; report
Wound dehiscence and evisceration	Day 5-10	A sudden 'give way' feeling, serosanguineous (pink) discharge, gaping, protruding bowel	Prevent by nutrition, splinting during coughing, avoiding straining and distension; if it happens — lie the patient flat with knees bent, cover with sterile saline-soaked gauze, do not push the bowel back, keep NPO, inform the surgeon immediately
Paralytic ileus	Day 1-3	Abdominal distension, absent bowel sounds, no flatus, vomiting	NPO and nasogastric decompression, correct electrolytes (especially potassium), early mobilisation, avoid excess opioids, flatus tube as ordered
Retention of urine	First 6-12 h	No voiding in 6-8 h, suprapubic fullness and tenderness, restlessness, frequent small voids	Privacy, upright position, warm bedpan, running water, ambulation; catheterise only if ordered
Urinary tract infection	Day 3-5	Fever, burning, frequency, cloudy urine	Early removal of the catheter, closed drainage, perineal hygiene, 2-3 L fluid

Complication	When it occurs	Signs	Prevention and nursing action
Deep vein thrombosis and pulmonary embolism	Day 3-10 (may be later)	DVT — unilateral calf pain, swelling, warmth, tenderness; PE — sudden dyspnoea, chest pain, tachycardia, collapse, falling SpO ₂ (an emergency)	Leg exercises, early ambulation, hydration, stockings, prophylactic heparin; if suspected — do not massage, keep the patient at rest, give oxygen, inform the doctor at once
Pressure sores, constipation, hiccups, parotitis, thrombophlebitis at the IV site	Any time	As described in earlier chapters	2-hourly turning, skin and mouth care, fluids and fibre, care of the IV site
Post-operative fever — a useful framework	'The 5 W's'	Wind (atelectasis/pneumonia, day 1-2) → Water (urinary infection, day 3-5) → Wound (infection, day 5-7) → Walking (DVT, day 5-10) → Wonder drugs (drug or transfusion reaction, any day)	Identify the cause; a low-grade fever in the first 24 h is usually atelectasis or the inflammatory response to surgery

CHAPTER AT A GLANCE

- Fasting: clear fluids 2 h, breast milk 4 h, light meal 6 h, heavy/fatty meal 8 h before anaesthesia.
- Hair removal with a clipper immediately before surgery — never a razor the night before; bath with soap/chlorhexidine.
- Consent: informed, written, voluntary, before sedation; the doctor explains, the nurse witnesses; 18 years and above; guardian for minors; emergency life-saving treatment may proceed without consent.
- Remove dentures, jewellery, nail polish, contact lenses; void before theatre; prophylactic antibiotic within 60 min of the incision.
- WHO checklist: Sign In, Time Out (before incision), Sign Out (counts and specimens).
- Post-op: unconscious → lateral/recovery position, no pillow; vital signs every 15 min until stable; ABC first.
- Report a urine output under 30 mL/h; expect voiding within 6-8 hours.
- Complications by day: atelectasis 1-3, wound infection 3-7, dehiscence 5-10, DVT 3-10; secondary haemorrhage 7-14 days.
- Post-op fever 5 W's: Wind, Water, Wound, Walking, Wonder drugs.
- Evisceration: flat with knees bent, sterile saline gauze, do not push back, NPO, call the surgeon.

Shock and Blood Transfusion

Types and stages of shock, its management, blood groups and components, transfusion procedure and reactions

19.1 Shock — Definition and Types

DEFINITION

Shock — a life-threatening state of **acute circulatory failure** in which the tissue perfusion is inadequate to meet the metabolic needs of the cells, leading to cellular hypoxia, anaerobic metabolism, lactic acidosis, cell death and multi-organ failure. *It is not merely low blood pressure — shock may exist with a normal BP (compensated shock).*

Table 19.1 Classification of shock

Type of shock	Mechanism	Causes	Distinguishing features
Hypovolaemic (the commonest)	Loss of circulating volume → reduced preload	Haemorrhage (trauma, surgery, gastro-intestinal bleeding, post-partum haemorrhage, ruptured ectopic), plasma loss (burns), fluid loss (severe vomiting, diarrhoea, cholera, diabetic ketoacidosis, heat stroke, third-space loss)	Cold clammy pale skin, collapsed veins, low JVP, thirst, rapid weak pulse, low BP, oliguria
Cardiogenic	Failure of the heart as a pump → reduced cardiac output	Myocardial infarction , arrhythmia, myocarditis, valve rupture, cardiomyopathy, drug overdose (beta blocker)	Cold skin with raised JVP , crepitations/pulmonary oedema , gallop rhythm , chest pain; <i>fluids must be given with great caution</i>
Obstructive	Mechanical obstruction to filling or outflow	Cardiac tamponade , massive pulmonary embolism , tension pneumothorax	Raised JVP, distant heart sounds, pulsus paradoxus, tracheal shift and absent breath sounds (pneumothorax) — needs immediate decompression
Distributive — septic	Widespread vasodilatation and capillary leak from infection	Gram-negative and gram-positive sepsis , pneumonia, urinary or biliary sepsis, meningococcaemia, post-abortion and puerperal sepsis	Warm, flushed, dry skin with a bounding pulse in the early ('warm') stage , fever or hypothermia, confusion, high or low white count, raised lactate; later cold and clammy
Distributive — anaphylactic	Type I hypersensitivity with massive histamine release	Drugs (penicillin, NSAIDs, contrast) , vaccines , insect sting , food (nuts, egg, shellfish) , latex , blood products	Sudden onset within minutes : itching, urticaria, angioedema of the lips and tongue, hoarseness, stridor, wheeze, abdominal cramps, collapse
Distributive — neurogenic	Loss of sympathetic tone → vasodilatation	Spinal cord injury above T6 , spinal anaesthesia, severe pain, fright, brain injury	Hypotension with a warm dry skin and a paradoxically slow pulse (bradycardia)
Endocrine	Hormone deficiency	Addisonian crisis, myxoedema coma, thyroid storm	Treated with the specific hormone

19.2 Stages and Clinical Features of Shock

Table 19.2 Stages of shock

Stage	Features
1. Initial / compensated (non-progressive)	Cardiac output falls but compensatory mechanisms (sympathetic drive, tachycardia, vasoconstriction, renin-angiotensin) maintain the BP: restlessness and anxiety , tachycardia , cool pale skin , slight rise in diastolic BP with a narrow pulse pressure , thirst , reduced urine output . <i>This is the stage to recognise and treat — the BP may still be normal.</i>
2. Progressive (decompensated)	Compensation fails: falling systolic BP (under 90 mmHg) , rapid thready pulse , cold clammy sweating skin , mottling , rapid shallow breathing , confusion or drowsiness , marked oliguria (under 30 mL/h) , metabolic acidosis
3. Refractory / irreversible	Prolonged hypoxia → cell death, multi-organ dysfunction (acute kidney injury, ARDS, liver failure, disseminated intravascular coagulation) , unresponsive hypotension despite treatment , coma and death

NUMBERS TO REMEMBER

Classes of haemorrhagic shock (adult, 70 kg with about 5 L blood volume):

- **Class I** — loss up to 15% (750 mL): pulse under 100, BP normal, slight anxiety.
- **Class II** — 15-30% (750-1500 mL): **pulse 100-120**, BP normal but narrow pulse pressure, respiration 20-30, urine 20-30 mL/h, anxious.
- **Class III** — 30-40% (1500-2000 mL): **pulse over 120, BP falls**, respiration 30-40, urine 5-15 mL/h, confused — needs blood.
- **Class IV** — over 40% (over 2000 mL): pulse over 140 and feeble, BP very low or unrecordable, negligible urine, lethargic/unconscious — immediate transfusion and surgery.

Shock index = $\text{pulse rate} \div \text{systolic BP}$ (normal 0.5-0.7; above 0.9-1.0 suggests significant shock).

19.3 Management of Shock

1. **Call for help** and treat it as an emergency; do not leave the patient.
2. **Airway and breathing** — clear the airway, give **high-flow oxygen (10-15 L/min by mask with a reservoir)**, assist ventilation if needed, monitor SpO₂.
3. **Position** — lay the patient **flat with the legs raised about 20-30 cm (passive leg raising)**, or in the shock position; *the head-down Trendelenburg position is no longer recommended*; **sit the patient up if there is pulmonary oedema (cardiogenic shock) or severe breathlessness**; recovery position if unconscious and breathing.
4. **Circulation** — secure **two large-bore IV cannulae (14-16 G)**; take blood for grouping and cross-matching, haemoglobin, electrolytes, sugar, lactate and culture; give **warmed crystalloid (normal saline or Ringer lactate) 500-1000 mL rapidly (20 mL/kg in children)** and reassess; give **blood for haemorrhagic shock** (and control the bleeding — direct pressure, elevation, tourniquet as a last resort, surgery).
5. **Treat the cause** — stop the bleeding; **adrenaline IM for anaphylaxis**; antibiotics within the first hour and source control for sepsis; needle decompression for tension pneumothorax; pericardiocentesis for tamponade; reperfusion for myocardial infarction; atropine and vasopressors for neurogenic shock; hydrocortisone for adrenal crisis.
6. **Drugs as prescribed** — vasopressors/inotropes (noradrenaline, dopamine, dobutamine, adrenaline) through a central line with continuous monitoring; sodium bicarbonate for severe acidosis; analgesia given cautiously in small IV doses.
7. **Monitor** — pulse, BP, respiration, SpO₂, temperature, level of consciousness and capillary refill every 5-15 minutes; ECG; **hourly urine output through a catheter (aim above 0.5 mL/kg/h)**; central venous pressure if available; blood gases and lactate.
8. **Keep the patient warm** (blankets, warmed fluids — *hypothermia worsens coagulopathy*) but do not overheat or use hot water bottles (they cause vasodilatation and burns).
9. **Nil by mouth**; nothing to drink even if the patient is thirsty (moisten the lips); insert a nasogastric tube if ordered.
10. **Reassure** the conscious patient, explain briefly, keep the family informed, and **record everything with times** (fluids, drugs, output, observations).

CAUTION / NEVER DO THIS

In shock, do NOT: give anything by mouth • give alcohol • apply external heat or a hot water bag • sit the patient up (except in cardiogenic shock/pulmonary oedema) • leave the patient alone • give a large dose of intramuscular or oral analgesia (absorption is unreliable) • delay treatment waiting for investigations.

Fainting (syncope) is NOT shock — it is a transient reflex fall in cerebral perfusion; the patient recovers within a minute or two when laid flat with the legs raised. Shock is progressive and needs active treatment.

19.4 Blood Groups and Compatibility

Table 19.3 The ABO system

Blood group	Antigen on the red cell	Antibody in the plasma	Can receive red cells from	Can donate red cells to	Frequency (India, approx.)
A	A	Anti-B	A, O	A, AB	22%
B	B	Anti-A	B, O	B, AB	33%
AB	A and B	None	A, B, AB, O (universal recipient)	AB only	7%

Blood group	Antigen on the red cell	Antibody in the plasma	Can receive red cells from	Can donate red cells to	Frequency (India, approx.)
O	None	Anti-A and anti-B	O only	A, B, AB, O (universal donor)	37%

- **ABO groups** were discovered by **Karl Landsteiner (1900, Nobel Prize 1930)**; the **Rh factor** (D antigen) by Landsteiner and Wiener (1940).
- **Rh positive** = D antigen present (about **94-95% of Indians**); **Rh negative** = absent. An Rh-negative person must receive Rh-negative blood.
- **O negative** is the **universal donor** of packed red cells (best for emergencies); **AB positive** is the universal recipient of red cells. For **plasma** the reverse is true — **AB plasma is the universal donor plasma**.
- **Haemolytic disease of the newborn (erythroblastosis fetalis)** — an Rh-negative mother carrying an Rh-positive fetus forms anti-D antibodies which destroy fetal red cells in a subsequent pregnancy; prevented by giving **anti-D immunoglobulin (Rh immunoglobulin) 300 micrograms IM within 72 hours** of delivery, abortion, ectopic pregnancy, amniocentesis or antepartum bleeding (and routinely at 28 weeks).
- Every unit must be **grouped and cross-matched** (compatibility testing); in a dire emergency, group-specific or O-negative blood may be given while the cross-match is completed.

19.5 Blood Components, Storage and Indications

Table 19.4 Blood components

Component	Storage and shelf life	Volume / effect	Indications
Whole blood	2-6 °C (ideally 4 °C) for 35-42 days in CPDA-1/additive solution	About 350-450 mL; raises haemoglobin by about 1 g/dL per unit	Massive acute haemorrhage with hypovolaemia, exchange transfusion
Packed red cells (PRBC)	2-6 °C, 35-42 days	About 200-300 mL; 1 unit raises Hb by about 1 g/dL or the haematocrit by 3%	The commonest component used — symptomatic anaemia, Hb under 7-8 g/dL, surgical loss, thalassaemia, chronic renal disease
Fresh frozen plasma (FFP)	-30 °C or below for 1 year ; once thawed, use within 4-6 hours (24 h at 2-6 °C)	200-250 mL; supplies all clotting factors	Multiple coagulation factor deficiency, liver disease, DIC, warfarin reversal with bleeding, massive transfusion
Platelet concentrate	20-24 °C (room temperature) with continuous gentle agitation, for only 5 days (never refrigerate)	50-70 mL per unit; raises the count by about 5000-10 000/ μ L	Platelet count under 10 000-20 000/ μ L, or under 50 000 with bleeding or before surgery; dengue with bleeding; leukaemia and chemotherapy
Cryoprecipitate	-30 °C, 1 year	10-20 mL; rich in factor VIII, von Willebrand factor, fibrinogen and factor XIII	Haemophilia A, von Willebrand disease, hypofibrinogenaemia, DIC
Others	As labelled	Factor VIII/IX concentrate, albumin, immunoglobulin, granulocytes	Haemophilia, hypoalbuminaemia, immunodeficiency

NUMBERS TO REMEMBER

Anticoagulant-preservative solutions — CPD (citrate-phosphate-dextrose) 21 days, CPDA-1 (with adenine) 35 days, and additive solutions (SAGM) 42 days; the citrate binds calcium to prevent clotting.

Blood must be transported in a **validated cold box/blood transport box at 2-10 °C** and must **never be kept in a domestic or ward refrigerator**; if a unit has been out of controlled storage for **more than 30 minutes** it should not be returned to stock.

Blood donor criteria (India) — age **18-65 years**, weight **45-50 kg or more**, **haemoglobin at least 12.5 g/dL**, pulse and BP normal, temperature normal, no infection or risk behaviour, no major surgery or tattoo in the past 6-12 months, not pregnant or lactating, **interval between whole-blood donations: 3 months (12 weeks) for men and women** (some guidelines: 90 days); one donation is 350 or 450 mL; the donor rests and takes fluids afterwards. **World Blood Donor Day — 14 June** (Karl Landsteiner's birthday); **National Voluntary Blood Donation Day — 1 October**.

Mandatory screening of every unit — **HIV 1 and 2, hepatitis B (HBsAg), hepatitis C, syphilis (VDRL) and malaria** (plus grouping, Rh typing and antibody screening).

19.6 Procedure of Blood Transfusion — Nursing Responsibilities

1. Confirm the **written order and the informed consent**; explain the procedure and the reason to the patient; record **baseline vital signs (temperature, pulse, respiration, BP) immediately before starting** — if the patient is febrile, inform the doctor before proceeding.
2. **Check the unit with a second qualified person at the bedside** and match against the patient and the records: **the patient's name and hospital number, the blood group and Rh of the patient and of the unit, the unit/bag number, the cross-match report, the expiry date, and the appearance of the bag** (*return the unit if there is any clot, discolouration, haemolysis (pink/brown plasma), gas bubbles, leakage or damage*). Both persons sign.
3. Secure a **large-bore cannula (18-20 G in an adult)**; use a **fresh blood administration set with a 170-200 micron filter**; prime with **normal saline only**.
4. **Never add any drug or solution to blood, and never use dextrose (haemolysis) or Ringer lactate (the calcium causes clotting) in the same line**; flush with normal saline before and after.
5. **Start the transfusion within 30 minutes of the unit leaving the blood bank**; do not warm it routinely (use an approved blood warmer only for massive or rapid transfusion, exchange transfusion or in the presence of cold agglutinins — *never warm blood in hot water, a microwave or on a radiator*).
6. **Run slowly for the first 15 minutes (about 1-2 mL/min / 20 drops per minute) and stay with the patient** — most severe reactions appear in this period; then adjust to the prescribed rate. **One unit is completed within 4 hours** (usually 2-3 hours); discard any unit hanging for more than 4 hours.
7. **Monitor** vital signs before, **15 minutes after starting, then every 30-60 minutes** during, at the end, and 1 hour after the transfusion; observe continuously for the signs of a reaction and record everything (time started, rate, unit number, volume, observations, time completed).
8. Teach the patient and family to **report immediately** any chills, itching, rash, back or chest pain, breathlessness, headache, giddiness or a feeling that 'something is wrong'.
9. After completion: flush with saline, remove and dispose of the bag and set as per policy (many hospitals retain the bag for 24 hours in case of a delayed reaction), record the final vital signs, and continue to observe.
10. **Special care** — in cardiac, renal, elderly and paediatric patients transfuse slowly with strict intake-output monitoring (risk of overload); in children the volume is calculated (usually **10-15 mL/kg of packed cells**) and given with a pump/burette.

19.7 Transfusion Reactions

Table 19.5 Transfusion reactions — recognise, stop, report

Reaction	Cause	Signs and time of onset	Management
Acute haemolytic reaction (the most dangerous)	ABO incompatibility — usually a clerical/identification error	Within minutes: fever with chills, burning along the vein, pain in the loin/back and chest, flushing, restlessness, hypotension, tachycardia, dyspnoea, bleeding from puncture sites (DIC), dark/red urine (haemoglobinuria), oliguria and renal failure	STOP the transfusion immediately • keep the line open with normal saline through a new set • maintain the airway and give oxygen • call the doctor • monitor vital signs and urine output • send the bag, the giving set, fresh blood samples and a urine specimen to the blood bank/laboratory • treat shock and protect the kidneys (fluids, diuretics as ordered) • complete an incident/reaction report
Febrile non-haemolytic reaction (the commonest)	Recipient antibodies against donor white cells/cytokines	Rise of temperature of 1 °C or more with chills and rigor, usually 30 min-2 h after starting; no haemolysis	Stop or slow the transfusion and inform the doctor; give an antipyretic (paracetamol, <i>not aspirin if the platelet count is low</i>); use leucocyte-depleted blood in future; rule out haemolysis and sepsis first
Allergic / urticarial reaction	Plasma proteins	Itching, urticaria (hives), flushing without fever, within minutes	Stop temporarily, give an antihistamine as ordered; if the symptoms settle completely the transfusion may be restarted slowly (for mild urticaria only)
Anaphylactic reaction	IgA deficiency with anti-IgA, severe hypersensitivity	Within seconds-minutes: angioedema, stridor, wheeze, severe dyspnoea, hypotension, collapse	Stop at once; adrenaline 1:1000, 0.5 mL IM; oxygen; fluids; hydrocortisone and antihistamine; airway support

Reaction	Cause	Signs and time of onset	Management
Transfusion-associated circulatory overload (TACO)	Too rapid or too large a volume, especially in the elderly, infants, and cardiac or renal patients	Dyspnoea, orthopnoea, cough with frothy sputum, raised JVP, crepitations, raised BP, headache, within hours	Stop or slow the transfusion, sit the patient upright with the legs dependent, give oxygen, give the prescribed diuretic (frusemide), monitor; prevent by slow transfusion and packed cells
TRALI (transfusion-related acute lung injury)	Donor antibodies against recipient leucocytes	Acute hypoxia with bilateral pulmonary infiltrates within 6 hours, but a normal JVP/no overload	Stop, give oxygen and ventilatory support; <i>diuretics do not help</i> ; report to the blood bank
Bacterial contamination / septic reaction	Contaminated unit (especially platelets stored at room temperature)	High fever with rigor, vomiting, severe hypotension and collapse soon after starting	Stop at once, take blood cultures from the patient and the bag, start broad-spectrum antibiotics and treat shock
Air embolism	Air entering the line	Sudden dyspnoea, chest pain, cyanosis, hypotension, 'mill-wheel' murmur	Clamp the line, place the patient on the left side with the head down (Durant's position) , give oxygen, call for help
Citrate toxicity / hypocalcaemia and hyperkalaemia	Massive or rapid transfusion (citrate binds calcium; stored cells leak potassium)	Perioral and finger tingling, tremor, muscle twitching, arrhythmia, prolonged QT	Slow the rate; calcium gluconate as prescribed; monitor ECG and electrolytes
Hypothermia and coagulopathy	Cold blood given rapidly in large volume	Shivering, arrhythmia, bleeding	Use a blood warmer for massive transfusion; keep the patient warm
Delayed reactions	Alloantibodies, iron overload, transmitted infection	Delayed haemolysis (5-10 days: falling Hb, mild jaundice), transfusion-transmitted infection (hepatitis B/C, HIV, syphilis, malaria, CMV), iron overload (haemosiderosis) in multiply transfused patients (thalassaemia), post-transfusion purpura, graft-versus-host disease	Screening of donors and units, leucodepletion and irradiation where indicated, iron chelation (deferasirox/deferrioxamine), follow-up testing

HIGH-YIELD / MOST ASKED

The first three actions in ANY suspected transfusion reaction: (1) STOP the transfusion immediately (2) keep the vein open with normal saline using a new administration set (3) inform the doctor and the blood bank, and monitor the vital signs.

Then send the bag, set, blood and urine samples for investigation and complete a reaction report.

The commonest cause of a fatal transfusion reaction is a clerical error — the wrong unit given to the wrong patient. Hence the **two-person bedside identity check** is the single most important safety step.

CHAPTER AT A GLANCE

- Shock = inadequate tissue perfusion; types: hypovolaemic (commonest), cardiogenic, obstructive, distributive (septic, anaphylactic, neurogenic), endocrine.
- Early/compensated shock: restlessness, tachycardia, cool skin, narrow pulse pressure, thirst, low urine — BP may still be normal.
- Warm dry skin with hypotension = early septic (or neurogenic, with bradycardia) shock; raised JVP with crepitations = cardiogenic.
- Management: oxygen, flat with legs raised, two large IV lines, crystalloid then blood, treat the cause, keep warm, NPO, hourly urine output; no external heat, no oral fluids.
- Class III haemorrhage = 30-40% loss (pulse over 120 with falling BP) and needs blood.
- O negative = universal donor of red cells; AB positive = universal recipient; AB plasma = universal donor plasma.
- Storage: whole blood/PRBC 2-6 °C for 35-42 days; platelets 20-24 °C with agitation for 5 days; FFP -30 °C for 1 year.
- Donor: 18-65 years, 45-50 kg, Hb 12.5 g/dL, every 3 months; mandatory screening for HIV, HBsAg, HCV, syphilis and malaria.
- Transfusion: two-person bedside check, normal saline only, filter set, start within 30 min, slow for the first 15 min, complete within 4 hours, vital signs at 15 min then half-hourly.
- Never add a drug to blood; never use dextrose or Ringer lactate in the blood line.
- Any reaction: STOP → saline through a new set → inform the doctor and blood bank; ABO incompatibility is the most dangerous and is usually a clerical error; TACO → sit up and give a diuretic.

Special Treatment

Oxygen therapy, suction, CPR, tracheostomy, catheter and stoma care, dialysis, physiotherapy and other special procedures

20.1 Oxygen Therapy

- **Oxygen is a drug** — it must be prescribed with the **device, flow rate/concentration, and the target saturation**, and its effect must be recorded.
- **Indications** — hypoxaemia (**SpO₂ below 90-94%**), respiratory distress, pneumonia, asthma, COPD exacerbation, pulmonary oedema, heart failure, myocardial infarction, shock, severe anaemia, carbon monoxide poisoning, post-anaesthesia, and during resuscitation.
- **Target saturation** — **94-98% for most patients**, but **88-92% for a patient with COPD/chronic type-II respiratory failure** (high-flow oxygen can abolish the hypoxic drive and cause carbon dioxide narcosis).

Table 20.1 Oxygen delivery devices

Device	Flow rate	Approximate FiO ₂	Notes
Nasal cannula (prongs)	1-6 L/min	24-44% (about 4% per litre)	Comfortable, the patient can eat and talk; drying above 4 L/min (humidify); check the nostrils and behind the ears for pressure
Simple face mask	5-10 L/min	40-60%	Never run below 5 L/min (rebreathing of CO ₂); interferes with eating and speaking
Venturi (air-entrainment) mask	As marked on the valve (colour-coded)	Fixed, accurate 24, 28, 31, 35, 40, 60%	The device of choice in COPD because the concentration is exact
Partial rebreathing mask	6-10 L/min	60-80%	Bag must not collapse completely on inspiration
Non-rebreathing mask (with reservoir)	10-15 L/min	60-95% (up to 100%)	For emergencies, trauma, shock, carbon monoxide poisoning; keep the reservoir bag inflated
Head box / oxygen hood / incubator	As ordered	Variable	Neonates and infants
High-flow nasal oxygen, CPAP, BiPAP, ventilator	Up to 60 L/min	21-100%	Severe hypoxia, type-II failure, sleep apnoea, ARDS, COVID-19; needs monitoring and skilled staff

CAUTION / NEVER DO THIS

Safety with oxygen (it supports combustion — it does not burn itself but makes everything else burn fiercely):

- **NO smoking, no naked flame, no spark, no electrical appliance, no oil, grease, petroleum jelly, alcohol rub or aerosol** near the patient or the cylinder; put up 'Oxygen in use — no smoking' signs.
- Cylinders **upright, chained/secured, in a cool ventilated place**; check the contents and the flow meter; never use a faulty regulator; do not use a lubricant on the valve.
- **Hazards of oxygen therapy** — drying and crusting of the mucosa, nasal and facial pressure sores, infection from a contaminated humidifier, **oxygen toxicity** (over 50% for over 24-48 h: substernal pain, cough, atelectasis, ARDS), **carbon dioxide narcosis in COPD, retinopathy of prematurity and bronchopulmonary dysplasia in newborns**, and absorption atelectasis.
- Change the humidifier water (sterile water) and the tubing as per policy; give frequent mouth and nasal care; monitor SpO₂ and the level of consciousness.

20.2 Suction, Nebulisation and Chest Physiotherapy

- **Suction** — to clear secretions from the mouth, pharynx, trachea or an artificial airway when the patient cannot cough effectively. **Pressure: adults 100-150 mmHg, children 95-110, infants 50-95 mmHg**. Use a **sterile catheter (not more than half the diameter of the airway)**, pre-oxygenate, insert **without suction**, apply suction **intermittently while withdrawing with a rotating movement, for not more than 10-15 seconds at a time**, allow the patient to recover for 30 seconds to 1 minute between passes, and **suction the mouth last**. Observe for hypoxia, bradycardia (vagal), arrhythmia, trauma and bleeding; record the amount and character of the secretions.
- **Nebulisation** — see Chapter 11; keep the mask snug, tap the chamber, complete in 10-15 minutes, do mouth care afterwards, and clean and dry the chamber and tubing after each use.

- **Chest physiotherapy — postural drainage** (positioning the affected lobe uppermost for 10-20 minutes, best done **before meals or 1-2 hours after, and never immediately after food**; contraindicated in raised intracranial pressure, unstable spine, haemoptysis and severe dyspnoea), **percussion (clapping with cupped hands) and vibration** over the affected segment during expiration (*not over the spine, breasts, kidneys or a fractured rib*), followed by **huffing and coughing**; used in bronchiectasis, cystic fibrosis, lung abscess, atelectasis and retained secretions.
- **Breathing exercises — diaphragmatic (abdominal) breathing, pursed-lip breathing (for COPD), segmental breathing, incentive spirometry (10 breaths every hour while awake)**, blowing bubbles/a balloon for children.

20.3 Cardiopulmonary Resuscitation (CPR) and Basic Life Support

1. **Check for danger, then check responsiveness** (shake and shout); if unresponsive, **shout for help/activate the emergency response** and get the defibrillator/crash cart.
2. **Check breathing and the carotid pulse simultaneously for not more than 10 seconds**; gasping (agonal breathing) counts as not breathing.
3. If there is no pulse, start **chest compressions immediately (C-A-B sequence)** on a firm flat surface: heel of the hand on the **lower half of the sternum (centre of the chest between the nipples)**, other hand on top, arms straight, shoulders over the hands; **depth 5-6 cm in an adult (about one-third of the chest depth — 5 cm in a child, 4 cm in an infant)**; **rate 100-120 per minute**; allow **complete recoil** and minimise interruptions (less than 10 seconds).
4. **Compression : ventilation ratio = 30 : 2** for adults (single or two rescuers) and for children/infants with a single rescuer; **15 : 2 for children and infants with two rescuers**; with an advanced airway, give continuous compressions with **1 breath every 6 seconds (10/min)**.
5. Open the airway with **head tilt-chin lift** (jaw thrust if a spinal injury is suspected); give each rescue breath over **1 second, just enough to make the chest rise** (avoid over-ventilation); use a bag-valve-mask with oxygen as soon as available.
6. **Attach the AED/defibrillator as soon as it arrives** and follow its prompts; **defibrillate shockable rhythms (ventricular fibrillation and pulseless ventricular tachycardia) at once**, then resume compressions immediately for 2 minutes before rechecking. *Asystole and pulseless electrical activity are not shockable.*
7. **Change the compressor every 2 minutes** (or every 5 cycles) to avoid fatigue; give **adrenaline 1 mg IV every 3-5 minutes** and other drugs as directed; treat the reversible causes (**the 4 H's and 4 T's — Hypoxia, Hypovolaemia, Hypo/hyperkalaemia and metabolic, Hypothermia; Tension pneumothorax, Tamponade, Toxins, Thrombosis**).
8. Continue until spontaneous circulation returns, a doctor decides to stop, or the rescuer is exhausted; after the return of circulation give **post-resuscitation care** (airway, oxygen targeted to 94-98%, 12-lead ECG, temperature management, transfer to intensive care) and **record the whole event with times**.
9. For an **infant**: 2 fingers (single rescuer) or the 2-thumb encircling technique (two rescuers) just below the inter-nipple line; check the **brachial** pulse. In **drowning and in children**, hypoxia is the usual cause, so begin with 5 rescue breaths (A-B-C) as per the paediatric/drowning guideline.
10. **Choking (foreign body obstruction)** — if the patient can cough, encourage coughing; if the obstruction is complete in a **conscious adult/child: 5 back blows between the shoulder blades then 5 abdominal thrusts (Heimlich manoeuvre)**, repeating until relieved; for an **infant: 5 back blows with the baby prone head-down on the forearm, then 5 chest thrusts (never abdominal thrusts)**; for a pregnant or obese person use chest thrusts; if the patient becomes unconscious, start CPR.

20.4 Tracheostomy and Airway Care

- **Tracheostomy** — a surgical opening in the trachea (usually between the 2nd and 3rd/4th tracheal rings) with a tube in place; done for **upper airway obstruction, prolonged ventilation, retained secretions, head and neck surgery, and to reduce dead space**.
- **Bedside articles that must always be ready** — suction apparatus with catheters, **a spare tracheostomy tube of the same size and one a size smaller**, tracheal dilator and spare tapes, sterile gloves and dressing tray, normal saline, humidified oxygen, resuscitation bag, and a pair of scissors.
- **Care** — strict asepsis; **suction only when needed** (see 20.2); keep the inner tube clean (remove, clean with saline and a brush, and replace as per policy); clean the stoma with saline and apply a keyhole dressing, changing it when soiled; secure the tapes with **one finger's space** and change them with a second person holding the tube; keep the air **humidified** (the upper airway is bypassed); mouth care 2-4 hourly; keep the patient in the semi-Fowler's position.

- **Communication** — the patient **cannot speak** with a cuffed tube: provide a pen and paper, a picture/alphabet board, a bell and agreed signs; explain and reassure; never leave the call bell out of reach.
- **Emergency** — if the tube is blocked, suction and, if necessary, remove the inner tube; **if the tube is accidentally displaced, hold the stoma open with the tracheal dilator, give oxygen, call for help and re-insert the spare tube**; watch for **bleeding, subcutaneous emphysema, infection, tube blockage, displacement, stenosis and tracheo-oesophageal fistula**.

20.5 Care of Tubes, Drains and Stomas

Table 20.2 Care of tubes, drains and stomas

Device	Key nursing points
Nasogastric (Ryle's) tube	See Chapter 9 — confirm the position before each use, flush, oral and nasal hygiene, secure the tape, measure the aspirate, keep the head elevated
Urinary catheter	See Chapter 11 — closed drainage, bag below bladder level, meatal care, adequate fluids, early removal
Intercostal (chest) drain with underwater seal	The bottle is kept always below the level of the chest and upright ; the tube tip must remain under water; look for swinging of the fluid level with respiration (means it is patent) and bubbling (means an air leak) ; never lift the bottle above the chest; never clamp a bubbling chest drain (risk of tension pneumothorax) except briefly to change the bottle as instructed; keep two clamps and a bottle of sterile water at the bedside for accidental disconnection (submerge the tube end in sterile water); if the tube comes out, cover the site with an occlusive dressing sealed on three sides and call for help at once; measure and record the drainage; encourage deep breathing and arm exercises
Wound drains (Redivac, Penrose, T-tube)	Chapter 8 — secure, keep below the wound, measure, asepsis, report a sudden change
Colostomy / ileostomy	Observe the stoma colour (should be pink/red and moist — report a dusky, black or retracted stoma) ; measure and cut the appliance to fit (about 2-3 mm larger than the stoma); protect the surrounding skin with a barrier; empty the bag when one-third to half full ; change the appliance when there is leakage or as scheduled; note the output (ileostomy output is liquid and continuous with high sodium and fluid loss; colostomy output is more formed); teach the patient self-care, diet (chew well, avoid gas-forming, odour-producing and obstructing foods such as nuts, popcorn and raw vegetables initially, drink 2-3 L), how to recognise problems, and where to obtain supplies; provide strong psychological support (body image, odour, work and sexual concerns) and refer to a stoma nurse/support group; colostomy irrigation as taught
Central venous catheter / PICC	Strict asepsis, transparent dressing, flush as per protocol, never leave a lumen open to air, observe for infection and thrombosis, measure CVP as ordered

20.6 Dialysis

Table 20.3 Haemodialysis and peritoneal dialysis

Point	Haemodialysis	Peritoneal dialysis
Principle	Blood is pumped through an artificial kidney (dialyser) where diffusion, osmosis and ultrafiltration occur across a semipermeable membrane	The patient's own peritoneum acts as the semipermeable membrane; dialysate is instilled into the peritoneal cavity, dwells, and is drained
Access	Arteriovenous fistula (the preferred long-term access), AV graft, or a temporary central venous catheter	Tenckhoff catheter in the abdominal wall
Schedule	3-4 hours, 2-3 times a week in a dialysis unit	CAPD — 4-5 exchanges a day at home, or by a machine at night (APD)
Nursing care	Never use the fistula arm for BP, injections, IV lines or blood sampling; check the bruit and thrill daily; no tight clothing, watch or heavy lifting on that arm ; weigh before and after (the weight difference = fluid removed); monitor BP and for cramps, hypotension, headache and disequilibrium; restrict fluid, sodium, potassium and phosphate; give drugs after dialysis if they are dialysable; check for bleeding from the puncture sites (heparin is used)	Strict asepsis at every exchange (peritonitis is the main complication — cloudy outflow, abdominal pain and fever) ; warm the dialysate to body temperature; record the inflow, dwell and outflow volumes and the fluid balance; the outflow should be clear; observe the exit site; higher protein loss so a high-protein diet is needed; watch for hyperglycaemia, constipation and hernia
Diet	Low protein is <i>not</i> required once on dialysis — 1.0-1.2 g/kg protein; restrict potassium, sodium, phosphate and fluid	1.2-1.5 g/kg protein (loss in dialysate); more liberal potassium

20.7 Physiotherapy, Traction and Rehabilitation Modalities

Table 20.4 Physiotherapy and rehabilitation

Modality	Description and nursing points
Exercises	Passive (done by the nurse/therapist for a paralysed or unconscious patient), active assisted , active , active resisted , and isometric (muscle contracted without joint movement — used inside a cast); range-of-motion exercises 2-3 times a day to all joints; <i>stop if there is pain or resistance</i>
Heat modalities	Infrared, short-wave diathermy, ultrasound, paraffin wax bath, hot packs — for chronic pain and stiffness; contraindicated over a metal implant/pacemaker (diathermy), in acute inflammation, malignancy, impaired sensation and pregnancy
Cold and hydrotherapy	Ice packs and cryotherapy for acute injury and spasticity; whirlpool bath and pool exercise for weight-relieved movement
Electrotherapy	TENS (transcutaneous electrical nerve stimulation) for pain, faradic and galvanic stimulation for denervated muscle, biofeedback
Massage	Effleurage, petrissage, tapotement, friction — improves circulation and relieves spasm; never massage over a suspected DVT, acute inflammation, infection, fracture, tumour or thrombophlebitis
Traction	Skin traction (Buck's extension, Russell, Bryant's — used in children under 2 years and under 12-14 kg, with both legs suspended and the buttocks just clear of the bed) and skeletal traction (Steinmann pin, Kirschner wire, Thomas splint, halo). Nursing care: the weights must hang free and never touch the floor or the bed and must never be removed or lifted without an order ; the ropes must run freely in the pulleys and not be frayed; maintain the correct line of pull and counter-traction (never let the foot piece rest against the pulley) ; keep the patient in correct alignment; check the pin site daily for infection and give pin-site care; assess neurovascular status (the 5 P's) at least every 4 hours ; prevent complications of immobility — pressure sores (especially at the heel and sacrum), foot drop (footboard/splint), constipation, chest infection and DVT; give a fracture bedpan; encourage exercises of the free joints and quadriceps drill
Walking aids and gait training	Chapter 15 — crutches, walker, stick; teach the correct height and gait; 'up with the good, down with the bad' on stairs
Occupational therapy, speech therapy, prosthetics and orthotics	Restore function and independence; splints to prevent contracture; artificial limb after amputation with stump care (bandaging, exercises, prevention of flexion contracture and phantom pain)

20.8 Other Special Treatments

Table 20.5 Miscellaneous special treatments

Treatment	Nursing points
ECG recording	Explain; the patient lies still and relaxed, not talking or moving; expose the chest, clean the skin, apply gel; limb leads — RA red, LA yellow, LL green, RL black ('Ride Your Green Bike') ; chest leads V1-V2 at the 4th intercostal space either side of the sternum, V4 at the 5th intercostal space in the mid-clavicular line, V3 between V2 and V4, V5 and V6 in the anterior and mid-axillary line at the level of V4; label the strip with the name, date and time; report <i>ST elevation, wide QRS, ventricular ectopics, VT/VF, absent P waves or bradycardia</i> at once
Radiotherapy	Do not wash off the skin markings ; wash the area gently with plain water only, pat dry, no soap, powder, deodorant, perfume, adhesive tape, hot water bag or ice on the treated area; wear loose cotton clothes; protect from sunlight; report skin breakdown; for a patient with an internal (sealed) source — limit the time spent close by, keep the maximum distance, use shielding, no pregnant staff or visitors or children, do not handle a dislodged source with the hands (use forceps and a lead container), and follow the radiation safety officer's instructions
Chemotherapy	See Chapter 17; antiemetics, mouth care, monitor blood counts, neutropenic precautions, protect from infection, report fever immediately
Mechanical ventilation	Explain and reassure; check the alarms, settings and the cuff pressure; ensure humidification; suction as needed; oral care with chlorhexidine; head of the bed 30-45° to prevent ventilator-associated pneumonia ; sedation and pain relief; DVT and stress-ulcer prophylaxis; communication board; never silence an alarm without checking the patient; keep a resuscitation bag ready
Defibrillation and cardiac monitoring	Ensure no one is touching the patient or the bed ('all clear'), the chest is dry, oxygen is moved away, and the pads/paddles are correctly placed (right of the sternum below the clavicle and the left mid-axillary line at the 5th space); resume compressions immediately after the shock
Phototherapy (neonatal jaundice)	Baby naked except for eye pads and a small nappy; lamp at the prescribed distance; turn the baby 2-hourly; monitor temperature, weight, hydration and stool; continue feeding; measure bilirubin as ordered

Treatment	Nursing points
Hot/cold and other applications	Chapter 11
CHAPTER AT A GLANCE • Oxygen is a prescribed drug; target SpO₂ 94-98% (88-92% in COPD, where a Venturi mask is preferred); non-rebreathing mask at 10-15 L/min for emergencies; simple mask never below 5 L/min. • Oxygen hazards: drying, infection, oxygen toxicity above 50%, CO₂ narcosis in COPD, retinopathy of prematurity; no flame, oil or spark. • Suction: sterile catheter, 100-150 mmHg in adults, 10-15 seconds per pass, suction only on withdrawal, mouth last. • CPR: C-A-B, compressions 5-6 cm at 100-120/min, ratio 30:2 (15:2 for two-rescuer paediatric), change compressor every 2 min, adrenaline 1 mg every 3-5 min, defibrillate VF/pulseless VT only. • Choking: 5 back blows + 5 abdominal thrusts (adult/child); back blows + chest thrusts for an infant (never abdominal). • Tracheostomy: keep a spare tube and tracheal dilator at the bedside; humidify; provide a means of communication. • Chest drain: keep below the chest, upright, tip under water; swinging = patent, bubbling = air leak; never clamp a bubbling drain; if it falls out, seal on three sides. • Stoma: pink and moist is normal; empty the bag at one-third to half full. • Fistula arm: no BP, no injection, no IV, no blood sampling; check the bruit and thrill. Peritoneal dialysis: cloudy outflow = peritonitis. • Traction weights hang free and are never removed; check the neurovascular status 4-hourly. • ECG leads: RA red, LA yellow, LL green, RL black. Radiotherapy: never wash off the markings; no soap, powder or heat on the site.	

Nursing in Special Diseases

Disease-wise nursing care of common medical, surgical, paediatric and emergency conditions

21.1 Respiratory Diseases

Table 21.1 Nursing in respiratory disease

Condition	Key features	Nursing care
Pneumonia	High fever with chills, cough with rusty sputum , pleuritic chest pain, rapid shallow breathing, crepitations; in children — fast breathing and chest in-drawing	Semi-Fowler's position , oxygen as prescribed, plenty of fluids (3 L), light nourishing diet, antibiotics on time, antipyretics, chest physiotherapy and deep breathing, teach cough etiquette, mouth care, monitor temperature, respiration, SpO ₂ and the character of the sputum, sputum for culture, prevent cross-infection and spread
Bronchial asthma	Episodic wheeze, dry cough (often at night), chest tightness, prolonged expiration, use of accessory muscles; a silent chest, exhaustion, cyanosis, inability to speak or a falling SpO₂ indicate a life-threatening attack	Sit the patient upright leaning forward, reassure and stay with him, give oxygen and nebulised salbutamol/ipratropium and steroids as prescribed, monitor peak flow and SpO ₂ , identify and remove the trigger (dust, smoke, pollen, cold air, exercise, aspirin/NSAIDs, beta blockers, stress), teach the correct inhaler technique with a spacer, the difference between reliever and preventer, rinse the mouth after steroid inhalers , and give a written action plan; avoid sedatives
COPD	Chronic productive cough, progressive breathlessness, barrel chest, pursed-lip breathing, cor pulmonale	Controlled oxygen (target 88-92%, Venturi mask) , pursed-lip and diaphragmatic breathing, stop smoking (the single most important measure) , energy conservation, small frequent high-calorie meals, pulmonary rehabilitation, influenza and pneumococcal vaccination, inhaler technique, watch for CO ₂ narcosis (drowsiness, headache, flapping tremor)
Pulmonary tuberculosis	Cough over 2 weeks, evening fever, night sweats, weight loss, haemoptysis	Airborne precautions (well-ventilated room, N95 for staff, mask for the patient, cough hygiene, sputum in a covered container), DOTS adherence (Chapter 17), high-protein high-calorie diet, side-effect education, contact screening, Ni-kshay registration, address stigma and nutritional support
Haemoptysis / pulmonary oedema	Coughing up blood / severe breathlessness with pink frothy sputum and crepitations	Haemoptysis: sit up, reassure, ice-cold sips, save and measure the blood, nil by mouth, inform the doctor. Pulmonary oedema: sit upright with the legs dependent, high-flow oxygen, IV furosemide, morphine, nitrates as ordered , strict output monitoring, no oral fluids

21.2 Cardiovascular Diseases

Table 21.2 Nursing in cardiovascular disease

Condition	Key features	Nursing care
Acute myocardial infarction	Severe crushing retrosternal pain lasting over 20 minutes, radiating to the left arm, jaw or back, not relieved by rest or nitrates , with sweating, nausea, vomiting, breathlessness, anxiety and a sense of impending death; may be silent in diabetics and the elderly	Absolute bed rest with a cardiac chair position; oxygen if hypoxic; chew soluble aspirin 300 mg , sublingual nitrate, and morphine/analgesia as prescribed; secure an IV line; 12-lead ECG within 10 minutes and continuous cardiac monitoring; prepare for thrombolysis or angioplasty; keep the defibrillator and emergency drugs ready; watch for arrhythmia (the commonest cause of early death), heart failure, shock and cardiac arrest ; keep the patient calm and undisturbed; light diet, avoid straining (stool softener), monitor intake-output; later — graded mobilisation and cardiac rehabilitation, and education on diet, exercise, smoking cessation, weight and drug adherence
Congestive heart failure	Breathlessness, orthopnoea, paroxysmal nocturnal dyspnoea, fatigue, pedal oedema, raised JVP, tender hepatomegaly, basal crepitations, rapid weight gain	Fowler's/cardiac position , rest with restricted activity, oxygen, low-salt diet and fluid restriction, daily weight at the same time (a gain of 1 kg = 1 L of fluid) , strict intake-output, small frequent meals, prevention of constipation and DVT, skin care of oedematous areas, give digoxin after checking the pulse (hold if under 60) , watch for digoxin toxicity (nausea, vomiting, visual halos, arrhythmia) and for hypokalaemia with diuretics, teach the patient to weigh daily and report a gain of 2 kg in 2-3 days

Condition	Key features	Nursing care
Hypertension	Usually symptomless ('the silent killer'); headache, giddiness, epistaxis; hypertensive emergency — BP over 180/120 with chest pain, breathlessness, visual or neurological symptoms	Correct BP measurement technique in both arms, low-salt DASH diet , weight reduction, regular exercise, stop smoking and limit alcohol, stress management, emphasise lifelong drug adherence even when feeling well , teach home monitoring and to report side effects (dry cough with ACE inhibitors, ankle swelling, postural giddiness — rise slowly), and screen for target-organ damage
Rheumatic fever / rheumatic heart disease	Follows streptococcal sore throat; migratory polyarthritis, carditis, chorea, subcutaneous nodules, erythema marginatum (Jones criteria)	Complete bed rest during active carditis, joint support and analgesia, long-term penicillin prophylaxis (secondary prophylaxis) , dental hygiene and antibiotic cover for procedures, treat sore throats promptly in the family
Varicose veins / DVT	Dilated tortuous veins, aching, pigmentation, venous ulcer / unilateral calf pain, swelling and warmth	Elevation, compression stockings, avoid prolonged standing , exercise, skin and ulcer care. DVT — bed rest as ordered, elevation, never massage the leg , anticoagulants with monitoring, measure the calf circumference, watch for pulmonary embolism (sudden dyspnoea and chest pain — an emergency)

21.3 Neurological Conditions

Table 21.3 Nursing in neurological conditions

Condition	Key features	Nursing care
Unconscious patient (any cause)	No response to stimuli; assess with the GCS; causes — head injury, stroke, poisoning, hypoglycaemia, uraemia, hepatic failure, epilepsy, meningitis, hypoxia	Priorities: airway, breathing, circulation. Lateral/recovery position with the head turned, suction ready, nothing by mouth; oxygen; 2-hourly position change and pressure-area care; eye care (pad and artificial tears to prevent corneal ulcer) ; mouth care 2-hourly; catheter or condom drainage with output charting; bowel care; nasogastric feeding; passive exercises and correct limb positioning with splints and a footboard; side rails up and never leave alone; observe and record GCS, pupils, vital signs and limb movement; talk to the patient by name and explain every procedure (hearing may be intact) ; never discuss the prognosis at the bedside; involve and support the family
Stroke (cerebrovascular accident)	Sudden weakness or numbness of one side, facial droop, slurred speech or aphasia, visual loss, severe headache, loss of balance; act FAST — Face, Arm, Speech, Time (thrombolysis is possible within 4.5 hours)	Urgent transfer for CT; maintain the airway and nil by mouth until a swallow assessment is done (aspiration risk) ; head of the bed elevated 30°; monitor neurological status, BP and blood sugar; position the affected limbs correctly with support to prevent contracture, shoulder subluxation and foot drop ; passive then active exercises from day 1; approach from the unaffected side and place articles there; feed on the unaffected side with thickened fluids; communicate patiently with an aphasic patient (simple questions, picture board, do not shout or complete his sentences); bladder and bowel training; early physiotherapy, speech and occupational therapy; prevent pressure sores and DVT; support the family and teach home care
Epilepsy / convulsion	Aura, then tonic-clonic movements with loss of consciousness, tongue bite, incontinence, cyanosis, followed by post-ictal confusion and sleep; status epilepticus = a seizure over 5 minutes or repeated seizures (an emergency)	During the fit: stay with the patient, note the time and describe the seizure, protect the head (a pillow), remove nearby objects, loosen tight clothing, turn the patient on the side, do NOT restrain, do NOT put anything (spoon, cloth, finger) in the mouth, do NOT give fluids; after the fit — recovery position, suction, oxygen, allow him to sleep, reorient, check for injury, record everything. Prevention: regular medication (never stop suddenly), adequate sleep, avoid alcohol and triggers (flashing lights, fever in children), padded side rails, avoid swimming alone, driving and climbing heights, MedicAlert identification, counselling on marriage, pregnancy and occupation; drug points — phenytoin gum hypertrophy, carbamazepine rash, valproate teratogenicity
Meningitis / raised intracranial pressure	Fever with severe headache, photophobia, neck stiffness, Kernig's and Brudzinski's signs , vomiting, rash (meningococcal); ICP — headache, projectile vomiting, papilloedema, falling consciousness, Cushing's reflex (rising BP with slow pulse), unequal pupils	Quiet, darkened room , head elevated 30° with the neck in neutral alignment, avoid straining, coughing and neck flexion, droplet precautions for meningococcal disease and chemoprophylaxis of contacts, neurological observation, seizure precautions, fluid balance, antibiotics on time, assist with lumbar puncture and position the patient flat afterwards, monitor for complications (deafness, fits, hydrocephalus)

Condition	Key features	Nursing care
Head injury	Loss of consciousness, vomiting, CSF leak from the nose or ear , battle sign, boggy swelling, amnesia	Airway, cervical spine protection, oxygen, GCS and pupil chart, head elevation, nothing packed in the ear or nose and no nasogastric tube if a base-of-skull fracture is suspected , avoid morphine (it masks pupils), report any deterioration at once
Paraplegia / spinal cord injury	Loss of power and sensation below the level, bladder and bowel involvement	Log-rolling with the spine in a straight line , pressure-area care, bladder (intermittent catheterisation) and bowel programme, prevention of contractures and DVT, watch for autonomic dysreflexia (sudden severe hypertension with headache and flushing, usually from a blocked catheter or constipation — sit the patient up and remove the cause), physiotherapy and rehabilitation, psychological support

21.4 Endocrine, Renal and Gastro-intestinal Conditions

Table 21.4 Nursing in endocrine, renal and gastro-intestinal disease

Condition	Key features	Nursing care
Diabetes mellitus	Polyuria, polydipsia, polyphagia, weight loss, fatigue, delayed healing, recurrent infection	Diet (fixed timing and carbohydrate), exercise, drugs, monitoring and education — the 5 pillars ; teach self-monitoring of blood glucose, insulin injection technique and site rotation, storage of insulin, sick-day rules, and above all foot care (Chapter 5); screen for retinopathy, nephropathy and neuropathy; never miss a meal after insulin
Hypoglycaemia (under 70 mg/dL) — an emergency	Sudden sweating, tremor, palpitation, hunger, pallor, confusion, irritability, slurred speech, fits, coma	Conscious: give 15-20 g of fast-acting glucose at once (3-4 teaspoons of sugar or glucose in water, or 100-150 mL of juice/soft drink), recheck after 15 minutes, repeat if needed, then give a complex carbohydrate snack. Unconscious: nothing by mouth — give IV 25% dextrose (or glucagon IM) and inform the doctor; find and correct the cause (missed meal, excess insulin, unusual exercise, alcohol)
Diabetic ketoacidosis / hyperglycaemia	Gradual onset: polyuria, thirst, vomiting, abdominal pain, Kussmaul breathing with an acetone smell , dehydration, drowsiness, high sugar with ketones	IV fluids (normal saline), insulin infusion , potassium replacement with monitoring, hourly sugar and output, watch the level of consciousness and the ECG, treat the precipitating infection, strict records
Thyroid disorders	Hyperthyroidism — weight loss with good appetite, tremor, palpitation, heat intolerance, sweating, goitre, exophthalmos, diarrhoea. Hypothyroidism — weight gain, cold intolerance, constipation, dry skin, hoarse voice, bradycardia, puffy face, slow thinking	Hyper: cool quiet room, high-calorie high-protein diet, no tea/coffee, eye care for proptosis, monitor pulse and weight, watch for thyroid storm ; after thyroidectomy — semi-Fowler's position, watch for bleeding, respiratory obstruction, hoarseness (recurrent laryngeal nerve) and tetany (parathyroid injury — keep calcium gluconate ready) . Hypo: warmth, high-fibre diet, gradual lifelong thyroxine on an empty stomach, monitor pulse and weight
Chronic kidney disease / acute kidney injury	Oliguria, oedema, hypertension, nausea, itching, sallow skin, uraemic breath, anaemia, fits in advanced uraemia	Strict intake-output and daily weight ; fluid, sodium, potassium, phosphate and protein restriction as ordered (Chapter 9); skin care for pruritus (no soap, emollients, short nails); oral care; watch for hyperkalaemia (arrhythmia) and fluid overload ; check the drug doses (many need adjustment) and avoid nephrotoxic drugs (NSAIDs, aminoglycosides, contrast); care of the dialysis access; anaemia and calcium management; psychological support and diet counselling
Urinary tract infection / renal stone	Burning micturition, frequency, urgency, suprapubic or loin pain, fever; stone — severe colicky loin-to-groin pain with haematuria and vomiting	Plenty of fluids (3 L/day) , complete the antibiotic course, perineal hygiene (front to back), void after intercourse, do not hold urine, warm applications and analgesia for colic, strain all urine for the stone , measure output, diet advice as per the stone type, and encourage follow-up
Peptic ulcer / gastritis	Epigastric burning pain related to meals, nausea, heartburn; complications — haematemesis, melaena, perforation (sudden severe pain with a rigid abdomen)	Small frequent bland meals, avoid chilli, spices, coffee, alcohol, smoking and NSAIDs, complete the anti- <i>H. pylori</i> regimen, teach the danger signs; for bleeding — nil by mouth, IV line, monitor vital signs and stool, save the vomitus

Condition	Key features	Nursing care
Viral hepatitis / cirrhosis with ascites	Jaundice, anorexia, nausea, dark urine, tender liver / distended abdomen, spider naevi, oedema, bleeding tendency, drowsiness (encephalopathy)	Rest, high-carbohydrate low-fat diet, absolutely no alcohol , standard precautions with blood and body fluids, skin care for pruritus, measure the abdominal girth and weight daily, low salt and fluid restriction for ascites, watch for bleeding and for encephalopathy (flapping tremor, confusion — restrict protein, give lactulose, prevent constipation) , avoid sedatives and hepatotoxic drugs, assist with paracentesis
Acute diarrhoea and cholera	Frequent loose stools, vomiting, cramps, dehydration	ORS and zinc, continue feeding/breast feeding (Chapter 13), IV Ringer lactate for severe dehydration, strict intake-output and weight, perineal care, enteric precautions and disinfection of excreta, hand hygiene teaching, notify as required

21.5 Surgical, Orthopaedic and Emergency Conditions

Table 21.5 Nursing in surgical, orthopaedic and emergency conditions

Condition	Key features	First aid and nursing care
Burns	Classified by depth (superficial/epidermal, superficial and deep partial-thickness, full-thickness) and by extent — ' rule of nines ' in adults (head 9%, each arm 9%, each leg 18%, front of trunk 18%, back 18%, perineum 1%; the palm of the patient's hand = about 1%); in children use the Lund-Browder chart	First aid: stop the burning, remove the person from danger, cool the burn with clean running water for 20 minutes (not ice), remove rings, watches and burnt non-adherent clothing, cover with a clean dry non-fluffy cloth or cling film; never apply toothpaste, ink, oil, butter, mud, kerosene, or burst blisters. Hospital care: airway assessment (suspect inhalation injury with facial burns, singed nostril hair, soot or hoarseness), IV fluid resuscitation by the Parkland formula — 4 mL x body weight (kg) x %TBSA of Ringer lactate in 24 hours, half in the first 8 hours from the time of the burn , urinary catheter with hourly output (aim 0.5-1 mL/kg/h), strict asepsis and barrier nursing, analgesia before dressings, tetanus prophylaxis, high-protein high-calorie diet (up to 3000-5000 kcal) , nasogastric feeding if needed, positioning and splinting to prevent contracture, early physiotherapy, watch for hypovolaemic shock (first 48 h), infection/sepsis, Curling's ulcer, acute kidney injury and compartment syndrome , and give strong psychological support for disfigurement
Fracture	Pain, tenderness, swelling, deformity, loss of function, abnormal mobility and crepitus; open (compound) fracture has a wound	First aid: do not move the part unnecessarily, control bleeding, cover an open wound with a sterile dressing, immobilise the joint above and below with a padded splint in the position found, elevate, apply cold, check the distal pulse and sensation, and transport carefully. Ward care: traction or cast care (Chapters 12, 20), neurovascular checks (the 5 P's) 2-4 hourly , elevation, analgesia, exercises of free joints and isometric exercises, prevention of the complications of immobility, high-protein calcium-rich diet, watch for compartment syndrome, fat embolism (dyspnoea, confusion, petechiae 24-72 h after a long-bone fracture), DVT, infection/osteomyelitis, delayed or non-union and avascular necrosis
Haemorrhage and wounds	Arterial (bright red spurting), venous (dark, steady flow), capillary (oozing); internal bleeding shows as shock without visible blood	Direct pressure over the wound with a clean pad, elevation of the part, rest, pressure bandage, tourniquet only for a life-threatening limb bleed (note the time); treat for shock, nil by mouth, IV line, group and cross-match, monitor vital signs; nose bleed — sit up leaning forward, pinch the soft part of the nose for 10-15 minutes, cold compress, do not tilt the head back
Poisoning	Depends on the agent; note the container, tablets, smell and vomitus	Remove the person from the source, maintain the airway (lateral position), keep the vomitus and container for identification, do NOT induce vomiting (never in corrosive, kerosene/petroleum or unconscious patients), gastric lavage only on the doctor's order within an hour with airway protection; specific antidotes (Chapter 10); for organophosphate (pesticide) poisoning — remove contaminated clothes, wash the skin, atropine and pralidoxime, watch secretions; for snake bite — reassure, immobilise the limb below heart level, do not cut, suck, apply a tourniquet or ice , transport urgently for antivenom
Heat stroke and hypothermia	Heat stroke: temperature above 40 °C, hot dry skin, confusion or coma. Hypothermia: below 35 °C, shivering (may be absent), confusion, bradycardia	Heat stroke — move to a cool place, remove clothes, rapid cooling with tepid sponging and fans, ice packs to the axillae and groins , IV fluids, monitor continuously. Hypothermia — gradual passive rewarming with blankets, warm drinks if conscious, warmed IV fluids, handle gently (risk of arrhythmia)

Condition	Key features	First aid and nursing care
Cancer (any site)	Warning signs — the 'CAUTION' list: Change in bowel or bladder habit, A sore that does not heal, Unusual bleeding or discharge, Thickening or lump, Indigestion or difficulty in swallowing, Obvious change in a wart or mole, Nagging cough or hoarseness	Pain relief (WHO ladder), nutrition, mouth care, care during chemotherapy and radiotherapy (Chapters 17, 20), prevention of infection, stoma and wound care, honest communication, body-image and psychological support, palliative and terminal care, family support, and screening/early detection education (self breast examination, Pap smear, oral examination, tobacco cessation)
HIV/AIDS	See Chapter 14	Standard precautions for all patients (not special isolation) , strict confidentiality and no discrimination , ART adherence counselling, nutrition, prevention and early treatment of opportunistic infections, skin and mouth care, safe sex and needle education, PPTCT, family and community counselling, and support for the care-giver
Anaemia / thalassaemia	Pallor, fatigue, breathlessness on exertion, palpitation, koilonychia, glossitis	Iron/folate therapy with teaching (take with vitamin C, not with tea, milk or antacids; black stools are normal), iron-rich diet, rest with graded activity, transfusion care and iron chelation in thalassaemia , deworming, education on menstrual and dietary causes, and antenatal supplementation
Arthritis (rheumatoid and osteoarthritis)	Joint pain and stiffness; RA — symmetrical small joints with morning stiffness over 30 min; OA — weight-bearing joints, pain on use	Rest during the acute phase with splints in a functional position, heat or cold application , gentle range-of-motion and isometric exercises, joint protection and energy conservation, weight reduction, assistive devices, correct posture, analgesics/NSAIDs after food (watch for gastric bleeding), teach about methotrexate being weekly , and maintain independence in daily activities

21.6 Paediatric Nursing Highlights

Table 21.6 Paediatric nursing highlights

Topic	Points
Normal newborn	Weight 2.5-3.5 kg (low birth weight under 2.5 kg), length 50 cm, head circumference 33-35 cm; pulse 120-160/min , respiration 30-60/min , temperature 36.5-37.5 °C; loses up to 10% of birth weight in the first week and regains it by 10-14 days; doubles the birth weight by 5 months, triples it by 1 year ; passes meconium within 24 h and urine within 48 h
Developmental milestones	Social smile 2 months • holds head steady/neck control 3 months • rolls over 5 months • sits without support 6-8 months • crawls 9 months • stands with support 9-10 months • walks alone 12-15 months • says 'mama/dada' 9-12 months • 2-3 word sentences 2 years • dry by day 2-3 years
Breast feeding	Exclusive breast feeding for the first 6 months (no water, honey, or 'ghutti'); initiate within 1 hour of a normal birth (colostrum is the first immunisation) ; feed on demand, 8-12 times a day; correct positioning and attachment (chin touching the breast, mouth wide open, lower lip turned out, more areola visible above than below); complementary feeding from 6 months with continued breast feeding up to 2 years; advantages — perfect nutrition, antibodies (IgA), bonding, cheap, prevents infection and allergy, contraceptive effect for the mother, reduces breast and ovarian cancer
Diarrhoea in children	ORS + zinc for 14 days + continued feeding (Chapter 13); assess dehydration by the WHO chart; do not give antidiarrhoeals or antibiotics routinely
Pneumonia in children	Fast breathing by age cut-off and chest in-drawing (Chapter 6); refer urgently if there is in-drawing, inability to drink, convulsions, lethargy or stridor at rest; oral amoxicillin for non-severe pneumonia
Malnutrition	Severe acute malnutrition: weight-for-height below -3 SD, MUAC under 11.5 cm (6-59 months), or bilateral pitting oedema ; treat at a Nutrition Rehabilitation Centre with F-75/F-100 therapeutic feeds, correction of hypoglycaemia, hypothermia, dehydration and infection, micronutrients, and catch-up growth with follow-up; growth monitoring with the Mother and Child Protection card
Care of the low birth weight/preterm baby	Warmth (kangaroo mother care, skin-to-skin), exclusive breast milk (expressed if needed by spoon/paladai/tube), infection prevention (hand washing), monitoring of weight, temperature and feeding , and early recognition of danger signs ; watch for hypothermia, hypoglycaemia, jaundice, apnoea and infection
Danger signs in a newborn/child (refer urgently)	Not feeding well or unable to drink or breast feed, lethargy or unconsciousness, convulsions, fast or difficult breathing with in-drawing or grunting, hypothermia or fever , persistent vomiting, umbilical redness or discharge with skin pustules, yellow palms and soles (jaundice), bulging fontanelle, blood in the stool

Topic	Points
Hospitalised child	Allow a parent to stay (prevent separation anxiety), explain in age-appropriate language, use play therapy, never threaten with an injection, allow choices, maintain the routine and schooling, minimise painful procedures and use topical anaesthesia, correct drug dose by weight, safety (cot sides, no small objects), and involve the mother in care
CHAPTER AT A GLANCE ● Pneumonia: rusty sputum, semi-Fowler's, fluids, oxygen; asthma: sit up, nebulise salbutamol, no sedatives, teach inhaler with spacer; COPD: target SpO₂ 88-92%. ● MI: chew aspirin 300 mg, ECG within 10 min, absolute rest, monitor for arrhythmia (commonest early cause of death). ● Heart failure: daily weight (1 kg = 1 L fluid), low salt, check pulse before digoxin (hold if under 60). ● Stroke: act FAST, thrombolysis within 4.5 h, nil by mouth until swallow assessed, approach from the unaffected side. ● Seizure: never restrain, never put anything in the mouth, time the fit, side position afterwards. ● Unconscious: lateral position, eye care, mouth care 2-hourly, 2-hourly turning, talk to the patient. ● Hypoglycaemia (under 70): conscious — 15-20 g glucose orally; unconscious — IV 25% dextrose. ● Burns: cool with running water for 20 min, cover; Parkland formula 4 mL x kg x %TBSA of Ringer lactate in 24 h, half in the first 8 h; rule of nines; palm = 1%. ● Fracture: splint the joints above and below in the position found; neurovascular checks; watch for compartment syndrome and fat embolism. ● Poisoning: airway, keep the container/vomitus, never induce vomiting in corrosive/kerosene/unconscious cases; snake bite — immobilise, no tourniquet or cutting. ● Newborn: 2.5-3.5 kg, pulse 120-160, respiration 30-60; exclusive breast feeding for 6 months, initiate within 1 hour; social smile 2 months, sits 6-8 months, walks 12-15 months.	

The Hospital Services

Types and organisation of hospitals, departments, pharmacy services, committees and the health-care delivery system of India

22.1 The Hospital — Definition and Functions

DEFINITION

Hospital (WHO) — 'an integral part of the social and medical organisation, the function of which is to provide for the population complete health care, both curative and preventive, and whose out-patient services reach out to the family in its home environment; the hospital is also a centre for the training of health workers and for bio-social research.'

Functions of a hospital

- **Curative** — diagnosis and treatment of the sick (out-patient, in-patient, emergency, intensive and surgical care).
- **Preventive and promotive** — immunisation, antenatal and child care, family planning, health education, screening, infection control, nutrition and rehabilitation services.
- **Training and education** — of doctors, nurses, pharmacists, technicians and other health workers, and of the community.
- **Research** — clinical, epidemiological and operational research; clinical trials.
- **Rehabilitation and referral** — physiotherapy, occupational therapy, prosthetics, and referral linkage with peripheral institutions.

22.2 Classification of Hospitals

Table 22.1 Classification of hospitals

Basis	Types
Ownership / control	Government (central, state, local body, railway, ESI, defence, public sector); private (for profit, corporate, nursing home); voluntary/charitable/mission trust ; autonomous (AIIMS, PGI)
System of medicine	Allopathic; AYUSH — Ayurveda, Yoga and Naturopathy, Unani, Siddha, Homoeopathy
Clinical scope	General (all common specialities); Specialised/special (paediatric, maternity, cancer, TB, mental, eye, ENT, infectious disease, orthopaedic); Teaching ; Isolation
Size (number of beds)	Small under 100 beds • Medium 100-300 • Large 300-500 • Very large/tertiary over 500 beds
Level of care	Primary (first contact — sub-centre, PHC, dispensary); Secondary (referral, basic specialities — CHC, sub-district and district hospital); Tertiary (super-speciality, teaching, research — medical college and regional institutes)
Length of stay / purpose	Short-stay and acute care; long-stay and chronic care; day-care/ambulatory surgery; hospice

NUMBERS TO REMEMBER

Recommended bed strength: about **1 bed per 1000 population** (WHO indicative norm); **85% bed occupancy** is considered efficient.

Bed distribution norm (approximate) — medicine 25-30%, surgery 20-25%, obstetrics and gynaecology 15-20%, paediatrics 10-15%, with the rest for other specialities; **ICU beds about 5-10%** of total beds.

Space norms: **7-10 sq m (about 80-100 sq ft) of floor space per bed in a general ward** with a **2.25 m (7 ft) centre-to-centre distance between beds** and 1.2-2.4 m of circulation space; about **100-150 sq ft per bed** including ancillary areas.

22.3 Organisation and Departments of a Hospital

Table 22.2 Hospital departments and services

Group	Departments / services
Clinical (patient care) services	Out-patient department (OPD) — the 'shop window' of the hospital, with registration, consultation rooms, injection room, dressing room, minor OT, sample collection and a pharmacy counter; In-patient (wards) — general, private, ICU/CCU/NICU/PICU, isolation, burns, post-operative; Emergency/casualty — 24 hours, with triage (red = immediate, yellow = urgent, green = non-urgent/walking, black = dead or expectant), resuscitation bay, observation beds, ambulance bay and a disaster plan; Operation theatre complex; Labour room and maternity; Speciality clinics (diabetes, ART, DOTS, immunisation, well-baby, family planning)
Diagnostic (supportive) services	Clinical laboratory (pathology, biochemistry, microbiology, haematology, serology, histopathology), radiology and imaging (X-ray, ultrasound, CT, MRI, mammography), blood bank , ECG/EEG/echocardiography, pulmonary function laboratory, nuclear medicine, endoscopy
Therapeutic support	Pharmacy , physiotherapy and occupational therapy, dietetics/kitchen, dialysis unit, radiotherapy, speech therapy, prosthetics and orthotics
Nursing service	Headed by the Nursing Superintendent → Deputy/Assistant Nursing Superintendent → Ward Sister/Sister-in-charge → Staff Nurse → Nursing Aide; responsible for 24-hour patient care, ward management, supplies, records, staff duty rosters and training
Administrative services	Medical Superintendent/Director, hospital administrator, personnel/human resources, accounts and billing, stores and purchase, medical records department (MRD), public relations, security, transport, legal cell
Utility / engineering services	CSSD (Central Sterile Supply Department) , laundry, kitchen, housekeeping, maintenance (electrical, plumbing, biomedical engineering), generator and UPS, water supply and storage, oxygen manifold/plant, medical gas pipeline, air conditioning, fire fighting, waste management and mortuary

- **CSSD** — receives, cleans, packs, sterilises, stores and issues all sterile supplies; the flow is strictly **one-way from the dirty (receiving) zone → cleaning → packing → sterilising → sterile store → issue**, with no crossing of clean and dirty traffic; it maintains sterilisation records, indicators and recall procedures.
- **Medical Records Department** — maintains, codes (**ICD**), stores and retrieves case records; supplies statistics and medico-legal documents; records are usually retained for **5-10 years (longer for medico-legal, paediatric and maternity records)**; confidentiality is paramount.

22.4 Hospital Pharmacy Services

- **Functions** — procurement, storage and inventory control of drugs; **dispensing to out-patients and in-patients**; sterile manufacturing where permitted (IV fluids, TPN, eye drops, cytotoxic reconstitution); drug distribution to wards; **drug information and patient counselling**; participation in ward rounds and in therapeutic drug monitoring; **ADR monitoring (pharmacovigilance)**; education and training; and maintaining statutory records and licences.
- **Drug distribution systems** — (1) **floor/ward stock system** (bulk supply to the ward; economical but with high wastage and error risk), (2) **individual prescription order system** (each order dispensed separately; safer but slow), (3) **unit dose dispensing system** (each dose separately packed and labelled for a patient for 24 hours — the **safest, least wasteful and most accurate system**, and the one that reduces medication errors most), and (4) combination/satellite pharmacy and automated dispensing cabinets.
- **Inventory control** — **ABC analysis** (A = 10% of items using 70% of the budget, needing tight control), **VED analysis** (Vital, Essential, Desirable), **EOQ** (economic order quantity), lead time, buffer/safety stock, reorder level, **FEFO/FIFO** issue, bin and stock cards, periodic physical stock verification, expiry monitoring and near-expiry redistribution, and cold-chain and narcotic records.
- **Hospital formulary** — a continuously revised, authoritative list of the drugs approved for use in the hospital with essential prescribing information, prepared and maintained by the **Pharmacy and Therapeutics Committee (PTC)** and based on the NLEM/WHO essential medicines concept; it promotes rational, economical and standardised prescribing.
- **Pharmacist's ward-level roles** — reconciling medication at admission and discharge, checking prescriptions for dose, interaction, duplication and allergy, advising on dilution, compatibility and rate of IV drugs, monitoring high-alert drugs and narrow-index drugs, educating nurses and patients, and reporting errors and reactions.

22.5 Hospital Committees

Table 22.3 Important hospital committees

Committee	Main functions
Pharmacy and Therapeutics Committee (PTC)	Develops and revises the hospital formulary and drug policy; evaluates requests for new drugs; promotes rational prescribing; reviews ADRs and medication errors; plans drug-use evaluation and staff education. Chaired usually by a senior physician with the pharmacist as secretary
Hospital Infection Control Committee (HICC)	Surveillance of hospital-acquired infection; infection-control policies, hand-hygiene and isolation protocols; antibiotic policy and stewardship; outbreak investigation; staff health and immunisation; waste management audit; training
Bio-medical Waste Management Committee	Segregation, colour coding, storage, transport, treatment and records as per the BMW Rules, 2016; staff training and annual returns
Other committees	Hospital management/governing council, quality assurance and patient safety, transfusion (blood bank) committee, ethics committee (for research and clinical trials), medical audit and mortality review, disaster management, purchase and condemnation, grievance/patient-welfare committee, fire and safety, and death review committee

22.6 Health Care Delivery System in India

Table 22.4 Health care infrastructure in rural India

Level / institution	Population covered (plain / hilly-tribal)	Staff and services
Village level	1000 / -	ASHA (Accredited Social Health Activist) — 1 per 1000 population, a village woman volunteer with performance-based incentives; Anganwadi worker (1 per 400-800 under ICDS); trained dai; village health, sanitation and nutrition committee
Sub-centre / Health and Wellness Centre (the most peripheral contact point)	5000 / 3000	1 ANM/MPW (Female) + 1 MPW (Male) (now also a Community Health Officer under Ayushman Bharat HWC); maternal and child health, immunisation, family planning, ORS and basic drugs, referral, and health education
Primary Health Centre (PHC)	30 000 / 20 000 (covering about 6 sub-centres)	1 medical officer + 14-15 paramedical staff, 4-6 beds ; OPD, basic laboratory, MCH and immunisation, family welfare, national programmes, referral and supervision of sub-centres
Community Health Centre (CHC) / First Referral Unit	1 20 000 / 80 000 (covering 4 PHCs)	4 specialists — physician, surgeon, obstetrician-gynaecologist and paediatrician — with 21 paramedical staff and 30 beds , operation theatre, labour room, X-ray, laboratory and blood storage; a First Referral Unit must provide 24-hour emergency obstetric care, newborn care and blood storage
Sub-divisional / Taluka hospital	About 5-6 lakh	50-100 beds with basic specialities
District hospital	Whole district (10-30 lakh)	200-500 beds (about 1 bed per 1000 population of the district); all basic and some super-specialities; the secondary-level apex of the district
Medical college / regional and national institutes	Region/state/country	Tertiary and super-speciality care, teaching and research (AIIMS, PGIMER, NIMHANS, JIPMER, CMC)

NUMBERS TO REMEMBER

Key ratios to remember: ASHA 1:1000 • Sub-centre 1:5000 (3000 hilly) • PHC 1:30 000 (20 000 hilly) • CHC 1:1 20 000 (80 000 hilly) • Anganwadi 1:400-800 • one sub-centre for every 3-5 villages, and 6 sub-centres per PHC, 4 PHCs per CHC.

Ayushman Bharat (2018) has two pillars: **(1) Health and Wellness Centres (HWCs)** — 1.5 lakh sub-centres and PHCs upgraded to provide comprehensive primary health care with a **Community Health Officer**, and **(2) Pradhan Mantri Jan Arogya Yojana (PM-JAY)** — health insurance cover of **Rs 5 lakh per family per year** for secondary and tertiary hospitalisation for the bottom 40% of the population (and for all aged 70+).

22.7 National Health Programmes (quick list)

Table 22.5 Major national health programmes

Programme	Focus
National Health Mission (NHM, 2013) = NRHM (2005) + NUHM (2013)	Umbrella mission for rural and urban health; ASHA, Janani Suraksha Yojana, mobile medical units, RBSK, RKSK
NTEP (formerly RNTCP)	Tuberculosis — DOTS, Ni-kshay, TB-free India by 2025
NACP	HIV/AIDS — ICTC, ART centres, blood safety, targeted intervention
NVBDCP	Vector-borne diseases — malaria, dengue, chikungunya, filariasis, kala-azar, Japanese encephalitis
NLEP	Leprosy — MDT, disability prevention
NPCB&VI	Blindness and visual impairment — cataract surgery, school eye screening, eye banks
RMNCH+A / RCH	Reproductive, maternal, newborn, child and adolescent health; JSY, JSSK, LaQshya, SUMAN, Mission Parivar Vikas
UIP / Mission Indradhanush	Universal immunisation coverage
NPCDCS / NP-NCD	Cancer, diabetes, cardiovascular disease and stroke — screening and NCD clinics
NMHP / DMHP	Mental health; Tele-MANAS
NPHCE	Health care of the elderly
POSHAN Abhiyaan, ICDS, Anaemia Mukht Bharat, NIDDCP, Mid-day meal	Nutrition, anaemia, iodine deficiency
Swachh Bharat Mission, Jal Jeevan Mission	Sanitation and safe drinking water
NTCP, National Deworming Day, Pulse Polio, IDSP, PMSMA, Ayushman Bharat Digital Mission	Tobacco control, deworming, polio, surveillance, antenatal care, digital health

22.8 Quality, Safety and Patient Rights

- **Accreditation** — NABH (National Accreditation Board for Hospitals and Healthcare Providers, under the Quality Council of India) for hospitals and blood banks; NABL for laboratories; JCI internationally; Kayakalp and LaQshya awards for cleanliness and labour-room quality; IPHS (Indian Public Health Standards) for public facilities.
- **Patient safety goals** — correct patient identification, effective communication (SBAR, read-back of verbal orders), safety of high-alert medications, correct-site surgery (WHO checklist), reduction of health-care-associated infection, reduction of the risk of falls, and reporting of incidents and near misses in a **blame-free culture**.
- **Patient's Charter of Rights (NHRC/Ministry of Health)** — right to information, records and reports, second opinion, emergency care, informed consent, confidentiality and privacy (a **female attendant during examination of a female patient**), transparency in rates and choice of pharmacy/laboratory, non-discrimination, safety and quality care, protection from unnecessary procedures, and a grievance-redressal mechanism.
- **Hospital statistics to know** — **bed occupancy rate** = (occupied bed days / available bed days) × 100; **average length of stay** = total patient days / number of discharges; **bed turnover rate**; **gross and net death rate**; **hospital infection rate**; and the maternal and perinatal mortality audit.

CHAPTER AT A GLANCE

- Levels of care: primary (sub-centre, PHC), secondary (CHC, district hospital), tertiary (medical college, institutes).
- Norms: ASHA 1:1000, sub-centre 1:5000 (3000 hilly), PHC 1:30 000 (20 000 hilly), CHC 1:1 20 000 (80 000 hilly); CHC has 4 specialists and 30 beds; PHC 1 doctor and 4-6 beds.
- Triage colours: red immediate, yellow urgent, green non-urgent, black dead/expectant.
- CSSD works one-way from dirty to sterile; MRD keeps records 5-10 years and uses ICD coding.
- Unit dose dispensing is the safest drug distribution system; ABC and VED analysis for inventory; FEFO for issue.
- PTC prepares the hospital formulary (pharmacist = secretary); HICC controls hospital infection and antibiotic policy.
- Ayushman Bharat = Health and Wellness Centres + PM-JAY (Rs 5 lakh per family per year).
- NABH accredits hospitals, NABL laboratories; bed occupancy rate = occupied bed days / available bed days × 100.

Preparation of the Patient for Special Treatment and Investigations

Patient preparation, articles, positioning and after care for common diagnostic and therapeutic procedures

23.1 General Principles of Preparing a Patient for Any Procedure

1. **Verify the order** and the correct patient, procedure, site and side; check that the **consent** form is signed where required.
2. **Explain** the purpose, the procedure, the sensations to expect, the duration, the position to be maintained and what is expected of the patient — a well-informed patient co-operates and needs less sedation.
3. Check for **allergies (especially to iodine, contrast media, latex and local anaesthetics), pregnancy, current medicines (anticoagulants, antiplatelets, metformin before contrast, insulin), renal function before contrast, and any implant/pacemaker/metal before MRI.**
4. Carry out the specific **preparation** — fasting, bowel preparation, bladder emptying or filling, skin preparation, hair removal, pre-medication, and stopping or adjusting drugs as ordered.
5. Record **baseline vital signs and weight**; remove dentures, jewellery and metal objects as appropriate; dress the patient in a gown.
6. Arrange the **articles/sterile tray** and the specimen containers with labels and forms; ensure good light, privacy and a warm room.
7. **Position** the patient correctly and support him; maintain privacy and drape properly; stay with the patient and observe him **during** the procedure (vital signs, colour, breathing, pain, fainting), and reassure him.
8. **After care** — the prescribed position and rest, observation of vital signs and of the puncture/biopsy site for bleeding, pain relief, food and fluids when allowed, prompt despatch of specimens, and **complete documentation** (procedure, time, operator, specimen sent, amount and character of fluid removed, complications and the patient's condition).
9. Know and watch for the **specific complications** of each procedure and report them at once.

23.2 Blood, Urine and Other Laboratory Investigations

Table 23.1 Common laboratory investigations

Investigation	Preparation	Points
Fasting blood sugar / lipid profile	8-12 hours of fasting (water allowed); no food, tea, coffee or smoking; usual drugs as advised	Collect early morning; for the oral glucose tolerance test take a fasting sample, give 75 g of glucose in 250-300 mL water, then sample at 1 and 2 hours; the patient must sit quietly and not eat, smoke or walk about
Post-prandial blood sugar	Sample exactly 2 hours after a meal or after the glucose load	Note the exact time of the meal
HbA1c, renal and liver function, electrolytes, complete blood count	Usually no fasting needed	Correct tube and order of draw (Chapter 13)
Blood culture	At the onset of fever/chill, before antibiotics , 2 sets from 2 sites	Meticulous skin antisepsis
24-hour urine, mid-stream urine, stool, sputum	See Chapter 13	Correct container, preservative and timing
Urea breath test / occult blood	Stop antibiotics and proton-pump inhibitors as advised / meat-free and iron-free diet for 3 days	Follow the laboratory instruction sheet

23.3 Radiological and Imaging Investigations

Table 23.2 Radiological investigations

Investigation	Preparation	After care / notes
Plain X-ray (chest, bone)	Remove metal objects, jewellery and clothing with buttons/zips from the field; protect the gonads and enquire about pregnancy ; the patient holds his breath as instructed	No special after care; lead apron for staff and for the escort
Plain X-ray abdomen (KUB)	Bowel preparation (laxative the night before) and an empty bladder	-
Barium meal / swallow (upper GI)	Nil by mouth for 6-8 hours (from midnight) ; no smoking; remove dentures; explain that the barium tastes chalky	Plenty of fluids and a laxative afterwards to clear the barium; warn the patient that the stools will be white/chalky for 1-3 days ; report constipation or abdominal pain (risk of impaction)
Barium enema (large bowel)	Low-residue diet for 1-2 days, clear liquids the day before, laxative and cleansing enema until the returns are clear ; nil by mouth after midnight	Barium is expelled with a cleansing enema/laxative afterwards; give fluids and rest; watch for perforation (severe pain), and never do it soon after a bowel biopsy
Intravenous pyelography/urography (IVP/IVU)	Consent ; check the creatinine and the history of iodine/contrast allergy; laxative the night before; nil by mouth for 6-8 hours (fluid restriction as ordered for a better picture) ; empty the bladder before	Push fluids afterwards to flush out the contrast ; observe for a contrast reaction (itching, rash, sneezing, wheeze, hypotension — keep the emergency tray and adrenaline ready) and for urine output; withhold metformin for 48 hours as per the local protocol
Ultrasound — abdomen	Nil by mouth for 6-8 hours (no gas-forming food) for the gall bladder and abdomen	No after care; harmless, no radiation
Ultrasound — pelvis (transabdominal) / obstetric	Full bladder — the patient drinks 3-4 glasses of water 1 hour before and does not void	Allow voiding immediately afterwards
CT scan	Consent for contrast; nil by mouth 4-6 hours if contrast is used; check the creatinine and allergy; remove metal; oral contrast may be given in divided doses before the scan	Push fluids if contrast was used; observe for a reaction; the patient must lie still; explain the noise and the enclosed space
MRI	The key question: any metal? Contraindicated with a cardiac pacemaker, implanted defibrillator, cochlear implant, metallic intra-ocular foreign body, or certain clips and pumps ; remove all metal — jewellery, hair pins, watch, coins, credit cards, hearing aid, dentures, patches, prosthesis; warn about the loud knocking noise (ear plugs are given) and the narrow tunnel (sedation for claustrophobia); the patient must lie absolutely still for 30-60 minutes	No radiation; contrast (gadolinium) needs a renal check; observe after sedation
Mammography	Best done in the week after menstruation; no talcum powder, deodorant or lotion on the breast or axilla on the day; warn about the compression discomfort	Screening from 40-50 years; teach breast self-examination
Radio-isotope (nuclear) scan, PET, DEXA	As per the department's instruction; fasting for some; the isotope is given orally or IV at a set time before the scan	The patient may be mildly radioactive for some hours — push fluids, flush the toilet twice, wash hands, and avoid close contact with pregnant women and infants for the stated period

23.4 Endoscopic Procedures

Table 23.3 Endoscopic procedures

Procedure	Preparation	After care
Upper GI endoscopy (gastroscopy)	Consent; nil by mouth 6-8 hours ; remove dentures; local anaesthetic throat spray and/or sedation; explain that he will not be able to talk during it and may gag	Nil by mouth until the gag reflex returns (1-2 hours) — then start with sips of water; watch for bleeding, perforation (severe pain, fever, surgical emphysema) and aspiration ; a sore throat is common; do not allow driving after sedation

Procedure	Preparation	After care
Colonoscopy / sigmoidoscopy	Consent; clear liquid diet for 1-2 days, full bowel preparation (polyethylene glycol) the evening before, nil by mouth 6-8 hours ; sedation; left lateral position	Rest; expect abdominal cramps and flatus (walking helps); resume diet when alert; report severe pain, distension, bleeding or fever (perforation)
Bronchoscopy	Consent; nil by mouth 6-8 hours ; remove dentures; pre-medication (atropine, sedative) and local anaesthesia; explain the procedure and mouth breathing	Nil by mouth 2 hours or until the cough and gag reflexes return ; semi-Fowler's position; observe for haemoptysis, dyspnoea, stridor, subcutaneous emphysema and fever; save the sputum for cytology/culture; mild blood-streaked sputum after a biopsy is expected
Cystoscopy	Consent; the bladder may need to be full or empty as instructed; fasting if under general anaesthesia; lithotomy position; strict asepsis	Push fluids; warn about burning on micturition and slightly blood-tinged urine for a day or two ; a warm sitz bath and analgesia help; report retention, heavy bleeding, clots or fever
Laparoscopy / hysteroscopy / ERCP	As for surgery — consent, fasting, skin and bowel preparation, anaesthetic assessment	Post-anaesthetic care; shoulder-tip pain from residual gas after laparoscopy is common and relieved by ambulation; watch for bleeding, and for pancreatitis after ERCP

23.5 Aspirations, Punctures and Biopsies

Table 23.4 Aspirations, punctures and biopsies

Procedure	Position and preparation	Articles	After care and complications
Lumbar puncture (CSF)	Consent; empty the bladder; lateral (side-lying) position at the edge of the bed with the back arched, knees drawn up to the chest and the chin tucked down ('fetal/knee-chest' position) , or sitting leaning forward; the site is the L3-L4 or L4-L5 interspace (below the end of the cord); local anaesthesia; the nurse supports the patient and holds the position, and helps him keep still	Sterile LP set with a spinal needle and manometer, local anaesthetic, antiseptic, gloves, drape, 3 numbered sterile tubes , dressing	Lie flat (prone or supine without a pillow) for 4-6 hours (some units 6-24 h) and push fluids to prevent a post-puncture headache (relieved by lying flat, fluids and analgesia); observe the site for leakage, and the vital signs, neurological status and movement of the limbs; complications — headache, backache, infection/meningitis, bleeding, nerve injury and herniation ('coning') if the intracranial pressure is raised — LP is contraindicated in raised ICP with papilloedema ; send the tubes at once
Pleural aspiration (thoracentesis) / intercostal drain	Consent; sitting up leaning forward over a table or a chair back with the arms supported (or lying on the unaffected side); chest X-ray or ultrasound first; the patient must not cough or move during needle insertion and must hold his breath when told	Sterile thoracentesis set, 50 mL syringe, three-way tap, specimen bottles, local anaesthetic, dressing, and a drainage bottle with underwater seal if a tube is to be inserted	Not more than 1-1.5 L should be removed at one sitting (risk of re-expansion pulmonary oedema and hypotension); observe the respiration, pulse, BP, SpO ₂ and for cough, chest pain, faintness, bleeding or subcutaneous emphysema ; chest X-ray afterwards to exclude pneumothorax ; position on the unaffected side; record the amount, colour and character of the fluid
Abdominal paracentesis (ascitic tap)	Consent; empty the bladder (essential — risk of bladder puncture) ; measure the abdominal girth and weight; upright/semi-Fowler's or sitting position ; the site is usually in the left iliac fossa/mid-line below the umbilicus	Sterile paracentesis set/trocar and cannula, drainage bag, measuring jar, specimen bottles, abdominal binder	Drain slowly (rapid removal of a large volume causes hypotension and shock; albumin cover is given for large-volume taps); measure the girth, weight, BP and pulse before, during and after; apply a sterile dressing and a binder; watch for leakage, bleeding, hypotension and peritonitis; record the volume and appearance
Pericardiocentesis	Emergency procedure for tamponade; semi-Fowler's; ECG monitoring	Sterile set, ECG monitor, defibrillator ready	Continuous cardiac monitoring; watch for arrhythmia and recurrence of tamponade
Bone marrow aspiration / trephine biopsy	Consent; site — posterior superior iliac spine (or sternum in adults, tibia in infants) ; prone or lateral position; local anaesthesia with sedation in children; explain the brief but sharp suction pain	Sterile marrow needle set, slides, specimen bottles	Pressure over the site for 5-10 minutes and a sterile pressure dressing; lie on the site for a while; observe for bleeding (especially if the platelets are low) and infection; analgesia

Procedure	Position and preparation	Articles	After care and complications
Liver / kidney / lung / lymph node biopsy	Consent; check platelets, PT/INR and bleeding time and the blood group ; fasting 6-8 h; position — supine with the right side slightly raised for the liver (prone for the kidney); the patient holds his breath in expiration for the liver biopsy as instructed	Sterile biopsy set, specimen containers with fixative, dressings	Bed rest 6-24 hours; liver biopsy — lie on the RIGHT side for 2 hours (to compress the site) then supine ; monitor pulse and BP every 15-30 minutes for 2 hours then hourly; watch for pain, bleeding/haemorrhage, shock, bile leak or peritonitis (liver), haematuria (kidney) and pneumothorax (lung) ; avoid lifting and straining for a week
Joint aspiration, pleural biopsy, fine-needle aspiration cytology, Pap smear, amniocentesis	Correct position, strict asepsis, consent and specimen labelling	As appropriate	Pap smear: not during menstruation, no douching, intercourse or vaginal medication for 48 hours before; lithotomy position. Amniocentesis: after 15 weeks, empty bladder, ultrasound guidance, anti-D for Rh-negative mothers, rest and observe for leakage, bleeding and contractions

23.6 Cardiac, Neurological and Functional Tests

Table 23.5 Cardiac, neurological and functional tests

Test	Preparation and points
ECG	No special preparation; the patient relaxes, does not talk or move; skin cleaned and gel applied; correct lead placement (Chapter 20)
Exercise (treadmill) stress test	Light meal 2-3 hours before, comfortable clothes and shoes ; withhold beta blockers/digoxin as advised; no smoking or caffeine; written consent; the emergency trolley and defibrillator must be ready; stop the test for chest pain, marked ST change, a fall in BP, arrhythmia or exhaustion
Echocardiography / Holter monitoring	No preparation; for Holter — keep a symptom diary, do not bathe or remove the electrodes
Cardiac catheterisation / coronary angiography	Consent, nil by mouth 6-8 hours, check the creatinine and contrast allergy, shave and prepare both groins/wrist, record the peripheral pulses, and withhold metformin ; explain the flushing sensation with contrast. After: bed rest with the limb straight and immobile (4-6 hours after a femoral puncture) , pressure dressing/compression device, check the puncture site for bleeding or haematoma and the distal pulse, colour, warmth and sensation every 15-30 minutes , push fluids to clear the contrast, monitor ECG and vital signs, and report bleeding, severe back pain, chest pain or a cold pulseless limb at once
EEG	Wash the hair, no oil, gel or spray ; light meal (do not starve — hypoglycaemia alters the record); avoid tea, coffee and cola; sedatives and antiepileptics withheld only if specifically ordered; sleep deprivation may be requested; explain that it is painless and that no current passes into the body; after — wash the hair
Pulmonary function test / spirometry	No heavy meal or tight clothing; withhold the bronchodilator for 4-6 hours if a reversibility test is planned; no smoking for 24 hours; demonstrate the maximum forced blow; sitting upright with a nose clip
Mantoux (tuberculin) test	0.1 mL of PPD (5 TU) intradermally on the flexor surface of the left forearm, raising a 6-10 mm wheal ; mark the site; read the induration (not the redness) after 48-72 hours; induration of 10 mm or more is positive (5 mm in HIV/immunocompromised); do not scratch, rub or cover the site
Skin allergy testing	Stop antihistamines 3-7 days before; keep adrenaline ready; observe for 30 minutes
Dialysis, radiotherapy and chemotherapy preparation	See Chapters 17 and 20

CHAPTER AT A GLANCE

- Always: consent, explanation, check allergies (iodine/contrast/latex), fasting, position, privacy, specimen labelling, vital signs and documentation.
- Barium meal: NPO 6-8 h, laxative and fluids afterwards, white stools for 1-3 days. Barium enema: full bowel prep.
- IVP/CT/angiography with contrast: check creatinine and allergy, withhold metformin, push fluids afterwards, watch for a contrast reaction.
- Pelvic/obstetric ultrasound: full bladder. Abdominal ultrasound: NPO 6-8 h.
- MRI: absolutely no metal; contraindicated with a pacemaker, cochlear implant or metallic eye foreign body; loud noise, lie still.
- Endoscopy/bronchoscopy: NPO until the gag reflex returns; watch for bleeding and perforation.
- Lumbar puncture: lateral fetal position, L3-L4/L4-L5, lie flat 4-6 h with fluids, headache is the commonest complication; contraindicated in raised ICP.

Test	Preparation and points
	• Thoracentesis: sitting leaning forward; do not remove more than 1-1.5 L; chest X-ray afterwards for pneumothorax. • Paracentesis: empty the bladder first, drain slowly, measure the girth and weight. • Liver biopsy: check clotting, hold breath in expiration, then lie on the RIGHT side for 2 hours; bed rest 6-24 h. • Angiography: bed rest with the limb straight, check the puncture site and distal pulses every 15-30 min. • Mantoux: 0.1 mL PPD intradermal, read the induration at 48-72 h; 10 mm or more is positive.

Child Birth and Its Management

Antenatal care, normal labour and delivery, care of the newborn, postnatal care, complications and family planning

24.1 Basic Obstetric Terms

Table 24.1 Obstetric terminology

Term	Meaning
Gravida / Para	Number of pregnancies (including the present one) / number of deliveries after viability. <i>Primigravida</i> = pregnant for the first time; <i>multigravida</i> ; <i>nullipara</i> = never delivered; <i>grand multipara</i> = 5 or more deliveries
Obstetric formula (GPLA)	Gravida, Para, Living children, Abortions
Antenatal / Intranatal / Postnatal (puerperium)	Before delivery / during labour / after delivery (the puerperium lasts 6 weeks)
Viability / Term / Preterm / Post-term	Viability from 28 weeks (WHO 22 weeks) ; term = 37-42 completed weeks (full term 39-40+6); preterm under 37 weeks; post-term over 42 weeks
Abortion / Still birth / Live birth	Loss of pregnancy before 20-28 weeks (or fetus under 500 g) / birth of a dead fetus after viability / birth showing any sign of life
Lie, presentation, position, attitude, engagement	Relation of the fetal long axis to the uterus / the part occupying the lower pole (cephalic/vertex is normal, about 96% ; breech, shoulder, face, brow) / relation of the denominator to the maternal pelvis / relation of the fetal parts to each other / passage of the widest diameter of the presenting part through the pelvic brim
Quickening / Lightening	First perception of fetal movement (18-20 weeks in a primigravida, 16-18 in a multigravida) / the sensation of relief as the head descends 2-3 weeks before term
Show / Liquor / Vernix / Caput / Moulding	Blood-stained mucus discharge at the onset of labour / amniotic fluid (about 1000 mL at term) / the greasy coat on the newborn's skin / oedematous swelling of the scalp / overlapping of the skull bones during descent
Lochia	Vaginal discharge in the puerperium: lochia rubra (red, days 1-4) , lochia serosa (pinkish-brown, days 5-9) , lochia alba (pale white, days 10-15 up to 3-6 weeks)
Involution	Return of the uterus to the non-pregnant size: the fundus falls about 1-1.25 cm (one finger breadth) per day — at the umbilicus immediately after delivery, midway on day 6, and not palpable per abdomen by day 12-14 ; weight falls from 1000 g to 60-80 g by 6 weeks
Colostrum	The thick yellowish first milk (first 2-4 days) — rich in protein, vitamin A and antibodies (IgA) ; the baby's 'first immunisation'

24.2 Diagnosis of Pregnancy and Calculation of the Due Date

- **Presumptive symptoms** — amenorrhoea, nausea and morning sickness (6-14 weeks), frequency of micturition, breast tenderness and tingling, fatigue, quickening.
- **Probable signs** — enlargement of the abdomen, **Hegar's sign** (softening of the isthmus, 6-10 weeks), **Goodell's sign** (softening of the cervix), **Chadwick's/Jacquemier's sign** (bluish discolouration of the vagina), **Osiander's sign**, **Braxton Hicks contractions**, **Piskacek's sign**, and a **positive urine hCG test**.
- **Positive (certain) signs** — **fetal heart sounds (audible with a stethoscope from 18-20 weeks, by Doppler from 10-12 weeks; rate 110-160/min)**, palpation of fetal parts and movements by the examiner, and **ultrasound visualisation (cardiac activity from 6-7 weeks)**.
- **Naegle's rule** — **expected date of delivery (EDD) = first day of the last menstrual period + 9 calendar months + 7 days** (or - 3 months + 7 days + 1 year); assumes a regular 28-day cycle. *Example: LMP 10 January 2026 → EDD 17 October 2026.*
- **Height of the fundus** — **12 weeks: just palpable above the symphysis • 16 weeks: midway between the symphysis and the umbilicus • 20-22 weeks: at the umbilicus • 28 weeks: 2-3 finger breadths above the umbilicus • 32 weeks: midway between the umbilicus and the xiphisternum • 36 weeks: at the level of the xiphisternum • 40 weeks: falls back to the 32-week level (as the head engages)**. After 24 weeks, the **symphysio-fundal height in cm is approximately equal to the weeks of gestation (± 2 cm)**.
- **Weight gain in pregnancy** — total **10-12 kg** (about 1 kg in the first trimester and then **350-400 g per week**); a gain of over 3 kg in a month or a sudden gain suggests fluid retention/pre-eclampsia.

24.3 Antenatal Care

- **Objectives** — to promote and maintain the health of mother and fetus, to detect and treat complications early ('**high-risk approach**'), to prepare the mother for labour, lactation and child care, to reduce maternal and perinatal mortality, and to motivate for family planning.
- **Schedule of visits** — the Government of India recommends **at least 4 antenatal visits** (1st within 12 weeks/as soon as pregnancy is suspected — registration, 2nd at 14-26 weeks, 3rd at 28-34 weeks, 4th at 36 weeks to term); the WHO 2016 model recommends **8 contacts**. Additionally a monthly visit up to 28 weeks, fortnightly to 36 weeks and weekly thereafter in the traditional schedule; **PMSMA** provides a fixed-day assured check-up on the 9th of every month.
- **At each visit** — **weight, blood pressure, pallor, oedema, urine for protein and sugar, abdominal palpation (fundal height, lie, presentation, fetal heart rate), fetal movements**, and enquiry about complaints and danger signs; counselling on diet, rest, hygiene, breast care, delivery planning and danger signs.
- **Investigations** — haemoglobin, blood group and Rh, blood sugar (**oral glucose tolerance test/DIPSI for gestational diabetes**), urine routine and culture, **HIV, VDRL/syphilis and HBsAg screening**, thyroid profile, ultrasound (dating in the 1st trimester, anomaly scan at 18-20 weeks, growth scan in the 3rd trimester), and malaria/other tests as indicated.
- **Prophylaxis and supplements** — **iron and folic acid: folic acid 400-500 micrograms daily from the pre-conception period through the first trimester (prevents neural tube defects); then 1 tablet of iron-folic acid (60 mg elemental iron + 500 mcg folic acid) daily for at least 180 days in pregnancy and 180 days postpartum (2 tablets daily if anaemic); calcium 500 mg twice daily (total 1 g) for 180 days in pregnancy and 180 days after; albendazole 400 mg once after the first trimester; Td (tetanus-diphtheria) 2 doses 4 weeks apart with a booster if the last dose was within 3 years.**
- **Advice** — a balanced diet with **an extra 350 kcal and 15-23 g of protein daily**, plenty of fluids and fibre; 8 hours of sleep with 2 hours of rest in the day; light exercise and walking; no heavy lifting or strenuous work; bathing daily with breast and perineal hygiene; loose clothing and low-heeled footwear; **no smoking, alcohol, tobacco or self-medication**; dental care; travel avoided in late pregnancy; sexual activity is permitted except with a history of abortion, bleeding or ruptured membranes; teach breast feeding and colostrum.
- **Minor ailments** — morning sickness (small dry frequent meals, avoid fatty food, get up slowly), heartburn (small meals, propped up, antacid), constipation (fibre, fluids, exercise), backache (posture, firm mattress), leg cramps (calcium, stretching), varicose veins and piles (elevation, stockings), oedema of the feet (rest with legs raised — but exclude pre-eclampsia), frequency of urine (exclude urinary infection), white discharge (hygiene; exclude infection).

CAUTION / NEVER DO THIS

Danger signs in pregnancy — the mother must report IMMEDIATELY:

- **Bleeding per vaginum** at any time • **leaking of watery fluid** • foul-smelling discharge
- **Severe headache, blurred vision, spots before the eyes, epigastric pain, convulsions** (pre-eclampsia/eclampsia) • **swelling of the face and hands**
- **Reduced or absent fetal movements** • severe abdominal pain • high fever with chills
- Persistent vomiting • severe breathlessness or palpitation • burning micturition or no urine • jaundice
- Labour pains before 37 weeks (preterm labour).

High-risk pregnancy — identify and refer

- Age **under 18 or over 35** years; height under 145 cm; weight under 45 kg; grand multipara (5 or more); short birth interval.
- **Anaemia (Hb under 11 g/dL; severe under 7), pre-eclampsia/hypertension**, diabetes, heart disease, tuberculosis, HIV, hypothyroidism, epilepsy, jaundice, renal disease, asthma, Rh-negative mother.
- **Malpresentation (breech, transverse), multiple pregnancy, polyhydramnios, antepartum haemorrhage (placenta praevia, abruptio), intrauterine growth restriction, post-term, preterm labour, premature rupture of membranes.**
- **Bad obstetric history** — previous caesarean section, stillbirth, neonatal death, abortion, prolonged or obstructed labour, post-partum haemorrhage, eclampsia, or a baby with a congenital anomaly.

24.4 Labour

DEFINITION

Labour — the series of physiological events by which the products of conception (fetus, placenta and membranes) are expelled from the uterus through the birth canal after 28 weeks of gestation. **Normal (eutocia) labour** is spontaneous in onset at term, with a single fetus in vertex presentation, completed vaginally without complication within 18 hours, and with no harm to mother or baby.

Signs of the onset of labour

- **True labour pains** — regular, rhythmic uterine contractions that **increase in frequency, duration and intensity**, felt from the back going to the front and down the thighs, **not relieved by rest or sedation**, associated with **progressive cervical dilatation and effacement**, and with a 'show'.
- **False labour** — irregular, brief, unpredictable pains felt mostly in the abdomen and groin, **relieved by walking or sedation**, with **no cervical change** (Braxton Hicks contractions).
- Other signs — **lightening, show (blood-stained mucus plug), and sometimes rupture of the membranes** with a gush of clear fluid.

Table 24.2 The stages of labour

Stage	Definition and duration	Nursing / midwifery care
First stage — dilatation	From the onset of true labour to full dilatation of the cervix (10 cm) . Latent phase (0-4 cm) and active phase (4-10 cm). Primigravida about 12 hours (6-18); multigravida about 6-8 hours (cervical dilatation in the active phase at least 1 cm/hour)	Admit and reassure; take the history and examine (vital signs, abdominal palpation, fetal heart rate every 30 minutes in the active phase , contractions every 30 minutes for 10 minutes, vaginal examination 4-hourly under strict asepsis); start the partograph ; encourage ambulation and an upright/left lateral position ; allow light fluids/oral intake as per protocol; encourage voiding every 2 hours (a full bladder delays labour) ; provide breathing and relaxation techniques, back rub and continuous companionship (birth companion); no routine enema, no routine shaving, no fundal pressure; give analgesia as ordered; keep the delivery tray, resuscitation equipment and radiant warmer ready
Second stage — expulsion	From full dilatation to the delivery of the baby . Primigravida up to 1-2 hours (2-3 with epidural); multigravida 30 minutes-1 hour	Fetal heart rate every 5 minutes (or after every contraction) ; the mother bears down only with the contractions once she has the urge; position of choice (dorsal, left lateral or squatting); clean the perineum and drape; support the perineum and control the delivery of the head between contractions to prevent a tear ; feel for a cord round the neck; wipe the baby's face, and clear the airway only if needed; allow restitution and external rotation, then deliver the shoulders (anterior first) and the trunk; note the exact time of birth ; episiotomy only if indicated (not routine) ; place the baby on the mother's abdomen/chest for skin-to-skin contact , dry and cover the head
Third stage — placental	From the birth of the baby to the expulsion of the placenta and membranes . 5-15 minutes (up to 30 minutes)	Active management of the third stage of labour (AMTSL) — (1) uterotonic within 1 minute of birth: oxytocin 10 IU IM (the drug of choice), (2) controlled cord traction with counter-pressure on the uterus (Brandt-Andrews), and (3) uterine massage after delivery of the placenta; delayed cord clamping (1-3 minutes) then cut with a sterile blade; examine the placenta and membranes for completeness and the cord for 3 vessels; measure the blood loss (normal up to 500 mL vaginally); inspect the perineum and repair any tear; check that the uterus is hard and well contracted
Fourth stage — immediate recovery	The first 1-2 hours after delivery — the period of greatest risk of post-partum haemorrhage	Check pulse, BP, uterine tone (fundal height and firmness), amount of bleeding and the pad, and the bladder every 15 minutes for the first hour and every 30 minutes for the second; massage the uterus if soft; encourage voiding; initiate breast feeding within the first hour ; keep the mother and baby together, warm and observed; give fluids and food; record everything

NUMBERS TO REMEMBER

Preparation for a home delivery — the '5 CLEANS' (to prevent sepsis and tetanus): clean hands, clean delivery surface, clean blade (for cutting the cord), clean cord tie, and clean cord stump (nothing applied).

A home delivery kit (Dai kit/DDK) contains: soap, a new blade, sterile cord ties/clamp, clean sheet, gauze, gloves, a mucus extractor, and a plastic sheet — plus a clean warm cloth for the baby, a bowl, and a torch.

Refer for institutional delivery: this is now the norm (**JSY/JSSK** provide cash assistance and free delivery, drugs, diagnostics, diet, blood and transport). Home delivery should be conducted only by a trained person with a clear referral plan.

24.5 Care of the Newborn

1. **Immediate care — the 'warm chain' and airway:** receive the baby in a **warm, clean, dry pre-warmed towel**; **dry thoroughly (including the head) and remove the wet cloth**; keep the baby **skin-to-skin on the mother's chest** and cover both; delay the bath for at least **24 hours (6 hours minimum)**; keep the room warm (25-28 °C) and free of draughts.
2. **Assess breathing and cry immediately**; clear the mouth then the nose gently only if there is obstruction; **if the baby does not breathe or cry within the 'golden minute', start bag-and-mask ventilation with room air/oxygen** (most babies need only drying and stimulation — flicking the soles and rubbing the back).
3. **Apgar score at 1 and 5 minutes** (each of 5 signs scored 0, 1 or 2 — **heart rate, respiratory effort, muscle tone, reflex irritability (grimace) and colour**; maximum 10): **7-10 normal, 4-6 moderate depression, 0-3 severe depression needing active resuscitation**.
4. **Cord care** — clamp/tie and cut with a sterile blade after 1-3 minutes; leave the stump **clean, dry and uncovered**; **apply nothing** (no ash, oil, powder or dung); it falls off in **5-10 days**; report redness, swelling, pus or bleeding (**omphalitis**).
5. **Eye care** — wipe each eye with a separate sterile swab from the inner to the outer canthus; prophylactic antibiotic eye drops/ointment as per policy.
6. **Vitamin K 1 mg intramuscularly** (0.5 mg if under 1 kg) into the antero-lateral thigh to prevent haemorrhagic disease of the newborn.
7. **Identification** — two labels/name bands with the mother's name and details, applied before the baby leaves the delivery area; record the **time of birth, sex, weight, length, head circumference and any anomaly**.
8. **Breast feeding** — initiate **within 1 hour** (skin-to-skin helps); give **colostrum**; exclusive breast feeding on demand; **no water, honey, ghutti, sugar water or formula**; teach correct position and attachment; expect **6-8 wet nappies a day** once the milk is established.
9. **Examination and observation** — check for congenital anomalies (cleft lip and palate, imperforate anus, spina bifida, talipes, undescended testis, hip dislocation), the passage of **urine within 48 hours and meconium within 24 hours**, temperature, colour (jaundice, cyanosis, pallor), respiration (30-60/min), activity and reflexes (rooting, sucking, swallowing, grasp, Moro, stepping, Babinski).
10. **Immunisation at birth** — **BCG, OPV-0 and hepatitis B birth dose** (Chapter 14); record in the immunisation card and register the birth.
11. **Teach the mother the newborn danger signs** (Chapter 21) and about warmth, hygiene, exclusive breast feeding, cord care, immunisation and follow-up (**home visits on days 1, 3, 7, 14, 21, 28 and 42 under HBNC**).

CLINICAL / PRACTICAL POINT

Physiological (normal) findings in a newborn that alarm parents: milia, Mongolian spots, erythema toxicum, vernix, caput succedaneum, swollen breasts and a little milk secretion, **pseudo-menstruation and vaginal discharge in girls**, physiological weight loss up to 10% in the first week, and **physiological jaundice appearing after 24 hours, peaking on days 3-5 and clearing by 10-14 days**.

Abnormal — report at once: **jaundice within the first 24 hours**, or jaundice of the palms and soles, or persisting beyond 2 weeks; cephalhaematoma that is enlarging; no urine in 48 h or no meconium in 24 h; a single umbilical artery; and any of the newborn danger signs.

24.6 Postnatal (Puerperal) Care

- **Observation** — for the first 2 hours: pulse, BP, uterine tone, bleeding and the bladder every 15-30 minutes; then temperature, pulse, respiration and BP **4-6 hourly for 24 hours and twice daily thereafter**; watch the **fundal height (involution), the character and odour of the lochia, the perineum/stitches, the breasts and nipples, the calves, and the bladder and bowel**.
- **Rest, mobility and hygiene** — **early ambulation** (within 6-8 hours of a normal delivery) to prevent thrombosis and to help bladder and bowel function; adequate sleep and rest; daily bath; **perineal care with front-to-back cleaning and a change of pad every 4-6 hours (or sooner)**; sitz bath for perineal discomfort.
- **Diet** — a normal balanced diet with **an extra 600 kcal and 19-25 g of protein** during the first 6 months of lactation, plenty of fluids (2-3 L), iron, folic acid and calcium continued for **180 days**, and no dietary taboos.
- **Breast care and feeding** — wash with plain water (no soap on the nipples), support with a well-fitting brassiere, correct attachment to prevent cracked nipples, feed on demand from both breasts, express milk if the breasts are full, and manage **engorgement (frequent feeding, warm compress before and cold after, expression) and cracked nipple (correct attachment, express a little milk over the nipple, continue feeding)**; **mastitis** (painful

red hot segment with fever) — continue feeding/expressing, warm compresses, analgesia and antibiotics; a **breast abscess** needs drainage.

- **Bladder and bowel** — encourage voiding within 6 hours (watch for retention and for urinary infection); prevent constipation with fluids, fibre and mobility.
- **Postnatal exercises and advice** — deep breathing, leg and abdominal exercises, **pelvic floor (Kegel) exercises**; avoid heavy lifting for 6 weeks; resume sexual activity when comfortable (usually after 6 weeks/after the lochia stops) **with contraception**; postnatal check-up at 6 weeks; register the birth; complete the baby's immunisation.
- **Psychological care** — 'postpartum blues' are common in the first week and transient; watch for **postnatal depression (persistent sadness, crying, inability to care for the baby)** and for **puerperal psychosis (confusion, delusions, risk to self and baby — a psychiatric emergency needing admission and never leaving mother and baby unsupervised)**.
- **Home visits** — the ASHA/ANM visits on **days 1, 3, 7, 14, 21, 28 and 42** under Home-Based Newborn Care to check the mother and the baby and to reinforce teaching.

24.7 Complications of Labour and the Puerperium

Table 24.3 Obstetric emergencies and complications

Complication	Features	Management / nursing action
Post-partum haemorrhage (PPH) — the leading cause of maternal death in India	Blood loss over 500 mL after vaginal delivery (over 1000 mL after caesarean) or any loss causing haemodynamic change; primary (within 24 h; causes — the '4 T's': Tone (atonic uterus — the commonest, about 70-80%), Trauma (tear, rupture), Tissue (retained placenta), Thrombin (coagulopathy)) and secondary (after 24 h to 6 weeks — retained bits, infection)	Call for help; massage/rub up the uterus with one hand and support it bimanually; empty the bladder (catheterise); give oxytocin 10 IU IM/IV infusion (or misoprostol, methylergometrine, carboprost as ordered); two large IV lines with rapid crystalloid; oxygen; keep the patient flat and warm; group and cross-match and arrange blood; monitor pulse, BP and blood loss continuously (weigh the pads); explore for tears and retained placenta; bimanual compression, aortic compression, uterine balloon tamponade or surgery if it continues; strict records
Pre-eclampsia and eclampsia	Pre-eclampsia: BP 140/90 or more after 20 weeks with proteinuria ($\pm$ headache, blurred vision, epigastric pain, oedema, hyper-reflexia). Eclampsia: convulsions	Regular BP and urine protein checks at every visit; a quiet, darkened room with padded side rails, nothing by mouth, airway and suction ready, lateral position during a fit, never restrain, never leave alone; magnesium sulphate as the drug of choice (monitor the respiratory rate above 12-16/min, urine output above 30 mL/h and the patellar/knee reflex before every dose; keep calcium gluconate 10% as the antidote); antihypertensives (labetalol, nifedipine, hydralazine); strict intake-output; prepare for delivery (the definitive treatment); continue observation for 48 hours after delivery (fits can occur postpartum)
Obstructed and prolonged labour	Labour over 18-24 hours with no progress, a Bandl's ring, a tense tender uterus, a distended bladder, maternal exhaustion and dehydration, and fetal distress	Recognise early with the partograph and refer immediately; do not give oxytocin at a peripheral centre; IV fluids, catheterisation, monitor the fetal heart, prepare for caesarean section; watch for rupture of the uterus, fistula, sepsis and PPH
Antepartum haemorrhage	Placenta praevia — painless, causeless, recurrent bright red bleeding with a soft uterus; abruptio placentae — painful bleeding with a tense tender uterus and fetal distress	Never do a vaginal examination at home or without ultrasound localisation in suspected placenta praevia; absolute bed rest, IV line, group and cross-match, monitor the mother and the fetal heart, and transfer urgently to a facility with blood and operating facilities
Puerperal sepsis	Fever of 38 °C or more on any 2 of the first 10 days after delivery (excluding the first 24 h), with offensive lochia, lower abdominal pain, a subinvolted tender uterus, and a discharging perineal wound	Prevented by the 5 cleans, aseptic technique, avoiding unnecessary vaginal examinations, treating anaemia, and early management of prolonged rupture of membranes; isolate, take swabs and blood cultures, give antibiotics, fluids and analgesia, continue breast feeding if possible, encourage the Fowler's position for drainage, and observe for septic shock and thrombophlebitis
Preterm labour and premature rupture of membranes	Contractions before 37 weeks / leaking of fluid before labour	Bed rest, sterile pad and no vaginal examination or douching, note the colour and odour of the fluid, monitor the temperature and the fetal heart, tocolytics, antenatal corticosteroids (dexamethasone/betamethasone) to mature the fetal lungs and antibiotics as ordered; refer where newborn care is available

Complication	Features	Management / nursing action
Other	Cord prolapse (knee-chest or exaggerated Sims' position, keep the cord moist and warm, do not handle it, give oxygen and transfer immediately for caesarean); breech and malpresentation; twin delivery; shoulder dystocia; uterine inversion; amniotic fluid embolism; deep vein thrombosis; retention of urine; abortion and ectopic pregnancy	Recognise, give first aid, and refer urgently — the key nursing skill is early recognition and prompt referral with a written note, an IV line running and an escort

24.8 Family Planning (Contraception)

Table 24.4 Methods of contraception

Method	Description	Points
Natural / fertility awareness	Calendar (rhythm), basal body temperature, cervical mucus (Billings), symptothermal, standard-days method with CycleBeads	No cost or side effects but a high failure rate; needs regular cycles and co-operation; the fertile period is about days 8-19 of a 28-day cycle (ovulation on day 14 ± 2)
Lactational amenorrhoea method (LAM)	Protective only if all three apply: the baby is under 6 months, is exclusively breast fed day and night , and the mother has not resumed menstruation	About 98% effective when all three conditions hold
Barrier	Condom (Nirodh) , female condom, diaphragm, cervical cap, spermicide	The condom is the only method that also protects against sexually transmitted infections and HIV ; use a new one each time, check the expiry, store away from heat, and hold the rim while withdrawing
Intrauterine device (IUD)	Copper T 380A (effective 10 years), Cu 375 (5 years), post-partum IUCD, and the hormonal LNG-IUS (Mirena, 5 years)	Long acting, reversible, highly effective; inserted preferably during menstruation, immediately post-partum (within 48 h) or 6 weeks after delivery; teach the patient to check the thread and to report the ' PAINS ' warning signs — Period late/abnormal bleeding, Abdominal pain or pain with intercourse, Infection/abnormal discharge, Not feeling well or fever and chills, String missing or longer/shorter ; side effects — heavier periods and cramps; contraindicated in pregnancy, sepsis, unexplained bleeding, distorted uterus and active pelvic infection
Hormonal — combined oral pills (Mala-N/Mala-D)	21 hormone tablets + 7 iron tablets ; start on the 5th day of the cycle and take one tablet daily at the same time	Highly effective if taken regularly; if one pill is missed, take it as soon as remembered and the next at the usual time; if 2 or more are missed, take one daily and use a condom for 7 days ; <i>contraindicated in pregnancy, breast feeding (under 6 months), over 35 with smoking, hypertension, heart disease, thrombosis, migraine with aura, liver disease and breast cancer</i> ; benefits — regular lighter periods, less dysmenorrhoea, protection against ovarian and endometrial cancer
Progestogen-only	Mini-pill, injectable DMPA (Antara — every 3 months), Chhaya (centchroman/ormeloxifene — a non-hormonal weekly pill: twice weekly for 12 weeks then once weekly), implants	Safe in lactation; irregular spotting or amenorrhoea is common and is not harmful — counsel beforehand; return of fertility may be delayed after DMPA
Emergency contraception	Levonorgestrel 1.5 mg as a single dose, as early as possible and within 72 hours (or a copper IUD within 5 days)	Not for regular use; does not protect against infection; may cause nausea and a change in the next period; <i>it is not an abortion pill</i>
Permanent (sterilisation)	Female: minilap or laparoscopic tubectomy (also post-partum); Male: vasectomy — simpler, safer and quicker (NSV — no-scalpel vasectomy)	Informed written consent, eligibility as per the standards (age, number of living children), and counselling that it is permanent ; after vasectomy, another method must be used for 3 months or 20 ejaculations (sperm are still present); a post-operative check and care of the site; the couple must be told it is not reversible

- **Eligible couple** — a married couple in which the wife is in the reproductive age group (**15-45 years**); a **couple protection rate** is the percentage of eligible couples effectively protected by any method.
- **Legal points** — **Medical Termination of Pregnancy Act, 1971 (amended 2021)**: termination up to **20 weeks on the opinion of one registered medical practitioner, 20-24 weeks for specified categories on the opinion of two**, and beyond 24 weeks only with a Medical Board's approval for substantial fetal abnormality; only the **woman's own consent** is needed (guardian's if she is under 18 or mentally ill); the provider's and the woman's identity are confidential. The **PCPNDT Act, 1994** prohibits **prenatal sex determination and disclosure of the sex of the fetus** — a punishable offence; every ultrasound centre must display the notice and maintain Form F records.
- **Schemes** — **Janani Suraksha Yojana (JSY)** — cash assistance for institutional delivery; **Janani Shishu Suraksha Karyakram (JSSK)** — free delivery, caesarean section, drugs, diagnostics, diet, blood and transport for the mother and the sick newborn; **PMSMA** — assured antenatal check-up on the 9th of each month; **SUMAN, LaQshya, Mission Parivar Vikas, Pradhan Mantri Matru Vandana Yojana** (maternity benefit).

CHAPTER AT A GLANCE

- EDD by Naegele's rule = LMP + 9 months + 7 days. Fundus: at the umbilicus at 20-22 weeks, xiphisternum at 36 weeks; SFH in cm = weeks after 24 weeks.
- Antenatal: at least 4 visits (WHO 8 contacts); IFA and calcium for 180 days + 180 days; albendazole after the first trimester; 2 doses of Td 4 weeks apart; extra 350 kcal and 15-23 g protein; weight gain 10-12 kg.
- Danger signs: bleeding, leaking, severe headache/blurred vision, convulsions, reduced fetal movements, fever, severe pain.
- Stages of labour: I dilatation (primi about 12 h, multi 6-8 h, active phase 1 cm/h); II expulsion (primi 1-2 h, multi 30-60 min, FHR every 5 min); III placental (5-15 min, AMTSL with oxytocin 10 IU IM within 1 minute); IV first 1-2 hours — watch for PPH.
- 5 cleans: clean hands, surface, blade, cord tie, cord stump. Normal blood loss up to 500 mL.
- Newborn: dry and keep warm, skin-to-skin, Apgar at 1 and 5 min (7-10 normal), delayed cord clamping, nothing on the cord, vitamin K 1 mg IM, breast feed within 1 hour, BCG + OPV-0 + hepatitis B at birth; bath after 24 hours.
- Physiological jaundice appears after 24 hours and clears by 10-14 days; jaundice in the first 24 hours is always abnormal.
- Lochia rubra 1-4 days, serosa 5-9, alba 10-15; fundus falls about 1 cm/day and is impalpable by day 12-14; puerperium = 6 weeks.
- PPH is the leading cause of maternal death — the 4 T's (Tone is commonest); massage the uterus, empty the bladder, oxytocin, IV fluids, arrange blood.
- Eclampsia: magnesium sulphate — check the respiratory rate, urine output and knee reflex; antidote calcium gluconate.
- Copper T 380A lasts 10 years; combined pill started on day 5; emergency contraception within 72 hours; vasectomy needs another method for 3 months.
- MTP Act 1971/2021 (20 weeks — one doctor; 20-24 weeks — two; woman's consent only); PCPNDT Act 1994 bans sex determination.

Rapid Revision — The Last-Night Sheet

The most frequently asked facts, figures and one-liners from the whole syllabus

A.1 Vital Signs and Normal Values

Table A.1 Numbers you must not forget

Parameter	Normal value
Temperature (oral / axillary / rectal)	37 °C (98.6 °F) / 0.5 °C less / 0.5 °C more; hyperpyrexia above 41 °C, hypothermia below 35 °C
Pulse — adult / newborn	60-100 (average 72) / 120-160 per minute
Respiration — adult / newborn	12-20 / 30-60 per minute; pulse : respiration = 4-5 : 1
Blood pressure / SpO ₂	Under 120/80 mmHg; pulse pressure 30-40; MAP 70-105 / SpO ₂ 95-100%
Urine output	1-2 mL/kg/h (30-60 mL/h; 1500 mL/day); oliguria under 400 mL/day
Fluid requirement	2-3 L/day (30-35 mL/kg)
Blood volume / Cardiac output	5-6 L (70-80 mL/kg) / 4-6 L per minute
Haemoglobin (men / women / pregnancy)	13-17 / 12-15 g/dL; anaemia in pregnancy below 11 g/dL
WBC / Platelets	4000-11 000 /mm ³ / 1.5-4.5 lakh /mm ³
Fasting blood sugar / HbA _{1c}	70-100 mg/dL (diabetes 126+) / under 5.7% (diabetes 6.5%+)
Sodium / Potassium	135-145 / 3.5-5.0 mEq/L
Urea / Creatinine / Bilirubin	15-40 / 0.6-1.2 / 0.3-1.2 mg/dL
GCS	3 (worst) to 15; 8 or less = coma

A.2 Timings and Intervals

- **Turn the patient** every 2 hours • **back care** 2-4 hourly • **mouth care of the unconscious** 2 hourly.
- **Hand wash** 20-40 seconds (alcohol rub 20-30 s) • **surgical scrub** 3-5 minutes.
- **Thermometer**: oral 2-3 min, axillary 3-5 min, rectal 1-2 min.
- **Autoclave** 121 °C/15 lb/15-20 min • **hot air oven** 160 °C/2 h • **boiling** 20-30 min.
- **Suction** 10-15 seconds per pass • **hot water bag** 50-60 °C (adult) • **sitz bath** 40-43 °C for 15-20 min.
- **Enema** 500-1000 mL at 40.5-43 °C, can 45-50 cm high, tube 7.5-10 cm, retain 5-15 min.
- **Flatus tube** 20-30 min • **ice bag / hot application** 20-30 min • **mustard plaster** 15-20 min.
- **Transfusion**: start within 30 min of issue, slow for the first 15 min, complete within 4 hours.
- **Fasting before anaesthesia**: clear fluids 2 h, breast milk 4 h, light meal 6 h, heavy meal 8 h.
- **PEP for HIV**: within 2 hours (max 72 h), for 28 days • **HBIG** within 24 hours.
- **CPR**: 30:2, 100-120 compressions/min, 5-6 cm deep, change compressor every 2 min, adrenaline every 3-5 min.
- **Mantoux** read at 48-72 hours; 10 mm induration = positive.
- **Suture removal**: face 3-5 days, trunk 7-10 days, over joints/leg 10-14 days.

A.3 'Firsts', 'Commonests' and 'Bests'

Table A.2 One-line answers that repeat every year

Question	Answer
Single most important measure to prevent infection	Hand hygiene
Commonest site of pressure sore / in a sitting patient	Sacrum / ischial tuberosity
Commonest hospital-acquired infection	Catheter-associated urinary tract infection
Most reliable and commonest method of hospital sterilisation	Autoclaving
Most heat-sensitive vaccine / most freeze-sensitive	OPV / hepatitis B, DPT, TT (adsorbed vaccines)
Safest and commonest route of drug administration / fastest	Oral / intravenous
Safest IM site in an adult / in an infant	Ventrogluteal / vastus lateralis
Commonest cause of a fatal transfusion reaction	Clerical error causing ABO incompatibility

Question	Answer
Commonest transfusion reaction	Febrile non-haemolytic reaction
Universal donor of red cells / of plasma	O negative / AB
Commonest type of shock	Hypovolaemic
Leading cause of maternal death in India	Post-partum haemorrhage
Commonest cause of PPH	Atonic uterus (Tone)
Commonest cause of delayed wound healing	Infection
Commonest early post-operative fever (day 1-2)	Atelectasis (Wind)
Commonest psychosis / commonest cause of dementia	Schizophrenia / Alzheimer's disease
Strongest predictor of suicide	A previous attempt
First antibody produced in infection / crosses the placenta	IgM / IgG
Reference protein with the highest biological value	Egg (100)
Commonest nutritional deficiency in the world	Iron deficiency anaemia
Best position for an unconscious patient / for dyspnoea	Lateral (recovery) / Fowler's or orthopnoeic
Best method to confirm a nasogastric tube position at the bedside	pH of the aspirate (5.5 or less); X-ray is the gold standard
Drug of choice in anaphylaxis / in eclampsia / in cholera	Adrenaline IM / magnesium sulphate / ORS and fluids
Safest drug distribution system in a hospital	Unit dose dispensing
Most strictly controlled drug schedule	Schedule X

A.4 Important Days and Landmarks

Table A.3 Health days

Day	Date	Day	Date
World Health Day	7 April	International Nurses Day	12 May
World Red Cross Day	8 May	World No Tobacco Day	31 May
World Blood Donor Day	14 June	World Population Day	11 July
World Breastfeeding Week	1-7 August	National Nutrition Week	1-7 September
World Pharmacists Day	25 September	World Heart Day	29 September
International Day of Older Persons	1 October	World Mental Health Day	10 October
Global Handwashing Day	15 October	World Diabetes Day	14 November
World AIDS Day	1 December	World Tuberculosis Day	24 March
World Cancer Day	4 February	World Kidney Day	2nd Thursday of March

- **Florence Nightingale** 1820-1910, Crimean War 1854-56, first nursing school 1860 • **Red Cross** (Henri Dunant) 1863, **Indian Red Cross** 1920 • **ICN** 1899 • **TNAI** 1908 • **INC Act** 1947 (constituted 1949) • **WHO** 1948 • **Pharmacy Act** 1948 • **Drugs and Cosmetics Act** 1940/Rules 1945 • **NDPS Act** 1985 • **UIP** 1985 • **NMHP** 1982 • **NRHM** 2005 • **Mental Healthcare Act** 2017 • **Ayushman Bharat** 2018 • **BMW Rules** 2016.
- **Smallpox eradicated** 1980 (last case in India 1975) • **India polio-free** 27 March 2014 • **maternal and neonatal tetanus eliminated** 2015.

A.5 Colour Codes and Mnemonics

- **Bio-medical waste** — **Yellow**: anatomical, soiled, expired drugs, laboratory waste (incinerate) • **Red**: recyclable plastics (autoclave and shred) • **White**: sharps • **Blue**: glass and metal implants • **Black**: general waste.
- **Gas cylinders** — **oxygen: black with a white shoulder** • nitrous oxide: blue • carbon dioxide: grey • entonox: blue with blue-and-white quarters.
- **ECG limb leads** — RA red, LA yellow, LL green, RL black ('Ride Your Green Bike').
- **Triage** — red immediate, yellow urgent, green minor, black dead/expectant.
- **ADPIE** — nursing process • **ABC** — airway, breathing, circulation • **RICE** — rest, ice, compression, elevation • **FAST** — face, arm, speech, time (stroke) • **SBAR** — situation, background, assessment, recommendation •

PQRST — pain assessment • **5 W's** — post-operative fever • **4 T's** — causes of PPH • **5 P's** — pain, pallor, paraesthesia, pulselessness, paralysis • **5 CLEANS** — home delivery • **3 D's** — pellagra (dermatitis, diarrhoea, dementia) • **5 moments** — hand hygiene • **PAINS** — IUD warning signs • **CAUTION** — cancer warning signs.

CHAPTER AT A GLANCE

Read this appendix the night before the examination, then re-read the '**Chapter at a Glance**' box at the end of every chapter. Those boxes together contain nearly every fact that has ever been asked from the Home Nursing section.

Pay special attention to: **numbers and timings, positions, colour codes, drug schedules, immunisation, the cold chain, sterilisation temperatures, transfusion rules, obstetric figures, and the 'commonest/best/first' one-liners.**